

FIGURE 6

The hypothesis that $\{a_n\}$ is bounded above is clearly essential in Theorem 2: if $\{a_n\}$ is not bounded above, then (whether or not $\{a_n\}$ is nondecreasing) $\{a_n\}$ clearly diverges. Upon first consideration, it might appear that there should be little trouble deciding whether or not a given nondecreasing sequence $\{a_n\}$ is bounded above, and consequently whether or not $\{a_n\}$ converges. In the next chapter such sequences will arise very naturally and, as we shall see, deciding whether or not they converge is hardly a trivial matter. For the present, you might try to decide whether or not the following (obviously increasing) sequence is bounded above:

$$1, 1 + \frac{1}{2}, 1 + \frac{1}{2} + \frac{1}{3}, 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, \dots$$

Although Theorem 2 treats only a very special class of sequences, it is more useful than might appear at first, because it is always possible to extract from an arbitrary sequence $\{a_n\}$ another sequence which is either nonincreasing or else nondecreasing. To be precise, let us define a **subsequence** of the sequence $\{a_n\}$ to be a sequence of the form

$$a_{n_1}, a_{n_2}, a_{n_3}, \dots,$$

where the n_j are natural numbers with

$$n_1 < n_2 < n_3 < \dots$$

Then every sequence contains a subsequence which is either nondecreasing or nonincreasing. It is possible to become quite befuddled trying to prove this assertion, although the proof is very short if you think of the right idea; it is worth recording as a lemma.

LEMMA Any sequence $\{a_n\}$ contains a subsequence which is either nondecreasing or nonincreasing.

PROOF Call a natural number n a “peak point” of the sequence $\{a_n\}$ if $a_m < a_n$ for all $m > n$ (Figure 6).

Case 1. The sequence has infinitely many peak points. In this case, if $n_1 < n_2 < n_3 < \dots$ are the peak points, then $a_{n_1} > a_{n_2} > a_{n_3} > \dots$, so $\{a_{n_k}\}$ is the desired (nonincreasing) subsequence.

Case 2. The sequence has only finitely many peak points. In this case, let n_1 be greater than all peak points. Since n_1 is not a peak point, there is some $n_2 > n_1$ such that $a_{n_2} \geq a_{n_1}$. Since n_2 is not a peak point (it is greater than n_1 , and hence greater than all peak points) there is some $n_3 > n_2$ such that $a_{n_3} \geq a_{n_2}$. Continuing in this way we obtain the desired (nondecreasing) subsequence. ■

If we assume that our original sequence $\{a_n\}$ is bounded, we can pick up an extra corollary along the way.

COROLLARY (THE BOLZANO-WEIERSTRASS THEOREM)

Every bounded sequence has a convergent subsequence.

Without some additional assumptions this is as far as we can go: it is easy to construct sequences having many, even infinitely many, subsequences converging to different numbers (see Problem 2). There is a reasonable assumption to add, which yields a necessary and sufficient condition for convergence of any sequence. Although this condition will not be crucial for our work, it does simplify many proofs. Moreover, this condition plays a fundamental role in more advanced investigations, and for this reason alone it is worth stating now.

If a sequence converges, so that the individual terms are eventually all close to the same number, then the difference of any two such individual terms should be very small. To be precise, if $\lim_{n \rightarrow \infty} a_n = l$ for some l , then for any $\varepsilon > 0$ there is an N such that $|a_n - l| < \varepsilon/2$ for $n > N$; now if both $n > N$ and $m > N$, then

$$|a_n - a_m| \leq |a_n - l| + |l - a_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This final inequality, $|a_n - a_m| < \varepsilon$, which eliminates mention of the limit l , can be used to formulate a condition (the Cauchy condition) which is clearly necessary for convergence of a sequence.

DEFINITION

A sequence $\{a_n\}$ is a **Cauchy sequence** if for every $\varepsilon > 0$ there is a natural number N such that, for all m and n ,

$$\text{if } m, n > N, \text{ then } |a_n - a_m| < \varepsilon.$$

(This condition is usually written $\lim_{m, n \rightarrow \infty} |a_m - a_n| = 0$.)

The beauty of the Cauchy condition is that it is also sufficient to ensure convergence of a sequence. After all our preliminary work, there is very little left to do in order to prove this.

THEOREM 3

A sequence $\{a_n\}$ converges if and only if it is a Cauchy sequence.

PROOF

We have already shown that $\{a_n\}$ is a Cauchy sequence if it converges. The proof of the converse assertion contains only one tricky feature: showing that every Cauchy sequence $\{a_n\}$ is bounded. If we take $\varepsilon = 1$ in the definition of a Cauchy sequence we find that there is some N such that

$$|a_m - a_n| < 1 \quad \text{for } m, n > N.$$

In particular, this means that

$$|a_m - a_{N+1}| < 1 \quad \text{for all } m > N.$$

Thus $\{a_m: m > N\}$ is bounded; since there are only finitely many other a_i 's the whole sequence is bounded.

The corollary to the Lemma thus implies that some subsequence of $\{a_n\}$ converges.

Only one point remains, whose proof will be left to you: if a subsequence of a Cauchy sequence converges, then the Cauchy sequence itself converges. ■

PROBLEMS

1. Verify each of the following limits.

$$(i) \quad \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

$$(ii) \quad \lim_{n \rightarrow \infty} \frac{n+3}{n^3+4} = 0.$$

$$(iii) \quad \lim_{n \rightarrow \infty} \sqrt[8]{n^2+1} - \sqrt[4]{n+1} = 0. \text{ Hint: You should at least be able to prove that } \lim_{n \rightarrow \infty} \sqrt[8]{n^2+1} - \sqrt[8]{n^2} = 0.$$

$$(iv) \quad \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0. \text{ Hint: } n! = n(n-1) \cdots k! \text{ for } k < n, \text{ in particular, for } k < n/2.$$

$$(v) \quad \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1, \quad a > 0.$$

$$(vi) \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$

$$(vii) \quad \lim_{n \rightarrow \infty} \sqrt[n]{n^2+n} = 1.$$

$$(viii) \quad \lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n} = \max(a, b).$$

$$(ix) \quad \lim_{n \rightarrow \infty} \frac{\alpha(n)}{n} = 0, \text{ where } \alpha(n) \text{ is the number of primes which divide } n. \\ \text{Hint: The fact that each prime is } \geq 2 \text{ gives a very simple estimate of how small } \alpha(n) \text{ must be.}$$

$$*(x) \quad \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k^p}{n^{p+1}} = \frac{1}{p+1}.$$

2. (a) What can be said about the sequence $\{a_n\}$ if it converges and each a_n is an integer?
- (b) Find all convergent subsequences of the sequence $1, -1, 1, -1, 1, -1, \dots$ (There are infinitely many, although there are only two limits which such subsequences can have.)
- (c) Find all convergent subsequences of the sequence $1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, \dots$ (There are infinitely many limits which such subsequences can have.)

(d) Consider the sequence

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{6}, \dots$$

For which numbers α is there a subsequence converging to α ?

3. (a) Prove that if a subsequence of a Cauchy sequence converges, then so does the original Cauchy sequence.
- (b) Prove that any subsequence of a convergent sequence converges.
4. (a) Prove that if $0 < a < 2$, then $a < \sqrt{2a} < 2$.
- (b) Prove that the sequence

$$\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots$$

converges.

- (c) Find the limit. Hint: Notice that if $\lim_{n \rightarrow \infty} a_n = l$, then $\lim_{n \rightarrow \infty} \sqrt{2a_n} = \sqrt{2l}$, by Theorem 1.

5. Identify the function $f(x) = \lim_{n \rightarrow \infty} (\lim_{k \rightarrow \infty} (\cos n! \pi x)^{2k})$. (It has been mentioned many times in this book.)
6. Many impressive looking limits can be evaluated easily (especially by the person who makes them up), because they are really lower or upper sums in disguise. With this remark as hint, evaluate each of the following. (Warning: the list contains one red herring which can be evaluated by elementary considerations.)

$$(i) \lim_{n \rightarrow \infty} \frac{\sqrt[n]{e} + \sqrt[n]{e^2} + \dots + \sqrt[n]{e^n}}{n}$$

$$(ii) \lim_{n \rightarrow \infty} \frac{\sqrt[n]{e} + \sqrt[n]{e^2} + \dots + \sqrt[n]{e^{2n}}}{n}$$

$$(iii) \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \dots + \frac{1}{2n} \right)$$

$$(iv) \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \dots + \frac{1}{(2n)^2} \right)$$

$$(v) \lim_{n \rightarrow \infty} \left(\frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(n+n)^2} \right)$$

$$(vi) \lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+2^2} + \dots + \frac{n}{n^2+n^2} \right)$$

7. Although limits like $\lim_{n \rightarrow \infty} \sqrt[n]{n}$ and $\lim_{n \rightarrow \infty} a^n$ can be evaluated using facts about the behavior of the logarithm and exponential functions, this approach is vaguely dissatisfying, because integral roots and powers can be defined without using the exponential function. Some of the standard “elementary” arguments for such limits are outlined here; the basic tools are inequalities derived from the binomial theorem, notably

$$(1+h)^n \geq 1 + nh, \text{ for } h > 0;$$

and, for part (c),

$$(1+h)^n \geq 1 + nh + \frac{n(n-1)}{2} h^2 \geq \frac{n(n-1)}{2} h^2, \quad \text{for } h > 0.$$

- (a) Prove that $\lim_{n \rightarrow \infty} a^n = \infty$ if $a > 1$, by setting $a = 1 + h$, where $h > 0$.
- (b) Prove that $\lim_{n \rightarrow \infty} a^n = 0$ if $0 < a < 1$.
- (c) Prove that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$ if $a > 1$, by setting $\sqrt[n]{a} = 1 + h$ and estimating h .
- (d) Prove that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$ if $0 < a < 1$.
- (e) Prove that $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.
8. (a) Prove that a convergent sequence is always bounded.
- (b) Suppose that $\lim_{n \rightarrow \infty} a_n = 0$, and that each $a_n > 0$. Prove that the set of all numbers a_n actually has a maximum member.
9. (a) Prove that

$$\frac{1}{n+1} < \log(n+1) - \log n < \frac{1}{n}.$$

(b) If

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n,$$

show that the sequence $\{a_n\}$ is decreasing, and that each $a_n \geq 0$. It follows that there is a number

$$\gamma = \lim_{n \rightarrow \infty} \left(1 + \cdots + \frac{1}{n} - \log n \right)$$

This number, known as Euler's number, has proved to be quite refractory; it is not even known whether γ is rational.

10. (a) Suppose that f is increasing on $[1, \infty)$. Show that

$$f(1) + \cdots + f(n-1) < \int_1^n f(x) dx < f(2) + \cdots + f(n).$$

(b) Now choose $f = \log$ and show that

$$\frac{n^n}{e^{n-1}} < n! < \frac{(n+1)^{n+1}}{e^n} < \frac{n^{n+1}}{e^{n-1}};$$

it follows that

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}.$$

This result shows that $\sqrt[n]{n!}$ is approximately n/e , in the sense that the ratio of these two quantities is close to 1 for large n . But we cannot conclude that $n!$ is close to $(n/e)^n$ in this sense; in fact, this is false. An estimate for $n!$ is very desirable, even for concrete computations, because $n!$ cannot be calculated easily even with logarithm tables. The standard (and difficult) theorem which provides the right information will be found in Problem 26-18.

- *11. This problem investigates for which $x > 0$ the symbol

$$x^{x^{x^{\cdots}}}$$

makes sense. In other words, if we define $a_1(x) = x$, $a_{n+1}(x) = x^{a_n(x)}$, when does $b(x) = \lim_{n \rightarrow \infty} a_n(x)$ exist?

- Prove that if $b(x)$ exists, then $x^{b(x)} = b(x)$.
(The situation is similar to that in Problem 4.)
- According to part (a), if $b(x)$ exists, then x can be written in the form $y^{1/y}$ for some y . Conclude that $0 < x \leq e^{1/e}$. Hint: Consider the graph of $f(y) = (\log y)/y$, as in Problem 17-24.
- Suppose, conversely, that $0 < x \leq e^{1/e}$. Show that each $a_n(x) \leq e$; since $\{a_n(x)\}$ is clearly increasing, this proves that $b(x)$ exists (and also that $b(x) \leq e$).
- Find $b(\sqrt{2})$ and $b(e^{1/e})$.
- Prove that b is differentiable on $(0, e^{1/e})$, and find a formula for $b'(x)$ in terms of $b(x)$. Hint: This part is similar to Problem 17-24(f).
- Derive a formula for $b'(x)$ by differentiating the equation $x^{b(x)} = b(x)$. (Of course, this method does not prove that b is differentiable.)

- *12. Prove that if $\lim_{n \rightarrow \infty} a_n = l$, then

$$\lim_{n \rightarrow \infty} \frac{a_1 + \cdots + a_n}{n} = l.$$

Hint: This problem is very similar to (in fact it is a special case of) Problem 13-34.

- *13. Suppose that $a_n \neq 0$ for each n and that $\lim_{n \rightarrow \infty} a_{n+1}/a_n = l$. Prove that

$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = l$. Hint: This requires the same sort of argument that works in Problem 12, together with the fact that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$, for $a > 0$.

- Suppose that $\{a_n\}$ is a convergent sequence of points all in $[0, 1]$. Prove that $\lim_{n \rightarrow \infty} a_n$ is also in $[0, 1]$.
- Find a convergent sequence $\{a_n\}$ of points all in $(0, 1)$ such that $\lim_{n \rightarrow \infty} a_n$ is not in $(0, 1)$.

15. Suppose that f is continuous and that the sequence

$$x, f(x), f(f(x)), f(f(f(x))), \dots$$

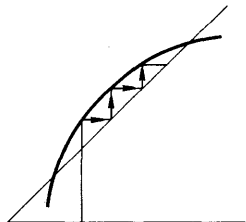


FIGURE 7

converges to l . Prove that l is a “fixed point” for f , i.e., $f(l) = l$. Hint: Two special cases have occurred already.

16. (a) Suppose that f is continuous on $[0, 1]$ and that $0 \leq f(x) \leq 1$ for all x in $[0, 1]$. Problem 7-11 shows that f has a fixed point (in the terminology of Problem 15). If f is *increasing*, a much stronger statement can be made: For any x in $[0, 1]$, the sequence

$$x, f(x), f(f(x)), \dots$$

has a limit (which is necessarily a fixed point, by Problem 15). Prove this assertion, by examining the behavior of the sequence for $f(x) > x$ and $f(x) < x$, or by looking at Figure 7. A diagram of this sort is used in Littlewood’s *Mathematician’s Miscellany* to preach the value of drawing pictures: “For the professional the only proof needed is [this Figure].”

- *(b) Suppose that f and g are two continuous functions on $[0, 1]$, with $0 \leq f(x) \leq 1$ and $0 \leq g(x) \leq 1$ for all x in $[0, 1]$, which satisfy $f \circ g = g \circ f$. Suppose, moreover, that f is increasing. Show that f and g have a common fixed point; in other words, there is a number l such that $f(l) = l = g(l)$. Hint: Begin by choosing a fixed point for g .
- *** (c) Does the conclusion of part (b) hold without the assumption that f is increasing?

The trick in Problem 15 is really much more valuable than Problem 15 might suggest, and some of the most important “fixed point theorems” depend upon looking at sequences of the form from $x, f(x), f(f(x)), \dots$. A special, but representative, case of one such theorem is treated in Problem 18 (for which the next problem is preparation).

17. (a) Use Problem 2-4 to show that if $c \neq 1$, then

$$c^m + c^{m+1} + \dots + c^n = \frac{c^m - c^{n+1}}{1 - c}.$$

- (b) Suppose that $|c| < 1$. Prove that

$$\lim_{m, n \rightarrow \infty} c^m + \dots + c^n = 0.$$

- (c) Suppose that $\{x_n\}$ is a sequence with $|x_n - x_{n+1}| \leq c^n$, where $c < 1$. Prove that $\{x_n\}$ is a Cauchy sequence.
- *18. Suppose that f is a function on $[a, b]$ such that

$$(*) \quad |f(x) - f(y)| \leq c|x - y|, \quad \text{for all } x \text{ and } y \text{ in } [a, b],$$

where $c < 1$. (Such a function is called a *contraction*.)

- (a) Prove that f is continuous.
- (b) Prove that f has at most one fixed point.

(c) By considering the sequence

$$x, f(x), f(f(x)), \dots,$$

for any x in $[a, b]$, prove that f does have a fixed point. (This result, in a more general setting, is known as the "contraction lemma.")

19. Let $\{x_n\}$ be a sequence which is bounded, and let

$$y_n = \sup \{x_n, x_{n+1}, x_{n+2}, \dots\}.$$

(a) Prove that the sequence $\{y_n\}$ converges. The limit $\lim_{n \rightarrow \infty} y_n$ is denoted by $\overline{\lim}_{n \rightarrow \infty} x_n$ or $\limsup_{n \rightarrow \infty} x_n$, and called the **limit superior**, or **upper limit**, of the sequence $\{x_n\}$.

(b) Find $\overline{\lim}_{n \rightarrow \infty} x_n$ for each of the following:

(i) $x_n = \frac{1}{n}$.

(ii) $x_n = (-1)^n \frac{1}{n}$.

(iii) $x_n = (-1)^n \left[1 + \frac{1}{n}\right]$.

(iv) $x_n = \sqrt[n]{n}$.

(c) Define $\lim_{n \rightarrow \infty} x_n$ (or $\liminf_{n \rightarrow \infty} x_n$) and prove that

$$\lim_{n \rightarrow \infty} x_n \leq \overline{\lim}_{n \rightarrow \infty} x_n.$$

(d) Prove that $\lim_{n \rightarrow \infty} x_n$ exists if and only if $\overline{\lim}_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n$ and that in this case $\lim_{n \rightarrow \infty} x_n = \overline{\lim}_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x_n$.

(e) Recall the definition, in Problem 8-18, of $\overline{\lim} A$ for a bounded set A . Prove that if the numbers x_n are distinct, then $\overline{\lim}_{n \rightarrow \infty} x_n = \overline{\lim} A$, where $A = \{x_n: n \text{ in } \mathbf{N}\}$.

*20. (a) Suppose that $\{b_n\}$ is nonincreasing, with $b_n \geq 0$ for each n , and that

$$m \leq a_1 + \dots + a_n \leq M$$

for all n . Prove that

$$b_1 m \leq a_1 b_1 + \dots + a_n b_n \leq b_1 M.$$

This result is known as Abel's Lemma. Although it may not look very interesting, it is the basis for several very interesting results which will appear in later problem sets; in addition, the proof

depends on a rather tricky way of writing $a_1b_1 + \cdots + a_nb_n$.

(b) Deduce that

$$b_k m \leq a_k b_k + \cdots + a_n b_n \leq b_k M$$

(a trivial consequence of part (a), inserted only because it is needed later).

***21.** The Bolzano-Weierstrass Theorem is usually stated, and also proved, quite differently than in the text—the classical statement uses the notion of limit points. A point x is a **limit point** of the set A if for every $\varepsilon > 0$ there is a point a in A with $|x - a| < \varepsilon$ but $x \neq a$.

(a) Find all limit points of the following sets.

(i) $\left\{\frac{1}{n} : n \text{ in } \mathbf{N}\right\}.$

(ii) $\left\{\frac{1}{n} + \frac{1}{m} : n \text{ and } m \text{ in } \mathbf{N}\right\}.$

(iii) $\left\{(-1)^n \left[1 + \frac{1}{n}\right] : n \text{ in } \mathbf{N}\right\}.$

(iv) $\mathbf{Z}.$

(v) $\mathbf{Q}.$

(b) Prove that x is a limit point of A if and only if for every $\varepsilon > 0$ there are infinitely many points a of A satisfying $|x - a| < \varepsilon$.

(c) Prove that $\lim A$ is the largest limit point of A , and $\underline{\lim} A$ the smallest.

The usual form of the Bolzano-Weierstrass Theorem states that if A is an infinite set of numbers contained in a closed interval $[a, b]$, then some point of $[a, b]$ is a limit point of A . Prove this in two ways:

(d) Using the form already proved in the text. Hint: Since A is infinite, there are distinct numbers $x_1, x_2, x_3, \dots$ in A .

(e) Using the Nested Intervals Theorem. Hint: If $[a, b]$ is divided into two intervals, at least one must contain infinitely many points of A .

***22.** Use the Bolzano-Weierstrass Theorem to prove that if f is continuous on $[a, b]$, then f is bounded above on $[a, b]$. Hint: If f is not bounded above, then there are points x_n in $[a, b]$ with $f(x_n) > n$.

****23.** (a) Let $\{a_n\}$ be the sequence

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{6}, \frac{2}{6}, \dots$$

Suppose that $0 \leq a < b \leq 1$. Let $N(n; a, b)$ be the number of integers $j \leq n$ such that a_j is in $[a, b]$. (Thus $N(2; \frac{1}{3}, \frac{2}{3}) = 2$, and

$N(4; \frac{1}{3}, \frac{2}{3}) = 3$.) Prove that

$$\lim_{n \rightarrow \infty} \frac{N(n; a, b)}{n} = b - a.$$

- (b) A sequence $\{a_n\}$ of numbers in $[0, 1]$ is called **uniformly distributed** in $[0, 1]$ if

$$\lim_{n \rightarrow \infty} \frac{N(n; a, b)}{n} = b - a$$

for all a and b with $0 \leq a < b \leq 1$. Prove that if s is a step function defined on $[0, 1]$, and $\{a_n\}$ is uniformly distributed in $[0, 1]$, then

$$\int_0^1 s = \lim_{n \rightarrow \infty} \frac{s(a_1) + \cdots + s(a_n)}{n}.$$

- (c) Prove that if $\{a_n\}$ is uniformly distributed in $[0, 1]$ and f is integrable on $[0, 1]$, then

$$\int_0^1 f = \lim_{n \rightarrow \infty} \frac{f(a_1) + \cdots + f(a_n)}{n}.$$

- **24.** (a) Let f be a function defined on $[0, 1]$ such that $\lim_{y \rightarrow a} f(y)$ exists for all a in $[0, 1]$. For any $\varepsilon > 0$ prove that there are only finitely many points a in $[0, 1]$ with $|\lim_{y \rightarrow a} f(y) - f(a)| > \varepsilon$. Hint: Show that the set of such points cannot have a limit point x , by showing that $\lim_{y \rightarrow x} f(y)$ could not exist.
- (b) Prove that, in the terminology of Problem 20-5, the set of points where f is discontinuous is countable. This finally answers the question of Problem 6-16: If f has only removable discontinuities, then f is continuous except at a countable set of points, and in particular, f cannot be discontinuous everywhere.

CHAPTER 22 INFINITE SERIES

Infinite sequences were introduced in the previous chapter with the specific intention of considering their “sums”

$$a_1 + a_2 + a_3 + \cdots$$

in this chapter. This is not an entirely straightforward matter, for the sum of infinitely many numbers is as yet completely undefined. What can be defined are the “partial sums”

$$s_n = a_1 + \cdots + a_n,$$

and the infinite sum must presumably be defined in terms of these partial sums. Fortunately, the mechanism for formulating this definition has already been developed in the previous chapter. If there is to be any hope of computing the infinite sum $a_1 + a_2 + a_3 + \cdots$, the partial sums s_n should represent closer and closer approximations as n is chosen larger and larger. This last assertion amounts to little more than a sloppy definition of limits: the “infinite sum” $a_1 + a_2 + a_3 + \cdots$ ought to be $\lim_{n \rightarrow \infty} s_n$. This approach will necessarily leave the “sum” of many sequences undefined, since the sequence $\{s_n\}$ may easily fail to have a limit. For example, the sequence

$$1, -1, 1, -1, \dots$$

with $a_n = (-1)^{n+1}$ yields the new sequence

$$\begin{aligned} s_1 &= a_1 = 1, \\ s_2 &= a_1 + a_2 = 0, \\ s_3 &= a_1 + a_2 + a_3 = 1, \\ s_4 &= a_1 + a_2 + a_3 + a_4 = 0, \\ &\cdots, \end{aligned}$$

for which $\lim_{n \rightarrow \infty} s_n$ does not exist. Although there happen to be some clever extensions of the definition suggested here (see Problems 8 and 23-11) it seems unavoidable that some sequences will have no sum. For this reason, an acceptable definition of the sum of a sequence should contain, as an essential component, terminology which distinguishes sequences for which sums can be defined from less fortunate sequences.

DEFINITION

The sequence $\{a_n\}$ is **summable** if the sequence $\{s_n\}$ converges, where

$$s_n = a_1 + \cdots + a_n.$$

In this case, $\lim_{n \rightarrow \infty} s_n$ is denoted by

$$\sum_{n=1}^{\infty} a_n \quad (\text{or, less formally, } a_1 + a_2 + a_3 + \cdots)$$

and is called the **sum** of the sequence $\{a_n\}$.

The terminology introduced in this definition is usually replaced by less precise expressions; indeed the title of this chapter is derived from such everyday language. An infinite sum $\sum_{n=1}^{\infty} a_n$ is usually called an *infinite series*, the word “series” emphasizing the connection with the infinite sequence $\{a_n\}$. The statement that $\{a_n\}$ is, or is not, summable is conventionally replaced by the statement that the series $\sum_{n=1}^{\infty} a_n$ does, or does not, converge. This terminology

is somewhat peculiar, because at best the symbol $\sum_{n=1}^{\infty} a_n$ denotes a number (so it can’t “converge”), and it doesn’t denote anything at all unless $\{a_n\}$ is summable. Nevertheless, this informal language is convenient, standard, and unlikely to yield to attacks on logical grounds.

Certain elementary arithmetical operations on infinite series are direct consequences of the definition. It is a simple exercise to show that if $\{a_n\}$ and $\{b_n\}$ are summable, then

$$\begin{aligned} \sum_{n=1}^{\infty} (a_n + b_n) &= \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n, \\ \sum_{n=1}^{\infty} c \cdot a_n &= c \cdot \sum_{n=1}^{\infty} a_n. \end{aligned}$$

As yet these equations are not very interesting, since we have no examples of summable sequences (except for the trivial examples in which the terms are eventually all 0). Before we actually exhibit a summable sequence, some general conditions for summability will be recorded.

There is one necessary and sufficient condition for summability which can be stated immediately. The sequence $\{a_n\}$ is summable if and only if the sequence $\{s_n\}$ converges, which happens, according to Theorem 21-3, if and only if $\lim_{m,n \rightarrow \infty} s_m - s_n = 0$; this condition can be rephrased in terms of the original

sequence as follows.

THE CAUCHY CRITERION

The sequence $\{a_n\}$ is summable if and only if

$$\lim_{m, n \rightarrow \infty} a_{n+1} + \cdots + a_m = 0.$$

Although the Cauchy criterion is of theoretical importance, it is not very useful for deciding the summability of any particular sequence. However, one simple consequence of the Cauchy criterion provides a *necessary* condition for summability which is too important not to be mentioned explicitly.

THE VANISHING CONDITION

If $\{a_n\}$ is summable, then

$$\lim_{n \rightarrow \infty} a_n = 0.$$

This condition follows from the Cauchy criterion by taking $m = n + 1$; it can also be proved directly as follows. If $\lim_{n \rightarrow \infty} s_n = l$, then

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} \\ &= l - l = 0. \end{aligned}$$

Unfortunately, this condition is far from sufficient. For example, $\lim_{n \rightarrow \infty} 1/n = 0$, but the sequence $\{1/n\}$ is not summable; in fact the following grouping of the numbers $1/n$ shows that the sequence $\{s_n\}$ is not bounded:

$$1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\substack{\geq \frac{1}{2} \\ (2 \text{ terms,} \\ \text{each} \geq \frac{1}{4})}} + \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}}_{\substack{\geq \frac{1}{2} \\ (4 \text{ terms,} \\ \text{each} \geq \frac{1}{8})}} + \underbrace{\frac{1}{9} + \cdots + \frac{1}{16}}_{\substack{\geq \frac{1}{2} \\ (8 \text{ terms,} \\ \text{each} \geq \frac{1}{16})}} + \cdots$$

The method of proof used in this example, a clever trick which one might never see, reveals the need for some more standard methods for attacking these problems. These methods shall be developed soon (one of them will give an alternate proof that $\sum_{n=1}^{\infty} 1/n$ does not converge) but it will be necessary to first procure a few examples of convergent series.

The most important of all infinite series are the “geometric series”

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \cdots$$

Only the cases $|r| < 1$ are interesting, since the individual terms do not approach 0 if $|r| \geq 1$. These series can be managed because the partial sums

$$s_n = 1 + r + \cdots + r^n$$

can be evaluated in simple terms. The two equations

$$\begin{aligned} s_n &= 1 + r + r^2 + \cdots + r^n \\ rs_n &= \quad r + r^2 + \cdots + r^n + r^{n+1} \end{aligned}$$

lead to

$$s_n(1 - r) = 1 - r^{n+1}$$

or

$$s_n = \frac{1 - r^{n+1}}{1 - r}$$

(division by $1 - r$ is valid since we are not considering the case $r = 1$). Now $\lim_{n \rightarrow \infty} r^n = 0$, since $|r| < 1$. It follows that

$$\sum_{n=0}^{\infty} r^n = \lim_{n \rightarrow \infty} \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r}, \quad |r| < 1.$$

In particular,

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n - 1 = \frac{1}{1 - \frac{1}{2}} - 1 = 1,$$

that is,

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1,$$

an infinite sum which can always be remembered from the picture in Figure 1.

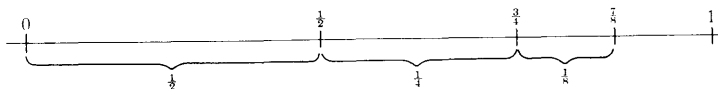


FIGURE 1

Special as they are, geometric series are standard examples from which important tests for summability will be derived.

For a while we shall consider only sequences $\{a_n\}$ with each $a_n \geq 0$; such sequences are called **nonnegative**. If $\{a_n\}$ is a nonnegative sequence, then the sequence $\{s_n\}$ is clearly nondecreasing. This remark, combined with Theorem 21-2, provides a simple-minded test for summability:

THE BOUNDEDNESS CRITERION

A nonnegative sequence $\{a_n\}$ is summable if and only if the set of partial sums s_n is bounded.

By itself, this criterion is not very helpful—deciding whether or not the set of all s_n is bounded is just what we are unable to do. On the other hand, if some convergent series are already available for comparison, this criterion can be used to obtain a result whose simplicity belies its importance (it is the basis for almost all other tests).

THEOREM 1 (THE COMPARISON TEST)

Suppose that

$$0 \leq a_n \leq b_n \quad \text{for all } n.$$

Then if $\sum_{n=1}^{\infty} b_n$ converges, so does $\sum_{n=1}^{\infty} a_n$.

PROOF If

$$\begin{aligned}s_n &= a_1 + \cdots + a_n, \\ t_n &= b_1 + \cdots + b_n,\end{aligned}$$

then

$$0 \leq s_n \leq t_n \quad \text{for all } n.$$

Now $\{t_n\}$ is bounded, since $\sum_{n=1}^{\infty} b_n$ converges. Therefore $\{s_n\}$ is bounded; consequently, by the boundedness criterion $\sum_{n=1}^{\infty} a_n$ converges.

Quite frequently the comparison test can be used to analyze very complicated looking series in which most of the complication is irrelevant. For example,

$$\sum_{n=1}^{\infty} \frac{2 + \sin^3(n+1)}{2^n + n^2}$$

converges because

$$0 \leq \frac{2 + \sin^3(n+1)}{2^n + n^2} < \frac{3}{2^n},$$

and

$$\sum_{n=1}^{\infty} \frac{3}{2^n} = 3 \sum_{n=1}^{\infty} \frac{1}{2^n}$$

is a convergent (geometric) series. Similarly, we would expect the series

$$\sum_{n=1}^{\infty} \frac{1}{2^n - 1 + \sin^2 n^3}$$

to converge, since the n th term of the series is practically $1/2^n$. This last assertion really means that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2^n - 1 + \sin^2 n^3} \right) / \frac{1}{2^n} = 1,$$

and this formulation shows us how to verify our conjecture. If we choose any number $c > 1$, then it is certainly true that

$$0 \leq \frac{1}{2^n - 1 + \sin^2 n^3} < c \frac{1}{2^n} \quad \text{for large enough } n.$$

This shows that

$$\sum_{n=N}^{\infty} \frac{1}{2^n - 1 + \sin^2 n^3}$$

converges for some N , and of course this implies convergence of the original series.

It is also possible to use the comparison test backwards. Consider, for example, the series

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2+1}.$$

We would expect this series to diverge, since $(n+1)/(n^2+1)$ is practically $1/n$ for large n . To prove this, choose any number c with $0 < c < 1$. Then

$$0 \leq \frac{1}{n} < c \frac{n+1}{n^2+1} \quad \text{for large enough } n.$$

This shows that the series $\sum_{n=1}^{\infty} (n+1)/(n^2+1)$ diverges, because if it converged, then by the comparison test $\sum_{n=1}^{\infty} 1/n$ would also converge, and this we know is false.

The comparison test yields other important tests when we use previously analyzed series as catalysts. Choosing the geometric series $\sum_{n=0}^{\infty} r^n$, the convergent series *par excellence*, we obtain the most important of all tests for summability.

THEOREM 2 (THE RATIO TEST)

Let $a_n > 0$ for all n , and suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r.$$

Then $\sum_{n=1}^{\infty} a_n$ converges if $r < 1$. On the other hand, if $r > 1$, then the terms a_n do not approach 0, so $\sum_{n=1}^{\infty} a_n$ diverges. (Notice that it is therefore essential to compute $\lim_{n \rightarrow \infty} a_{n+1}/a_n$ and not $\lim_{n \rightarrow \infty} a_n/a_{n+1}$!)

PROOF Suppose first that $r < 1$. Choose any number s with $r < s < 1$. The hypothesis

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r < 1$$

implies that there is some N such that

$$\frac{a_{n+1}}{a_n} \leq s \quad \text{for } n \geq N.$$

This can be written

$$a_{n+1} \leq sa_n \quad \text{for } n \geq N.$$

Thus

$$\begin{aligned} a_{N+1} &\leq sa_N, \\ a_{N+2} &\leq sa_{N+1} \leq s^2a_N, \\ &\vdots \\ &\vdots \\ a_{N+k} &\leq s^ka_N. \end{aligned}$$

Since $\sum_{k=0}^{\infty} a_N s^k = a_N \sum_{k=0}^{\infty} s^k$ converges, the comparison test shows that

$$\sum_{n=N}^{\infty} a_n = \sum_{k=0}^{\infty} a_{N+k}$$

converges. This implies the convergence of the series $\sum_{n=1}^{\infty} a_n$, which has only finitely many terms not included in $\sum_{n=N}^{\infty} a_n$.

The case $r > 1$ is even easier. If $1 < s < r$, then there is a number N such that

$$\frac{a_{n+1}}{a_n} \geq s \quad \text{for } n \geq N,$$

which means that

$$a_{N+k} \geq a_N s^k \geq a_N \quad k = 0, 1, \dots$$

This shows that the individual terms of $\{a_n\}$ do not approach 0, so $\{a_n\}$ is not summable. ■

As a simple application of the ratio test, consider the series $\sum_{n=1}^{\infty} 1/n!$. Letting $a_n = 1/n!$ we obtain

$$\frac{a_{n+1}}{a_n} = \frac{1}{(n+1)!} \cdot \frac{n!}{1} = \frac{n!}{(n+1)!} = \frac{1}{n+1}.$$

Thus

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 0,$$

which shows that the series $\sum_{n=1}^{\infty} 1/n!$ converges. If we consider instead the series $\sum_{n=1}^{\infty} r^n/n!$, where r is some fixed positive number, then

$$\lim_{n \rightarrow \infty} \frac{\frac{r^{n+1}}{(n+1)!}}{\frac{r^n}{n!}} = \lim_{n \rightarrow \infty} \frac{r}{n+1} = 0,$$

so $\sum_{n=1}^{\infty} r^n/n!$ converges. It follows that

$$\lim_{n \rightarrow \infty} \frac{r^n}{n!} = 0,$$

a result already proved in Chapter 16 (the proof given there was based on the same ideas as those used in the ratio test). Finally, if we consider the series

$\sum_{n=1}^{\infty} nr^n$ we have

$$\lim_{n \rightarrow \infty} \frac{(n+1)r^{n+1}}{nr^n} = \lim_{n \rightarrow \infty} r \cdot \frac{n+1}{n} = r,$$

since $\lim_{n \rightarrow \infty} (n+1)/n = 1$. This proves that if $0 \leq r < 1$, then $\sum_{n=1}^{\infty} nr^n$ converges, and consequently

$$\lim_{n \rightarrow \infty} nr^n = 0.$$

(This result clearly holds for $-1 < r \leq 0$, also.) It is a useful exercise to provide a direct proof of this limit, without using the ratio test as an intermediary.

Although the ratio test will be of the utmost theoretical importance, as a practical tool it will frequently be found disappointing. One drawback of the ratio test is the fact that $\lim_{n \rightarrow \infty} a_{n+1}/a_n$ may be quite difficult to determine, and

may not even exist. A more serious deficiency, which appears with maddening regularity, is the fact that the limit might equal 1. The case $\lim_{n \rightarrow \infty} a_{n+1}/a_n = 1$

is precisely the one which is inconclusive: $\{a_n\}$ might not be summable (for example, if $a_n = 1/n$), but then again it might be. In fact, our very next test

will show that $\sum_{n=1}^{\infty} (1/n)^2$ converges, even though

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n+1}\right)^2}{\left(\frac{1}{n}\right)^2} = 1.$$

This test provides a quite different method for determining convergence or divergence of infinite series—like the ratio test, it is an immediate consequence of the comparison test, but the series chosen for comparison is quite novel.

THEOREM 3 (THE INTEGRAL TEST)

Suppose that f is positive and decreasing on $[1, \infty)$, and that $f(n) = a_n$ for all n . Then $\sum_{n=1}^{\infty} a_n$ converges if and only if the limit

$$\int_1^{\infty} f = \lim_{A \rightarrow \infty} \int_1^A f$$

exists.

PROOF The existence of $\lim_{A \rightarrow \infty} \int_1^A f$ is equivalent to convergence of the series

$$\int_1^2 f + \int_2^3 f + \int_3^4 f + \cdots$$

Now, since f is decreasing we have (Figure 2)

$$f(n+1) < \int_n^{n+1} f < f(n).$$

The first half of this double inequality shows that the series $\sum_{n=1}^{\infty} a_{n+1}$ may be compared to the series $\sum_{n=1}^{\infty} \int_n^{n+1} f$, proving that $\sum_{n=1}^{\infty} a_{n+1}$ (and hence $\sum_{n=1}^{\infty} a_n$) converges if $\lim_{A \rightarrow \infty} \int_1^A f$ exists.

The second half of the inequality shows that the series $\sum_{n=1}^{\infty} \int_n^{n+1} f$ may be compared to the series $\sum_{n=1}^{\infty} a_n$, proving that $\lim_{A \rightarrow \infty} \int_1^A f$ must exist if $\sum_{n=1}^{\infty} a_n$ converges. ■

Only one example using the integral test will be given here, but it settles the question of convergence for infinitely many series at once. If $p > 0$, the convergence of $\sum_{n=1}^{\infty} 1/n^p$ is equivalent, by the integral test, to the existence of

$$\int_1^{\infty} \frac{1}{x^p} dx.$$

Now

$$\int_1^A \frac{1}{x^p} dx = \begin{cases} -\frac{1}{(p-1)} \cdot \frac{1}{A^{p-1}} + \frac{1}{p-1}, & p \neq 1 \\ \log A, & p = 1. \end{cases}$$

This shows that $\lim_{A \rightarrow \infty} \int_1^A 1/x^p dx$ exists if $p > 1$, but not if $p \leq 1$. Thus $\sum_{n=1}^{\infty} 1/n^p$

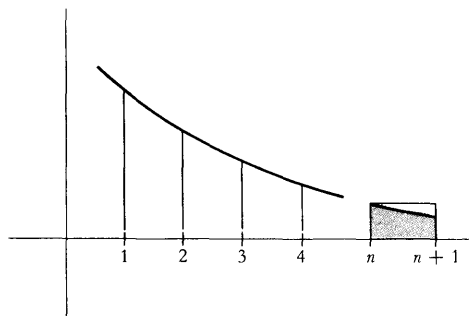


FIGURE 2

converges precisely for $p > 1$. In particular, $\sum_{n=1}^{\infty} 1/n$ diverges.

The tests considered so far apply only to nonnegative sequences, but non-positive sequences may be handled in precisely the same way. In fact, since

$$\sum_{n=1}^{\infty} a_n = -\left(\sum_{n=1}^{\infty} -a_n\right),$$

all considerations about nonpositive sequences can be reduced to questions involving nonnegative sequences. Sequences which contain both positive and negative terms are quite another story.

If $\sum_{n=1}^{\infty} a_n$ is a sequence with both positive and negative terms, one can consider instead the sequence $\sum_{n=1}^{\infty} |a_n|$, all of whose terms are nonnegative. Cheerfully ignoring the possibility that we may have thrown away all the interesting information about the original sequence, we proceed to eulogize those sequences which are converted by this procedure into convergent sequences.

DEFINITION

The series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** if the series $\sum_{n=1}^{\infty} |a_n|$ is convergent. (In more formal language, the sequence $\{a_n\}$ is **absolutely summable** if the sequence $\{|a_n|\}$ is summable.)

Although we have no right to expect this definition to be of any interest, it turns out to be exceedingly important. The following theorem shows that the definition is at least not entirely useless.

THEOREM 4

Every absolutely convergent series is convergent. Moreover, a series is absolutely convergent if and only if the series formed from its positive terms and the series formed from its negative terms both converge.

PROOF

If $\sum_{n=1}^{\infty} |a_n|$ converges, then, by the Cauchy criterion,

$$\lim_{m, n \rightarrow \infty} |a_{n+1}| + \cdots + |a_m| = 0.$$

Since

$$|a_{n+1} + \cdots + a_m| \leq |a_{n+1}| + \cdots + |a_m|,$$

it follows that

$$\lim_{m, n \rightarrow \infty} a_{n+1} + \cdots + a_m = 0,$$

which shows that $\sum_{n=1}^{\infty} a_n$ converges.

To prove the second part of the theorem, let

$$a_n^+ = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n \leq 0, \end{cases}$$

$$a_n^- = \begin{cases} a_n, & \text{if } a_n \leq 0 \\ 0, & \text{if } a_n \geq 0, \end{cases}$$

so that $\sum_{n=1}^{\infty} a_n^+$ is the series formed from the positive terms of $\sum_{n=1}^{\infty} a_n$, and

$\sum_{n=1}^{\infty} a_n^-$ is the series formed from the negative terms.

If $\sum_{n=1}^{\infty} a_n^+$ and $\sum_{n=1}^{\infty} a_n^-$ both converge, then

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} [a_n^+ - (a_n^-)] = \sum_{n=1}^{\infty} a_n^+ - \sum_{n=1}^{\infty} a_n^-$$

also converges, so $\sum_{n=1}^{\infty} a_n$ converges absolutely.

On the other hand, if $\sum_{n=1}^{\infty} |a_n|$ converges, then, as we have just shown,

$\sum_{n=1}^{\infty} a_n$ also converges. Therefore

$$\sum_{n=1}^{\infty} a_n^+ = \frac{1}{2} \left(\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} |a_n| \right)$$

and

$$\sum_{n=1}^{\infty} a_n^- = \frac{1}{2} \left(\sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} |a_n| \right)$$

both converge. ■

It follows from Theorem 4 that every convergent series with positive terms can be used to obtain infinitely many other convergent series, simply by putting in minus signs at random. Not every convergent series can be obtained in this way, however—there are series which are convergent but not absolutely convergent (such series are called **conditionally convergent**). In order to prove this statement we need a test for convergence which applies specifically to series with positive and negative terms.

THEOREM 5 (LEIBNIZ'S THEOREM)

Suppose that

$$a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0,$$

and that

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Then the series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + a_5 - \cdots$$

converges.

PROOF Figure 3 illustrates relationships between the partial sums which we will establish:

- (1) $s_2 \leq s_4 \leq s_6 \leq \cdots$,
- (2) $s_1 \geq s_3 \geq s_5 \geq \cdots$,
- (3) $s_k \leq s_l$ if k is even and l is odd.



FIGURE 3

To prove the first two inequalities, observe that

- (1) $s_{2n+2} = s_{2n} + a_{2n+1} - a_{2n}$
 $\geq s_{2n}$, since $a_{2n+1} \geq a_{2n}$;
- (2) $s_{2n+3} = s_{2n+1} - a_{2n+2} + a_{2n+3}$
 $\geq s_{2n+1}$, since $a_{2n+2} \geq a_{2n+3}$.

To prove the third inequality, notice first that

$$\begin{aligned} s_{2n} &= s_{2n-1} - a_{2n} \\ &\leq s_{2n-1}, \text{ since } a_{2n} \geq 0. \end{aligned}$$

This proves only a special case of (3), but in conjunction with (1) and (2) the general case is easy: if k is even and l is odd, choose n such that

$$2n \geq k \quad \text{and} \quad 2n - 1 \geq l;$$

then

$$s_k \leq s_{2n} \leq s_{2n-1} \leq s_l,$$

which proves (3).

Now, the sequence $\{s_{2n}\}$ converges, because it is nondecreasing and is bounded above (by s_l for any odd l). Let

$$\alpha = \sup \{s_{2n}\} = \lim_{n \rightarrow \infty} s_{2n}.$$

Similarly, let

$$\beta = \inf \{s_{2n+1}\} = \lim_{n \rightarrow \infty} s_{2n+1}.$$

It follows from (3) that $\alpha \leq \beta$; since

$$s_{2n+1} - s_{2n} = a_{2n+1} \quad \text{and} \quad \lim_{n \rightarrow \infty} a_n = 0$$

it is actually the case that $\alpha = \beta$. This proves that $\alpha = \beta = \lim_{n \rightarrow \infty} s_n$. ■

$\sum_{n=1}^{\infty} (-1)^{n+1}/n$ is not special in this regard; indeed, its behavior is typical of series which are not absolutely convergent—the following result (really more of a grand counterexample than a theorem) shows how bad conditionally convergent series are.

THEOREM 6 If $\sum_{n=1}^{\infty} a_n$ converges, but does not converge absolutely, then for any number α there is a rearrangement $\{b_n\}$ of $\{a_n\}$ such that $\sum_{n=1}^{\infty} b_n = \alpha$.

PROOF Let $\sum_{n=1}^{\infty} p_n$ denote the series formed from the positive terms of $\{a_n\}$ and let $\sum_{n=1}^{\infty} q_n$ denote the series of negative terms. It follows from Theorem 4 that at least one of these series does not converge. As a matter of fact, both must fail to converge, for if one had bounded partial sums, and the other had unbounded partial sums, then the original series $\sum_{n=1}^{\infty} a_n$ would also have unbounded partial sums, contradicting the assumption that it converges.

Now let α be any number. Assume, for simplicity, that $\alpha > 0$ (the proof for $\alpha < 0$ will be a simple modification). Since the series $\sum_{n=1}^{\infty} p_n$ is not convergent, there is a number N such that

$$\sum_{n=1}^N p_n > \alpha.$$

We will choose N_1 to be the *smallest* N with this property. This means that

$$(1) \quad \sum_{n=1}^{N_1-1} p_n \leq \alpha,$$

$$\text{but } (2) \quad \sum_{n=1}^{N_1} p_n > \alpha.$$

Then if

$$S_1 = \sum_{n=1}^{N_1} p_n,$$

we have

$$S_1 - \alpha \leq p_{N_1}.$$

This relation, which is clear from Figure 4, follows immediately from equation (1):

$$S_1 - \alpha \leq S_1 - \sum_{n=1}^{N_1-1} p_n = p_{N_1}.$$

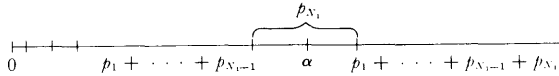


FIGURE 4

To the sum S_1 we now add on just enough negative terms to obtain a new sum T_1 which is less than α . In other words, we choose the smallest integer M_1 for which

$$T_1 = S_1 + \sum_{n=1}^{M_1} q_n < \alpha.$$

As before, we have

$$\alpha - T_1 \leq -q_{M_1}.$$

We now continue this procedure indefinitely, obtaining sums alternately larger and smaller than α , each time choosing the smallest N_k or M_k possible. The sequence

$$p_1, \dots, p_{N_1}, q_1, \dots, q_{M_1}, p_{N_1+1}, \dots, p_{N_2}, \dots$$

is a rearrangement of $\{a_n\}$. The partial sums of this rearrangement increase to S_1 , then decrease to T_1 , then increase to S_2 , then decrease to T_2 , etc. To complete the proof we simply note that $|S_k - \alpha|$ and $|T_k - \alpha|$ are less than or equal to p_{N_k} or $-q_{M_k}$, respectively, and that these terms, being members of the original sequence $\{a_n\}$, must decrease to 0, since $\sum_{n=1}^{\infty} a_n$ converges. ■

Together with Theorem 6, the next theorem establishes conclusively the distinction between conditionally convergent and absolutely convergent series.

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $\{b_n\}$ is any rearrangement of $\{a_n\}$, then $\sum_{n=1}^{\infty} b_n$ also converges (absolutely), and

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n.$$

PROOF Let us denote the partial sums of $\{a_n\}$ by s_n , and the partial sums of $\{b_n\}$ by t_n .

Suppose that $\epsilon > 0$. Since $\sum_{n=1}^{\infty} a_n$ converges, there is some N such that

$$\left| \sum_{n=1}^{\infty} a_n - s_N \right| < \epsilon.$$

Moreover, since $\sum_{n=1}^{\infty} |a_n|$ converges, we can also choose N so that

$$\sum_{n=1}^{\infty} |a_n| - (|a_1| + \cdots + |a_N|) < \varepsilon,$$

i.e., so that

$$|a_{N+1}| + |a_{N+2}| + |a_{N+3}| + \cdots < \varepsilon.$$

Now choose M so large that each of $a_1, \dots, a_N$ appear among $b_1, \dots, b_M$. Then whenever $m > M$, the difference $t_m - s_N$ is the sum of certain a_i , where $a_1, \dots, a_N$ are definitely excluded. Consequently,

$$|t_m - s_N| \leq |a_{N+1}| + |a_{N+2}| + |a_{N+3}| + \cdots$$

Thus, if $m > M$, then

$$\begin{aligned} \left| \sum_{n=1}^{\infty} a_n - t_m \right| &= \left| \sum_{n=1}^{\infty} a_n - s_N + t_m - s_N \right| \\ &\leq \left| \sum_{n=1}^{\infty} a_n - s_N \right| + |t_m - s_N| \\ &< \varepsilon + \varepsilon. \end{aligned}$$

Since this is true for every $\varepsilon > 0$, the series $\sum_{n=1}^{\infty} b_n$ converges to $\sum_{n=1}^{\infty} a_n$. To prove that it converges absolutely, notice that the above argument can be applied to the positive terms of $\sum_{n=1}^{\infty} a_n$ and the negative terms separately; this shows that the series formed from the positive and negative terms of $\sum_{n=1}^{\infty} b_n$ each converge, so $\sum_{n=1}^{\infty} b_n$ converges absolutely. ■

The theorems of this chapter have been concerned with summability of sequences, but not with the actual sums. Generally speaking, there is no reason to presume that a given infinite sum can be “evaluated” in any simpler terms. However, many simple expressions can be equated to infinite sums by using Taylor’s Theorem. Chapter 19 provides many examples of functions for which

$$f(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + R_{n,a}(x),$$

where $\lim_{n \rightarrow \infty} R_{n,a}(x) = 0$. This is precisely equivalent to

$$f(x) = \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i,$$

which means, in turn, that

$$f(x) = \sum_{i=0}^{\infty} \frac{f^{(i)}(a)}{i!} (x-a)^i.$$

As particular examples we have

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots,$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots,$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \quad |x| \leq 1,$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots, \quad -1 < x \leq 1.$$

(Notice that the series for $\arctan x$ and $\log(1+x)$ do not even converge for $|x| > 1$; in addition, when $x = -1$, the series for $\log(1+x)$ becomes

$$-1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} - \cdots$$

which does not converge.)

Some pretty impressive results are obtained with particular values of x :

$$0 = \pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!} - \frac{\pi^7}{7!} + \cdots,$$

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots,$$

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots,$$

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

More significant developments may be anticipated if we compare the series for $\sin x$ and $\cos x$ a little more carefully. The series for $\cos x$ is just the one we would have obtained if we had enthusiastically differentiated both sides of the equation

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

term-by-term, ignoring the fact that we have never proved anything about the derivatives of infinite sums. Likewise, if we differentiate both sides of the formula for $\cos x$ formally (i.e., without justification) we obtain the formula $\cos'(x) = -\sin x$, and if we differentiate the formula for e^x we obtain $\exp'(x) = \exp(x)$. In the next chapter we shall see that such term-by-term differentiation of infinite sums is indeed valid in certain important cases.

PROBLEMS

1. Decide whether each of the following infinite series is convergent or divergent. The tools which you will need are Leibniz's Theorem and the comparison, ratio, and integral tests. A few examples have been picked with malice aforethought; two series which look quite similar may require different tests (and then again, they may not). The hint below indicates which tests may be used.

$$(i) \quad \sum_{n=1}^{\infty} \frac{\sin n\theta}{n^2}.$$

$$(ii) \quad 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

$$(iii) \quad 1 - \frac{1}{2} + \frac{2}{3} - \frac{1}{3} + \frac{2}{4} - \frac{1}{4} + \frac{2}{5} - \frac{1}{5} + \cdots$$

$$(iv) \quad \sum_{n=1}^{\infty} (-1)^n \frac{\log n}{n}.$$

$$(v) \quad \sum_{n=2}^{\infty} \frac{1}{\sqrt[3]{n^2 - 1}}. \quad (\text{The summation begins with } n = 2 \text{ simply to avoid the meaningless term obtained for } n = 1.)$$

$$(vi) \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2 + 1}}.$$

$$(vii) \quad \sum_{n=1}^{\infty} \frac{n^2}{n!}.$$

$$(viii) \quad \sum_{n=1}^{\infty} \frac{\log n}{n}.$$

$$(ix) \quad \sum_{n=2}^{\infty} \frac{1}{\log n}.$$

$$(x) \quad \sum_{n=2}^{\infty} \frac{1}{(\log n)^k}.$$

$$(xi) \quad \sum_{n=1}^{\infty} \frac{1}{(\log n)^n}.$$

$$(xii) \quad \sum_{n=2}^{\infty} (-1)^n \frac{1}{(\log n)^n}.$$

$$(xiii) \quad \sum_{n=1}^{\infty} \frac{n^2}{n^3 + 1}.$$

$$(xiv) \quad \sum_{n=1}^{\infty} \sin \frac{1}{n}.$$

$$(xv) \quad \sum_{n=2}^{\infty} \frac{1}{n \log n}.$$

$$(xvi) \quad \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}.$$

$$(xvii) \quad \sum_{n=2}^{\infty} \frac{1}{n^2(\log n)}.$$

$$(xviii) \quad \sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

$$(xix) \quad \sum_{n=1}^{\infty} \frac{2^n n!}{n^n}.$$

$$(xx) \quad \sum_{n=1}^{\infty} \frac{3^n n!}{n^n}.$$

Hint: Use the comparison test for (i), (ii), (v), (vi), (ix), (x), (xi), (xiii), (xiv), (xvii); the ratio test for (vii), (xviii), (xix), (xx); the integral test for (viii), (xv), (xvi).

The next two problems examine, with hints, some infinite series that require more delicate analysis than those in Problem 1.

- *2. (a) If you have successfully solved examples (xix) and (xx) from Problem 1, it should be clear that $\sum_{n=1}^{\infty} a^n n! / n^n$ converges for $a < e$ and diverges for $a > e$. For $a = e$ the ratio test fails; show that $\sum_{n=1}^{\infty} e^n n! / n^n$ actually diverges, by using Problem 21-10.

- (b) Decide when $\sum_{n=1}^{\infty} n^n / a^n n!$ converges, again resorting to Problem 21-10 when the ratio test fails.

- *3. Problem 1 presented the two series $\sum_{n=2}^{\infty} (\log n)^{-k}$ and $\sum_{n=2}^{\infty} (\log n)^{-n}$, of which the first diverges while the second converges. The series

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}},$$

which lies between these two, is analyzed in parts (a) and (b).

- (a) Show that $\int_0^{\infty} e^y / y^y dy$ exists, by considering the series $\sum_{n=1}^{\infty} (e/n)^n$.

(b) Show that

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log n}}$$

converges, by using the integral test. Hint: Use an appropriate substitution and part (a).

(c) Show that

$$\sum_{n=2}^{\infty} \frac{1}{(\log n)^{\log(\log n)}}$$

diverges, by using the integral test. Hint: Use the same substitution as in part (b), and show directly that the resulting integral diverges.

4. (a) Let $\{a_n\}$ be a sequence of integers with $0 \leq a_n \leq 9$. Prove that $\sum_{n=1}^{\infty} a_n 10^{-n}$ exists (and lies between 0 and 1). (This, of course, is the number which we usually denote by $0.a_1a_2a_3a_4 \dots$)
- (b) Suppose that $0 \leq x \leq 1$. Prove that there is a sequence of integers $\{a_n\}$ with $0 \leq a_n \leq 9$ and $\sum_{n=1}^{\infty} a_n 10^{-n} = x$. Hint: For example, $a_1 = [10x]$ (where $[y]$ denotes the greatest integer which is $\leq y$).
- (c) Show that if $\{a_n\}$ is repeating, i.e., is of the form $a_1, a_2, \dots, a_k, a_1, a_2, \dots, a_k, a_1, a_2, \dots$, then $\sum_{n=1}^{\infty} a_n 10^{-n}$ is a rational number (and find it). The same result naturally holds if $\{a_n\}$ is eventually repeating, i.e., if the sequence $\{a_{N+k}\}$ is repeating for some N .
- (d) Prove that if $x = \sum_{n=1}^{\infty} a_n 10^{-n}$ is rational, then $\{a_n\}$ is eventually repeating. (Just look at the process of finding the decimal expansion of p/q —dividing q into p by long division.)
5. Suppose that $\{a_n\}$ satisfies the hypothesis of Leibniz's Theorem. Use the proof of Leibniz's Theorem to obtain the following estimate:

$$\left| \sum_{n=1}^{\infty} (-1)^{n+1} a_n - [a_1 - a_2 + \dots \pm a_N] \right| < a_N.$$

6. Prove the following theorem (another kind of "comparison test"). If $a_n, b_n > 0$ and $\lim_{n \rightarrow \infty} a_n/b_n = c \neq 0$, then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.

7. Prove that if $a_n \geq 0$ and $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r$, then $\sum_{n=1}^{\infty} a_n$ converges if $r < 1$, and diverges if $r > 1$. (The proof is very similar to that of the ratio test.)

This result is known as the “root test.” It is easy to construct series for which the ratio test fails, while the root test works. For example, the root test shows that the series

$$\frac{1}{2} + \frac{1}{3} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{3}\right)^3 + \cdots$$

converges, even though the ratios of successive terms do not approach a limit. Most examples are of this rather artificial nature, but the root test is nevertheless quite an important theoretical tool, and if the ratio test works the root test will also (by Problem 21-13). It is possible to eliminate limits from the root test; a simple modification of the proof shows

that $\sum_{n=1}^{\infty} a_n$ converges if there is some $s < 1$ such that all but finitely many

$\sqrt[n]{a_n}$ are $\leq s$, and that $\sum_{n=1}^{\infty} a_n$ diverges if infinitely many $\sqrt[n]{a_n}$ are ≥ 1 .

This result is known as the “delicate root test” (there is a similar delicate ratio test). It follows, using the notation of Problem 21-19, that $\sum_{n=1}^{\infty} a_n$

converges if $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} < 1$ and diverges if $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} > 1$; no conclusion

is possible if $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1$.

8. A sequence $\{a_n\}$ is called **Cesaro summable**, with Cesaro sum l , if

$$\lim_{n \rightarrow \infty} \frac{s_1 + \cdots + s_n}{n} = l$$

(where $s_k = a_1 + \cdots + a_k$). Problem 21-12 shows that a summable sequence is automatically Cesaro summable, with sum equal to its Cesaro sum. Find a sequence which is *not* summable, but which *is* Cesaro summable.

9. This problem outlines an alternative proof of Theorem 7 which does not rely on the Cauchy criterion.

(a) Suppose that $a_n \geq 0$ for each n . Let $\{b_n\}$ be a rearrangement of $\{a_n\}$, and let $s_n = a_1 + \cdots + a_n$ and $t_n = b_1 + \cdots + b_n$. Show that for each n there is some m with $s_n \leq t_m$.

(b) Show that $\sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n$.

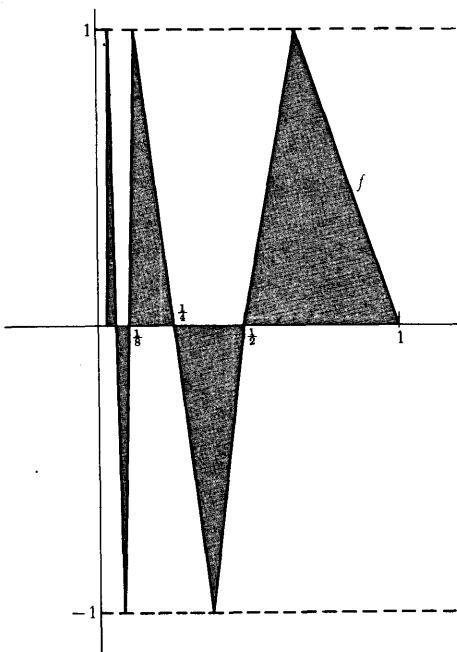


FIGURE 5

(c) Show that $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n$.

(d) Now replace the condition $a_n \geq 0$ by the hypothesis that $\sum_{n=1}^{\infty} a_n$ converges absolutely, using the second part of Theorem 4.

10. (a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $\{b_n\}$ is any subsequence of $\{a_n\}$, then $\sum_{n=1}^{\infty} b_n$ converges (absolutely).

(b) Show that this is false if $\sum_{n=1}^{\infty} a_n$ does not converge absolutely.

* (c) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then

$$\sum_{n=1}^{\infty} a_n = (a_1 + a_3 + a_5 + \cdots) + (a_2 + a_4 + a_6 + \cdots).$$

11. Prove that if $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, then $\left| \sum_{n=1}^{\infty} a_n \right| \leq \sum_{n=1}^{\infty} |a_n|$.

*12. Problem 18-28 shows that $\int_0^{\infty} (\sin x)/x \, dx$ converges. Prove that $\int_0^{\infty} |(\sin x)/x| \, dx$ diverges.

*13. Find a continuous function f with $f(x) \geq 0$ for all x , such that $\int_0^{\infty} f(x) \, dx$ exists, but $\lim_{x \rightarrow \infty} f(x)$ does not exist.

*14. Let $f(x) = x \sin 1/x$ for $0 < x \leq 1$, and let $f(0) = 0$. Recall the definition of $\ell(f, P)$ from Problem 15-32. Show that the set of all $\ell(f, P)$ for P a partition of $[0, 1]$ is not bounded (thus f has “infinite length”). Hint: Try partitions of the form

$$P = \left\{ 0, \frac{1}{2n}, \frac{1}{n}, \frac{1}{2(n-1)}, \frac{1}{n-1}, \dots, \frac{1}{2}, 1 \right\}.$$

15. Let f be the function shown in Figure 5. Find $\int_0^1 f$, and also the area of the shaded region in Figure 5.

*16. In this problem we will establish the “binomial series”

$$(1+x)^{\alpha} = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k, \quad |x| < 1,$$

for any α , by showing that $\lim_{n \rightarrow \infty} R_{n,0}(x) = 0$. The proof is in several steps, and uses the Cauchy and Lagrange forms as found in Problem 19-6.

- (a) Use the ratio test to show that the series $\sum_{k=0}^{\infty} \binom{\alpha}{k} r^k$ does indeed converge for $|r| < 1$ (this is not to say that it necessarily converges to $(1+r)^\alpha$). It follows in particular that $\lim_{n \rightarrow \infty} \binom{\alpha}{n} r^n = 0$ for $|r| < 1$.
- (b) Suppose first that $0 \leq x < 1$. Show that $\lim_{n \rightarrow \infty} R_{n,0}(x) = 0$, by using Lagrange's form of the remainder, noticing that $(1+t)^{\alpha-n-1} \leq 1$ for $n+1 > \alpha$.
- (c) Now suppose that $-1 < x < 0$; the number t in Cauchy's form of the remainder satisfies $-1 < x < t \leq 0$. Show that
- $$|x(1+t)^{\alpha-1}| \leq |x|M, \quad \text{where } M = \max(1, (1+x)^{\alpha-1}),$$
- and

$$\left| \frac{x-t}{1+t} \right| = |x| \left(\frac{1-t/x}{1+t} \right) \leq |x|.$$

Using Cauchy's form of the remainder, and the fact that

$$(n+1) \binom{\alpha}{n+1} = \alpha \binom{\alpha-1}{n},$$

show that $\lim_{n \rightarrow \infty} R_{n,0}(x) = 0$.

- *17.** (a) Suppose that the partial sums of the sequence $\{a_n\}$ are bounded, and that $\{b_n\}$ is a decreasing sequence with $\lim_{n \rightarrow \infty} b_n = 0$. Prove that

$\sum_{n=1}^{\infty} a_n b_n$ converges. Hint: Use Abel's Lemma (Problem 21-20) to check the Cauchy criterion.

- (b) Derive Leibniz's Theorem from this result.

- (c) Prove, using Problem 15-31, that the series $\sum_{n=1}^{\infty} (\cos nx)/n$ converges if x is not of the form $2k\pi$ for any integer k (in which case it clearly diverges).

- *18.** Suppose $\{a_n\}$ is decreasing and $\lim_{n \rightarrow \infty} a_n = 0$. Prove that if $\sum_{n=1}^{\infty} a_n$ con-

verges, then $\sum_{n=1}^{\infty} 2^n a_{2^n}$ also converges (the "Cauchy Condensation Theorem"). Notice that the divergence of $\sum_{n=1}^{\infty} 1/n$ is a special case, for if

$\sum_{n=1}^{\infty} 1/n$ converged, then $\sum_{n=1}^{\infty} 2^n (1/2^n)$ would also converge; this remark may serve as a hint.

- *19. (a) Prove that if $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} b_n^2$ converge, then $\sum_{n=1}^{\infty} a_n b_n$ converges.
 (b) Prove that if $\sum_{n=1}^{\infty} a_n^2$ converges, then $\sum_{n=1}^{\infty} a_n/n$ converges.
- *20. Suppose $\{a_n\}$ is decreasing and each $a_n \geq 0$. Prove that if $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} n a_n = 0$. Hint: Write down the Cauchy criterion and be sure to use the fact that $\{a_n\}$ is decreasing.
- *21. If $\sum_{n=1}^{\infty} a_n$ converges, then the partial sums s_n are bounded, and $\lim_{n \rightarrow \infty} a_n = 0$. It is tempting to conjecture that boundedness of the partial sums, together with the condition $\lim_{n \rightarrow \infty} a_n = 0$, implies convergence of $\sum_{n=1}^{\infty} a_n$. This is *not* true, but finding a counterexample requires a little ingenuity. As a hint, notice that some *subsequence* of the partial sums will have to converge; you must somehow allow this to happen, without letting the sequence itself converge.
- *22. The divergence of $\sum_{n=1}^{\infty} 1/n$ is a particular consequence of the following remarkable fact: Any positive rational number x can be written as a *finite* sum of *distinct* numbers of the form $1/n$. Such a representation can always be found by the straightforward method of continually adding on the next number in the sequence which is not too large. For example, the calculation

$$\begin{aligned} \frac{19}{15} - \frac{1}{2} &= \frac{23}{30} \\ \frac{23}{30} - \frac{1}{3} &= \frac{13}{30} \\ \frac{13}{30} - \frac{1}{4} &= \frac{11}{60} \\ \frac{11}{60} &< \frac{1}{5} \\ \frac{11}{60} - \frac{1}{6} &= \frac{1}{60} \end{aligned}$$

shows that

$$\frac{19}{15} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{60}.$$

Notice that the numerators 23, 13, 11, 1 are decreasing. Prove that every positive rational x can be written in such a way by showing that the numerators in this sort of calculation must always decrease, so that eventually the remainder has numerator 1. Hint: You just have to check that if $p/q < 1/n$, but $p/q > 1/(n+1)$, then the numerator of $p/q - 1/(n+1)$ is less than p .

The considerations at the end of the previous chapter suggest an entirely new way of looking at infinite series. Our attention will shift from particular infinite sums to equations like

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots$$

which concern sums of quantities that depend on x . In other words, we are interested in *functions* defined by equations of the form

$$f(x) = f_1(x) + f_2(x) + f_3(x) + \cdots$$

(in the previous example $f_n(x) = x^{n-1}/(n-1)!$). In such a situation $\{f_n\}$ will be some sequence of functions; for each x we obtain a sequence of numbers $\{f_n(x)\}$, and $f(x)$ is the sum of this sequence. In order to analyze such functions it will certainly be necessary to remember that each sum

$$f_1(x) + f_2(x) + f_3(x) + \cdots$$

is, by definition, the limit of the sequence

$$f_1(x), f_1(x) + f_2(x), f_1(x) + f_2(x) + f_3(x), \dots$$

If we define a new sequence of functions $\{s_n\}$ by

$$s_n = f_1 + \cdots + f_n,$$

then we can express this fact more succinctly by writing

$$f(x) = \lim_{n \rightarrow \infty} s_n(x).$$

For some time we shall therefore concentrate on functions defined as limits,

$$f(x) = \lim_{n \rightarrow \infty} f_n(x),$$

rather than on functions defined as infinite sums. The total body of results about such functions can be summed up very easily: nothing one would hope to be true actually is—instead we have a splendid collection of counterexamples. The first of these shows that even if each f_n is continuous, the function f may not be! Contrary to what you may expect, the functions f_n will be very simple. Figure 1 shows the graphs of the functions

$$f_n(x) = \begin{cases} x^n, & 0 \leq x \leq 1 \\ 1, & x \geq 1. \end{cases}$$

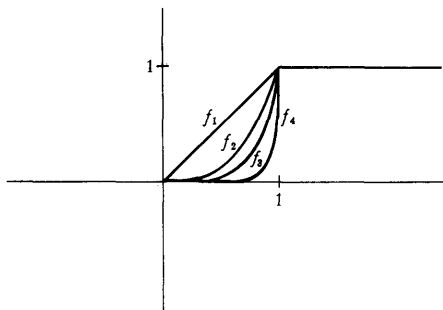


FIGURE 1

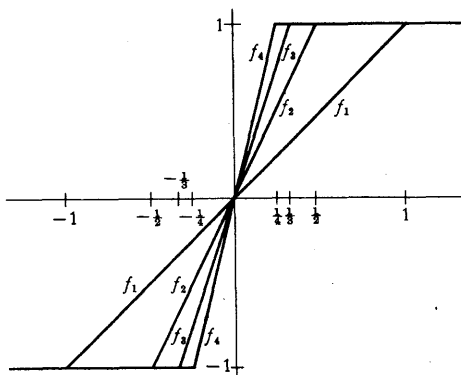


FIGURE 2

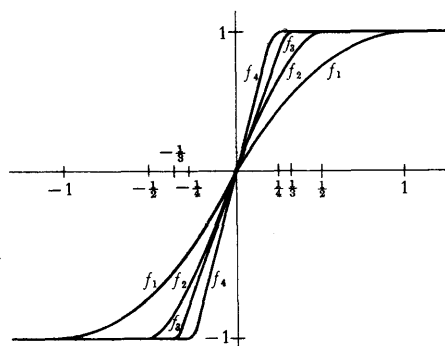


FIGURE 3

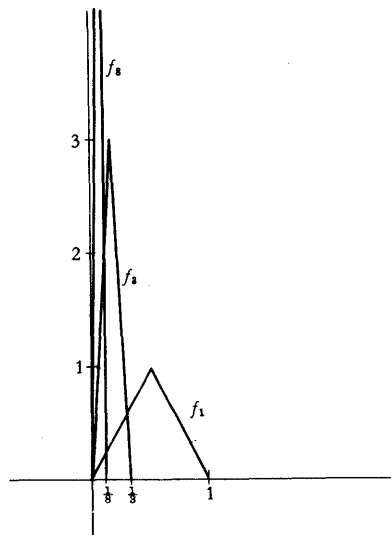


FIGURE 4

These functions are all continuous, but the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is not continuous; in fact,

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & 0 \leq x < 1 \\ 1, & x \geq 1. \end{cases}$$

Another example of this same phenomenon is illustrated in Figure 2; the functions f_n are defined by

$$f_n(x) = \begin{cases} -1, & x \leq -\frac{1}{n} \\ nx, & -\frac{1}{n} \leq x \leq \frac{1}{n} \\ 1, & \frac{1}{n} \leq x. \end{cases}$$

In this case, if $x < 0$, then $f_n(x)$ is eventually (i.e., for large enough n) equal to -1 , and if $x > 0$, then $f_n(x)$ is eventually 1 , while $f_n(0) = 0$ for all n . Thus

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0; \end{cases}$$

so, once again, the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is not continuous.

By rounding off the corners in the previous examples it is even possible to produce a sequence of *differentiable* functions $\{f_n\}$ for which the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is not continuous. One such sequence is easy to define explicitly:

$$f_n(x) = \begin{cases} -1, & x \leq -\frac{1}{n} \\ \sin\left(\frac{n\pi x}{2}\right), & -\frac{1}{n} \leq x \leq \frac{1}{n} \\ 1, & \frac{1}{n} \leq x. \end{cases}$$

These functions are differentiable (Figure 3), but we still have

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0. \end{cases}$$

Continuity and differentiability are, moreover, not the only properties for which problems arise. Another difficulty is illustrated by the sequence $\{f_n\}$ shown in Figure 4; on the interval $[0, 1/n]$ the graph of f_n forms an isosceles triangle of altitude n , while $f_n(x) = 0$ for $x \geq 1/n$. These functions may be defined explicitly as follows:

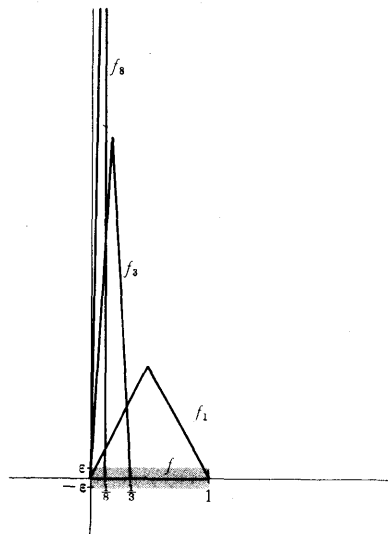


FIGURE 5

$$f_n(x) = \begin{cases} 2n^2x, & 0 \leq x \leq \frac{1}{2n} \\ 2n - 2n^2x, & \frac{1}{2n} \leq x \leq \frac{1}{n} \\ 0, & \frac{1}{n} \leq x \leq 1. \end{cases}$$

Because this sequence varies so erratically near 0, our primitive mathematical instincts might suggest that $\lim_{n \rightarrow \infty} f_n(x)$ does not always exist. Nevertheless, this limit does exist for all x , and the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is even continuous. In fact, if $x > 0$, then $f_n(x)$ is eventually 0, so $\lim_{n \rightarrow \infty} f_n(x) = 0$; moreover, $f_n(0) = 0$ for all n , so that we certainly have $\lim_{n \rightarrow \infty} f_n(0) = 0$. In other words, $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$ for all x . On the other hand, the integral quickly reveals the strange behavior of this sequence; we have

$$\int_0^1 f_n(x) dx = \frac{1}{2},$$

but

$$\int_0^1 f(x) dx = 0.$$

Thus,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \neq \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx.$$

This particular sequence of functions behaves in a way that we really never imagined when we first considered functions defined by limits. Although it is true that

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad \text{for each } x \text{ in } [0, 1],$$

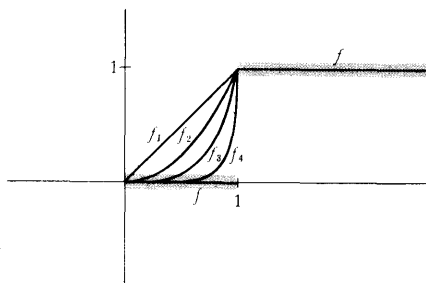


FIGURE 6

the graphs of the functions f_n do not “approach” the graph of f in the sense of lying close to it—if, as in Figure 5, we draw a strip around f of total width 2ϵ (allowing a width of ϵ above and below), then the graphs of f_n do not lie completely within this strip, no matter how large an n we choose. Of course, for each x there is some N such that the point $(x, f_n(x))$ lies in this strip for $n > N$; this assertion just amounts to the fact that $\lim_{n \rightarrow \infty} f_n(x) = f(x)$. But it is necessary to choose larger and larger N 's as x is chosen closer and closer to 0, and no one N will work for all x at once.

The same situation actually occurs, though less blatantly, for each of the other examples given previously. Figure 6 illustrates this point for the sequence

$$f_n(x) = \begin{cases} x^n, & 0 \leq x \leq 1 \\ 1, & x \geq 1. \end{cases}$$

A strip of total width 2ϵ has been drawn around the graph of $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. If $\epsilon < \frac{1}{2}$, this strip consists of two pieces, which contain no points with second

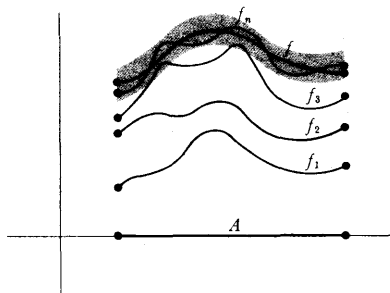


FIGURE 7

coordinate equal to $\frac{1}{2}$; since each function f_n takes on the value $\frac{1}{2}$, the graph of each f_n fails to lie within this strip. Once again, for each point x there is some N such that $(x, f_n(x))$ lies in the strip for $n > N$; but it is not possible to pick one N which works for all x at once.

It is easy to check that precisely the same situation occurs for each of the other examples. In each case we have a function f , and a sequence of functions $\{f_n\}$, all defined on some set A , such that

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad \text{for all } x \text{ in } A.$$

This means that

for all $\varepsilon > 0$, and for all x in A , there is some N such that if $n > N$, then $|f(x) - f_n(x)| < \varepsilon$.

But in each case different N 's must be chosen for different x 's, and in each case it is *not* true that

for all $\varepsilon > 0$ there is some N such that for all x in A , if $n > N$, then $|f(x) - f_n(x)| < \varepsilon$.

Although this condition differs from the first only by a minor displacement of the phrase "for all x in A ," it has a totally different significance. If a sequence $\{f_n\}$ satisfies this second condition, then the graphs of f_n eventually lie close to the graph of f , as illustrated in Figure 7. This condition turns out to be just the one which makes the study of limit functions feasible.

DEFINITION

Let $\{f_n\}$ be a sequence of functions defined on A , and let f be a function which is also defined on A . Then f is called the **uniform limit of $\{f_n\}$ on A** if for every $\varepsilon > 0$ there is some N such that for all x in A ,

$$\text{if } n > N, \text{ then } |f(x) - f_n(x)| < \varepsilon.$$

We also say that $\{f_n\}$ **converges uniformly to f on A** , or that f_n **approaches f uniformly on A** .

As a contrast to this definition, if we know only that

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad \text{for each } x \text{ in } A,$$

then we say that $\{f_n\}$ **converges pointwise to f on A** . Clearly, uniform convergence implies pointwise convergence (but not conversely!).

Evidence for the usefulness of uniform convergence is not at all difficult to amass. Integrals represent a particularly easy topic; Figure 7 makes it almost obvious that if $\{f_n\}$ converges uniformly to f , then the integral of f_n can be made as close to the integral of f as desired. Expressed more precisely, we have the following theorem.

THEOREM 1

Suppose that $\{f_n\}$ is a sequence of functions which are integrable on $[a, b]$, and that $\{f_n\}$ converges uniformly on $[a, b]$ to a function f which is integrable

on $[a, b]$. Then

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

PROOF Let $\varepsilon > 0$. There is some N such that for all $n > N$ we have

$$|f(x) - f_n(x)| < \varepsilon \quad \text{for all } x \text{ in } [a, b].$$

Thus, if $n > N$ we have

$$\begin{aligned} \left| \int_a^b f(x) dx - \int_a^b f_n(x) dx \right| &= \left| \int_a^b [f(x) - f_n(x)] dx \right| \\ &\leq \int_a^b |f(x) - f_n(x)| dx \\ &\leq \int_a^b \varepsilon dx \\ &= \varepsilon(b - a). \end{aligned}$$

Since this is true for any $\varepsilon > 0$, it follows that

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n. \blacksquare$$

The treatment of continuity is only a little more difficult, involving an “ $\varepsilon/3$ -argument,” a three-step estimate of $|f(x) - f(x + h)|$. If $\{f_n\}$ is a sequence of continuous functions which converges uniformly to f , then there is some n such that

$$(1) \quad |f(x) - f_n(x)| < \frac{\varepsilon}{3},$$

$$(2) \quad |f(x + h) - f_n(x + h)| < \frac{\varepsilon}{3}.$$

Moreover, since f_n is continuous, for sufficiently small h we have

$$(3) \quad |f_n(x) - f_n(x + h)| < \frac{\varepsilon}{3}.$$

It will follow from (1), (2), and (3) that $|f(x) - f(x + h)| < \varepsilon$. In order to obtain (3), however, we must restrict the size of $|h|$ in a way that cannot be predicted until n has already been chosen; it is therefore quite essential that there be some fixed n which makes (2) true, no matter how small $|h|$ may be—it is precisely at this point that uniform convergence enters the proof.

THEOREM 2 Suppose that $\{f_n\}$ is a sequence of functions which are continuous on $[a, b]$, and that $\{f_n\}$ converges uniformly on $[a, b]$ to f . Then f is also continuous on $[a, b]$.

PROOF For each x in $[a, b]$ we must prove that f is continuous at x . We will deal only with x in (a, b) ; the cases $x = a$ and $x = b$ require the usual simple modifications.

Let $\varepsilon > 0$. Since $\{f_n\}$ converges uniformly to f on $[a, b]$, there is some n such that

$$|f(y) - f_n(y)| < \frac{\varepsilon}{3} \quad \text{for all } y \text{ in } [a, b].$$

In particular, for all h such that $x + h$ is in $[a, b]$, we have

$$(1) \quad |f(x) - f_n(x)| < \frac{\varepsilon}{3},$$

$$(2) \quad |f(x+h) - f_n(x+h)| < \frac{\varepsilon}{3}.$$

Now f_n is continuous, so there is some $\delta > 0$ such that for $|h| < \delta$ we have

$$(3) \quad |f_n(x) - f_n(x+h)| < \frac{\varepsilon}{3}.$$

Thus, if $|h| < \delta$, then

$$\begin{aligned} |f(x+h) - f(x)| &= |f(x+h) - f_n(x+h) + f_n(x+h) - f_n(x) + f_n(x) - f(x)| \\ &\leq |f(x+h) - f_n(x+h)| + |f_n(x+h) - f_n(x)| + |f_n(x) - f(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

This proves that f is continuous at x . ■

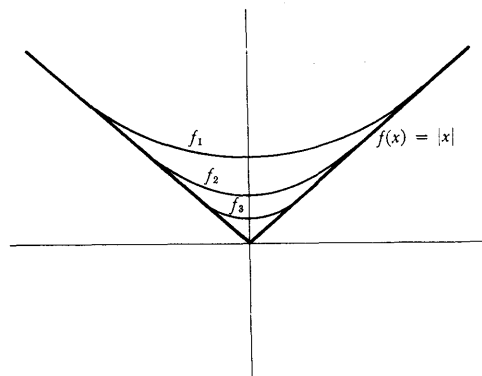


FIGURE 8

After the two noteworthy successes provided by Theorem 1 and Theorem 2, the situation for differentiability turns out to be very disappointing. If each f_n is differentiable, and if $\{f_n\}$ converges uniformly to f , it is still not necessarily true that f is differentiable. For example, Figure 8 shows that there is a sequence of differentiable functions $\{f_n\}$ which converges uniformly to the

function $f(x) = |x|$. Even if f is differentiable, it may not be true that

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x);$$

this is not at all surprising if we reflect that a smooth function can be approximated by very rapidly oscillating functions. For example (Figure 9), if

$$f_n(x) = \frac{1}{n} \sin(n^2 x),$$

then $\{f_n\}$ converges uniformly to the function $f(x) = 0$, but

$$f'_n(x) = n \cos(n^2 x),$$

and $\lim_{n \rightarrow \infty} n \cos(n^2 x)$ does not always exist (for example, it does not exist if $x = 0$).

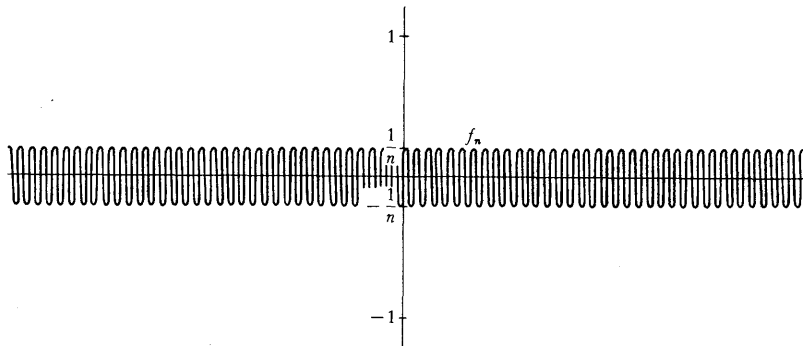


FIGURE 9

Despite such examples, the Fundamental Theorem of Calculus practically guarantees that some sort of theorem about derivatives will be a consequence of Theorem 1; the crucial hypothesis is that $\{f'_n\}$ converge uniformly (to some continuous function).

THEOREM 3 Suppose that $\{f_n\}$ is a sequence of functions which are differentiable on $[a, b]$, and that $\{f_n\}$ converges (pointwise) to f . Suppose, moreover, that $\{f'_n\}$ converges uniformly on $[a, b]$ to some continuous function g . Then f is differentiable and

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x).$$

PROOF Applying Theorem 1 to the interval $[a, x]$, we see that for each x we have

$$\begin{aligned} \int_a^x g &= \lim_{n \rightarrow \infty} \int_a^x f'_n \\ &= \lim_{n \rightarrow \infty} [f_n(x) - f_n(a)] \\ &= f(x) - f(a). \end{aligned}$$

Since g is continuous, it follows that $f'(x) = g(x) = \lim_{x \rightarrow \infty} f_n(x)$ for all x in the interval $[a, b]$. ■

Now that the basic facts about uniform limits have been established, it is clear how to treat functions defined as infinite sums,

$$f(x) = f_1(x) + f_2(x) + f_3(x) + \cdots$$

This equation means that

$$f(x) = \lim_{n \rightarrow \infty} f_1(x) + \cdots + f_n(x);$$

our previous theorems apply when the new sequence

$$f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$$

converges uniformly to f . Since this is the only case we shall ever be interested in, we single it out with a definition.

DEFINITION

The series $\sum_{n=1}^{\infty} f_n$ **converges uniformly** (more formally: the sequence $\{f_n\}$ is **uniformly summable**) **to f on A** , if the sequence

$$f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$$

converges uniformly to f on A .

We can now apply each of Theorems 1, 2, and 3 to uniformly convergent series; the results may be stated in one common corollary.

COROLLARY

Let $\sum_{n=1}^{\infty} f_n$ converge uniformly to f on $[a, b]$.

- (1) If each f_n is continuous on $[a, b]$, then f is continuous on $[a, b]$.
- (2) If f and each f_n is integrable on $[a, b]$, then

$$\int_a^b f = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Moreover, if $\sum_{n=1}^{\infty} f_n$ converges (pointwise) to f on $[a, b]$, and $\sum_{n=1}^{\infty} f_n'$ converges uniformly on $[a, b]$ to some continuous function, then

$$(3) \quad f'(x) = \sum_{n=1}^{\infty} f_n'(x) \quad \text{for all } x \text{ in } [a, b].$$

PROOF

- (1) If each f_n is continuous, then so is each $f_1 + \cdots + f_n$, and f is the uniform limit of the sequence $f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$, so f is continuous by Theorem 2.
- (2) Since $f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$ converges uniformly to f , it follows from Theorem 1 that

$$\begin{aligned}\int_a^b f &= \lim_{n \rightarrow \infty} \int_a^b (f_1 + \cdots + f_n) \\ &= \lim_{n \rightarrow \infty} \left(\int_a^b f_1 + \cdots + \int_a^b f_n \right) \\ &= \sum_{n=1}^{\infty} \int_a^b f_n.\end{aligned}$$

- (3) Each function $f_1 + \cdots + f_n$ is differentiable, with derivative $f_1' + \cdots + f_n'$, and $f_1', f_1' + f_2', f_1' + f_2' + f_3', \dots$ converges uniformly to a continuous function, by hypothesis. It follows from Theorem 3 that

$$\begin{aligned}f'(x) &= \lim_{n \rightarrow \infty} [f_1'(x) + \cdots + f_n'(x)] \\ &= \sum_{n=1}^{\infty} f_n'(x). \blacksquare\end{aligned}$$

At the moment this corollary is not very useful, since it seems quite difficult to predict when the sequence $f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$ will converge uniformly. The most important condition which ensures such uniform convergence is provided by the following theorem; the proof is almost a triviality because of the cleverness with which the very simple hypotheses have been chosen.

THEOREM 4
(THE WEIERSTRASS M-TEST)

Let $\{f_n\}$ be a sequence of functions defined on A , and suppose that $\{M_n\}$ is a sequence of numbers such that

$$|f_n(x)| \leq M_n \quad \text{for all } x \text{ in } A.$$

Suppose moreover that $\sum_{n=1}^{\infty} M_n$ converges. Then for each x in A the series

$\sum_{n=1}^{\infty} f_n(x)$ converges (in fact, it converges absolutely), and $\sum_{n=1}^{\infty} f_n$ converges uniformly on A to the function

$$f(x) = \sum_{n=1}^{\infty} f_n(x).$$

PROOF For each x in A the series $\sum_{n=1}^{\infty} |f_n(x)|$ converges, by the comparison test;

consequently $\sum_{n=1}^{\infty} f_n(x)$ converges (absolutely). Moreover, for all x in A we have

$$\begin{aligned} |f(x) - [f_1(x) + \cdots + f_N(x)]| &= \left| \sum_{n=N+1}^{\infty} f_n(x) \right| \\ &\leq \sum_{n=N+1}^{\infty} |f_n(x)| \\ &\leq \sum_{n=N+1}^{\infty} M_n. \end{aligned}$$

Since $\sum_{n=1}^{\infty} M_n$ converges, the number $\sum_{n=N+1}^{\infty} M_n$ can be made as small as desired, by choosing N sufficiently large. ■

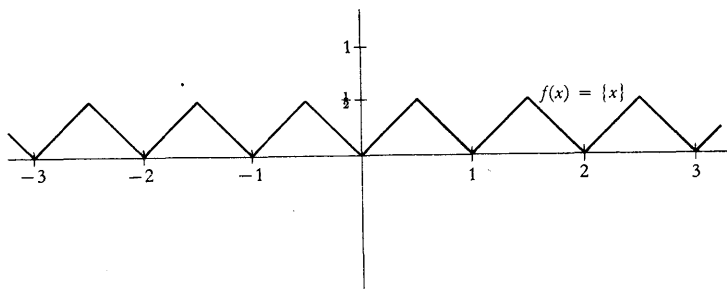


FIGURE 10

The following sequence $\{f_n\}$ illustrates a simple application of the Weierstrass M -test. Let $\{x\}$ denote the distance from x to the nearest integer (the graph of $f(x) = \{x\}$ is illustrated in Figure 10). Now define

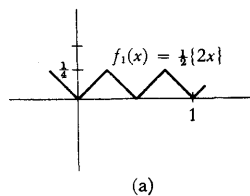
$$f_n(x) = \frac{1}{10^n} \{10^n x\}.$$

The functions f_1 and f_2 are shown in Figure 11 (but to make the drawings simpler, 10^n has been replaced by 2^n). This sequence of functions has been defined so that the Weierstrass M -test automatically applies: clearly

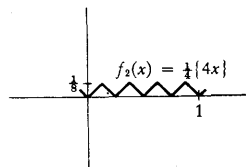
$$|f_n(x)| \leq \frac{1}{10^n} \quad \text{for all } x,$$

and $\sum_{n=1}^{\infty} 1/10^n$ converges. Thus $\sum_{n=1}^{\infty} f_n$ converges uniformly; since each f_n is continuous, the corollary implies that the function

$$f(x) = \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} \frac{1}{10^n} \{10^n x\}$$



(a)



(b)

FIGURE 11

is also continuous. Figure 12 shows the graph of the first few partial sums $f_1 + \cdots + f_n$. As n increases, the graphs become harder and harder to draw, and the infinite sum $\sum_{n=1}^{\infty} f_n$ is quite undrawable, as shown by the following theorem (included mainly as an interesting sidelight, to be skipped if you find the going too rough).

THEOREM 5 The function

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{10^n} \{10^n x\}$$

is continuous everywhere and differentiable nowhere!

PROOF We have just shown that f is continuous; this is the only part of the proof which uses uniform convergence. We will prove that f is not differentiable at a , for any a , by the straightforward method of exhibiting a particular sequence $\{h_m\}$ approaching 0 for which

$$\lim_{m \rightarrow \infty} \frac{f(a + h_m) - f(a)}{h_m}$$

does not exist. It obviously suffices to consider only those numbers a satisfying $0 < a \leq 1$.

Suppose that the decimal expansion of a is

$$a = 0.a_1a_2a_3a_4 \dots$$

Let $h_m = 10^{-m}$ if $a_m \neq 4$ or 9, but let $h_m = -10^{-m}$ if $a_m = 4$ or 9 (the reason for these two exceptions will appear soon). Then

$$\begin{aligned} \frac{f(a + h_m) - f(a)}{h_m} &= \sum_{n=1}^{\infty} \frac{1}{10^n} \cdot \frac{\{10^n(a + h_m)\} - \{10^n a\}}{\pm 10^{-m}} \\ &= \sum_{n=1}^{\infty} \pm 10^{m-n} [\{10^n(a + h_m)\} - \{10^n a\}]. \end{aligned}$$

This infinite series is really a finite sum, because if $n \geq m$, then $10^n h_m$ is an integer, so

$$\{10^n(a + h_m)\} - \{10^n a\} = 0.$$

On the other hand, for $n < m$ we can write

$$\begin{aligned} 10^n a &= \text{integer} + 0.a_{n+1}a_{n+2}a_{n+3} \dots a_m \dots \\ 10^n(a + h_m) &= \text{integer} + 0.a_{n+1}a_{n+2}a_{n+3} \dots (a_m \pm 1) \dots \end{aligned}$$

(in order for the second equation to be true it is essential that we choose $h_m = -10^{-m}$ when $a_m = 9$). Now suppose that

$$0.a_{n+1}a_{n+2}a_{n+3} \dots a_m \dots \leq \frac{1}{2}.$$

Then we also have

$$0.a_{n+1}a_{n+2}a_{n+3} \dots (a_m \pm 1) \dots \leq \frac{1}{2}$$

(in the special case $m = n + 1$ the second equation is true because we chose $h_m = -10^{-m}$ when $a_m = 4$). This means that

$$\{10^n(a + h_m)\} - \{10^n a\} = \pm 10^{n-m},$$

and exactly the same equation can be derived when $0.a_{n+1}a_{n+2}a_{n+3} \dots > \frac{1}{2}$. Thus, for $n < m$ we have

$$10^{m-n}[\{10^n(a + h_m)\} - \{10^n a\}] = \pm 1.$$

In other words,

$$\frac{f(a + h_m) - f(a)}{h_m}$$

is the sum of $m - 1$ numbers, each of which is ± 1 . Now adding $+1$ or -1 to a number changes it from odd to even, and vice versa. The sum of $m - 1$ numbers each ± 1 is therefore an *even integer* if m is odd, and an *odd integer* if m is even. Consequently the sequence of ratios

$$\frac{f(a + h_m) - f(a)}{h_m}$$

cannot possibly converge, since it is a sequence of integers which are alternately odd and even. ■

In addition to its role in the previous theorem, the Weierstrass *M*-test is an ideal tool for analyzing functions which are very well behaved. We will give special attention to functions of the form

$$f(x) = \sum_{n=0}^{\infty} a_n(x - a)^n,$$

which can also be described by the equation

$$f(x) = \sum_{n=0}^{\infty} f_n(x),$$

for $f_n(x) = a_n(x - a)^n$. Such an infinite sum, of functions which depend only on powers of $(x - a)$, is called a **power series centered at a** . For the sake of simplicity, we will usually concentrate on power series centered at 0,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

One especially important group of power series are those of the form

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

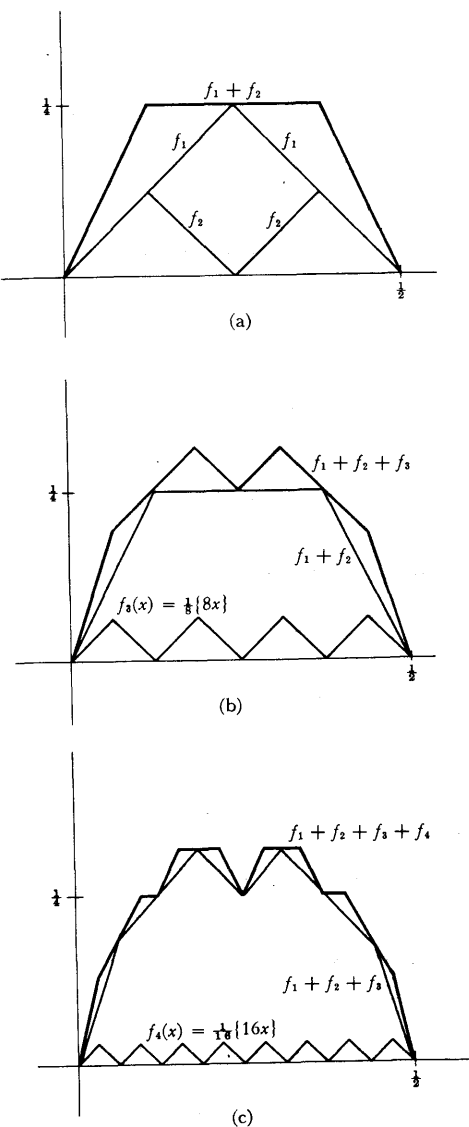


FIGURE 12

where f is some function which has derivatives of all orders at a ; this series is called the **Taylor series for f at a** . Of course, it is not necessarily true that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n;$$

this equation holds only when the remainder terms satisfy $\lim_{n \rightarrow \infty} R_{n,a}(x) = 0$.

We already know that a power series $\sum_{n=0}^{\infty} a_n x^n$ does not necessarily converge for all x . For example the power series

$$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

converges only for $|x| \leq 1$, while the power series

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots$$

converges only for $-1 < x \leq 1$. It is even possible to produce a power series which converges only for $x = 0$. For example, the power series

$$\sum_{n=0}^{\infty} n! x^n$$

does not converge for $x \neq 0$; indeed, the ratios

$$\frac{(n+1)!(x^{n+1})}{n!x^n} = (n+1)x$$

are unbounded for any $x \neq 0$. If a power series $\sum_{n=0}^{\infty} a_n x^n$ does converge for some $x_0 \neq 0$ however, then a great deal can be said about the series $\sum_{n=0}^{\infty} a_n x^n$ for $|x| < |x_0|$.

THEOREM 6 Suppose that the series

$$f(x_0) = \sum_{n=0}^{\infty} a_n x_0^n$$

converges, and let a be any number with $0 < a < |x_0|$. Then on $[-a, a]$ the series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges uniformly (and absolutely). Moreover, the same is true for the

series

$$g(x) = \sum_{n=1}^{\infty} na_n x^{n-1}.$$

Finally, f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} na_n x^{n-1}$$

for all x with $|x| < |x_0|$.

PROOF Since $\sum_{n=0}^{\infty} a_n x_0^n$ converges, the terms $a_n x_0^n$ approach 0. Hence they are surely bounded: there is some number M such that

$$|a_n x_0^n| = |a_n| \cdot |x_0^n| \leq M \quad \text{for all } n.$$

Now if x is in $[-a, a]$, then $|x| \leq |a|$, so

$$\begin{aligned} |a_n x^n| &= |a_n| \cdot |x^n| \\ &\leq |a_n| \cdot |a^n| \\ &= |a_n| \cdot |x_0|^n \cdot \left| \frac{a}{x_0} \right|^n \quad (\text{this is the clever step}) \\ &\leq M \left| \frac{a}{x_0} \right|^n. \end{aligned}$$

But $|a/x_0| < 1$, so the (geometric) series

$$\sum_{n=0}^{\infty} M \left| \frac{a}{x_0} \right|^n = M \sum_{n=0}^{\infty} \left| \frac{a}{x_0} \right|^n$$

converges. Choosing $M \cdot |a/x_0|^n$ as the number M_n in the Weierstrass M -test, it follows that $\sum_{n=0}^{\infty} a_n x^n$ converges uniformly on $[-a, a]$.

To prove the same assertion for $g(x) = \sum_{n=1}^{\infty} na_n x^{n-1}$ notice that

$$\begin{aligned} |na_n x^{n-1}| &= n|a_n| \cdot |x^{n-1}| \\ &\leq n|a_n| \cdot |a^{n-1}| \\ &= \frac{|a_n|}{|a|} \cdot |x_0|^n n \left| \frac{a}{x_0} \right|^n \\ &\leq \frac{M}{|a|} n \left| \frac{a}{x_0} \right|^n. \end{aligned}$$

Since $|a/x_0| < 1$, the series

$$\sum_{n=1}^{\infty} \frac{M}{|a|} n \left| \frac{a}{x_0} \right|^n = \frac{M}{|a|} \sum_{n=1}^{\infty} n \left| \frac{a}{x_0} \right|^n$$

converges (this fact was proved in Chapter 22 as an application of the ratio

test). Another appeal to the Weierstrass M -test proves that $\sum_{n=1}^{\infty} na_n x^{n-1}$ converges uniformly on $[-a, a]$.

Finally, our corollary proves, first that g is continuous, and then that

$$f'(x) = g(x) = \sum_{n=1}^{\infty} na_n x^{n-1} \quad \text{for } x \text{ in } [-a, a].$$

Since we could have chosen any number a with $0 < a < |x_0|$, this result holds for all x with $|x| < |x_0|$. ■

Theorem 6, applied to the infinite series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots,$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots,$$

yields precisely the results which are expected. Each of these converges for any x_0 , hence the conclusions of Theorem 6 apply for any x :

$$\sin'(x) = 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \cdots = \cos x,$$

$$\cos'(x) = -\frac{2x}{2!} + \frac{4x^3}{4!} - \frac{6x^5}{6!} + \cdots = -\sin x,$$

$$\exp'(x) = 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \cdots = \exp(x).$$

For the functions $\arctan$ and $f(x) = \log(1+x)$ the situation is only slightly more complicated. Since the series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

converges for $x_0 = 1$, it also converges for $|x| < 1$, and

$$\arctan'(x) = 1 - x^2 + x^4 - x^6 + \cdots = \frac{1}{1+x^2} \quad \text{for } |x| < 1.$$

In this case, the series happens to converge for $x = -1$ also. However, the formula for the derivative is not correct for $x = 1$ or $x = -1$; indeed the series

$$1 - x^2 + x^4 - x^6 + \cdots$$

diverges for $x = 1$ and $x = -1$. Notice that this does not contradict Theorem 6, which proves that the derivative is given by the expected formula only for $|x| < |x_0|$.

Since the series

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$$

converges for $x_0 = 1$, it also converges for $|x| < 1$, and

$$\frac{1}{1+x} = \log'(1+x) = 1 - x + x^2 - x^3 + \cdots \quad \text{for } |x| < 1.$$

In this case, the original series does not converge for $x = -1$; moreover, the differentiated series does not converge for $x = 1$.

All the considerations which apply to a power series will automatically apply to its derivative, at the points where the derivative is represented by a power series. If

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges for all x in some interval $(-R, R)$, then Theorem 6 implies that

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

for all x in $(-R, R)$. Applying Theorem 6 once again we find that

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2},$$

and proceeding by induction we find that

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n x^{n-k}.$$

Thus, a function defined by a power series which converges in some interval $(-R, R)$ is automatically infinitely differentiable in that interval. Moreover, the previous equation implies that

$$f^{(k)}(0) = k! a_k,$$

so that

$$a_k = \frac{f^{(k)}(0)}{k!}.$$

In other words, a convergent power series centered at 0 is always the Taylor series at 0 of the function which it defines.

On this happy note we could easily end our study of power series and Taylor series. A careful assessment of our situation will reveal some unexplained facts however.

The Taylor series of $\sin$, $\cos$, and $\exp$ are as satisfactory as we could desire; they converge for all x , and can be differentiated term-by-term for all x . The Taylor series of the function $f(x) = \log(1+x)$ is slightly less pleasing, because

it converges only for $-1 < x \leq 1$, but this deficiency is a necessary consequence of the basic nature of power series. If the Taylor series for f converged for any x_0 with $|x_0| > 1$, then it would converge on the interval $(-|x_0|, |x_0|)$, and on this interval the function which it defines would be differentiable, and thus continuous. But this is impossible, since it is unbounded on the interval $(-1, 1)$, where it equals $\log(1 + x)$.

The Taylor series for arctan is more difficult to comprehend—there seems to be no possible excuse for the refusal of this series to converge when $|x| > 1$. This mysterious behavior is exemplified even more strikingly by the function $f(x) = 1/(1 + x^2)$, an infinitely differentiable function which is the next best thing to a polynomial function. The Taylor polynomial of f is given by

$$f(x) = \frac{1}{1 + x^2} = 1 - x^2 + x^4 - x^6 + x^8 - \cdots$$

If $|x| \geq 1$ the Taylor series does not converge at all. Why? What unseen obstacle prevents the Taylor series from extending past 1 and -1 ? Asking this sort of question is always dangerous, since we may have to settle for an unsympathetic answer: it happens because it happens—that's the way things are! In this case there does happen to be an explanation, but this explanation is impossible to give at the present time; although the question is about real numbers, it can be answered intelligently only when placed in a broader context. It will therefore be necessary to devote two chapters to quite new material before completing our discussion of Taylor series in Chapter 26.

PROBLEMS

- For each of the following sequences $\{f_n\}$, determine the pointwise limit of $\{f_n\}$ (if it exists) on the indicated interval, and decide whether $\{f_n\}$ converges uniformly to this function.

(i) $f_n(x) = \sqrt[n]{x}$, on $[0, 1]$.

(ii) $f_n(x) = \begin{cases} 0, & x \leq n \\ x - n, & x \geq n \end{cases}$ on $[a, b]$, and on $\mathbf{R}$.

(iii) $f_n(x) = \frac{e^x}{x^n}$, on $(1, \infty)$.

(iv) $f(x) = e^{-nx^2}$, on $[-1, 1]$.

(v) $f(x) = \frac{e^{-x^2}}{n}$, on $\mathbf{R}$.

- Find the Taylor series at 0 for each of the following functions.

(i) $f(x) = \frac{1}{x - a}$, $a \neq 0$.

(ii) $f(x) = \log(x - a)$, $a \neq 0$.

$$(iii) f(x) = \frac{1}{\sqrt{1-x}} = (1-x)^{-1/2}. \quad (\text{Use Problem 19-6.})$$

$$(iv) f(x) = \frac{1}{\sqrt{1-x^2}}.$$

$$(v) f(x) = \arcsin x.$$

3. Find each of the following infinite sums.

$$(i) 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots$$

$$(ii) 1 - x^3 + x^6 - x^9 + \cdots. \text{ Hint: What is } 1 - x + x^2 - x^3 + \cdots ?$$

$$(iii) \frac{x^2}{2} - \frac{x^3}{3 \cdot 2} + \frac{x^4}{4 \cdot 3} - \frac{x^5}{5 \cdot 4} + \cdots \text{ for } |x| < 1. \text{ Hint: Differentiate.}$$

$$*(iv) \left. \frac{1}{2 \cdot 1} - \frac{1}{3 \cdot 2} + \frac{1}{4 \cdot 3} - \frac{1}{5 \cdot 4} + \cdots \right\} (\text{See Problem 10.})$$

$$*(v) 1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \cdots$$

4. If $f(x) = (\sin x)/x$ for $x \neq 0$ and $f(0) = 1$, find $f^{(k)}(0)$. Hint: Find the power series for f .

5. In this problem we deduce the binomial series $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$, $|x| < 1$ without all the work of Problem 22-16, although we will use a fact established in part (a) of that problem—the series $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$ does converge for $|x| < 1$.

(a) Prove that $(1+x)f'(x) = \alpha f(x)$ for $|x| < 1$.

(b) Now show that any function f satisfying part (a) is of the form $f(x) = c(1+x)^\alpha$ for some constant c , and use this fact to establish the binomial series. Hint: Consider $g(x) = f(x)/(1+x)^\alpha$.

6. (a) Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges for all x in some interval $(-R, R)$ and that $f(x) = 0$ for all x in $(-R, R)$. Prove that each $a_n = 0$. (If you remember the formula for a_n this is easy.)
 (b) Suppose we know only that $f(x_n) = 0$ for some sequence $\{x_n\}$ with $\lim_{n \rightarrow \infty} x_n = 0$. Prove again that each $a_n = 0$. Hint: First show that $f(0) = a_0 = 0$; then that $f'(0) = a_1 = 0$, etc.

This result shows that if $f(x) = e^{-1/x^2} \sin 1/x$ for $x \neq 0$, then f cannot possibly be written as a power series. It also shows that a function defined by a power series cannot be 0 for $x \leq 0$ but nonzero for $x > 0$ —thus a power series cannot describe the motion of a particle which has remained at rest until time 0, and then begins to move!

- (c) Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ converge for all x in some interval containing 0 and that $f(t_m) = g(t_m)$ for some sequence $\{t_m\}$ converging to 0. Show that $a_n = b_n$ for each n . In particular, a function can have only one representation as a power series centered at 0.
7. Prove that if $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is an even function, then $a_n = 0$ for n odd, and if f is an odd function, then $a_n = 0$ for n even.
- *8. Recall that the Fibonacci sequence $\{a_n\}$ is defined by $a_1 = a_2 = 1$, $a_{n+1} = a_n + a_{n-1}$.
- (a) Show that $a_{n+1}/a_n \leq 2$.
- (b) Let

$$f(x) = \sum_{n=1}^{\infty} a_n x^{n-1} = 1 + x + 2x^2 + 3x^3 + \cdots$$

Use the ratio test to prove that $f(x)$ converges if $|x| < 1/2$.

- (c) Prove that if $|x| < 1/2$, then

$$f(x) = \frac{-1}{x^2 + x - 1}.$$

Hint: This equation can be written $f(x) - xf(x) - x^2f(x) = 1$.

- (d) Use the partial fraction decomposition for $1/(x^2 + x - 1)$, and the power series for $1/(x - a)$, to obtain another power series for f .
- (e) It follows from Problem 6 that the two power series obtained for f must be the same. Use this fact to show that

$$a_n = \frac{\left(\frac{1 + \sqrt{5}}{2}\right)^n - \left(\frac{1 - \sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

9. Show that the power series for $f(x) = \log(1 - x)$ converges only for $-1 \leq x < 1$, and that the power series for $g(x) = \log[(1 + x)/(1 - x)]$ converges only for x in $(-1, 1)$.
- *10. Suppose that $\sum_{n=0}^{\infty} a_n$ converges. We know that the series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ must converge uniformly on $[-a, a]$ for $0 < a < 1$, but it may not converge uniformly on $[-1, 1]$; in fact, it may not even converge at the point -1 (for example, if $f(x) = \log(1 + x)$). However, a beautiful theorem of Abel shows that the series *does* converge uniformly on $[0, 1]$. Consequently, f is continuous on $[0, 1]$ and, in particular, $\sum_{n=0}^{\infty} a_n =$

$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n$. Prove Abel's Theorem by noticing that if $|a_m + \cdots + a_n| < \epsilon$, then $|a_m x^m + \cdots + a_n x^n| < \epsilon$, by Abel's Lemma (Problem 21-20).

11. A sequence $\{a_n\}$ is called **Abel summable** if $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n$ exists; Problem 10 shows that a summable sequence is necessarily Abel summable. Find a sequence which is Abel summable, but which is not summable. Hint: Look over the list of Taylor series until you find one which does not converge at 1, even though the function it represents is continuous at 1.
12. In Theorem 3 we assumed only that $\{f_n\}$ converges pointwise to f . Show that the remaining hypotheses ensure that $\{f_n\}$ actually converges uniformly to f .
13. (a) Suppose that $\{f_n\}$ is a sequence of bounded (not necessarily continuous) functions on $[a, b]$ which converge uniformly to f on $[a, b]$. Prove that f is bounded on $[a, b]$.
(b) Find a sequence of continuous functions on $[a, b]$ which converge pointwise to an unbounded function on $[a, b]$.
- *14. Suppose that f is differentiable. Prove that the function f' is the pointwise limit of a sequence of continuous functions. (Since we already know examples of discontinuous derivatives, this provides another example where the pointwise limit of continuous functions is not continuous.)
15. Find a sequence of integrable functions $\{f_n\}$ which converges to the (nonintegrable) function f that is 1 on the rationals and 0 on the irrationals. Hint: Each f_n will be 0 except at a few points.
16. This problem outlines a completely different approach to the integral; consequently, it is unfair to use any facts about integrals learned previously.
 - (a) Let s be a step function on $[a, b]$, so that s is constant on (t_{i-1}, t_i) for some partition $\{t_0, \dots, t_n\}$ of $[a, b]$. Define $\int_a^b s$ as $\sum_{i=1}^n s_i(t_i - t_{i-1})$ where s_i is the (constant) value of s on (t_{i-1}, t_i) . Show that this definition does not depend on the partition $\{t_0, \dots, t_n\}$.
 - (b) A function f is called a **regulated** function on $[a, b]$ if it is the uniform limit of a sequence of step functions $\{s_n\}$ on $[a, b]$. Show that in this case there is, for every $\epsilon > 0$, some N such that for $m, n > N$ we have $|s_n(x) - s_m(x)| < \epsilon$ for all x in $[a, b]$.
 - (c) Show that the sequence of numbers $\{\int_a^b s_n\}$ will be a Cauchy sequence.
 - (d) Suppose that $\{t_n\}$ is another sequence of step functions on $[a, b]$ which converges uniformly to f . Show that for every $\epsilon > 0$ there is an N such that for $n > N$ we have $|s_n(x) - t_n(x)| < \epsilon$ for x in $[a, b]$.
 - (e) Conclude that $\lim_{n \rightarrow \infty} \int_a^b s_n = \lim_{n \rightarrow \infty} \int_a^b t_n$. This means that we can

define $\int_a^b f$ to be $\lim_{n \rightarrow \infty} \int_a^b s_n$ for any sequence of step functions $\{s_n\}$ converging uniformly to f . The only remaining question is: Which functions are regulated? Here is a partial answer.

- *(f) Prove that a continuous function is regulated. Hint: To find a step function s on $[a, b]$ with $|f(x) - s(x)| < \varepsilon$ for all x in $[a, b]$, consider all y for which there is such a step function on $[a, y]$.
- *17. Find a sequence $\{f_n\}$ approaching f uniformly on $[0, 1]$ for which $\lim_{n \rightarrow \infty} (\text{length of } f_n \text{ on } [0, 1]) \neq \text{length of } f \text{ on } [0, 1]$. (Length is defined in Problem 15-32, but the simplest example will involve functions the length of whose graphs will be obvious.)

CHAPTER 24 COMPLEX NUMBERS

With the exception of the last few paragraphs of the previous chapter, this book has presented unremitting propaganda for the real numbers. Nevertheless, the real numbers do have a great deficiency—not every polynomial function has a root. The simplest and most notable example is the fact that no number x can satisfy $x^2 + 1 = 0$. This deficiency is so severe that long ago mathematicians felt the need to “invent” a number i with the property that $i^2 + 1 = 0$. For a long time the status of the “number” i was quite mysterious: since there is no number x satisfying $x^2 + 1 = 0$, it is nonsensical to say “let i be the number satisfying $i^2 + 1 = 0$.” Nevertheless, admission of the “imaginary” number i to the family of numbers seemed to simplify greatly many algebraic computations, especially when “complex numbers” $a + bi$ (for a and b in $\mathbf{R}$) were allowed, and all the laws of arithmetical computation enumerated in Chapter 1 were assumed to be valid. For example, every quadratic equation

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

can be solved formally to give

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{or} \quad x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

If $b^2 - 4ac \geq 0$, these formulas give correct solutions; when complex numbers are allowed the formulas seem to make sense in all cases. For example, the equation

$$x^2 + x + 1 = 0$$

has no real root, since

$$x^2 + x + 1 = (x + \frac{1}{2})^2 + \frac{3}{4} > 0, \quad \text{for all } x.$$

But the formula for the roots of a quadratic equation suggest the “solutions”

$$x = \frac{-1 + \sqrt{-3}}{2} \quad \text{and} \quad x = \frac{-1 - \sqrt{-3}}{2};$$

if we understand $\sqrt{-3}$ to mean $\sqrt{3 \cdot (-1)} = \sqrt{3} \cdot \sqrt{-1} = \sqrt{3}i$, then these numbers would be

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}i \quad \text{and} \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

It is not hard to check that these, as yet purely formal, numbers do indeed

satisfy the equation

$$x^2 + x + 1 = 0.$$

It is even possible to “solve” quadratic equations whose coefficients are themselves complex numbers. For example, the equation

$$x^2 + x + 1 + i = 0$$

ought to have the solutions

$$x = \frac{-1 \pm \sqrt{1 - 4(1+i)}}{2} = \frac{-1 \pm \sqrt{-3 - 4i}}{2},$$

where the symbol $\sqrt{-3 - 4i}$ means a complex number $\alpha + \beta i$ whose square is $-3 - 4i$. In order to have

$$(\alpha + \beta i)^2 = \alpha^2 - \beta^2 + 2\alpha\beta i = -3 - 4i$$

we need

$$\begin{aligned}\alpha^2 - \beta^2 &= -3, \\ 2\alpha\beta &= -4.\end{aligned}$$

These two equations can easily be solved for real α and β ; in fact, there are two possible solutions:

$$\begin{aligned}\alpha &= 1 & \text{and} & & \alpha &= -1 \\ \beta &= -2 & & & \beta &= 2.\end{aligned}$$

Thus the two “square roots” of $-3 - 4i$ are $1 - 2i$ and $-1 + 2i$. There is no reasonable way to decide which one of these should be called $\sqrt{-3 - 4i}$, and which $-\sqrt{-3 - 4i}$; the conventional usage of $\sqrt{x}$ makes sense only for real $x \geq 0$, in which case $\sqrt{x}$ denotes the (real) nonnegative root. For this reason, the solution

$$x = \frac{-1 \pm \sqrt{-3 - 4i}}{2}$$

must be understood as an abbreviation for:

$$x = \frac{-1 + r}{2}, \quad \text{where } r \text{ is one of the square roots of } -3 - 4i.$$

With this understanding we arrive at the solutions

$$\begin{aligned}x &= \frac{-1 + 1 - 2i}{2} = -i, \\ x &= \frac{-1 - 1 + 2i}{2} = -1 + i;\end{aligned}$$

as you can easily check, these numbers do provide formal solutions for the equation

$$x^2 + x + 1 + i = 0.$$

For cubic equations complex numbers are equally useful. Every cubic equation

$$ax^3 + bx^2 + cx + d = 0 \quad (a \neq 0)$$

with real coefficients a, b, c , and d , has, as we know, a real root α , and if we divide $ax^3 + bx^2 + cx + d$ by $x - \alpha$ we obtain a second-degree polynomial whose roots are the other roots of $ax^3 + bx^2 + cx + d = 0$; the roots of this second-degree polynomial may be complex numbers. Thus a cubic equation will have either three real roots or one real root and 2 complex roots. The existence of the real root is guaranteed by our theorem that every odd degree equation has a real root, but it is not really necessary to appeal to this theorem (which is of no use at all if the coefficients are complex); in the case of a cubic equation we can, with sufficient cleverness, actually find a formula for all the roots. The following derivation is presented not only as an interesting illustration of the ingenuity of early mathematicians, but as further evidence for the importance of complex numbers (whatever they may be).

To solve the most general cubic equation, it obviously suffices to consider only equations of the form

$$x^3 + bx^2 + cx + d = 0.$$

It is even possible to eliminate the term involving x^2 , by a fairly straightforward manipulation. If we let

$$x = y - \frac{b}{3},$$

then

$$x^3 = y^3 - by^2 + \frac{b^2y}{3} - \frac{b^3}{27},$$

$$x^2 = y^2 - \frac{2by}{3} + \frac{b^2}{9},$$

so

$$\begin{aligned} 0 &= x^3 + bx^2 + cx + d \\ &= \left(y^3 - by^2 + \frac{b^2y}{3} - \frac{b^3}{27}\right) + \left(by^2 - \frac{2b^2y}{3} + \frac{b^3}{9}\right) + \left(cy - \frac{bc}{3}\right) + d \\ &= y^3 + \left(\frac{b^2}{3} - \frac{2b^2}{3} + c\right)y + \left(\frac{b^3}{9} - \frac{b^3}{27} - \frac{bc}{3} + d\right). \end{aligned}$$

The right-hand side now contains no term with y^2 . If we can solve the equation for y we can find x ; this shows that it suffices to consider in the first place only equations of the form

$$x^3 + px + q = 0.$$

In the special case $p = 0$ we obtain the equation $x^3 = -q$. We shall see later on that every complex number does have a cube root, in fact it has three, so that this equation has three solutions. The case $p \neq 0$, on the other hand,

requires quite an ingenious step. Let

$$(*) \quad x = w - \frac{p}{3w} \quad \left(w = \frac{p}{1 - 3x} \neq 0 \right).$$

Then

$$\begin{aligned} 0 &= x^3 + px + q = \left(w - \frac{p}{3w} \right)^3 + p \left(w - \frac{p}{3w} \right) + q \\ &= w^3 - \frac{3w^2p}{3w} + \frac{3wp^2}{9w^2} - \frac{p^3}{27w^3} + pw - \frac{p^2}{3w} + q \\ &= w^3 - \frac{p^3}{27w^3} + q. \end{aligned}$$

This equation can be written

$$27(w^3)^2 + 27q(w^3) - p^3 = 0,$$

which is a quadratic equation in w^3 (!).

Thus

$$\begin{aligned} w^3 &= \frac{-27q \pm \sqrt{(27)^2q^2 + 4 \cdot 27p^3}}{2 \cdot 27} \\ &= -\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}. \end{aligned}$$

Remember that this really means:

$$w^3 = -\frac{q}{2} + r, \quad \text{where } r \text{ is a square root of } \frac{q^2}{4} + \frac{p^3}{27}.$$

We can therefore write

$$w = \sqrt[3]{-\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}};$$

this equation means that w is some cube root of $-q/2 + r$, where r is some square root of $q^2/4 + p^3/27$. This allows six possibilities for w , but when these are substituted into (*), yielding

$$x = \sqrt[3]{-\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} - \frac{p}{3 \cdot \sqrt[3]{-\frac{q}{2} \pm \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}},$$

it turns out that only 3 different values for x will be obtained! An even more surprising feature of this solution arises when we consider a cubic equation all of whose roots are real; the formula derived above may still involve complex numbers in an essential way. For example, the roots of

$$x^3 - 15x - 4 = 0$$

are 4, $-2 + \sqrt{3}$, and $-2 - \sqrt{3}$. On the other hand, the formula derived

above (with $p = -15$, $q = -4$) gives as one solution

$$\begin{aligned} x &= \sqrt[3]{2 + \sqrt{4 - 125}} - \frac{-15}{3 \cdot \sqrt[3]{2 + \sqrt{4 - 125}}} \\ &= \sqrt[3]{2 + 11i} + \frac{15}{3 \cdot \sqrt[3]{2 + 11i}}. \end{aligned}$$

Now,

$$\begin{aligned} (2 + i)^3 &= 2^3 + 3 \cdot 2^2 i + 3 \cdot 2 \cdot i^2 + i^3 \\ &= 8 + 12i - 6 - i \\ &= 2 + 11i, \end{aligned}$$

so one of the cube roots of $2 + 11i$ is $2 + i$. Thus, for one solution of the equation we obtain

$$\begin{aligned} x &= 2 + i + \frac{15}{6 + 3i} \\ &= 2 + i + \frac{15}{6 + 3i} \cdot \frac{6 - 3i}{6 - 3i} \\ &= 2 + i + \frac{90 - 45i}{36 + 9} \\ &= 4(!). \end{aligned}$$

The other roots can also be found if the other cube roots of $2 + 11i$ are known. The fact that even one of these real roots is obtained from an expression which depends on complex numbers is impressive enough to suggest that the use of complex numbers cannot be entirely nonsense. As a matter of fact, the formulas for the solutions of the quadratic and cubic equations can be interpreted entirely in terms of real numbers.

Suppose we agree, for the moment, to write all complex numbers as $a + bi$, writing the real number a as $a + 0i$ and the number i as $0 + 1i$. The laws of ordinary arithmetic and the relation $i^2 = -1$ show that

$$\begin{aligned} (a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi) \cdot (c + di) &= (ac - bd) + (ad + bc)i. \end{aligned}$$

Thus, an equation like

$$(1 + 2i) \cdot (3 + 1i) = 1 + 7i$$

may be regarded as simply an abbreviation for the *two* equations

$$\begin{aligned} 1 \cdot 3 - 2 \cdot 1 &= 1, \\ 1 \cdot 1 + 2 \cdot 3 &= 7. \end{aligned}$$

The solution of the quadratic equation $ax^2 + bx + c = 0$ with real coefficients could be paraphrased as follows:

$$\text{If } \begin{cases} u^2 - v^2 = b^2 - 4ac, \\ uv = 0, \end{cases} \quad (\text{i.e., if } (u + vi)^2 = b^2 - 4ac),$$

$$\text{then } \begin{cases} a \left[\left(\frac{-b+u}{2a} \right)^2 - \left(\frac{v}{2a} \right)^2 \right] + b \left[\frac{-b+u}{2a} \right] + c = 0, \\ a \left[2 \left(\frac{-b+u}{2a} \right) \left(\frac{v}{2a} \right) \right] + b \left[\frac{v}{a} \right] = 0, \end{cases}$$

$$\left(\text{i.e., then } a \left(\frac{-b+u+vi}{2a} \right)^2 + b \left(\frac{-b+u+vi}{2a} \right) + c = 0 \right).$$

It is not very hard to check this assertion about real numbers without writing down a single “ i ,” but the complications of the statement itself should convince you that equations about complex numbers are worthwhile as abbreviations for pairs of equations about real numbers. (If you are still not convinced, try paraphrasing the solution of the cubic equation.) If we really intend to use complex numbers consistently, however, it is going to be necessary to present some reasonable definition.

One possibility has been implicit in this whole discussion. All mathematical properties of a complex number $a + bi$ are determined completely by the real numbers a and b ; any mathematical object with this same property may reasonably be used to define a complex number. The obvious candidate is the ordered pair (a, b) of real numbers; we shall accordingly *define* a complex number to be a pair of real numbers, and likewise *define* what addition and multiplication of complex numbers is to mean.

DEFINITION

A **complex number** is an ordered pair of real numbers; if $z = (a, b)$ is a complex number, then a is called the **real part** of z , and b is called the **imaginary part** of z . The set of all complex numbers is denoted by **C**. If (a, b) and (c, d) are two complex numbers we define

$$\begin{aligned} (a, b) + (c, d) &= (a + c, b + d) \\ (a, b) \cdot (c, d) &= (a \cdot c - b \cdot d, a \cdot d + b \cdot c). \end{aligned}$$

(The $+$ and $\cdot$ appearing on the left side are new symbols being defined, while the $+$ and $\cdot$ appearing on the right side are the familiar addition and multiplication for real numbers.)

When complex numbers were first introduced, it was understood that real numbers were, in particular, complex numbers; if our definition is taken seriously this is not true—a real number is not a pair of real numbers, after all. This difficulty is only a minor annoyance, however. Notice that

$$\begin{aligned} (a, 0) + (b, 0) &= (a + b, 0 + 0) = (a + b, 0), \\ (a, 0) \cdot (b, 0) &= (a \cdot b - 0 \cdot 0, a \cdot 0 + 0 \cdot b) = (a \cdot b, 0); \end{aligned}$$

this shows that the complex numbers of the form $(a, 0)$ behave precisely the same with respect to addition and multiplication of complex numbers as real

numbers do with their own addition and multiplication. For this reason we will adopt the convention that $(a, 0)$ will be denoted simply by a . The familiar $a + bi$ notation for complex numbers can now be recovered if one more definition is made.

DEFINITION

$$i = (0, 1).$$

Notice that $i^2 = (0, 1) \cdot (0, 1) = (-1, 0) = -1$ (the last equality sign depends on our convention). Moreover

$$\begin{aligned}(a, b) &= (a, 0) + (0, b) \\ &= (a, 0) + (b, 0) \cdot (0, 1) \\ &= a + bi.\end{aligned}$$

You may feel that our definition was merely an elaborate device for defining complex numbers as “expressions of the form $a + bi$.” That is essentially correct; it is a firmly established prejudice of modern mathematics that new objects must be defined as something specific, not as “expressions.” Nevertheless, it is interesting to note that mathematicians were sincerely worried about using complex numbers until the modern definition was proposed. Moreover, the precise definition emphasizes one important point. Our aim in introducing complex numbers was to avoid the necessity of paraphrasing statements about complex numbers in terms of their real and imaginary parts. This means that we wish to work with complex numbers in the same way that we worked with rational or real numbers. For example, the solution of the cubic equation required writing $x = w - p/3w$, so we want to know that $1/w$ makes sense. Moreover, w^2 was found by solving a quadratic equation, which requires numerous other algebraic manipulations. In short, we are likely to use, at some time or other, any manipulations performed on real numbers. We certainly do not want to stop each time and justify every step. Fortunately this is not necessary. Since all algebraic manipulations performed on real numbers can be justified by the properties listed in Chapter 1, it is only necessary to check that these properties are also true for complex numbers. In most cases this is quite easy, and these facts will not be listed as formal theorems. For example, the proof of P1,

$$[(a, b) + (c, d)] + (e, f) = (a, b) + [(c, d) + (e, f)]$$

requires only the application of the definition of addition for complex numbers. The left side becomes

$$([a + c] + e, [b + d] + f),$$

and the right side becomes

$$(a + [c + e], b + [d + f]);$$

these two are equal because P1 is true for real numbers. It is a good idea to check P2–P7 and P9. Notice that the complex numbers playing the role of 0 and 1 in P2 and P6 are $(0, 0)$ and $(1, 0)$, respectively. It is not hard to figure out what $-(a, b)$ is, but the multiplicative inverse for (a, b) required in P8 is a little trickier: if $(a, b) \neq (0, 0)$, then $a^2 + b^2 \neq 0$ and

$$(a, b) \cdot \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = (1, 0).$$

This fact could have been guessed in two ways. To find (x, y) with

$$(a, b) \cdot (x, y) = (1, 0)$$

it is only necessary to solve the equations

$$\begin{aligned} ax - by &= 1, \\ bx + ay &= 0. \end{aligned}$$

The solutions are $x = a/(a^2 + b^2)$, $y = -b/(a^2 + b^2)$. It is also possible to reason that if $1/(a + bi)$ means anything, then it should be true that

$$\frac{1}{a + bi} = \frac{1}{a + bi} \cdot \frac{a - bi}{a - bi} = \frac{a - bi}{a^2 + b^2}.$$

Once the existence of inverses has actually been proved (after guessing the inverse by some method), it follows that this manipulation is really valid; it is the easiest one to remember when the inverse of a complex number is actually being sought—it was precisely this trick which we used to evaluate

$$\begin{aligned} \frac{15}{6 + 3i} &= \frac{15}{6 + 3i} \cdot \frac{6 - 3i}{6 - 3i} \\ &= \frac{90 - 45i}{36 + 9}. \end{aligned}$$

Unlike P1–P9, the rules P10–P12 do not have analogues: it is easy to prove that there is *no* set P of *complex* numbers such that P10–P12 are satisfied for all *complex* numbers. In fact, if there were, then P would have to contain 1 (since $1 = 1^2$) and also -1 (since $-1 = i^2$), and this would contradict P10. The absence of P10–P12 will not have disastrous consequences, but it does mean that we cannot define $z < w$ for complex z and w . Also, you may remember that for the real numbers, P10–P12 were used to prove that $1 + 1 \neq 0$. Fortunately, the corresponding fact for complex numbers can be reduced to this one: clearly $(1, 0) + (1, 0) \neq (0, 0)$.

Although we will usually write complex numbers in the form $a + bi$, it is worth remembering that the set of all complex numbers $\mathbf{C}$ is just the collection of all pairs of real numbers. Long ago this collection was identified with the plane, and for this reason the plane is often called the “complex plane.” The horizontal axis, which consists of all points $(a, 0)$ for a in $\mathbf{R}$, is often called the

real axis, and the vertical axis is called the *imaginary axis*. Two important definitions are also related to this geometric picture.

DEFINITION

If $z = x + iy$ is a complex number (with x and y real), then the **conjugate** $\bar{z}$ of z is defined as

$$\bar{z} = x - iy,$$

and the **absolute value** or **modulus** $|z|$ of z is defined as

$$|z| = \sqrt{x^2 + y^2}.$$

(Notice that $x^2 + y^2 \geq 0$, so that $\sqrt{x^2 + y^2}$ is defined unambiguously; it denotes the nonnegative real square root of $x^2 + y^2$.)

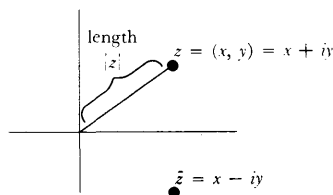


FIGURE 1

Geometrically, $\bar{z}$ is simply the reflection of z in the real axis, while $|z|$ is the distance from z to $(0, 0)$ (Figure 1). Notice that the absolute value notation for complex numbers is consistent with that for real numbers. The **distance** between two complex numbers z and w can be defined quite easily as $|z - w|$. The following theorem lists all the important properties of conjugates and absolute values.

THEOREM 1

Let z and w be complex numbers. Then

- (1) $\bar{\bar{z}} = z$.
- (2) $\bar{z} = z$ if and only if z is real (i.e., is of the form $a + 0i$, for some real number a).
- (3) $\overline{z + w} = \bar{z} + \bar{w}$.
- (4) $\overline{-z} = -(\bar{z})$.
- (5) $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$.
- (6) $\overline{z^{-1}} = (\bar{z})^{-1}$, if $z \neq 0$.
- (7) $|z|^2 = z \cdot \bar{z}$.
- (8) $|z \cdot w| = |z| \cdot |w|$.
- (9) $|z + w| \leq |z| + |w|$.

PROOF

Assertions (1) and (2) are obvious. Equations (3) and (5) may be checked by straightforward calculations and (4) and (6) may then be proved by a trick:

$$\begin{aligned} 0 &= \bar{0} = \overline{z + (-z)} = \bar{z} + \overline{-z}, & \text{so } \overline{-z} &= -(\bar{z}), \\ 1 &= \bar{1} = \overline{z \cdot (z^{-1})} = \bar{z} \cdot \overline{z^{-1}}, & \text{so } \overline{z^{-1}} &= (\bar{z})^{-1}. \end{aligned}$$

Equations (7) and (8) may also be proved by a straightforward calculation. The only difficult part of the theorem is (9). This inequality has, in fact, already occurred (Problem 4-8), but the proof will be repeated here, using slightly different terminology.

It is clear that equality holds in (9) if $z = 0$ or $w = 0$. It is also easy to see that (9) is true if $z = \lambda w$ for any real number λ (consider separately the cases

$\lambda > 0$ and $\lambda < 0$). Suppose, on the other hand, that $z \neq \lambda w$ for any real number λ , and that $w \neq 0$. Then, for all real numbers λ ,

$$\begin{aligned}
 (*) \quad 0 < |z - \lambda w|^2 &= (z - \lambda w) \cdot \overline{(z - \lambda w)} \\
 &= (z - \lambda w) \cdot (\bar{z} - \lambda \bar{w}) \\
 &= z\bar{z} + \lambda^2 w\bar{w} - \lambda(w\bar{z} + z\bar{w}) \\
 &= \lambda^2 |w|^2 + |z|^2 - \lambda(w\bar{z} + z\bar{w}).
 \end{aligned}$$

Notice that $w\bar{z} + z\bar{w}$ is real, since

$$\overline{w\bar{z} + z\bar{w}} = \bar{w}\bar{\bar{z}} + \bar{z}\bar{\bar{w}} = \bar{w}z + \bar{z}w = w\bar{z} + z\bar{w}.$$

Thus the right side of (*) is a quadratic equation in λ with real coefficients and no real solutions; its discriminant must therefore be negative. Thus

$$(w\bar{z} + z\bar{w})^2 - 4|w|^2 \cdot |z|^2 < 0;$$

it follows, since $w\bar{z} + z\bar{w}$ and $|w| \cdot |z|$ are real numbers, and $|w| \cdot |z| \geq 0$, that

$$(w\bar{z} + z\bar{w}) < 2|w| \cdot |z|.$$

From this inequality it follows that

$$\begin{aligned}
 |z + w|^2 &= (z + w) \cdot (\bar{z} + \bar{w}) \\
 &= |z|^2 + |w|^2 + (w\bar{z} + z\bar{w}) \\
 &< |z|^2 + |w|^2 + 2|w| \cdot |z| \\
 &= (|z| + |w|)^2,
 \end{aligned}$$

which implies that

$$|z + w| < |z| + |w|. \blacksquare$$

The operations of addition and multiplication of complex numbers both have important geometric interpretations. The picture for addition is very simple (Figure 2). Two complex numbers $z = (a, b)$ and $w = (c, d)$ determine a parallelogram having for two of its sides the line segment from $(0, 0)$ to z , and the line segment from $(0, 0)$ to w ; the vertex opposite $(0, 0)$ is $z + w$ (a proof of this geometric fact is left to you).

The interpretation of multiplication is more involved. If $z = 0$ or $w = 0$, then $z \cdot w = 0$ (a one-line computational proof can be given, but even this is unnecessary—the assertion has already been shown to follow from P1–P9), so we may restrict our attention to nonzero complex numbers. We begin by putting every nonzero complex number into a special form.

For any complex number $z \neq 0$ we can write

$$z = |z| \frac{z}{|z|};$$

in this expression, $|z|$ is a positive real number, while

$$\left| \frac{z}{|z|} \right| = \frac{|z|}{|z|} = 1,$$

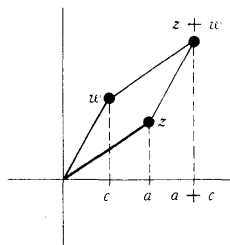


FIGURE 2

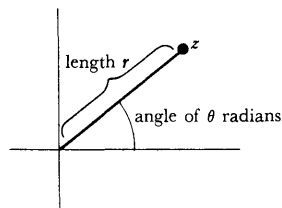


FIGURE 3

so that $z/|z|$ is a complex number of absolute value 1. Now any complex number $a = x + iy$ with $1 = |a| = x^2 + y^2$ can be written in the form

$$a = (\cos \theta, \sin \theta) = \cos \theta + i \sin \theta$$

for some number θ . Thus every nonzero complex number z can be written

$$z = r(\cos \theta + i \sin \theta)$$

for some $r > 0$ and some number θ . The number r is unique (it equals $|z|$), but θ is not unique; if θ_0 is one possibility, then the others are $\theta_0 + 2k\pi$ for k in $\mathbf{Z}$ —any one of these numbers is called an **argument** of z . Figure 3 shows z in terms of r and θ . (To find an argument θ for $z = x + iy$ we may note that the equation

$$x + iy = z = |z|(\cos \theta + i \sin \theta)$$

means that

$$\begin{aligned} x &= |z| \cos \theta, \\ y &= |z| \sin \theta. \end{aligned}$$

If $x \neq 0$ we can therefore take $\theta = \arctan y/x$, and if $x = 0$, we can take $\theta = \pi/2$ when $y > 0$ and $\theta = 3\pi/2$ when $y < 0$.)

Now the product of two nonzero complex numbers

$$\begin{aligned} z &= r(\cos \theta + i \sin \theta), \\ w &= s(\cos \phi + i \sin \phi), \end{aligned}$$

is

$$\begin{aligned} z \cdot w &= rs(\cos \theta + i \sin \theta)(\cos \phi + i \sin \phi) \\ &= rs[(\cos \theta \cos \phi - \sin \theta \sin \phi) + i(\sin \theta \cos \phi + \cos \theta \sin \phi)] \\ &= rs[\cos(\theta + \phi) + i \sin(\theta + \phi)]. \end{aligned}$$

Thus, the absolute value of a product is the product of the absolute values of the factors, while the sum of any argument for each of the factors will be an argument for the product. For a nonzero complex number

$$z = r(\cos \theta + i \sin \theta)$$

it is now an easy matter to prove by induction the following very important formula (sometimes known as De Moivre's Theorem):

$$z^n = |z|^n(\cos n\theta + i \sin n\theta), \text{ for any argument } \theta \text{ of } z.$$

This formula describes z^n so explicitly that it is easy to decide just when $z^n = w$:

THEOREM 2 Every nonzero complex number has exactly n complex n th roots. More precisely, for any complex number $w \neq 0$, and any natural number n , there are precisely n different complex numbers z satisfying $z^n = w$.

PROOF Let

$$w = s(\cos \phi + i \sin \phi)$$

for $s = |w|$ and some number ϕ . Then a complex number

$$z = r(\cos \theta + i \sin \theta)$$

satisfies $z^n = w$ if and only if

$$r^n(\cos n\theta + i \sin n\theta) = s(\cos \phi + i \sin \phi),$$

which happens if and only if

$$\begin{aligned} r^n &= s, \\ \cos n\theta + i \sin n\theta &= \cos \phi + i \sin \phi. \end{aligned}$$

From the first equation it follows that

$$r = \sqrt[n]{s},$$

where $\sqrt[n]{s}$ denotes the positive real n th root of s . From the second equation it follows that for some integer k we have

$$\theta = \theta_k = \frac{\phi}{n} + \frac{2k\pi}{n}.$$

Conversely, if we choose $r = \sqrt[n]{s}$ and $\theta = \theta_k$ for some k , then the number $z = r(\cos \theta + i \sin \theta)$ will satisfy $z^n = w$. To determine the number of n th roots of w , it is therefore only necessary to determine which such z are distinct. Now any integer k can be written

$$k = nq + k'$$

for some integer q , and some integer k' between 0 and $n - 1$. Then

$$\cos \theta_k + i \sin \theta_k = \cos \theta_{k'} + i \sin \theta_{k'}.$$

This shows that every z satisfying $z^n = w$ can be written

$$z = \sqrt[n]{s} (\cos \theta_k + i \sin \theta_k) \quad k = 0, \dots, n - 1.$$

Moreover, it is easy to see that these numbers are all different, since any two θ_k for $k = 0, \dots, n - 1$ differ by less than 2π . ■

In the course of proving Theorem 2, we have actually developed a method for finding the n th roots of a complex number. For example, to find the cube roots of i (Figure 4) note that $|i| = 1$ and that $\pi/2$ is an argument for i . The cube roots of i are therefore

$$\begin{aligned} &1 \cdot \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right], \\ &1 \cdot \left[\cos \left(\frac{\pi}{6} + \frac{2\pi}{3} \right) + i \sin \left(\frac{\pi}{6} + \frac{2\pi}{3} \right) \right] = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}, \\ &1 \cdot \left[\cos \left(\frac{\pi}{6} + \frac{4\pi}{3} \right) + i \sin \left(\frac{\pi}{6} + \frac{4\pi}{3} \right) \right] = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}. \end{aligned}$$

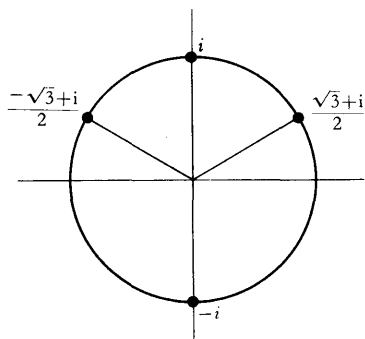


FIGURE 4

Since

$$\begin{aligned}\cos \frac{\pi}{6} &= \frac{\sqrt{3}}{2}, & \sin \frac{\pi}{6} &= \frac{1}{2}, \\ \cos \frac{5\pi}{6} &= -\frac{\sqrt{3}}{2}, & \sin \frac{5\pi}{6} &= \frac{1}{2}, \\ \cos \frac{3\pi}{2} &= 0, & \sin \frac{3\pi}{2} &= -1,\end{aligned}$$

the cube roots of i are

$$\frac{\sqrt{3} + i}{2}, \quad \frac{-\sqrt{3} + i}{2}, \quad -i.$$

In general, we cannot expect to obtain such simple results. For example, to find the cube roots of $2 + 11i$, note that $|2 + 11i| = \sqrt{2^2 + 11^2} = \sqrt{125}$ and that $\arctan \frac{11}{2}$ is an argument for $2 + 11i$. One of the cube roots of $2 + 11i$ is therefore

$$\begin{aligned}\sqrt[3]{125} \left[\cos \left(\frac{\arctan \frac{11}{2}}{3} \right) + i \sin \left(\frac{\arctan \frac{11}{2}}{3} \right) \right] \\ = \sqrt{5} \left[\cos \left(\frac{\arctan \frac{11}{2}}{3} \right) + i \sin \left(\frac{\arctan \frac{11}{2}}{3} \right) \right].\end{aligned}$$

Previously we noted that $2 + i$ is also a cube root of $2 + 11i$. Since $|2 + i| = \sqrt{2^2 + 1^2} = \sqrt{5}$, and since $\arctan \frac{1}{2}$ is an argument of $2 + i$, we can write this cube root as

$$2 + i = \sqrt{5} (\cos \arctan \tfrac{1}{2} + i \sin \arctan \tfrac{1}{2}).$$

These two cube roots are actually the same number, because

$$\frac{\arctan \frac{11}{2}}{3} = \arctan \frac{1}{2}$$

(you can check this by using the formula in Problem 15-8), but this is hardly the sort of thing one might notice!

The fact that every complex number has an n th root for all n is just a special case of a very important theorem. The number i was originally introduced in order to provide a solution for the equation $x^2 + 1 = 0$. The *Fundamental Theorem of Algebra* states the remarkable fact that this one addition automatically provides solutions for all other polynomial equations: every equation

$$z^n + a_{n-1}z^{n-1} + \cdots + a_0 = 0 \quad a_0, \dots, a_{n-1} \text{ in } \mathbb{C}$$

has a complex root!

In the next chapter we shall give an almost complete proof of the Fundamental Theorem of Algebra; the slight gap left in the text can be filled in as an

exercise (Problem 25-5). The proof of the theorem will rely on several new concepts which come up quite naturally in a more thorough investigation of complex numbers.

PROBLEMS

1. Find the absolute value and argument of each of the following.
 - (i) $3 + 4i$.
 - (ii) $(3 + 4i)^{-1}$.
 - (iii) $(1 + i)^5$.
 - (iv) $\sqrt[7]{3 + 4i}$.
 - (v) $|3 + 4i|$.
2. Solve the following equations.
 - (i) $x^2 + ix + 1 = 0$.
 - (ii) $x^4 + x^2 + 1 = 0$.
 - (iii) $x^2 + 2ix - 1 = 0$.
 - (iv) $\begin{cases} ix - (1 + i)y = 3, \\ (2 + i)x + iy = 4. \end{cases}$
 - (v) $x^3 - x^2 - x - 2 = 0$.
3. Describe the set of all complex numbers z such that
 - (i) $\bar{z} = -z$.
 - (ii) $\bar{z} = z^{-1}$.
 - (iii) $|z - a| = |z - b|$.
 - (iv) $|z - a| + |z - b| = c$.
 - (v) $|z| < 1 - \text{real part of } z$.
4. Prove that $|z| = |\bar{z}|$, and that the real part of z is $(z + \bar{z})/2$, while the imaginary part is $(z - \bar{z})/2i$.
5. Prove that $|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$, and interpret this statement geometrically.
6. What is the pictorial relation between z and $\sqrt{i} \cdot z \sqrt{-i}$? Hint: Which line goes into the real axis under multiplication by $\sqrt{-i}$?
7. (a) Prove that if $a_0, \dots, a_{n-1}$ are real and $a + bi$ (for a and b real) satisfies the equation $z^n + a_{n-1}z^{n-1} + \dots + a_0 = 0$, then $a - bi$ also satisfies this equation. (Thus the nonreal roots of such an equation always occur in pairs, and the number of such roots is even.)
 (b) Conclude that $z^n + a_{n-1}z^{n-1} + \dots + a_0$ is divisible by $z^2 - 2az + (a^2 + b^2)$ (whose coefficients are real).
- *8. (a) Let c be an integer which is not the square of another integer. If a and b are integers we define the **conjugate** of $a + b\sqrt{c}$, denoted by $\overline{a + b\sqrt{c}}$, as $a - b\sqrt{c}$. Show that the conjugate is well defined

- by showing that a number can be written $a + b\sqrt{c}$, for integers a and b , in only one way.
- (b) Show that for all α and β of the form $a + b\sqrt{c}$, we have $\overline{\overline{\alpha}} = \alpha$, $\overline{\alpha} = \alpha$ if and only if α is an integer, $\overline{\alpha + \beta} = \overline{\alpha} + \overline{\beta}$, $\overline{-\alpha} = -\overline{\alpha}$, $\overline{\alpha \cdot \beta} = \overline{\alpha} \cdot \overline{\beta}$, and $\overline{\alpha^{-1}} = (\overline{\alpha})^{-1}$ if $\alpha \neq 0$.
- (c) Prove that if $a_0, \dots, a_{n-1}$ are integers and $z = a + b\sqrt{c}$ satisfies the equation $z^n + a_{n-1}z^{n-1} + \dots + a_0 = 0$, then $\bar{z} = a - b\sqrt{c}$ also satisfies this equation.
9. Find all the 4th roots of i ; express the one having smallest argument in a form that does not involve any trigonometric functions.
- *10. (a) Prove that if ω is an n th root of 1, then so is ω^k .
 (b) A number ω is called a **primitive n th root** of 1 if $\{1, \omega, \omega^2, \dots, \omega^{n-1}\}$ is the set of all n th roots of 1. How many primitive n th roots of 1 are there for $n = 3, 4, 5, 9$?
 (c) Let ω be an n th root of 1, with $\omega \neq 1$. Prove that $\sum_{k=0}^{n-1} \omega^k = 0$.
- *11. (a) Prove that if $z_1, \dots, z_k$ lie on one side of some straight line through 0, then $z_1 + \dots + z_k \neq 0$. Hint: This is obvious from the geometric interpretation of addition, but an analytic proof is also easy: the assertion is clear if the line is the real axis, and a trick will reduce the general case to this one.
 (b) Show further that $z_1^{-1}, \dots, z_k^{-1}$ all lie on one side of a straight line through 0, so that $z_1^{-1} + \dots + z_k^{-1} \neq 0$.
- *12. Prove that if $|z_1| = |z_2| = |z_3|$ and $z_1 + z_2 + z_3 = 0$, then z_1, z_2 , and z_3 are the vertices of an equilateral triangle. Hint: It will help to assume that z_1 is real, and this can be done with no loss of generality. Why?

CHAPTER 25 COMPLEX FUNCTIONS

You will probably not be surprised to learn that a deeper investigation of complex numbers depends on the notion of functions. Until now a function was (intuitively) a rule which assigned real numbers to certain other real numbers. But there is no reason why this concept should not be extended; we might just as well consider a rule which assigns complex numbers to certain other complex numbers. A rigorous definition presents no problems (we will not even accord it the full honors of a formal definition): a function is a collection of pairs of complex numbers which does not contain two distinct pairs with the same first element. Since we consider real numbers to be certain complex numbers, the old definition is really a special case of the new one. Nevertheless, we will sometimes resort to special terminology in order to clarify the context in which a function is being considered. A function f is called **real-valued** if $f(z)$ is a real number for all z in the domain of f , and **complex-valued** to emphasize that it is not necessarily real-valued. Similarly, we will usually state explicitly that a function f is defined on [a subset of] $\mathbf{R}$ in those cases where the domain of f is [a subset of] $\mathbf{R}$; in other cases we sometimes mention that f is defined on [a subset of] $\mathbf{C}$ to emphasize that $f(z)$ is defined for complex z as well as real z .

Among the multitude of functions defined on $\mathbf{C}$, certain ones are particularly important. Foremost among these are the functions of the form

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0,$$

where $a_0, \dots, a_n$ are complex numbers. These functions are called, as in the real case, polynomial functions; they include the function $f(z) = z$ (the “identity function”) and functions of the form $f(z) = a$ for some complex number a (“constant functions”). Another important generalization of a familiar function is the “absolute value function” $f(z) = |z|$ for all z in $\mathbf{C}$.

Two functions of particular importance for complex numbers are Re (the “real part function”) and Im (the “imaginary part function”), defined by

$$\begin{aligned} \text{Re}(x + iy) &= x, \\ \text{Im}(x + iy) &= y, \end{aligned} \quad \text{for } x \text{ and } y \text{ real.}$$

The “conjugate function” is defined by

$$f(z) = \bar{z} = \text{Re}(z) - i\text{Im}(z).$$

Familiar real-valued functions defined on $\mathbf{R}$ may be combined in many ways to produce new complex-valued functions defined on $\mathbf{C}$ —an example is the function

$$f(x + iy) = e^y \sin(x - y) + ix^3 \cos y.$$

The formula for this particular function illustrates a decomposition which is always possible. Any complex-valued function f can be written in the form

$$f = u + iv$$

for some real-valued functions u and v —simply define $u(z)$ as the real part of $f(z)$, and $v(z)$ as the imaginary part. This decomposition is often very useful, but not always; for example, it would be inconvenient to describe a polynomial function in this way.

One other function will play an important role in this chapter. Recall that an *argument* of a nonzero complex number z is a (real) number θ such that

$$z = |z|(\cos \theta + i \sin \theta).$$

There are infinitely many arguments for z , but just one which satisfies $0 \leq \theta < 2\pi$. If we call this unique argument $\theta(z)$, then θ is a (real-valued) function (the “argument function”) on $\{z \text{ in } \mathbb{C}: z \neq 0\}$.

“Graphs” of complex-valued functions defined on $\mathbb{C}$, since they lie in 4-dimensional space, are presumably not very useful for visualization. The alternative picture of a function mentioned in Chapter 4 can be used instead: we draw two copies of $\mathbb{C}$, and arrows from z in one copy, to $f(z)$ in the other (Figure 1).

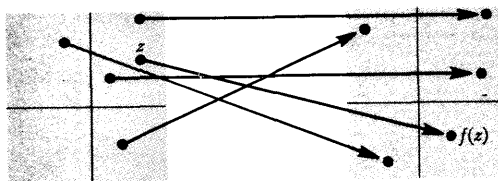


FIGURE 1

The most common pictorial representation of a complex-valued function is produced by labeling a point in the plane with the value $f(z)$, instead of with z (which can be estimated from the position of the point in the picture). Figure 2 shows this sort of picture for several different functions. Certain features of the function are illustrated very clearly by such a “graph.” For example, the absolute value function is constant on concentric circles around 0, the functions Re and Im are constant on the vertical and horizontal lines, respectively, and the function $f(z) = z^2$ wraps the circle of radius r twice around the circle of radius r^2 .

Despite the problems involved in visualizing complex-valued functions in general, it is still possible to define analogues of important properties previously defined for real-valued functions on $\mathbb{R}$, and in some cases these properties may be easier to visualize in the complex case. For example, the notion of limit can be defined as follows:

$\lim_{z \rightarrow a} f(z) = l$ means that for every (real) number $\epsilon > 0$ there is a (real) number $\delta > 0$ such that, for all z , if $0 < |z - a| < \delta$, then $|f(z) - l| < \epsilon$.

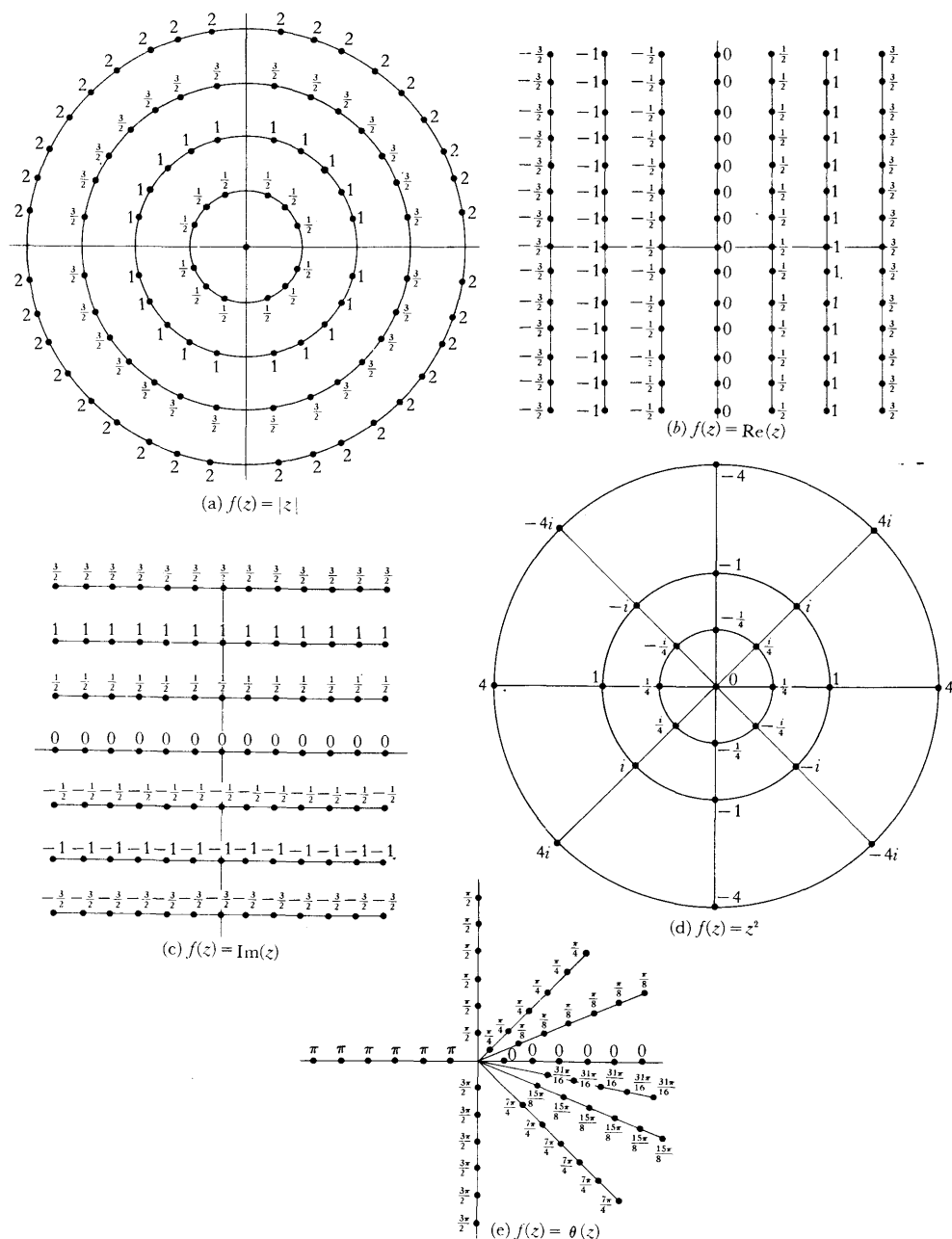


FIGURE 2

Although the definition reads precisely as before, the interpretation is slightly different. Since $|z - w|$ is the distance between the complex numbers z and w , the equation $\lim_{z \rightarrow a} f(z) = l$ means that the values of $f(z)$ can be made to lie inside any given circle around l , provided that z is restricted to lie inside a sufficiently small circle around a . This assertion is particularly easy to visualize using the “two copy” picture of a function (Figure 3).

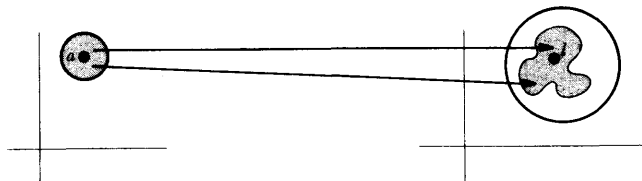


FIGURE 3

Certain facts about limits can be proved exactly as in the real case. In particular,

$$\begin{aligned}\lim_{z \rightarrow a} c &= c, \\ \lim_{z \rightarrow a} z &= a, \\ \lim_{z \rightarrow a} [f(z) + g(z)] &= \lim_{z \rightarrow a} f(z) + \lim_{z \rightarrow a} g(z), \\ \lim_{z \rightarrow a} f(z) \cdot g(z) &= \lim_{z \rightarrow a} f(z) \cdot \lim_{z \rightarrow a} g(z), \\ \lim_{z \rightarrow a} \frac{1}{g(z)} &= \frac{1}{\lim_{z \rightarrow a} g(z)}, \quad \text{if } \lim_{z \rightarrow a} g(z) \neq 0.\end{aligned}$$

The essential property of absolute values upon which these results are based is the inequality $|z + w| \leq |z| + |w|$, and this inequality holds for complex numbers as well as for real numbers. These facts already provide quite a few limits, but many more can be obtained from the following theorem.

THEOREM 1 Let $f(z) = u(z) + iv(z)$ for real-valued functions u and v , and let $l = \alpha + i\beta$ for real numbers α and β . Then $\lim_{z \rightarrow a} f(z) = l$ if and only if

$$\begin{aligned}\lim_{z \rightarrow a} u(z) &= \alpha, \\ \lim_{z \rightarrow a} v(z) &= \beta.\end{aligned}$$

PROOF Suppose first that $\lim_{z \rightarrow a} f(z) = l$. If $\varepsilon > 0$, there is $\delta > 0$ such that, for all z ,

$$\text{if } 0 < |z - a| < \delta, \text{ then } |f(z) - l| < \varepsilon.$$

The second inequality can be written

$$|[u(z) - \alpha] + i[v(z) - \beta]| < \varepsilon,$$

or

$$[u(z) - \alpha]^2 + [v(z) - \beta]^2 < \varepsilon^2.$$

Since $u(z) - \alpha$ and $v(z) - \beta$ are both real numbers, their squares are positive; this inequality therefore implies that

$$[u(z) - \alpha]^2 < \varepsilon^2 \quad \text{and} \quad [v(z) - \beta]^2 < \varepsilon^2,$$

which implies that

$$|u(z) - \alpha| < \varepsilon \quad \text{and} \quad |v(z) - \beta| < \varepsilon.$$

Since this is true for all $\varepsilon > 0$, it follows that

$$\lim_{z \rightarrow a} u(z) = \alpha \quad \text{and} \quad \lim_{z \rightarrow a} v(z) = \beta.$$

Now suppose that these two equations hold. If $\varepsilon > 0$, there is a $\delta > 0$ such that, for all z , if $0 < |z - a| < \delta$, then

$$|u(z) - \alpha| < \frac{\varepsilon}{2} \quad \text{and} \quad |v(z) - \beta| < \frac{\varepsilon}{2},$$

which implies that

$$\begin{aligned} |f(z) - l| &= |[u(z) - \alpha] + i[v(z) - \beta]| \\ &\leq |u(z) - \alpha| + |i| \cdot |v(z) - \beta| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This proves that $\lim_{z \rightarrow a} f(z) = l$. ■

In order to apply Theorem 1 fruitfully, notice that since we already know the limit $\lim_{z \rightarrow a} z = a$, we can conclude that

$$\begin{aligned} \lim_{z \rightarrow a} \operatorname{Re}(z) &= \operatorname{Re}(a), \\ \lim_{z \rightarrow a} \operatorname{Im}(z) &= \operatorname{Im}(a). \end{aligned}$$

A limit like

$$\lim_{z \rightarrow a} \sin(\operatorname{Re}(z)) = \sin(\operatorname{Re}(a))$$

follows easily, using continuity of $\sin$. Many applications of these principles prove such limits as the following:

$$\begin{aligned} \lim_{z \rightarrow a} \bar{z} &= \bar{a}, \\ \lim_{z \rightarrow a} |z| &= |a|, \\ \lim_{(x+iy) \rightarrow a+bi} e^y \sin x + ix^3 \cos y &= e^b \sin a + ia^3 \cos b. \end{aligned}$$

Now that the notion of limit has been extended to complex functions, the notion of continuity can also be extended: f is **continuous at a** if $\lim_{z \rightarrow a} f(z) =$

$f(a)$, and f is **continuous** if f is continuous at a for all a in the domain of f . The previous work on limits shows that all the following functions are continuous:

$$\begin{aligned} f(z) &= a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0, \\ f(z) &= \bar{z}, \\ f(z) &= |z|, \\ f(x + iy) &= e^y \sin x + ix^3 \cos y. \end{aligned}$$

Examples of discontinuous functions are easy to produce, and certain ones come up very naturally. One particularly frustrating example is the “argument function” θ , which is discontinuous at all nonnegative real numbers (see the “graph” in Figure 2). By suitably redefining θ it is possible to change the discontinuities; for example (Figure 4), if $\theta'(z)$ denotes the unique argument of z with $\pi/2 \leq \theta'(z) < 5\pi/2$, then θ' is discontinuous at ai for every nonnegative real number a . But, no matter how θ is redefined, some discontinuities will always occur.

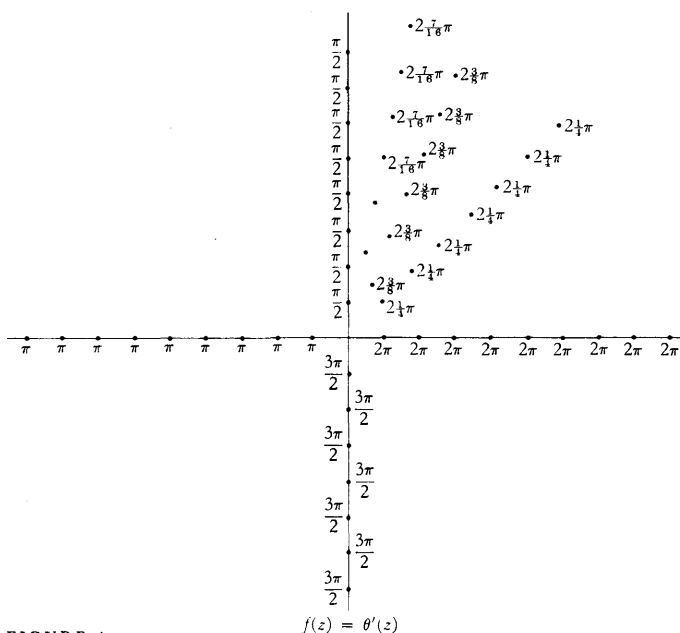


FIGURE 4

The discontinuity of θ has important bearing on the problem of defining a “square-root function,” that is, a function f such that $(f(z))^2 = z$ for all z . For real numbers the function $\sqrt{}$ had as domain only the nonnegative real numbers. If complex numbers are allowed, then every number has two square roots (except 0, which has only one). Although this situation may seem better, it is in some ways worse; since the square roots of z are complex numbers, there is no clear criterion for selecting one root to be $f(z)$, in preference to the other.

One way to define f is the following. We set $f(0) = 0$, and for $z \neq 0$ we set

$$f(z) = \sqrt{|z|} \left(\cos \frac{\theta(z)}{2} + i \sin \frac{\theta(z)}{2} \right).$$

Clearly $(f(z))^2 = z$, but the function f is discontinuous, since θ is discontinuous. As a matter of fact, it is impossible to find a continuous f such that $(f(z))^2 = z$ for all z . Although you may be willing to accept this assertion without any fuss, the argument, given below, is rather tricky.

We will actually show that a continuous square root function f cannot even be defined on the set of all z with $|z| = 1$. Suppose there were such a function f . We must have $f(1) = 1$ or $f(1) = -1$, and we can therefore assume that $f(1) = -1$, since $-f$ would also be a continuous square root function. In this case, since f does not have the value 1 at 1, it *never* takes on the value 1. This means that if we define g by

$$g(z) = \theta(f(z)) \quad \text{for } |z| = 1,$$

then g will be continuous, since θ is continuous at all w with $|w| = 1$ except for $w = 1$, which is not $f(z)$ for any z . But the equation

$$(f(z))^2 = z$$

implies that

$$\theta(z) = 2\theta(f(z)) = 2g(z),$$

which would imply that θ is continuous at z for all z with $|z| = 1$. Since this is false, no such function f can exist. Similar arguments show that it is impossible to define continuous “ n th-root functions” for any $n \geq 2$.

For continuous complex functions there are important analogues of certain theorems which describe the behavior of real-valued functions on closed intervals. A natural analogue of the interval $[a, b]$ is the set of all complex numbers $z = x + iy$ with $a \leq x \leq b$ and $c \leq y \leq d$ (Figure 5). This set is called a **closed rectangle**, and is denoted by $[a, b] \times [c, d]$.

If f is a continuous complex-valued function whose domain is $[a, b] \times [c, d]$, then it seems reasonable, and is indeed true, that f is bounded on $[a, b] \times [c, d]$. That is, there is some real number M such that

$$|f(z)| \leq M \quad \text{for all } z \text{ in } [a, b] \times [c, d].$$

It does not make sense to say that f has a maximum and a minimum value on $[a, b] \times [c, d]$, since there is no notion of order for complex numbers. If f is a real-valued function, however, then this assertion does make sense, and is true. In particular, if f is any complex-valued continuous function on $[a, b] \times [c, d]$, then $|f|$ is also continuous, so there is some z_0 in $[a, b] \times [c, d]$ such that

$$|f(z_0)| \leq |f(z)| \quad \text{for all } z \text{ in } [a, b] \times [c, d];$$

a similar statement is true with the inequality reversed. It is sometimes said that “ f attains its maximum and minimum modulus on $[a, b] \times [c, d]$.”

The various facts listed in the previous paragraph will not be proved here,

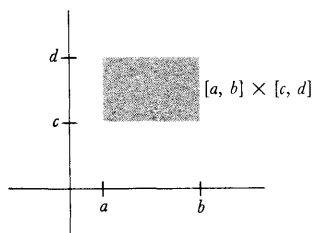


FIGURE 5

although proofs are outlined in Problem 5. Assuming these facts, however, we can now give a proof of the Fundamental Theorem of Algebra, which is really quite surprising, since we have not yet said much to distinguish polynomial functions from other continuous functions.

THEOREM 2 (THE FUNDAMENTAL THEOREM OF ALGEBRA)

Let $a_0, \dots, a_{n-1}$ be any complex numbers. Then there is a complex number z such that

$$z^n + a_{n-1}z^{n-1} + a_{n-2}z^{n-2} + \dots + a_0 = 0.$$

PROOF Let

$$f(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0.$$

Then f is continuous, and so is the function $|f|$ defined by

$$|f|(z) = |f(z)| = |z^n + a_{n-1}z^{n-1} + \dots + a_0|.$$

Our proof is based on the observation that a point z_0 with $f(z_0) = 0$ would clearly be a minimum point for $|f|$. To prove the theorem we will first show that $|f|$ does indeed have a smallest value on the *whole complex plane*. The proof will be almost identical to the proof, in Chapter 7, that a polynomial function of even degree (with real coefficients) has a smallest value on all of $\mathbf{R}$; both proofs depend on the fact that if $|z|$ is large, then $|f(z)|$ is large.

We begin by writing, for $z \neq 0$,

$$f(z) = z^n \left(1 + \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right),$$

so that

$$|f(z)| = |z|^n \cdot \left| 1 + \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right|.$$

Let

$$M = \max(1, 2n|a_{n-1}|, \dots, 2n|a_0|).$$

Then for all z with $|z| \geq M$, we have $|z^k| \geq |z|$ and

$$\frac{|a_{n-k}|}{|z^k|} \leq \frac{|a_{n-k}|}{|z|} \leq \frac{|a_{n-k}|}{2n|a_{n-k}|} = \frac{1}{2n},$$

so

$$\left| \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right| \leq \left| \frac{a_{n-1}}{z} \right| + \dots + \left| \frac{a_0}{z^n} \right| \leq \frac{1}{2},$$

which implies that

$$\left| 1 - \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right| \geq 1 - \left| \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right| \geq \frac{1}{2}.$$

This means that

$$|f(z)| \geq \frac{|z|^n}{2} \quad \text{for } |z| \geq M.$$

In particular, if $|z| \geq M$ and also $|z| \geq \sqrt[n]{2|f(0)|}$, then

$$|f(z)| \geq |f(0)|.$$

Now let $[a, b] \times [c, d]$ be a closed rectangle (Figure 6) which contains $\{z: |z| \leq \max(M, \sqrt[n]{2|f(0)|})\}$, and suppose that the minimum of $|f|$ on $[a, b] \times [c, d]$ is attained at z_0 , so that

$$(1) \quad |f(z_0)| \leq |f(z)| \quad \text{for } z \text{ in } [a, b] \times [c, d].$$

It follows, in particular, that $|f(z_0)| \leq |f(0)|$. Thus

$$(2) \quad \text{if } |z| \geq \max(M, \sqrt[n]{2|f(0)|}), \text{ then } |f(z)| \geq |f(0)| \geq |f(z_0)|.$$

Combining (1) and (2) we see that $|f(z_0)| \leq |f(z)|$ for all z , so that $|f|$ attains its minimum value on the whole complex plane at z_0 .

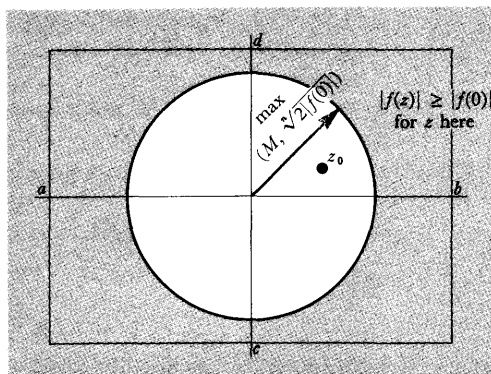


FIGURE 6

To complete the proof of the theorem we now show that $f(z_0) = 0$. It is convenient to introduce the function g defined by

$$g(z) = f(z + z_0).$$

Then g is a polynomial function of degree n , whose minimum absolute value occurs at 0. We want to show that $g(0) = 0$.

Suppose instead that $g(0) = \alpha \neq 0$. If m is the smallest positive power of z which occurs in the expression for g , we can write

$$g(z) = \alpha + \beta z^m + c_{m+1} z^{m+1} + \cdots + c_n z^n,$$

where $\beta \neq 0$. Now, according to Theorem 24-2 there is a complex number γ such that

$$\gamma^m = -\frac{\alpha}{\beta}$$

Then, setting $d_k = c_k \gamma^k$, we have

$$\begin{aligned}
 |g(\gamma z)| &= |\alpha + \beta \gamma^m z^m + d_{m+1} z^{m+1} + \cdots + d_n z^n| \\
 &= |\alpha - \alpha z^m + d_{m+1} z^{m+1} + \cdots| \\
 &= \left| \alpha \left(1 - z^m + \frac{d_{m+1}}{\alpha} z^{m+1} + \cdots \right) \right| \\
 &= \left| \alpha \left(1 - z^m + z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right) \right| \\
 &= |\alpha| \cdot \left| 1 - z^m + z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right|.
 \end{aligned}$$

This expression, so tortuously arrived at, will enable us to reach a quick contradiction. Notice first that if $|z|$ is chosen small enough, we will have

$$\left| \frac{d_{m+1}}{\alpha} z + \cdots \right| < 1.$$

If we choose, from among all z for which this inequality holds, some z which is *real and positive*, then

$$\left| z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right| < |z^m| = z^m.$$

Consequently, if $0 < z < 1$ we have

$$\begin{aligned}
 \left| 1 - z^m + z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right| &\leq |1 - z^m| + \left| z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right| \\
 &= 1 - z^m + \left| z^m \left[\frac{d_{m+1}}{\alpha} z + \cdots \right] \right| \\
 &< 1 - z^m + z^m \\
 &= 1.
 \end{aligned}$$

This is the desired contradiction: for such a number z we have

$$|g(\gamma z)| < |\alpha|,$$

contradicting the fact that $|\alpha|$ is the minimum of $|g|$ on the whole plane. Hence, the original assumption must be incorrect, and $g(0) = 0$. This implies, finally, that $f(z_0) = 0$. ■

Even taking into account our omission of the proofs for the basic facts about continuous complex functions, this proof verified a deep fact with surprisingly little work. It is only natural to hope that other interesting developments will arise if we pursue further the analogues of properties of real functions. The next obvious step is to define derivatives: a function f is **differentiable at a** if

$$\lim_{z \rightarrow 0} \frac{f(a+z) - f(a)}{z} \text{ exists,}$$

in which case the limit is denoted by $f'(a)$. It is easy to prove that

$$\begin{aligned} f'(a) &= 0 & \text{if } f(z) &= c, \\ f'(a) &= 1 & \text{if } f(z) &= z, \\ (f+g)'(a) &= f'(a) + g'(a), \\ (f \cdot g)'(a) &= f'(a)g(a) + f(a)g'(a), \\ \left(\frac{1}{g}\right)'(a) &= \frac{-g'(a)}{[g(a)]^2} & \text{if } g(a) \neq 0, \\ (f \circ g)'(a) &= f'(g(a)) \cdot g'(a); \end{aligned}$$

the proofs of all these formulas are exactly the same as before. It follows, in particular, that if $f(z) = z^n$, then $f'(z) = nz^{n-1}$. These formulas only prove the differentiability of rational functions however. Many other obvious candidates are *not* differentiable. Suppose, for example, that

$$f(x + iy) = x - iy \quad (\text{i.e., } f(z) = \bar{z}).$$

If f is to be differentiable at 0, then the limit

$$\lim_{(x+iy) \rightarrow 0} \frac{f(x+iy) - f(0)}{x+iy} = \lim_{(x+iy) \rightarrow 0} \frac{x-iy}{x+iy}$$

must exist. Notice however, that

$$\text{if } y = 0, \text{ then } \frac{x-iy}{x+iy} = 1,$$

and

$$\text{if } x = 0, \text{ then } \frac{x-iy}{x+iy} = -1;$$

therefore this limit cannot possibly exist, since the quotient has both the values 1 and -1 for $x+iy$ arbitrarily close to 0.

In view of this example, it is not at all clear where other differentiable functions are to come from. If you recall the definitions of $\sin$ and $\exp$, you will see that there is no hope at all of generalizing these definitions to complex numbers. At the moment the outlook is bleak, but all our problems will soon be solved.

PROBLEMS

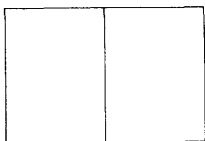
1. (a) For any real number y , define $\alpha(x) = x + iy$ (so that α is a complex-valued function defined on $\mathbf{R}$). Show that α is continuous. (This follows immediately from a theorem in this chapter.) Show similarly that $\beta(y) = x + iy$ is continuous.
- (b) Let f be a continuous function defined on $\mathbf{C}$. For fixed y , let $g(x) = f(x + iy)$. Show that g is a continuous function (defined on $\mathbf{R}$). Show similarly that $h(y) = f(x + iy)$ is continuous. Hint: Use part (a).
2. (a) Suppose that f is a continuous real-valued function defined on a closed rectangle $[a, b] \times [c, d]$. Prove that if f takes on the values $f(z)$

and $f(w)$ for z and w in $[a, b] \times [c, d]$, then f also takes all values between $f(z)$ and $f(w)$. Hint: Consider $g(t) = f(tz + (1 - t)w)$ for t in $[0, 1]$.

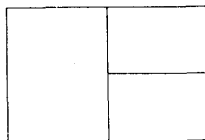
- (b) If f is a continuous complex-valued function defined on $[a, b] \times [c, d]$, the assertion in part (a) no longer makes any sense, since we cannot talk of complex numbers between $f(z)$ and $f(w)$. We might conjecture that f takes on all values on the line segment between $f(z)$ and $f(w)$, but even this is false. Find an example which shows this.
3. (a) Prove that if $a_0, \dots, a_{n-1}$ are any complex numbers, then there are complex numbers $z_1, \dots, z_n$ (not necessarily distinct) such that

$$z^n + a_{n-1}z^{n-1} + \dots + a_0 = \prod_{i=1}^n (z - z_i).$$

- (b) Prove that if $a_0, \dots, a_{n-1}$ are *real*, then $z^n + a_{n-1}z^{n-1} + \dots + a_0$ can be written as a product of linear factors $z + a$ and quadratic factors $z^2 + az + b$ all of whose coefficients are real. (Use Problem 24-7.)
4. In this problem we will consider only polynomials with real coefficients. Such a polynomial is called a **sum of squares** if it can be written as $h_1^2 + \dots + h_n^2$ for polynomials h_i with real coefficients.
- (a) Prove that if f is a sum of squares, then $f(x) \geq 0$ for all x .
- (b) Prove that if f and g are sums of squares, then so is $f \cdot g$.
- (c) Suppose that $f(x) \geq 0$ for all x . Show that f is a sum of squares. Hint: First write $f(x) = x^k g(x)$, where $g(x) \neq 0$ for all x . Then k must be even (why?), and $g(x) > 0$ for all x . Now use Problem 3(b).



(a)



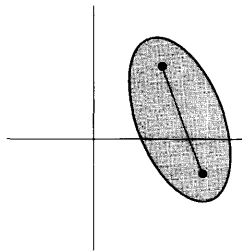
(b)

FIGURE 7

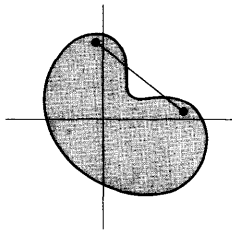
5. (a) Let A be a set of complex numbers. A number z is called, as in the real case, a **limit point** of the set A if for every (real) $\epsilon > 0$, there is a point a in A with $|z - a| < \epsilon$ but $z \neq a$. Prove the two-dimensional version of the Bolzano-Weierstrass Theorem: If A is an infinite subset of $[a, b] \times [c, d]$, then A has a limit point in $[a, b] \times [c, d]$. Hint: First divide $[a, b] \times [c, d]$ in half by a vertical line as in Figure 7(a). Since A is infinite, at least one half contains infinitely many points of A . Divide this in half by a horizontal line, as in Figure 7(b). Continue in this way, alternately dividing by vertical and horizontal lines.

(The two-dimensional bisection argument outlined in this hint is so standard that the title "Bolzano-Weierstrass" often serves to describe the method of proof, in addition to the theorem itself. See, for example, H. Petard, "A Contribution to the Mathematical Theory of Big Game Hunting," *Amer. Math. Monthly*, **45** (1938), 446-447.)

- (b) Prove that a continuous (complex-valued) function on $[a, b] \times [c, d]$ is bounded on $[a, b] \times [c, d]$. (Imitate Problem 21-22.)
- (c) Prove that if f is a real-valued continuous function on $[a, b] \times [c, d]$, then f takes on a maximum and minimum value on $[a, b] \times [c, d]$. (You can use the same trick that works for Theorem 7-3.)



(a) a convex subset of the plane



(b) a nonconvex subset of the plane

FIGURE 8

*6. The proof of Theorem 2 cannot be considered to be completely elementary because the possibility of choosing γ with $\gamma^m = -\alpha/\beta$ depends on Theorem 24-2, and thus on the trigonometric functions. It is therefore of some interest to provide an elementary proof that there is a solution for the equation $z^n - c = 0$.

- Make an explicit computation to show that solutions of $z^2 - c = 0$ can be found for any complex number c .
- Explain why the solution of $z^n - c = 0$ can be reduced to the case where n is odd.
- Let z_0 be the point where the function $f(z) = z^n - c$ has its minimum absolute value. If $z_0 \neq 0$, show that the integer m in the proof of Theorem 2 is equal to 1; since we can certainly find γ with $\gamma^1 = -\alpha/\beta$, the remainder of the proof works for f . It therefore suffices to show that the minimum absolute value of f does not occur at 0.
- Suppose instead that f has its minimum absolute value at 0. Since n is odd, the points $\pm\delta$, $\pm\delta i$ go under f into $-c \pm \delta^n$, $-c \pm \delta^n i$. Show that for small δ at least one of these points has smaller absolute value than $-c$, thereby obtaining a contradiction.

7. Let $f(z) = (z - z_1)^{m_1} \cdots (z - z_k)^{m_k}$.

(a) Show that $f'(z) = (z - z_1)^{m_1} \cdots (z - z_k)^{m_k} \cdot \sum_{\alpha=1}^k m_\alpha (z - z_\alpha)^{-1}$.

(b) Let $g(z) = \sum_{\alpha=1}^k m_\alpha (z - z_\alpha)^{-1}$. Show that if $g(z) = 0$, then $z_1, \dots, z_k$ cannot all lie on the same side of a straight line through z . Hint: Use Problem 24-11.

(c) A subset K of the plane is **convex** if K contains the line segment joining any two points in it (Figure 8). For any set A , there is a smallest convex set containing it, which is called the **convex hull** of A (Figure 9); if a point P is not in the convex hull of A , then all of A is contained

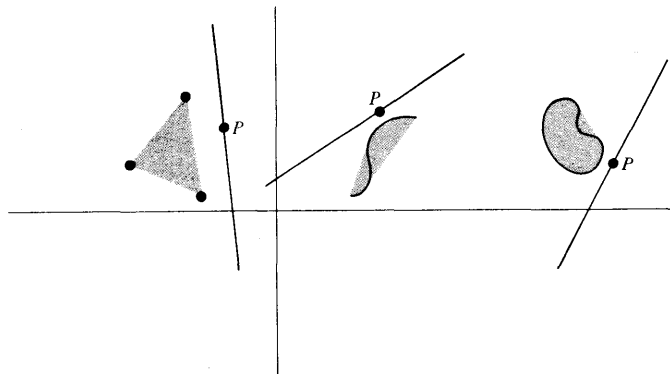


FIGURE 9

on one side of some straight line through P . Using this information, prove that the roots of $f'(z) = 0$ lie within the convex hull of the set $\{z_1, \dots, z_k\}$. (The definition of convexity in the Appendix to Chapter 11 is related to the one given here—a function f is convex if the set of points lying above or on the graph of f is a convex set, and if the graph contains no straight line segments. Further details about convex sets will be found in reference [19] of the Suggested Reading.)

8. Prove that if f is differentiable at z , then f is continuous at z .
- *9. Suppose that $f = u + iv$ where u and v are real-valued functions.
- For fixed y_0 let $g(x) = u(x + iy_0)$ and $h(x) = v(x + iy_0)$. Show that if $f'(x_0 + iy_0) = \alpha + i\beta$ for real α and β , then $g'(x_0) = \alpha$ and $h'(x_0) = \beta$.
 - On the other hand, suppose that $k(y) = u(x_0 + iy)$ and $l(y) = v(x_0 + iy)$. Show that $l'(y_0) = \alpha$ and $k'(y_0) = -\beta$.
 - Suppose that $f'(z) = 0$ for all z . Show that f is a constant function.
10. (a) Using the expression

$$f(x) = \frac{1}{1+x^2} = \frac{1}{2i} \left(\frac{1}{x-i} - \frac{1}{x+i} \right),$$

find $f^{(k)}(x)$ for all k .

- (b) Use this result to find $\arctan^{(k)}(0)$ for all k .

CHAPTER 26 COMPLEX POWER SERIES

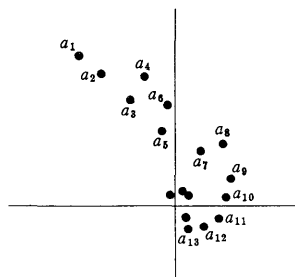


FIGURE 1

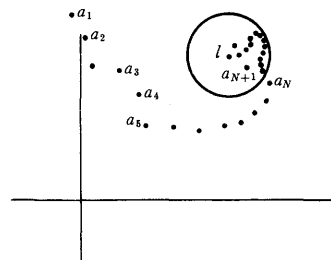


FIGURE 2

If you have not already guessed where differentiable complex functions are going to come from, the title of this chapter should give the secret away: we intend to define functions by means of infinite series. This will necessitate a discussion of infinite sequences of complex numbers, and sums of such sequences, but (as was the case with limits and continuity) the basic definitions are almost exactly the same as for real sequences and series.

An **infinite sequence** of complex numbers is, formally, a complex-valued function whose domain is $\mathbf{N}$; the convenient subscript notation for sequences of real numbers will also be used for sequences of complex numbers. A sequence $\{a_n\}$ of complex numbers is most conveniently pictured by labeling the points a_n in the plane (Figure 1).

The sequence shown in Figure 1 converges to 0, “convergence” of complex sequences being defined precisely as for real sequences: the sequence $\{a_n\}$ **converges** to l , in symbols

$$\lim_{n \rightarrow \infty} a_n = l,$$

if for every $\varepsilon > 0$ there is a natural number N such that, for all n ,

$$\text{if } n > N, \text{ then } |a_n - l| < \varepsilon.$$

This condition means that any circle drawn around l will contain a_n for all sufficiently large n (Figure 2); expressed more colloquially, the sequence is eventually inside any circle drawn around l .

Convergence of complex sequences is not only defined precisely for real sequences, but can even be reduced to this familiar case.

THEOREM 1 Let

$$a_n = b_n + ic_n \quad \text{for real } b_n \text{ and } c_n,$$

and let

$$l = \beta + i\gamma \quad \text{for real } \beta \text{ and } \gamma.$$

Then $\lim_{n \rightarrow \infty} a_n = l$ if and only if

$$\lim_{n \rightarrow \infty} b_n = \beta \quad \text{and} \quad \lim_{n \rightarrow \infty} c_n = \gamma.$$

PROOF

The proof is left as an easy exercise. If there is any doubt as to how to proceed, consult the similar Theorem 1 of Chapter 25. ■

The **sum** of a sequence $\{a_n\}$ is defined, once again, as $\lim_{n \rightarrow \infty} s_n$, where

$$s_n = a_1 + \cdots + a_n.$$

Sequences for which this limit exists are **summable**; alternatively, we may say that the infinite series $\sum_{n=1}^{\infty} a_n$ **converges** if this limit exists, and **diverges** otherwise. It is unnecessary to develop any new tests for convergence of infinite series, because of the following theorem.

THEOREM 2 Let

$$a_n = b_n + ic_n \quad \text{for real } b_n \text{ and } c_n.$$

Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ both converge, and in this case

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} b_n + i \left(\sum_{n=1}^{\infty} c_n \right).$$

PROOF This is an immediate consequence of Theorem 1 applied to the sequence of partial sums of $\{a_n\}$. ■

There is also a notion of absolute convergence for complex series: the series $\sum_{n=1}^{\infty} a_n$ **converges absolutely** if the series $\sum_{n=1}^{\infty} |a_n|$ converges (this is a series of real numbers, and consequently one to which our earlier tests may be applied). The following theorem is not quite so easy as the preceding two.

THEOREM 3 Let

$$a_n = b_n + ic_n \quad \text{for real } b_n \text{ and } c_n.$$

Then $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ both converge absolutely.

PROOF Suppose first that $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ both converge absolutely, i.e., that $\sum_{n=1}^{\infty} |b_n|$ and $\sum_{n=1}^{\infty} |c_n|$ both converge. It follows that $\sum_{n=1}^{\infty} |b_n| + |c_n|$ converges. Now,

$$|a_n| = |b_n + ic_n| \leq |b_n| + |c_n|.$$

It follows from the comparison test that $\sum_{n=1}^{\infty} |a_n|$ converges (the numbers $|a_n|$ and $|b_n| + |c_n|$ are real and nonnegative). Thus $\sum_{n=1}^{\infty} a_n$ converges absolutely.

Now suppose that $\sum_{n=1}^{\infty} |a_n|$ converges. Since

$$|a_n| = \sqrt{b_n^2 + c_n^2},$$

it is clear that

$$|b_n| \leq |a_n| \quad \text{and} \quad |c_n| \leq |a_n|.$$

Once again, the comparison test shows that $\sum_{n=1}^{\infty} |b_n|$ and $\sum_{n=1}^{\infty} |c_n|$ converge. ■

Two consequences of Theorem 3 are particularly noteworthy. If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ also converge absolutely; consequently $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ converge, by Theorem 22-4, so $\sum_{n=1}^{\infty} a_n$ converges by Theorem 2. In other words, absolute convergence implies convergence. Similar reasoning shows that any rearrangement of an absolutely convergent series has the same sum. These facts can also be proved directly, without using the corresponding theorems for real numbers, by first establishing an analogue of the Cauchy criterion (see Problem 12).

With these preliminaries safely disposed of, we can now consider **complex power series**, that is, functions of the form

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n = a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots$$

Here the numbers a and a_n are allowed to be complex, and we are naturally interested in the behavior of f for complex z . As in the real case, we shall usually consider power series centered at 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n;$$

in this case, if $f(z_0)$ converges, then $f(z)$ will also converge for $|z| < |z_0|$. The proof of this fact will be similar to the proof of Theorem 23-6, but, for reasons that will soon become clear, we will not use all the paraphernalia of uniform convergence and the Weierstrass M -test, even though they have complex analogues. Our next theorem consequently generalizes only a small part of Theorem 23-6.

THEOREM 4 Suppose that

$$\sum_{n=0}^{\infty} a_n z_0^n = a_0 + a_1 z_0 + a_2 z_0^2 + \cdots$$

converges for some $z_0 \neq 0$. Then if $|z| < |z_0|$, the two series

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots$$

$$\sum_{n=1}^{\infty} n a_n z^{n-1} = a_1 + 2a_2 z + 3a_3 z^2 + \cdots$$

both converge absolutely.

PROOF As in the proof of Theorem 23-6, we will need only the fact that the set of numbers $a_n z_0^n$ is bounded: there is a number M such that

$$|a_n z_0^n| \leq M \quad \text{for all } n.$$

We then have

$$|a_n z^n| = |a_n z_0^n| \cdot \left| \frac{z}{z_0} \right|^n$$

$$\leq M \left| \frac{z}{z_0} \right|^n,$$

and, for $z \neq 0$,

$$|n a_n z^{n-1}| = \frac{1}{|z|} n |a_n z_0^n| \cdot \left| \frac{z}{z_0} \right|^n$$

$$\leq \frac{M}{|z|} n \left| \frac{z}{z_0} \right|^n.$$

Since the series $\sum_{n=0}^{\infty} |z/z_0|^n$ and $\sum_{n=1}^{\infty} n |z/z_0|^{n-1}$ converge, this shows that both $\sum_{n=0}^{\infty} a_n z^n$ and $\sum_{n=1}^{\infty} n a_n z^{n-1}$ converge absolutely (the argument for $\sum_{n=1}^{\infty} n a_n z^{n-1}$ assumed that $z \neq 0$, but this series certainly converges for $z = 0$ also). ■

Theorem 4 evidently restricts greatly the possibilities for the set

$$\{z: \sum_{n=0}^{\infty} a_n z^n \text{ converges}\}.$$

For example, the shaded set A in Figure 3 cannot be the set of all z where $\sum_{n=0}^{\infty} a_n z^n$ converges, since it contains z , but not the number w satisfying $|w| < |z|$.

It seems quite unlikely that the set of points where a power series converges could be anything except the set of points inside a circle. If we allow “circles of radius 0” (when the power series converges only at 0) and “circles of radius ∞ ” (when the power series converges at all points), then this assertion is true (with one complication which we will soon mention); the proof requires only Theorem 4 and a knack for good organization.

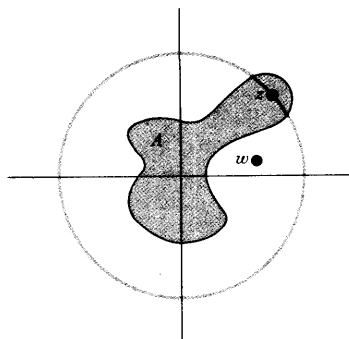


FIGURE 3

THEOREM 5 For any power series

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$$

one of the following three possibilities must be true:

- (1) $\sum_{n=0}^{\infty} a_n z^n$ converges only for $z = 0$.
- (2) $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely for all z in $\mathbf{C}$.
- (3) There is a number $R > 0$ such that $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely if $|z| < R$ and diverges if $|z| > R$. (Notice that we do not mention what happens when $|z| = R$.)

PROOF Let

$$S = \{x \text{ in } \mathbf{R} : \sum_{n=0}^{\infty} a_n w^n \text{ converges for some } w \text{ with } |w| = x\}.$$

Suppose first that S is unbounded. Then for any complex number z , there is a number x in S such that $|z| < x$. By definition of S , this means that $\sum_{n=0}^{\infty} a_n w^n$ converges for some w with $|w| = x > |z|$. It follows from Theorem 4 that $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely. Thus, in this case possibility (2) is true.

Now suppose that S is bounded, and let R be the least upper bound of S . If $R = 0$, then $\sum_{n=0}^{\infty} a_n z^n$ converges only for $z = 0$, so possibility (1) is true. Suppose, on the other hand, that $R > 0$. Then if z is a complex number with $|z| < R$, there is a number x in S with $|z| < x$. Once again, this means that $\sum_{n=0}^{\infty} a_n w^n$ converges for some w with $|z| < |w|$, so that $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely. Moreover, if $|z| > R$, then $\sum_{n=0}^{\infty} a_n z^n$ does not converge, since $|z|$ is not in S . ■

The number R which occurs in case (3) is called the **radius of convergence** of $\sum_{n=0}^{\infty} a_n z^n$. In cases (1) and (2) it is customary to say that the radius of convergence is 0 and ∞ , respectively. When $0 < R < \infty$, the circle $\{z : |z| = R\}$ is called the **circle of convergence** of $\sum_{n=0}^{\infty} a_n z^n$. If z is outside the circle, then,

of course, $\sum_{n=0}^{\infty} a_n z^n$ does not converge, but actually a much stronger statement can be made: the terms $a_n z^n$ are not even bounded. To prove this, let w be any number with $|z| > |w| > R$; if the terms $a_n z^n$ were bounded, then the proof of Theorem 4 would show that $\sum_{n=0}^{\infty} a_n w^n$ converges, which is false. Thus

(Figure 4), inside the circle of convergence the series $\sum_{n=0}^{\infty} a_n z^n$ converges in the best possible way (absolutely) and outside the circle the series diverges in the worst possible way (the terms $a_n z^n$ are not bounded).

What happens *on* the circle of convergence is a much more difficult question. We will not consider that question at all, except to mention that there are power series which converge everywhere on the circle of convergence, power series which converge nowhere on the circle of convergence, and power series that do just about anything in between. (See Problem 5.)

Remember that our goal in this chapter is to produce differentiable functions. We therefore want to generalize the result proved for real power series in Chapter 23, that a function defined by a power series can be differentiated term-by-term inside the circle of convergence. At this point we can no longer imitate the proof of Chapter 23, even if we were willing to introduce uniform convergence, because no analogue of Theorem 23-3 seems available. Instead we will use a direct argument (which could also have been used in Chapter 23). Before beginning the proof, we notice that at least there is no problem about the convergence of the series produced by term-by-term differentiation. If the series $\sum_{n=0}^{\infty} a_n z^n$ has radius of convergence R , then Theorem 4 immediately

implies that the series $\sum_{n=1}^{\infty} n a_n z^{n-1}$ also converges for $|z| < R$. Moreover, if $|z| > R$, so that the terms $a_n z^n$ are unbounded, then the terms $n a_n z^{n-1}$ are surely unbounded, so $\sum_{n=1}^{\infty} n a_n z^{n-1}$ does not converge. This shows that the radius of convergence of $\sum_{n=1}^{\infty} n a_n z^{n-1}$ is also exactly R .

THEOREM 6 If the power series

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

has radius of convergence $R > 0$, then f is differentiable at z for all z with $|z| < R$, and

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

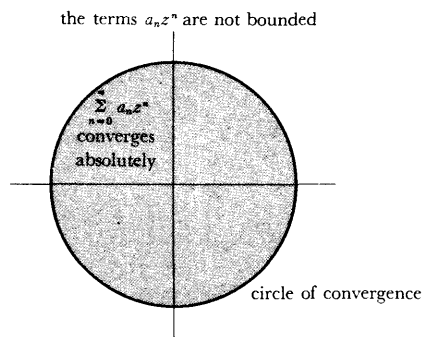


FIGURE 4

PROOF We will use another “ $\varepsilon/3$ argument.” The fact that the theorem is clearly true for polynomial functions suggests writing

$$\begin{aligned}
 (*) \quad \left| \frac{f(z+h) - f(z)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} \right| &= \left| \sum_{n=0}^{\infty} a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} \right| \\
 &\leq \left| \sum_{n=0}^{\infty} a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} \right| \\
 &\quad + \left| \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=1}^N na_n z^{n-1} \right| \\
 &\quad + \left| \sum_{n=1}^N na_n z^{n-1} - \sum_{n=1}^{\infty} na_n z^{n-1} \right|.
 \end{aligned}$$

We will show that for any $\varepsilon > 0$, each absolute value on the right side can be made $< \varepsilon/3$ by choosing N sufficiently large and h sufficiently small. This will clearly prove the theorem.

Only the first term in the right side of $(*)$ will present any difficulties. To begin with, choose some z_0 with $|z| < |z_0| < R$; henceforth we will consider only h with $|z+h| \leq |z_0|$. The expression $((z+h)^n - z^n)/h$ can be written in a more convenient way if we remember that

$$\frac{x^n - y^n}{x - y} = x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1}.$$

Applying this to

$$\frac{(z+h)^n - z^n}{h} = \frac{(z+h)^n - z^n}{(z+h) - z},$$

we obtain

$$\frac{(z+h)^n - z^n}{h} = (z+h)^{n-1} + z(z+h)^{n-2} + \cdots + z^{n-1}.$$

Since

$$|(z+h)^{n-1} + z(z+h)^{n-2} + \cdots + z^{n-1}| \leq n|z_0|^{n-1},$$

we have

$$\left| a_n \frac{((z+h)^n - z^n)}{h} \right| \leq n|a_n| \cdot |z_0|^{n-1}.$$

But the series $\sum_{n=1}^{\infty} n|a_n| \cdot |z_0|^{n-1}$ converges, so if N is sufficiently large, then

$$\sum_{n=N+1}^{\infty} n|a_n| \cdot |z_0|^{n-1} < \frac{\varepsilon}{3}.$$

This means that

$$\begin{aligned} & \left| \sum_{n=0}^{\infty} a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} \right| \\ &= \left| \sum_{n=N+1}^{\infty} a_n \frac{((z+h)^n - z^n)}{h} \right| \leq \sum_{n=N+1}^{\infty} \left| a_n \frac{((z+h)^n - z^n)}{h} \right| \\ &\leq \sum_{n=N+1}^{\infty} n|a_n| \cdot |z_0|^{n-1} < \frac{\varepsilon}{3}. \end{aligned}$$

In short, if N is sufficiently large, then

$$(1) \left| \sum_{n=0}^{\infty} a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} \right| < \frac{\varepsilon}{3}$$

for all h with $|z+h| \leq |z_0|$.

It is easy to deal with the third term on the right side of (*): Since $\sum_{n=1}^{\infty} na_n z^{n-1}$ converges, it follows that if N is sufficiently large, then

$$(2) \left| \sum_{n=1}^{\infty} na_n z^{n-1} - \sum_{n=1}^N na_n z^{n-1} \right| < \frac{\varepsilon}{3}.$$

Finally, choosing an N such that (1) and (2) are true, we note that

$$\lim_{h \rightarrow 0} \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} = \sum_{n=1}^N na_n z^{n-1},$$

since the polynomial function $g(z) = \sum_{n=0}^N a_n z^n$ is certainly differentiable.

Therefore

$$(3) \left| \sum_{n=0}^N a_n \frac{((z+h)^n - z^n)}{h} - \sum_{n=1}^N na_n z^{n-1} \right| < \frac{\varepsilon}{3}$$

for sufficiently small h .

As we have already indicated, (1), (2), and (3) prove the theorem. ■

Theorem 6 has an obvious corollary: a function represented by a power series is infinitely differentiable inside the circle of convergence, and the power series is its Taylor series at 0. It follows, in particular, that f is continuous inside the circle of convergence, since a function differentiable at z is continuous at z (Problem 25-8).

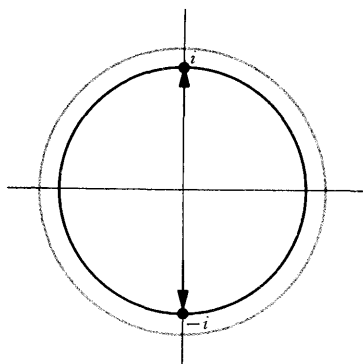


FIGURE 5

The continuity of a power series inside its circle of convergence helps explain the behavior of certain Taylor series obtained for real functions, and gives the promised answers to the questions raised at the end of Chapter 23. We have already seen that the Taylor series for the function $f(z) = 1/(1 + z^2)$, namely,

$$1 - z^2 + z^4 - z^6 + \cdots,$$

converges for real z only when $|z| < 1$, and consequently has radius of convergence 1. It is no accident that the circle of convergence contains the two points i and $-i$ at which f is undefined. If this power series converged in a circle of radius greater than 1, then (Figure 5) it would represent a function which was continuous in that circle, in particular at i and $-i$. But this is impossible, since it equals $1/(1 + z^2)$ inside the unit circle, and $1/(1 + z^2)$ does not approach a limit as z approaches i or $-i$ from inside the unit circle.

The use of complex numbers also sheds some light on the strange behavior of the Taylor series for the function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Although we have not yet defined e^z for complex z , it will presumably be true that if y is real and unequal to 0, then

$$f(iy) = e^{-1/(iy)^2} = e^{1/y^2}.$$

The interesting fact about this expression is that it becomes large as y becomes small. Thus f will not even be continuous at 0 when defined for complex numbers, so it is hardly surprising that it is equal to its Taylor series only for $z = 0$.

The method by which we will actually define e^z (as well as $\sin z$ and $\cos z$) for complex z should by now be clear. For real x we know that

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots,$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots$$

For complex z we therefore *define*

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots,$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots,$$

$$\exp(z) = e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} - \cdots$$

Then $\sin'(z) = \cos z$, $\cos'(z) = -\sin z$, and $\exp'(z) = \exp(z)$ by Theorem 6.

Moreover, if we replace z by iz in the series for e^z , something particularly interesting happens:

$$\begin{aligned} e^{iz} &= 1 + iz + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \frac{(iz)^4}{4!} + \frac{(iz)^5}{5!} + \cdots \\ &= 1 + iz - \frac{z^2}{2!} - \frac{iz^3}{3!} + \frac{z^4}{4!} + \frac{iz^5}{5!} + \cdots \\ &= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots\right) + i\left(z - \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots\right) \end{aligned}$$

(this step can be justified by the fact that the series converges absolutely), so

$$e^{iz} = \cos z + i \sin z.$$

It is clear from the definitions (i.e., the power series) that

$$\begin{aligned} \sin(-z) &= -\sin z, \\ \cos(-z) &= \cos z, \end{aligned}$$

so we also have

$$e^{-iz} = \cos z - i \sin z.$$

From the equations for e^{iz} and e^{-iz} we can derive the formulas

$$\begin{aligned} \sin z &= \frac{e^{iz} - e^{-iz}}{2i}, \\ \cos z &= \frac{e^{iz} + e^{-iz}}{2}. \end{aligned}$$

The development of complex power series thus places the exponential function at the very core of the development of the elementary functions—it reveals a connection between the trigonometric and exponential functions which was never imagined when these functions were first defined, and which could never have been discovered without the use of complex numbers. As a by-product of this relationship, we obtain a hitherto unsuspected connection between the numbers e and π : if in the formula

$$e^{iz} = \cos z + i \sin z$$

we take $z = \pi$, we obtain the remarkable result

$$e^{i\pi} = -1.$$

With these remarks we will bring to a close our investigation of complex functions. And yet there are still several basic facts about power series which have not been mentioned. Thus far, we have seldom considered power series centered at a ,

$$f(z) = \sum_{n=0}^{\infty} a_n(z - a)^n,$$

except for $a = 0$. This omission was adopted partly to simplify the exposition.

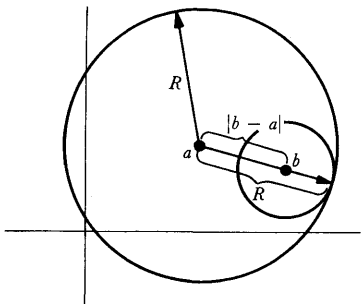


FIGURE 6

For power series centered at a there are obvious versions of all the theorems in this chapter (the proofs require only trivial modifications): there is a number R (possibly 0 or " ∞ ") such that the series $\sum_{n=0}^{\infty} a_n(z - a)^n$ converges absolutely for z with $|z - a| < R$, and has unbounded terms for z with $|z - a| > R$; moreover, for all z with $|z - a| < R$ the function

$$f(z) = \sum_{n=0}^{\infty} a_n(z - a)^n$$

has derivative

$$f'(z) = \sum_{n=1}^{\infty} n a_n(z - a)^{n-1}.$$

It is less straightforward to investigate the possibility of representing a function as a power series centered at b , if it is already written as a power series centered at a . If

$$f(z) = \sum_{n=0}^{\infty} a_n(z - a)^n$$

has radius of convergence R , and b is a point with $|b - a| < R$ (Figure 6), then it is true that $f(z)$ can also be written as a power series centered at b ,

$$f(z) = \sum_{n=0}^{\infty} b_n(z - b)^n$$

(the numbers b_n are necessarily $f^{(n)}(b)/n!$); moreover, this series has radius of convergence at least $R - |b - a|$ (it may be larger).

We will *not* prove the facts mentioned in the previous paragraph, and there are several other important facts we shall not prove. If

$$f(z) = \sum_{n=0}^{\infty} a_n(z - a)^n \quad \text{and} \quad g(z) = \sum_{n=0}^{\infty} b_n(z - a)^n$$

converge in some circle around a , it is certainly to be hoped that $f + g$ and $f \cdot g$ will also converge in a circle around a ; moreover, if $g(a) \neq 0$, the function $1/g$ should have a power series representation. Finally, if

$$f(z) = \sum_{n=0}^{\infty} a_n(z - a)^n \quad \text{and} \quad g(z) = \sum_{n=0}^{\infty} b_n(z - b)^n,$$

and $g(b) \neq 0$, then it should be possible to write $f \circ g$ as a power series centered at b .

All of these facts could be proved now without introducing any basic new ideas. The treatment of $f + g$ is easy, while $f \cdot g$ presents a few problems, and $1/g$ presents even more problems. The possibility of changing a power series centered at a into one centered at b becomes yet more involved, and the treatment of $f \circ g$ requires real skill. Rather than end this section with a *tour*

de force of computations, we will instead give a preview of “complex analysis,” one of the most beautiful branches of mathematics, where all these facts are derived as straightforward consequences of some fundamental results.

Power series were introduced in this chapter in order to provide complex functions which are differentiable. Since these functions are actually infinitely differentiable, it is natural to suppose that we have therefore selected only a very special collection of differentiable complex functions. The basic theorems of complex analysis show that this is not at all true:

If a complex function is defined in some region A of the plane and is differentiable in A , then it is automatically infinitely differentiable in A . Moreover, for each point a in A the Taylor series for f at a will converge to f in any circle contained in A (Figure 7).

These facts are among the first to be proved in complex analysis. It is impossible to give any idea of the proofs themselves—the methods used are quite different from anything in elementary calculus. If these facts are granted, however, then the facts mentioned before can be proved very easily.

Suppose, for example, that f and g are functions which can be written as power series. Then, as we have shown, f and g are differentiable—it then follows from easy general theorems that $f + g$, $f \cdot g$, $1/g$ and $f \circ g$ are also differentiable. Appealing to the results from complex analysis, it follows that they can be written as power series.

Similarly, if

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$$

has radius of convergence R , then f is differentiable in the region $A = \{z: |z-a| < R\}$. Thus, if b is in A , it is possible to write f as a power series centered at b , which will converge in the circle of radius $R - |b-a|$. This

series may actually converge in a larger circle, because $\sum_{n=0}^{\infty} a_n(z-a)^n$ may be the series for a function differentiable in a larger region than A . For example, suppose that $f(z) = 1/(1+z^2)$. Then f is differentiable, except at i and $-i$,

where it is not defined. Thus $f(z)$ can be written as a power series $\sum_{n=0}^{\infty} a_n z^n$ with radius of convergence 1 (as a matter of fact, we know that $a_{2n} = (-1)^n$ and $a_k = 0$ if k is odd). It is also possible to write

$$f(z) = \sum_{n=0}^{\infty} b_n(z - \tfrac{1}{2})^n,$$

where the numbers b_n are necessarily $b_n = f^{(n)}(\frac{1}{2})/n!$. We can easily predict the radius of convergence of this series: it is $\sqrt{1 + (\frac{1}{2})^2}$, the distance from $\frac{1}{2}$ to i or $-i$ (Figure 8).

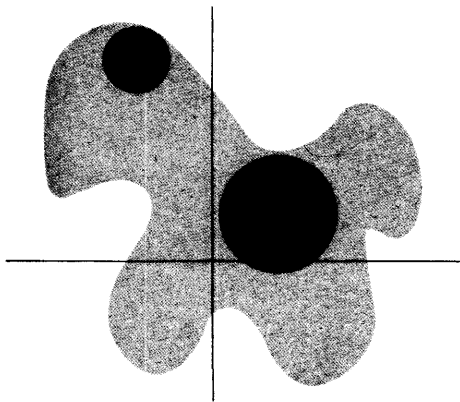


FIGURE 7

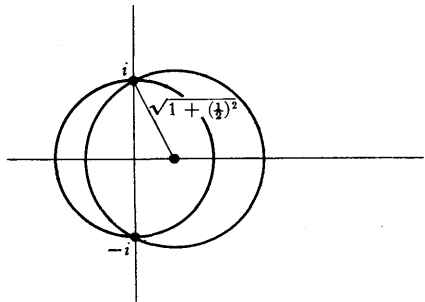


FIGURE 8

As an added incentive to investigate complex analysis further, one more result will be mentioned, which lies quite near the surface, and which will be found in any treatment of the subject.

For real z the values of $\sin z$ always lie between -1 and 1 , but for complex z this is not at all true. In fact, if $z = iy$, for y real, then

$$\sin iy = \frac{e^{i(iy)} - e^{-i(iy)}}{2i} = \frac{e^{-y} - e^y}{2i}.$$

If y is large, then $\sin iy$ is also large in absolute value. This behavior of $\sin$ is typical of functions which are defined and differentiable on the whole complex plane (such functions are called *entire*). A result which comes quite early in complex analysis is the following:

Liouville's Theorem: The only bounded entire functions are the constant functions.

As a simple application of Liouville's Theorem, consider a polynomial function

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n,$$

where $n > 1$, so that f is not a constant. We already know that $f(z)$ is large for large z , so Liouville's Theorem tells us nothing interesting about f . But consider the function

$$g(z) = \frac{1}{f(z)}.$$

If $f(z)$ were never 0, then g would be entire; since $f(z)$ becomes large for large z , the function g would also be bounded, contradicting Liouville's Theorem. Thus $f(z) = 0$ for some z , and we have proved the Fundamental Theorem of Algebra.

PROBLEMS

1. Decide whether each of the following series converges, and whether it converges absolutely.

$$(i) \quad \sum_{n=1}^{\infty} \frac{(1+i)^n}{n!}.$$

$$(ii) \quad \sum_{n=1}^{\infty} \frac{1+2i}{2^n}.$$

$$(iii) \quad \sum_{n=1}^{\infty} \frac{i^n}{n}.$$

$$(iv) \sum_{n=1}^{\infty} \left(\frac{1}{2} + \frac{1}{2}i\right)^n.$$

$$(v) \sum_{n=2}^{\infty} \frac{\log n}{n} + i^n \frac{\log n}{n}.$$

2. Use the ratio test to show that the radius of convergence of each of the following power series is 1. (In each case the ratios of successive terms will approach a limit < 1 if $|z| < 1$, but for $|z| > 1$ the ratios will tend to ∞ or to a limit > 1 .)

$$(i) \sum_{n=1}^{\infty} \frac{z^n}{n^2}.$$

$$(ii) \sum_{n=1}^{\infty} \frac{z^n}{n}.$$

$$(iii) \sum_{n=1}^{\infty} z^n.$$

$$(iv) \sum_{n=1}^{\infty} (n + 2^{-n})z^n.$$

$$(v) \sum_{n=1}^{\infty} 2^n z^{n!}.$$

3. Use the root test (Problem 22-7) to find the radius of convergence of each of the following power series.

$$(i) \frac{z}{2} + \frac{z^2}{3} + \frac{z^3}{2^2} + \frac{z^4}{3^2} + \frac{z^5}{2^3} + \frac{z^6}{3^3} + \cdots$$

$$(ii) \sum_{n=1}^{\infty} \frac{n! z^n}{n^n}.$$

$$(iii) \sum_{n=1}^{\infty} \frac{n}{2^n} z^n.$$

$$(iv) \sum_{n=1}^{\infty} \frac{n^2}{2^n} z^n.$$

$$(v) \sum_{n=1}^{\infty} 2^n z^{n!}.$$

4. The root test can always be used, in theory at least, to find the radius of convergence of a power series; in fact, a close analysis of the situation

leads to a formula for the radius of convergence, known as the “Cauchy-Hadamard formula.” Suppose first that the set of numbers $\sqrt[n]{|a_n|}$ is bounded.

(a) Use Problem 22-7 to show that if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} z < 1$, then $\sum_{n=0}^{\infty} a_n z^n$ converges.

(b) Also show that if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} z > 1$, then $\sum_{n=0}^{\infty} a_n z^n$ has unbounded terms.

(c) Parts (a) and (b) show that the radius of convergence of $\sum_{n=0}^{\infty} a_n z^n$ is $1/\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ (where “ $1/0$ ” means “ ∞ ”). To complete the formula, define $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$ if the set of all $\sqrt[n]{|a_n|}$ is unbounded. Prove that in this case, $\sum_{n=0}^{\infty} a_n z^n$ diverges for $z \neq 0$, so that the radius of convergence is 0 (which may be considered as “ $1/\infty$ ”).

5. Consider the following three series from Problem 2:

$$\sum_{n=1}^{\infty} \frac{z^n}{n^2}, \quad \sum_{n=1}^{\infty} \frac{z^n}{n}, \quad \sum_{n=1}^{\infty} z^n.$$

Prove that the first series converges everywhere on the unit circle; that the third series converges nowhere on the unit circle; and that the second series converges for at least one point on the unit circle and diverges for at least one point on the unit circle.

6. (a) Prove that $e^z \cdot e^w = e^{z+w}$ for all complex z and w . (You just have to imitate the relevant parts of Problem 17-35.)
 (b) Show that $\sin(z+w) = \sin z \cos w + \cos z \sin w$ and $\cos(z+w) = \cos z \cos w - \sin z \sin w$ for all complex z and w .
7. (a) Prove that every complex number of absolute value 1 can be written e^{iy} for some real number y .
 (b) Prove that $|e^{x+iy}| = e^x$ for real x and y .
8. (a) Prove that $\exp$ takes on every complex value except 0.
 (b) Prove that $\sin$ takes on every complex value.
9. (a) Show that a complex-valued function $f = u + iv$ satisfies the equation

$$(*) f^{(n)} + a_{n-1}f^{(n-1)} + \cdots + a_0 f = 0,$$

if and only if u and v do. Hint: Use Problem 25-9.

- (b) Show that if $a = b + ci$ is a complex root of the equation $z^n + a_{n-1}z^{n-1} + \cdots + a_0 = 0$, then $f(x) = e^{bx} \sin cx$ and $f(x) = e^{bx} \cos cx$ are both solutions of (*).

10. (a) Show that $\exp$ is *not* one-one on $\mathbf{C}$.
 (b) Given $w \neq 0$, show that $e^z = w$ if and only if $z = x + iy$ with $x = \log |w|$ (here $\log$ denotes the real logarithm function), and y an argument of w .
 *(c) Show that there does not exist a continuous function $\log$ defined for nonzero complex numbers, such that $\exp(\log(z)) = z$ for all $z \neq 0$. (Show that $\log$ cannot even be defined continuously for $|z| = 1$.)

Since there is no way to define a continuous logarithm function we cannot speak of *the* logarithm of a complex number, but only of "a logarithm for w ," meaning one of the infinitely many numbers z with $e^z = w$.

- (d) Find all logarithms for i .
 (e) Find all values of i^i , that is, of e^{iz} where z is a logarithm of i . (The answers will be real numbers.)
 (f) Show that $(1^i)^i$ has infinitely many possible values, while 1^{ii} has only one.
11. (a) For real x show that we can choose $\log(x + i)$ and $\log(x - i)$ to be

$$\log(x + i) = \log(1 + x^2) + i\left(\frac{\pi}{2} - \arctan x\right),$$

$$\log(x - i) = \log(1 + x^2) - i\left(\frac{\pi}{2} - \arctan x\right).$$

(It will help to note that $\pi/2 - \arctan x = \arctan 1/x$ for $x \neq 0$.)

- (b) The expression

$$\frac{1}{1 + x^2} = \frac{1}{2i} \left(\frac{1}{x - i} - \frac{1}{x + i} \right)$$

yields, formally,

$$\int \frac{dx}{1 + x^2} = \frac{1}{2i} [\log(x - i) - \log(x + i)].$$

Use part (a) to check that this answer agrees with the usual one.

12. (a) A sequence $\{a_n\}$ of complex numbers is called a **Cauchy sequence** if $\lim_{m, n \rightarrow \infty} |a_m - a_n| = 0$. Suppose that $a_n = b_n + ic_n$, where b_n and c_n are real. Prove that $\{a_n\}$ is a Cauchy sequence if and only if $\{b_n\}$ and $\{c_n\}$ are Cauchy sequences.
 (b) Prove that every Cauchy sequence of complex numbers converges.
 (c) Give direct proofs, without using theorems about real series, that an absolutely convergent series is convergent and that any rearrangement has the same sum. (It is permitted, and in fact advisable, to use the *proofs* of the corresponding theorems for real series.)

13. Prove the formula

$$\frac{1}{2} + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{\sin(n + \frac{1}{2})\theta}{2 \sin \frac{\theta}{2}},$$

by writing $\cos n\theta = (e^{in\theta} + e^{-in\theta})/2$, summing the resulting geometric series, and multiplying top and bottom by $e^{-i\theta/2}$.

14. Let
- $\{a_n\}$
- be the Fibonacci sequence,
- $a_1 = a_2 = 1$
- ,
- $a_{n+2} = a_n + a_{n+1}$
- .

(a) If $r_n = a_{n+1}/a_n$, show that $r_{n+1} = 1 + 1/r_n$.

(b) Show that $r = \lim_{n \rightarrow \infty} r_n$ exists, and $r = 1 + 1/r$. Conclude that

$$r = (1 + \sqrt{5})/2.$$

(c) Show that $\sum_{n=1}^{\infty} a_n z^n$ has radius of convergence $2/(1 + \sqrt{5})$. (Using

the unproved theorems in this chapter, and the fact that $\sum_{n=1}^{\infty} a_n z^n = -1/(z^2 + z - 1)$ from Problem 23-8, we could have predicted that the radius of convergence is the smallest absolute value of the roots of $z^2 + z - 1 = 0$; since the roots are $(-1 \pm \sqrt{5})/2$, the radius of convergence should be $(-1 + \sqrt{5})/2$. Notice that this number is indeed equal to $2/(1 + \sqrt{5})$.)

15. Since $(e^z - 1)/z$ can be written as the power series $1 + z/2! + z^2/3! + \cdots$ which is nonzero at 0, it follows that if $g(z) = z/(e^z - 1)$ for $z \neq 0$, and $g(0) = 1$, then g is differentiable inside some circle around 0. It then follows from the unproved theorem in this chapter that there is a power series

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{b_n}{n!} z^n$$

with nonzero radius of convergence. We can even predict the radius of convergence; it is 2π , since this is the smallest absolute value of the numbers $z = 2k\pi i$ for which $e^z - 1 = 0$. The numbers b_n appearing here are called the **Bernoulli numbers**.*

(a) Clearly $b_0 = g(0) = 1$. Now show that

$$\begin{aligned} \frac{z}{e^z - 1} &= -\frac{z}{2} + \frac{z}{2} \cdot \frac{e^z + 1}{e^z - 1}, \\ \frac{e^{-z} + 1}{e^{-z} - 1} &= -\frac{e^z + 1}{e^z - 1}, \end{aligned}$$

* Sometimes the numbers $B_n = (-1)^{n-1}b_{2n}$ are called the Bernoulli numbers, because $b_n = 0$ if n is odd and > 1 (see part ((a))) and because the numbers b_{2n} alternate in sign, although we will not prove this. Other modifications of this nomenclature are also in use.

and deduce that

$$b_1 = -\frac{1}{2}, \quad b_n = 0 \quad \text{if } n \text{ is odd and } n > 1.$$

(b) By finding the coefficient of z^n in the right side of the equation

$$z = \left(\sum_{k=0}^{\infty} \frac{b_k}{k!} z^k \right) \left(z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \right),$$

show that

$$\sum_{i=0}^{n-1} \binom{n}{i} b_i = 0 \quad \text{for } n > 1.$$

This formula allows us to compute any b_k in terms of previous ones, and shows that each is rational. Calculate two or three of the following:

$$b_2 = \frac{1}{6}, \quad b_4 = -\frac{1}{30}, \quad b_6 = \frac{1}{42}, \quad b_8 = -\frac{1}{30}.$$

*(c) Part (a) shows that

$$\sum_{n=0}^{\infty} \frac{b_{2n}}{(2n)!} z^{2n} = \frac{z}{2} \cdot \frac{e^z + 1}{e^z - 1} = \frac{z}{2} \cdot \frac{e^{z/2} + e^{-z/2}}{e^{z/2} - e^{-z/2}}.$$

Replace z by $2iz$ and show that

$$z \cot z = \sum_{n=0}^{\infty} \frac{b_{2n}}{(2n)!} (-1)^n 2^{2n} z^{2n}.$$

*(d) Show that

$$\tan z = \cot z - 2 \cot 2z.$$

*(e) Show that

$$\tan z = \sum_{n=1}^{\infty} \frac{b_{2n}}{(2n)!} (-1)^{n-1} 2^{2n} (2^{2n} - 1) z^{2n-1}.$$

(This series converges for $|z| < \pi/2$.)

16. The Bernoulli numbers play an important role in a theorem which is best introduced by some notational nonsense. Let us use D to denote the “differentiation operator,” so that Df denotes f' . Then $D^k f$ will mean $f^{(k)}$ and $e^D f$ will mean $\sum_{n=0}^{\infty} f^{(n)}/n!$ (of course this series makes no sense in general, but it will make sense if f is a polynomial function, for example). Finally, let Δ denote the “difference operator” for which $\Delta f(x) = f(x+1) - f(x)$. Now Taylor’s Theorem implies, disregarding questions

of convergence, that

$$f(x+1) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!},$$

or

$$(*) \quad f(x+1) - f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x)}{n!};$$

we may write this symbolically as $\Delta f = (e^D - 1)f$, where 1 is the "identity operator." Even more symbolically this can be written $\Delta = e^D - 1$, which suggests that

$$D = \frac{D}{e^D - 1} \Delta.$$

Thus we obviously ought to have

$$D = \sum_{k=0}^{\infty} \frac{b_k}{k!} D^k \Delta,$$

i.e.,

$$(**) \quad f'(x) = \sum_{k=0}^{\infty} \frac{b_k}{k!} [f^{(k)}(x+1) - f^{(k)}(x)].$$

The beautiful thing about all this nonsense is that it works!

- (a) Prove that (**) is literally true if f is a polynomial function (in which case the infinite sum is really a finite sum). Hint: By applying (*) to $f^{(k)}$, find a formula for $f^{(k)}(x+1) - f^{(k)}(x)$; then use the formula in Problem 15(b) to find the coefficient of $f^{(j)}(x)$ in the right side of (**).
- (b) Deduce from (**) that

$$f'(0) + \cdots + f'(n) = \sum_{k=0}^{\infty} \frac{b_k}{k!} [f^{(k)}(n+1) - f^{(k)}(0)].$$

- (c) Show that for any polynomial function g we have

$$g(0) + \cdots + g(n) = \int_0^{n+1} g(t) dt + \sum_{k=1}^{\infty} \frac{b_k}{k!} [g^{(k-1)}(n+1) - g^{(k-1)}(0)].$$

- (d) Apply this to $g(x) = x^p$ to show that

$$\sum_{k=1}^{n-1} k^p = \frac{n^{p+1}}{p+1} + \sum_{k=1}^{p+1} \frac{b_k}{k} \binom{p}{k-1} n^{p-k+1}.$$

Using the fact that $b_1 = -\frac{1}{2}$, show that

$$\sum_{k=1}^n k^p = \frac{n^{p+1}}{p+1} + \frac{n^p}{2} + \sum_{k=2}^{p+1} \frac{b_k}{k} \binom{p}{k-1} n^{p-k+1}.$$

The first ten instances of this formula were written out in Problem 2-6, which offered as a challenge the discovery of the general pattern. This may now seem to be a preposterous suggestion, but the Bernoulli numbers were actually discovered in precisely this way! After writing out these 10 formulas, Bernoulli claims (in his posthumously printed work *Ars Conjectandi*, 1713): "Whoever will examine the series as to their regularity may be able to continue the table." He then writes down the above formula, offering no proof at all, merely noting that the coefficients b_k (which he denoted simply by $A, B, C, \dots$) satisfy the equation in Problem 15(b). The relation between these numbers and the coefficients in the power series for $z/(e^z - 1)$ was discovered by Euler.

- *17. The formula in Problem 16(c) can be generalized to the case where g is not a polynomial function; the infinite sum must be replaced by a finite sum plus a remainder term. In order to find an expression for the remainder, it is useful to introduce some new functions.

(a) The Bernoulli polynomials φ_n are defined by

$$\varphi_n(x) = \sum_{k=0}^n \binom{n}{k} b_{n-k} x^k.$$

The first three are

$$\begin{aligned}\varphi_1(x) &= x - \frac{1}{2}, \\ \varphi_2(x) &= x^2 - x + \frac{1}{6}, \\ \varphi_3(x) &= x^3 - \frac{3x^2}{2} + \frac{x}{2}.\end{aligned}$$

Show that

$$\begin{aligned}\varphi_n(0) &= b_n, \\ \varphi_n(1) &= b_n \quad \text{if } n > 1, \\ \varphi_n'(x) &= n\varphi_{n-1}(x), \\ \varphi_n(x) &= (-1)^n \varphi_n(1-x) \quad \text{for } n > 1.\end{aligned}$$

Hint: Prove the last equation by induction on n , starting with $n = 2$.

- (b) Let $R_N^k(x)$ be the remainder term in Taylor's formula for $f^{(k)}$, on the interval $[x, x+1]$, so that

$$(*) \quad f^{(k)}(x+1) - f^{(k)}(x) = \sum_{n=0}^N \frac{f^{(k+n)}(x)}{n!} + R_N^k(x).$$

Prove that

$$f'(x) = \sum_{k=0}^N \frac{b_k}{k!} [f^{(k)}(x+1) - f^{(k)}(x)] - \sum_{k=0}^N \frac{b_k}{k!} R_{N-k}^k(x).$$

Hint: Imitate Problem 16(a). Notice the subscript $N-k$ on R .

- (c) Use the integral form of the remainder to show that

$$\sum_{k=0}^N \frac{b_k}{k!} R_{N-k}^k(x) = \int_x^{x+1} \frac{\varphi_N(x+1-t)}{N!} f^{(N+1)}(t) dt.$$

- (d) Deduce the "Euler-Maclaurin Summation Formula":

$$\begin{aligned} g(x) + g(x+1) + \cdots + g(x+n) \\ = \int_x^{x+n+1} g(t) dt + \sum_{k=1}^N \frac{b_k}{k!} [g^{(k-1)}(x+n-1) - g^{(k-1)}(x)] + S_N(x, n), \end{aligned}$$

where

$$S_N(x, n) = - \sum_{j=0}^n \int_{x+j}^{x+j+1} \frac{\varphi_N(x+j+1-t)}{N!} g^{(N)}(t) dt.$$

- (e) Let ψ_n be the periodic function, with period 1, which satisfies $\psi_n(t) = \varphi_n(t)$ for $0 \leq t < 1$. (Part (a) implies that if $n > 1$, then ψ_n is continuous, since $\varphi_n(1) = \varphi_n(0)$, and also that φ_n is even if n is even and odd if n is odd.) Show that

$$\begin{aligned} S_N(x, n) &= - \int_x^{x+n+1} \frac{\psi_N(x-t)}{N!} g^{(N)}(t) dt \\ &\left(= (-1)^{N+1} \int_x^{x+n+1} \frac{\psi_N(t)}{N!} g^{(N)}(t) dt \text{ if } x \text{ is an integer} \right). \end{aligned}$$

Unlike the remainder in Taylor's Theorem, the remainder $S_N(x, n)$ usually does not satisfy $\lim_{N \rightarrow \infty} S_N(x, n) = 0$, because the Bernoulli numbers and functions become large very rapidly (although the first few examples do not suggest this). Nevertheless, important information can often be obtained from the summation formula. The general situation is best discussed within the context of a specialized study ("asymptotic series"), but the next problem shows one particularly important example.

****18.** (a) Use the Euler-Maclaurin Formula, with $N = 2$, to show that

$$\log 1 + \cdots + \log (n-1) = \int_1^n \log t \, dt - \frac{1}{2} \log n + \int_1^n \frac{\psi_2(t)}{2t^2} \, dt.$$

(b) Show that

$$\log \left(\frac{n!}{n^{n+1/2} e^{-n}} \right) = \int_1^n \frac{\psi_2(t)}{2t^2} \, dt.$$

(c) Explain why the improper integral $\beta = \int_1^\infty \psi_2(t)/2t^2 \, dt$ exists, and show that if $\alpha = e^\beta$, then

$$\log \left(\frac{n!}{\alpha n^{n+1/2} e^{-n}} \right) = - \int_n^\infty \frac{\psi_2(t)}{2t^2} \, dt.$$

(d) Problem 18-26(d) shows that

$$\sqrt{\pi} = \lim_{n \rightarrow \infty} \frac{(n!)^{2^{2n}}}{(2n)! \sqrt{n}}.$$

Use part (c) to show that

$$\sqrt{\pi} = \lim_{n \rightarrow \infty} \frac{\alpha^2 n^{2n+1} e^{-2n} 2^{2n}}{\alpha (2n)^{2n+1/2} e^{-2n} \sqrt{n}},$$

and conclude that $\alpha = \sqrt{2\pi}$.

(e) Show that the maximum value of $|\varphi_2(x)|$ for x in $[0, 1]$ is $\frac{1}{6}$, so that

$$\left| \int_n^\infty \frac{\psi_2(t)}{2t^2} \, dt \right| \leq \frac{1}{12n}.$$

Conclude that

$$\sqrt{2\pi} n^{n+1/2} e^{-n-1/12n} \leq n! \leq \sqrt{2\pi} n^{n+1/2} e^{-n+1/12n}.$$

The final result of Problem 18, a strong form of Stirling's Formula, shows that $n!$ is approximately $\sqrt{2\pi} n^{n+1/2} e^{-n}$, in the sense that this expression differs from $n!$ by an amount which is small compared to n when n is large. For example, for $n = 10$ we obtain 3598696 instead of 3628800, with an error $< 1\%$.

A more general form of Stirling's Formula illustrates the "asymptotic" nature of the summation formula. The same argument which was used in Problem 18 can now be used to show that for $N \geq 2$ we have

$$\log \left(\frac{n!}{\sqrt{2\pi} n^{n+1/2} e^{-n}} \right) = \sum_{k=2}^N \frac{b_k}{k(k-1)n^{k-1}} \pm \int_n^\infty \frac{\psi_N(t)}{N! t^N} \, dt.$$

Since ψ_N is bounded, we can obtain estimates of the form

$$\left| \int_n^\infty \frac{\psi_N(t)}{N! t^N} dt \right| \leq \frac{M_N}{n^{N-1}}.$$

If N is large, the constant M_N will also be large; but for very large n the factor n^{1-N} will make the product very small. Thus, the expression

$$\sqrt{2\pi} n^{n+1/2} e^{-n} \cdot \exp \left(\sum_{k=2}^N \frac{b_k}{k(k-1)n^{k-1}} \right)$$

may be a very bad approximation for $n!$ when n is small, but for large n (*how* large depends on N) it will be an extremely good one (*how* good depends on N).

*There was a most ingenious Architect
who had contrived a new Method
for building Houses,
by beginning at the Roof, and working
downwards to the Foundation.*

JONATHAN SWIFT

PART 5
EPILOGUE

CHAPTER 27 FIELDS

Throughout this book a conscientious attempt has been made to define all important concepts, even terms like “function,” for which an intuitive definition is often considered sufficient. But $\mathbf{Q}$ and $\mathbf{R}$, the two main protagonists of this story, have only been named, never defined. What has never been defined can never be analyzed thoroughly, and “properties” P1–P13 must be considered assumptions, not theorems, about numbers. Nevertheless, the term “axiom” has been purposely avoided, and in this chapter the logical status of P1–P13 will be scrutinized more carefully.

Like $\mathbf{Q}$ and $\mathbf{R}$, the sets $\mathbf{N}$ and $\mathbf{Z}$ have also remained undefined. True, some talk about all four was inserted in Chapter 2, but those rough descriptions are far from a definition. To say, for example, that $\mathbf{N}$ consists of 1, 2, 3, etc., merely names some elements of $\mathbf{N}$ without identifying them (and the “etc.” is useless). The natural numbers *can* be defined, but the procedure is involved and not quite pertinent to the rest of the book. The Suggested Reading list contains references to this problem, as well as to the other steps that are required if one wishes to develop calculus from its basic logical starting point. The further development of this program would proceed with the definition of $\mathbf{Z}$, in terms of $\mathbf{N}$, and the definition of $\mathbf{Q}$ in terms of $\mathbf{Z}$. This program results in a certain well-defined set $\mathbf{Q}$, certain explicitly defined operations $+$ and $\cdot$, and properties P1–P12 as *theorems*. The final step in this program is the construction of $\mathbf{R}$, in terms of $\mathbf{Q}$. It is this last construction which concerns us. Assuming that $\mathbf{Q}$ has been defined, and that P1–P12 have been proved for $\mathbf{Q}$, we shall ultimately *define* $\mathbf{R}$ and *prove* all of P1–P13 for $\mathbf{R}$.

Our intention of proving P1–P13 means that we must define not only real numbers, but also addition and multiplication of real numbers. Indeed, the real numbers are of interest only as a set together with these operations: how the real numbers behave with respect to addition and multiplication is crucial; what the real numbers may actually be is quite irrelevant. This assertion can be expressed in a meaningful mathematical way, by using the concept of a “field,” which includes as special cases the three important number systems of this book. This extraordinarily important abstraction of modern mathematics incorporates the properties P1–P9 common to $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$. A **field** is a set F (of objects of any sort whatsoever), together with two “binary operations” $+$ and $\cdot$ defined on F (that is, two rules which associate to elements a and b in F , other elements $a + b$ and $a \cdot b$ in F) for which the following conditions are satisfied:

- (1) $(a + b) + c = a + (b + c)$ for all a , b , and c in F .
- (2) There is some element 0 in F such that
 - (i) $a + 0 = a$ for all a in F ,
 - (ii) for every a in F , there is some element b in F such that $a + b = 0$.

- (3) $a + b = b + a$ for all a and b in F .
 (4) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all a, b , and c in F .
 (5) There is some element 1 in F such that $1 \neq 0$ and
 (i) $a \cdot 1 = a$ for all a in F ,
 (ii) For every a in F with $a \neq 0$, there is some element b in F such that $a \cdot b = 1$.
 (6) $a \cdot b = b \cdot a$ for all a and b in F .
 (7) $a \cdot (b + c) = a \cdot b + a \cdot c$ for all a, b , and c in F .

The familiar examples of fields are, as already indicated, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$, with $+$ and $\cdot$ being the familiar operations of $+$ and $\cdot$. It is probably unnecessary to explain why these are fields, but the explanation is, at any rate, quite brief. When $+$ and $\cdot$ are understood to mean the ordinary $+$ and $\cdot$, the rules 1, 3, 4, 6, 7 are simply restatements of P1, P4, P5, P8, P9; the elements which play the role of 0 and 1 are the numbers 0 and 1 (which accounts for the choice of the symbols $0, 1$); and the number b in (2) or (5) is $-a$ or a^{-1} , respectively. (For this reason, in an arbitrary field F we denote by $-a$ the element such that $a + (-a) = 0$, and by a^{-1} the element such that $a \cdot a^{-1} = 1$, for $a \neq 0$.)

In addition to $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$, there are several other fields which can be described easily. One example is the collection F_1 of all numbers $a + b\sqrt{2}$ for a, b in $\mathbf{Q}$. The operations $+$ and $\cdot$ will, once again, be the usual $+$ and $\cdot$ for real numbers. It is necessary to point out that these operations really do produce new elements of F_1 :

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2}, \text{ which is in } F_1;$$

$$(a + b\sqrt{2}) \cdot (c + d\sqrt{2}) = (ac + 2bd) + (bc + ad)\sqrt{2}, \text{ which is in } F_1.$$

Conditions (1), (3), (4), (6), (7) for a field are obvious for F_1 : since these hold for all real numbers, they certainly hold for all real numbers of the form $a + b\sqrt{2}$. Condition (2) holds because the number $0 = 0 + 0\sqrt{2}$ is in F_1 and, for $\alpha = a + b\sqrt{2}$ in F_1 the number $\beta = (-a) + (-b)\sqrt{2}$ in F_1 satisfies $\alpha + \beta = 0$. Similarly, $1 = 1 + 0\sqrt{2}$ is in F_1 , so (5i) is satisfied. The verification of (5ii) is the only slightly difficult point. If $a + b\sqrt{2} \neq 0$, then

$$a + b\sqrt{2} \cdot \frac{1}{a + b\sqrt{2}} = 1;$$

it is therefore necessary to show that $1/(a + b\sqrt{2})$ is in F_1 . This is true because

$$\frac{1}{a + b\sqrt{2}} = \frac{a - b\sqrt{2}}{(a - b\sqrt{2})(a + b\sqrt{2})} = \frac{a}{a^2 - 2b^2} + \frac{(-b)}{a^2 - 2b^2}\sqrt{2}.$$

(The division by $a - b\sqrt{2}$ is valid because the relation $a - b\sqrt{2} = 0$ could

be true only if $a = b = 0$ (since $\sqrt{2}$ is irrational) which is ruled out by the hypothesis $a + b\sqrt{2} \neq 0$.)

The next example of a field, F_2 , is considerably simpler in one respect: it contains only two elements, which we might as well denote by **0** and **1**. The operations $+$ and $\cdot$ are described by the following tables.

$+$	0	1	$\cdot$	0	1
0	0	1	0	0	0
1	1	0	1	0	1

The verification of conditions (1)–(7) are straightforward, case-by-case checks. For example, condition (1) may be proved by checking the 8 equations obtained by setting $a, b, c = 0$ or **1**. Notice that in this field $\mathbf{1} + \mathbf{1} = \mathbf{0}$; this equation may also be written $\mathbf{1} = -\mathbf{1}$.

Our final example of a field is rather silly: F_3 consists of all pairs (a, a) for a in $\mathbf{R}$, and $+$ and $\cdot$ are defined by

$$\begin{aligned}(a, a) + (b, b) &= (a + b, a + b), \\ (a, a) \cdot (b, b) &= (a \cdot b, a \cdot b).\end{aligned}$$

(The $+$ and $\cdot$ appearing on the right side are ordinary addition and multiplication for $\mathbf{R}$.) The verification that F_3 is a field is left to you as a simple exercise.

A detailed investigation of the properties of fields is a study in itself, but for our purposes, fields provide an ideal framework in which to discuss the properties of numbers in the most economical way. For example, the consequences of P1–P9 which were derived for “numbers” in Chapter 1 actually hold for any field; in particular, they are true for the fields $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$.

Notice that certain common properties of $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$ do not hold for all fields. For example, it is possible for the equation $\mathbf{1} + \mathbf{1} = \mathbf{0}$ to hold in some fields, and consequently $a - b = b - a$ does not necessarily imply that $a = b$. For the field $\mathbf{C}$ the assertion $\mathbf{1} + \mathbf{1} \neq \mathbf{0}$ was derived from the explicit description of $\mathbf{C}$; for the fields $\mathbf{Q}$ and $\mathbf{R}$, however, this assertion was derived from further properties which do not have analogues in the conditions for a field. There is a related concept which does use these properties. An **ordered field** is a field F (with operations $+$ and $\cdot$) together with a certain subset $\mathbf{P}$ of F (the “positive” elements) with the following properties:

(8) For all a in F , one and only one of the following is true:

- (i) $a = \mathbf{0}$,
- (ii) a is in $\mathbf{P}$,
- (iii) $-a$ is in $\mathbf{P}$.

(9) If a and b are in $\mathbf{P}$, then $a + b$ is in $\mathbf{P}$.

(10) If a and b are in $\mathbf{P}$, then $a \cdot b$ is in $\mathbf{P}$.

We have already seen that the field $\mathbf{C}$ cannot be made into an ordered field. The field F_2 , with only two elements, likewise cannot be made into an ordered field: in fact, condition (8), applied to $\mathbf{1} = -\mathbf{1}$, shows that $\mathbf{1}$ must be in $\mathbf{P}$; then (9) implies that $\mathbf{1} + \mathbf{1} = \mathbf{0}$ is in $\mathbf{P}$, contradicting (8). On the other hand, the field F_1 , consisting of all numbers $a + b\sqrt{2}$ with a, b in $\mathbf{Q}$, certainly can be made into an ordered field: let $\mathbf{P}$ be the set of all $a + b\sqrt{2}$ which are positive real numbers (in the ordinary sense). The field F_3 can also be made into an ordered field; the description of $\mathbf{P}$ is left to you.

It is natural to introduce notation for an arbitrary ordered field which corresponds to that used for $\mathbf{Q}$ and $\mathbf{R}$: we define

$$\begin{aligned} a > b & \text{ if } a - b \text{ is in } \mathbf{P}, \\ a < b & \text{ if } b > a, \\ a \leq b & \text{ if } a < b \text{ or } a = b, \\ a \geq b & \text{ if } a > b \text{ or } a = b. \end{aligned}$$

Using these definitions we can reproduce, for an arbitrary ordered field F , the definitions of Chapter 7:

A set A of elements of F is **bounded above** if there is some x in F such that $x \geq a$ for all a in A . Any such x is called an **upper bound** for A . An element x of F is a **least upper bound** for A if x is an upper bound for A and $x \leq y$ for every y in F which is an upper bound for A .

Finally, it is possible to state an analogue of property P13 for $\mathbf{R}$; this leads to the last abstraction of this chapter:

A **complete ordered field** is an ordered field in which every nonempty set which is bounded above has a least upper bound.

The consideration of fields may seem to have taken us far from the goal of constructing the real numbers. However, we are now provided with an intelligible means of formulating this goal. There are two questions which will be answered in the remaining two chapters:

1. Is there a complete ordered field?
2. Is there only one complete ordered field?

Our starting point for these considerations will be $\mathbf{Q}$, assumed to be an ordered field, containing $\mathbf{N}$ and $\mathbf{Z}$ as certain subsets. At one crucial point it will be necessary to assume another fact about $\mathbf{Q}$:

Let x be an element of $\mathbf{Q}$ with $x > 0$. Then for any y in $\mathbf{Q}$ there is some n in $\mathbf{N}$ such that $nx > y$.

This assumption, which asserts that the rational numbers have the Archimedian property of the reals, does not follow from the other properties of an ordered field (for the example that demonstrates this conclusively see [17]). The important point for us is that when $\mathbf{Q}$ is explicitly constructed, proper-

ties P1–P12 appear as theorems, and so does this additional assumption; if we really began from the beginning, no assumptions about $\mathbf{Q}$ would be necessary.

PROBLEMS

1. Let F be the set $\{0, 1, 2\}$ and define operations $+$ and $\cdot$ on F by the following table. (The rule for constructing this table is as follows: add or multiply in the usual way, and then subtract the highest possible multiple of 3; thus $2 \cdot 2 = 4 = 3 + 1$, so $2 \cdot 2 = 1$.)

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Show that F is a field, and prove that it cannot be made into an ordered field.

2. Suppose now that we try to construct a field F having elements 0, 1, 2, 3 with operations $+$ and $\cdot$ defined as in the previous example, by adding or multiplying in the usual way, and then subtracting the highest possible multiple of 4. Show that F will *not* be a field.
3. Let $F = \{0, 1, \alpha, \beta\}$ and define operations $+$ and $\cdot$ on F by the following tables.

$+$	0	1	α	β
0	0	1	α	β
1	1	0	β	α
α	α	β	0	1
β	β	α	1	0

$\cdot$	0	1	α	β
0	0	0	0	0
1	0	1	α	β
α	0	α	β	1
β	0	β	1	α

Show that F is a field.

4. (a) Let F be a field in which $1 + 1 = 0$. Show that $a + a = 0$ for all a (this can also be written $a = -a$).
 (b) Suppose that $a + a = 0$ for some $a \neq 0$. Show that $1 + 1 = 0$ (and consequently $b + b = 0$ for all b).

5. (a) Show that in any field we have

$$\underbrace{(1 + \cdots + 1)}_{m \text{ times}} \cdot \underbrace{(1 + \cdots + 1)}_{n \text{ times}} = \underbrace{1 + \cdots + 1}_{mn \text{ times}}$$

for all natural numbers m and n .

- (b) Suppose that in the field F we have

$$\underbrace{1 + \cdots + 1}_m = 0$$

for some natural number m . Show that the smallest m with this property must be a prime number (this prime number is called the **characteristic** of F).

6. Let F be any field with only finitely many elements.

- (a) Show that there must be distinct natural numbers m and n with

$$\underbrace{1 + \cdots + 1}_m = \underbrace{1 + \cdots + 1}_n.$$

- (b) Conclude that there is some natural number k with

$$\underbrace{1 + \cdots + 1}_k = 0.$$

7. Let a, b, c , and d be elements of a field F with $a \cdot d - b \cdot c \neq 0$. Show that the equations

$$\begin{aligned} a \cdot x + b \cdot y &= \alpha, \\ c \cdot x + d \cdot y &= \beta, \end{aligned}$$

can be solved for x and y .

8. Let a be an element of a field F . A “square root” of a is an element b of F with $b^2 = b \cdot b = a$.

- (a) How many square roots does 0 have?
 (b) Suppose $a \neq 0$. Show that a has two square roots, unless $1 + 1 = 0$, in which case a has only one.

9. (a) Consider an equation $x^2 + b \cdot x + c = 0$, where b and c are elements of a field F . Suppose that $b^2 - 4 \cdot c$ has a square root r in F . Show that $(-b + r)/2$ is a solution of this equation.

- (b) In the field F_2 of the text, both elements clearly have a square root. On the other hand, it is easy to check that neither element satisfies the equation $x^2 + x + 1 = 0$. Thus some detail in part (a) must be incorrect. What is it?

10. Let F be a field and a an element of F which does *not* have a square root. This problem shows how to construct a bigger field F' , containing F , in which a does have a square root. (This construction has already been carried through in a special case, namely, $F = \mathbf{R}$ and $a = -1$; this special case should guide you through this example.)

Let F' consist of all pairs (x, y) with x and y in F . If the operations on F are $+$ and $\cdot$, define operations $\oplus$ and $\odot$ on F' as follows:

$$\begin{aligned}(x, y) \oplus (z, w) &= (x + z, y + w), \\ (x, y) \odot (z, w) &= (x \cdot z + a \cdot y \cdot w, y \cdot z + x \cdot w).\end{aligned}$$

- (a) Prove that F' , with the operations $\oplus$ and $\odot$, is a field.
(b) Prove that

$$\begin{aligned}(x, 0) \oplus (y, 0) &= (x + y, 0), \\ (x, 0) \odot (y, 0) &= (x \cdot y, 0),\end{aligned}$$

so that we may agree to abbreviate $(x, 0)$ by x .

- (c) Find a square root of $a = (a, 0)$ in F' .

11. Let F be the set of all four-tuples (w, x, y, z) of real numbers. Define $+$ and $\cdot$ by

$$\begin{aligned}(s, t, u, v) + (w, x, y, z) &= (s + w, t + x, u + y, v + z), \\ (s, t, u, v) \cdot (w, x, y, z) &= (sw - tx - uy - vz, sx + tw + uz - vy, \\ &\quad sy + uv + vx - tz, sz + vw + ty - ux).\end{aligned}$$

- (a) Show that F satisfies all conditions for a field, except (6). At times the algebra will become quite ornate, but the existence of multiplicative inverses is the only point requiring any thought.
(b) It is customary to denote

$$\begin{aligned}(0, 1, 0, 0) &\text{ by } i, \\ (0, 0, 1, 0) &\text{ by } j, \\ (0, 0, 0, 1) &\text{ by } k.\end{aligned}$$

Find all 9 products of pairs i, j , and k . The results will show in particular that condition (6) is definitely false. This “skew field” F is known as the **quaternions**.

The mass of drudgery which this chapter necessarily contains is relieved by one truly first-rate idea. In order to prove that a complete ordered field exists we will have to explicitly describe one in detail; verifying conditions (1)–(10) for an ordered field will be a straightforward ordeal, but the description of the field itself, of the elements in it, is ingenious indeed.

At our disposal is the set of rational numbers, and from this raw material it is necessary to produce the field which will ultimately be called the real numbers. To the uninitiated this must seem utterly hopeless—if only the rational numbers are known, where are the others to come from? By now we have had enough experience to realize that the situation may not be quite so hopeless as that casual consideration suggests. The strategy to be adopted in our construction has already been used effectively for defining functions and complex numbers. Instead of trying to determine the “real nature” of these concepts, we settled for a definition that described enough about them to determine their mathematical properties completely.

A similar proposal for defining real numbers requires a description of real numbers in terms of rational numbers. The observation, that a real number ought to be determined completely by the set of rational numbers less than it, suggests a strikingly simple and quite attractive possibility: a real number might (and in fact eventually will) be described as a collection of rational numbers. In order to make this proposal effective, however, some means must be found for describing “the set of rational numbers less than a real number” without mentioning real numbers, which are still nothing more than heuristic figments of our mathematical imagination.

If A is to be regarded as the set of rational numbers which are less than the real number α , then A ought to have the following property: If x is in A and y is a rational number satisfying $y < x$, then y is in A . In addition to this property, the set A should have a few others. Since there should be some rational number $x < \alpha$, the set A should not be empty. Likewise, since there should be some rational number $x > \alpha$, the set A should not be all of $\mathbf{Q}$. Finally, if $x < \alpha$, then there should be another rational number y with $x < y < \alpha$, so A should not contain a greatest member.

If we temporarily regard the real numbers as known, then it is not hard to check (Problem 8-17) that a set A with these properties is indeed the set of rational numbers less than some real number α . Since the real numbers are presently in limbo, your proof, if you supply one, must be regarded only as an unofficial comment on these proceedings. It will serve to convince you, however, that we have not failed to notice any crucial property of the set A . There appears to be no reason for hesitating any longer.

DEFINITION

A **real number** is a set α , of rational numbers, with the following four properties:

- (1) If x is in α and y is a rational number with $y < x$, then y is also in α .
- (2) $\alpha \neq \emptyset$.
- (3) $\alpha \neq \mathbf{Q}$.
- (4) There is no greatest element in α ; in other words, if x is in α , then there is some y in α with $y > x$.

The set of all real numbers is denoted by $\mathbf{R}$.

Just to remind you of the philosophy behind our definition, here is an explicit example of a real number:

$$\alpha = \{x \text{ in } \mathbf{Q}: x < 0 \text{ or } x^2 < 2\}.$$

It should be clear that α is the real number which will eventually be known as $\sqrt{2}$, but it is not an entirely trivial exercise to show that α actually is a real number. The whole point of such an exercise is to prove this using only facts about $\mathbf{Q}$; the hard part will be checking condition (4), but this has already appeared as a problem in a previous chapter (finding out which one is up to you). Notice that condition (4), although quite bothersome here, is really essential in order to avoid ambiguity; without it both

$$\{x \text{ in } \mathbf{Q}: x < 1\}$$

and

$$\{x \text{ in } \mathbf{Q}: x \leq 1\}$$

would be candidates for the “real number 1.”

The shift from A to α in our definition indicates both a conceptual and a notational concern. Henceforth, a real number *is*, by definition, a set of rational numbers. This means, in particular, that a rational number (a member of $\mathbf{Q}$) is *not* a real number; instead every rational number x has a natural counterpart which is a real number, namely, $\{y \text{ in } \mathbf{Q}: y < x\}$. After completing the construction of the real numbers, we can mentally throw away the elements of $\mathbf{Q}$ and agree that $\mathbf{Q}$ will henceforth denote these special sets. For the moment, however, it will be necessary to work at the same time with rational numbers, real numbers (sets of rational numbers) and even sets of real numbers (sets of sets of rational numbers). Some confusion is perhaps inevitable, but proper notation should keep this to a minimum. Rational numbers will be denoted by lower case Roman letters (x, y, z, a, b, c) and real numbers by lower case Greek letters (α, β, γ); capital Roman letters (A, B, C) will be used to denote sets of real numbers.

The remainder of this chapter is devoted to the definition of $+$, $\cdot$, and $\mathbf{P}$ for $\mathbf{R}$, and a proof that with these structures $\mathbf{R}$ is indeed a complete ordered field.

We shall actually begin with the definition of $\mathbf{P}$, and even here we shall work backwards. We first define $\alpha < \beta$; later, when $+$, $\cdot$, and 0 are available, we shall define $\mathbf{P}$ as the set of all α with $0 < \alpha$, and prove the necessary properties for $\mathbf{P}$. The reason for beginning with the definition of $<$ is the simplicity of this concept in our present setup:

Definition. If α and β are real numbers, then $\alpha < \beta$ means that α is contained in β (that is, every element of α is also an element of β), but $\alpha \neq \beta$.

A repetition of the definitions of $\leq$, $>$, $\geq$ would be stultifying, but it is interesting to note that $\leq$ can now be expressed more simply than $<$; if α and β are real numbers, then $\alpha \leq \beta$ if and only if α is contained in β .

If A is a bounded collection of real numbers, it is almost obvious that A should have a least upper bound. Each α in A is a collection of rational numbers; if these rational numbers are all put in one collection β , then β is presumably $\sup A$. In the proof of the following theorem we check all the little details which have not been mentioned, not least of which is the assertion that β is a real number. (We will not bother numbering theorems in this chapter, since they all add up to one big Theorem: There is a complete ordered field.)

THEOREM If A is a set of real numbers and $A \neq \emptyset$ and A is bounded above, then A has a least upper bound.

PROOF Let $\beta = \{x: x \text{ is in some } \alpha \text{ in } A\}$. Then β is certainly a collection of rational numbers; the proof that β is a real number requires checking four facts.

- (1) Suppose that x is in β and $y < x$. The first condition means that x is in α for some α in A . Since α is a real number, the assumption $y < x$ implies that y is in α . Therefore it is certainly true that y is in β .
- (2) Since $A \neq \emptyset$, there is some α in A . Since α is a real number, there is some x in α . This means that x is in β , so $\beta \neq \emptyset$.
- (3) Since A is bounded above, there is some real number γ such that $\alpha < \gamma$ for every α in A . Since γ is a real number, there is some rational number x which is not in γ . Now $\alpha < \gamma$ means that α is contained in γ , so it is also true that x is not in α for any α in A . This means that x is not in β ; so $\beta \neq \mathbf{Q}$.
- (4) Suppose that x is in β . Then x is in α for some α in A . Since α does not have a greatest member, there is some rational number y with $x < y$ and y in α . But this means that y is in β ; thus β does not have a greatest member.

These four observations prove that β is a real number. The proof that β is the least upper bound of A is easier. If α is in A , then clearly α is contained in β ; this means that $\alpha \leq \beta$, so β is an upper bound for A . On the other hand, if γ is an upper bound for A , then $\alpha \leq \gamma$ for every α in A ; this means

that α is contained in γ , for every α in A , and this surely implies that β is contained in γ . This, in turn, means that $\beta \leq \gamma$; thus β is the least upper bound of A . ■

The definition of $+$ is both obvious and easy, but it must be complemented with a proof that this “obvious” definition makes any sense at all.

Definition. If α and β are real numbers, then

$$\alpha + \beta = \{x: x = y + z \text{ for some } y \text{ in } \alpha \text{ and some } z \text{ in } \beta\}.$$

THEOREM If α and β are real numbers, then $\alpha + \beta$ is a real number.

PROOF Once again four facts must be verified.

- (1) Suppose $w < x$ for some x in $\alpha + \beta$. Then $x = y + z$ for some y in α and some z in β , which means that $w < y + z$, and consequently, $w - y < z$. This shows that $w - y$ is in β (since z is in β , and β is a real number). Since $w = y + (w - y)$, it follows that w is in $\alpha + \beta$.
- (2) It is clear that $\alpha + \beta \neq \emptyset$, since $\alpha \neq \emptyset$ and $\beta \neq \emptyset$.
- (3) Since $\alpha \neq \mathbf{Q}$ and $\beta \neq \mathbf{Q}$, there are rational numbers a and b with a not in α and b not in β . Any x in α satisfies $x < a$ (for if $a < x$, then condition (1) for a real number would imply that a is in α); similarly any y in β satisfies $y < b$. Thus $x + y < a + b$ for any x in α and y in β . This shows that $a + b$ is not in $\alpha + \beta$, so $\alpha + \beta \neq \mathbf{Q}$.
- (4) If x is in $\alpha + \beta$, then $x = y + z$ for y in α and z in β . There are y' in α and z' in β with $y < y'$ and $z < z'$; then $x < y' + z'$ and $y' + z'$ is in $\alpha + \beta$. Thus $\alpha + \beta$ has no greatest member. ■

By now you can see how tiresome this whole procedure is going to be. Every time we mention a new real number, we must prove that it *is* a real number; this requires checking four conditions, and even when trivial they require concentration. There is really no help for this (except that it will be less boring if you check the four conditions for yourself). Fortunately, however, a few points of interest will arise now and then, and some of our theorems will be easy. In particular, two properties of $+$ present no problems.

THEOREM If α , β , and γ are real numbers, then $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.

PROOF Since $(x + y) + z = x + (y + z)$ for all rational numbers x , y , and z , every member of $(\alpha + \beta) + \gamma$ is also a member of $\alpha + (\beta + \gamma)$, and vice versa. ■

THEOREM If α and β are real numbers, then $\alpha + \beta = \beta + \alpha$.

PROOF Left to you (even easier). ■

To prove the other properties of $+$ we first define 0 .

Definition. $0 = \{x \text{ in } \mathbf{Q}: x < 0\}$.

It is, thank goodness, obvious that 0 is a real number, and the following theorem is also simple.

THEOREM If α is a real number, then $\alpha + 0 = \alpha$.

PROOF If x is in α and y is in 0 , then $y < 0$, so $x + y < x$. This implies that $x + y$ is in α . Thus every member of $\alpha + 0$ is also a member of α .

On the other hand, if x is in α , then there is a rational number y in α such that $y > x$. Since $x = y + (x - y)$, where y is in α , and $x - y < 0$ (so that $x - y$ is in 0), this shows that x is in $\alpha + 0$. Thus every member of α is also a member of $\alpha + 0$. ■

The reasonable candidate for $-\alpha$ would seem to be the set

$$\{x \text{ in } \mathbf{Q}: -x \text{ is not in } \alpha\}$$

(since $-x$ not in α means, intuitively, that $-x > \alpha$, so that $x < -\alpha$). But in certain cases this set will not even be a real number. Although a real number α does not have a greatest member, the set

$$\mathbf{Q} - \alpha = \{x \text{ in } \mathbf{Q}: x \text{ is not in } \alpha\}$$

may have a *least* element x_0 ; when α is a real number of this kind, the set $\{x: -x \text{ is not in } \alpha\}$ will have a greatest element $-x_0$. It is therefore necessary to introduce a slight modification into the definition of $-\alpha$, which comes equipped with a theorem.

Definition. If α is a real number, then

$$-\alpha = \{x \text{ in } \mathbf{Q}: -x \text{ is not in } \alpha, \text{ but } -x \text{ is not the least element of } \mathbf{Q} - \alpha\}.$$

THEOREM If α is a real number, then $-\alpha$ is a real number.

PROOF

- (1) Suppose that x is in $-\alpha$ and $y < x$. Then $-y > -x$. Since $-x$ is not in α , it is also true that $-y$ is not in α . Moreover, it is clear that $-y$ is not the smallest element of $\mathbf{Q} - \alpha$, since $-x$ is a smaller element. This shows that y is in $-\alpha$.
- (2) Since $\alpha \neq \mathbf{Q}$, there is some rational number y which is not in α . We can assume that y is not the smallest rational number in $\mathbf{Q} - \alpha$ (since y can always be replaced by any $y' > y$). Then $-y$ is in $-\alpha$. Thus $-\alpha \neq \emptyset$.
- (3) Since $\alpha \neq \emptyset$, there is some x in α . Then $-x$ cannot possibly be in $-\alpha$, so $-\alpha \neq \mathbf{Q}$.

- (4) If x is in $-\alpha$, then $-x$ is not in α , and there is a rational number $y < -x$ which is also not in α . Let z be a rational number with $y < z < -x$. Then z is also not in α , and z is clearly not the smallest element of $\mathbf{Q} - \alpha$. So $-z$ is in $-\alpha$. Since $-z > x$, this shows that $-\alpha$ does not have a greatest element. ■

The proof that $\alpha + (-\alpha) = 0$ is not entirely straightforward. The difficulties are not caused, as you might presume, by the finicky details in the definition of $-\alpha$. Rather, at this point we require the Archimedian property of $\mathbf{Q}$ stated on page 490, which does not follow from P1–P12. This property is needed to prove the following lemma, which plays a crucial role in the next theorem.

LEMMA Let α be a real number, and z a positive rational number. Then there are (Figure 1) rational numbers x in α , and y not in α , such that $y - x = z$. Moreover, we may assume that y is not the smallest element of $\mathbf{Q} - \alpha$.

PROOF Suppose first that z is in α . If the numbers

$$z, 2z, 3z, \dots$$

were all in α , then every rational number would be in α , since every rational number w satisfies $w < nz$ for some n , by the additional assumption on page 490. This contradicts the fact that α is a real number, so there is some k such that $x = kz$ is in α and $y = (k+1)z$ is not in α . Clearly $y - x = z$.

Moreover, if y happens to be the smallest element of $\mathbf{Q} - \alpha$, let $x' > x$ be an element of α , and replace x by x' , and y by $y + (x' - x)$.

If z is not in α , there is a similar proof, based on the fact that the numbers $(-n)z$ cannot all fail to be in α . ■

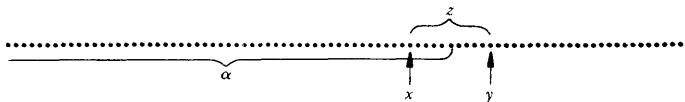


FIGURE 1

THEOREM If α is a real number, then

$$\alpha + (-\alpha) = 0.$$

PROOF Suppose x is in α and y is in $-\alpha$. Then $-y$ is not in α , so $-y > x$. Hence $x + y < 0$, so $x + y$ is in 0 . Thus every member of $\alpha + (-\alpha)$ is in 0 .

It is a little more difficult to go in the other direction. If z is in 0 , then $-z > 0$. According to the lemma, there is some x in α , and some y not in α , with y not the smallest element of $\mathbf{Q} - \alpha$, such that $y - x = -z$. This equation can be written $x + (-y) = z$. Since x is in α , and $-y$ is in $-\alpha$, this proves that z is in $\alpha + (-\alpha)$. ■

Before proceeding with multiplication, we define the “positive elements” and prove a basic property:

Definition. $\mathbf{P} = \{\alpha \text{ in } \mathbf{R}: \alpha > 0\}$.

Notice that $\alpha + \beta$ is clearly in $\mathbf{P}$ if α and β are.

THEOREM If α is a real number, then one and only one of the following conditions holds:

- (i) $\alpha = 0$,
- (ii) α is in $\mathbf{P}$,
- (iii) $-\alpha$ is in $\mathbf{P}$.

PROOF If α contains any positive rational number, then α certainly contains all negative rational numbers, so α contains 0 and $\alpha \neq 0$, i.e., α is in $\mathbf{P}$. If α contains no positive rational numbers, then one of two possibilities must hold:

- (1) α contains all negative rational numbers; then $\alpha = 0$.
- (2) there is some negative rational number x which is not in α ; it can be assumed that x is not the least element of $\mathbf{Q} - \alpha$ (since x could be replaced by $x/2 > x$); then $-\alpha$ contains the positive rational number $-x$, so, as we have just proved, $-\alpha$ is in $\mathbf{P}$.

This shows that *at least one* of (i)–(iii) must hold. If $\alpha = 0$, it is clearly impossible for condition (ii) or (iii) to hold. Moreover, it is impossible that $\alpha > 0$ and $-\alpha > 0$ both hold, since this would imply that $0 = \alpha + (-\alpha) > 0$. ■

Recall that $\alpha > \beta$ was defined to mean that α contains β , but is unequal to β . This definition was fine for proving completeness, but now we have to show that it is equivalent to the definition which would be made in terms of $\mathbf{P}$. Thus, we must show that $\alpha - \beta > 0$ is equivalent to $\alpha > \beta$. This is clearly a consequence of the next theorem.

THEOREM If α , β , and γ are real numbers and $\alpha > \beta$, then $\alpha + \gamma > \beta + \gamma$.

PROOF The hypothesis $\alpha > \beta$ implies that β is contained in α ; it follows immediately from the definition of $+$ that $\beta + \gamma$ is contained in $\alpha + \gamma$. This shows that $\alpha + \gamma \geq \beta + \gamma$. We can easily rule out the possibility of equality, for if

$$\alpha + \gamma = \beta + \gamma,$$

then

$$\alpha = (\alpha + \gamma) + (-\gamma) = (\beta + \gamma) + (-\gamma) = \beta,$$

which is false. Thus $\alpha + \gamma > \beta + \gamma$. ■

Multiplication presents difficulties of its own. If α , $\beta > 0$, then $\alpha \cdot \beta$ can be defined as follows.

Definition. If α and β are real numbers and $\alpha, \beta > 0$, then

$$\alpha \cdot \beta = \{z: z \leq 0 \text{ or } z = x \cdot y \text{ for some } x \text{ in } \alpha \text{ and } y \text{ in } \beta \text{ with } x, y > 0\}.$$

THEOREM If α and β are real numbers with $\alpha, \beta > 0$, then $\alpha \cdot \beta$ is a real number.

PROOF As usual, we must check four conditions.

- (1) Suppose $w < z$, where z is in $\alpha \cdot \beta$. If $w \leq 0$, then w is automatically in $\alpha \cdot \beta$. Suppose that $w > 0$. Then $z > 0$, so $z = x \cdot y$ for some positive x in α and positive y in β . Now

$$w = \frac{wz}{z} = \frac{wxy}{z} = \left(\frac{w}{z} \cdot x\right) \cdot y.$$

Since $0 < w < z$, we have $w/z < 1$, so $(w/z) \cdot x$ is in α . Thus w is in $\alpha \cdot \beta$.

- (2) Clearly $\alpha \cdot \beta \neq \emptyset$.
 (3) If x is not in α , and y is not in β , then $x > x'$ for all x' in α , and $y > y'$ for all y' in β . Hence $xy > x'y'$ for all such positive x' and y' . So xy is not in $\alpha \cdot \beta$; thus $\alpha \cdot \beta \neq \mathbf{Q}$.
 (4) Suppose w is in $\alpha \cdot \beta$, and $w \leq 0$. There is some x in α with $x > 0$ and some y in β with $y > 0$. Then $z = xy$ is in $\alpha \cdot \beta$ and $z > w$. Now suppose $w > 0$. Then $w = xy$ for some positive x in α and some positive y in β . Moreover, α contains some $x' > x$; if $z = x'y$, then $z > xy = w$, and z is in $\alpha \cdot \beta$. Thus $\alpha \cdot \beta$ does not have a greatest element. ■

Notice that $\alpha \cdot \beta$ is clearly in $\mathbf{P}$ if α and β are. This completes the verification of all properties of $\mathbf{P}$. To complete the definition of $\cdot$ we first define $|\alpha|$.

Definition. If α is a real number, then

$$|\alpha| = \begin{cases} \alpha, & \text{if } \alpha \geq 0 \\ -\alpha, & \text{if } \alpha \leq 0. \end{cases}$$

Definition. If α and β are real numbers, then

$$\alpha \cdot \beta = \begin{cases} 0, & \text{if } \alpha = 0 \text{ or } \beta = 0 \\ |\alpha| \cdot |\beta|, & \text{if } \alpha > 0, \beta > 0 \text{ or } \alpha < 0, \beta < 0 \\ -(|\alpha| \cdot |\beta|), & \text{if } \alpha > 0, \beta < 0 \text{ or } \alpha < 0, \beta > 0. \end{cases}$$

As one might suspect, the proofs of the properties of multiplication usually involve reduction to the case of positive numbers.

THEOREM If α, β , and γ are real numbers, then $\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$.

PROOF This is clear if $\alpha, \beta, \gamma > 0$. The proof for the general case requires considering separate cases (and is simplified slightly if one uses the following theorem). ■

THEOREM If α and β are real numbers, then $\alpha \cdot \beta = \beta \cdot \alpha$.

PROOF This is clear if $\alpha, \beta > 0$, and the other cases are easily checked. ■

Definition. $\mathbf{1} = \{x \text{ in } \mathbf{Q}: x < 1\}$.
(It is clear that $\mathbf{1}$ is a real number.)

THEOREM If α is a real number, then $\alpha \cdot \mathbf{1} = \alpha$.

PROOF Let $\alpha > 0$. It is easy to see that every member of $\alpha \cdot \mathbf{1}$ is also a member of α . On the other hand, suppose x is in α . If $x \leq 0$, then x is automatically in $\alpha \cdot \mathbf{1}$. If $x > 0$, then there is some rational number y in α such that $x < y$. Then $x = y \cdot (x/y)$, and x/y is in $\mathbf{1}$, so x is in $\alpha \cdot \mathbf{1}$. This proves that $\alpha \cdot \mathbf{1} = \alpha$ if $\alpha > 0$.

If $\alpha < 0$, then, applying the result just proved, we have

$$\alpha \cdot \mathbf{1} = -(|\alpha| \cdot |\mathbf{1}|) = -(|\alpha|) = \alpha.$$

Finally, the theorem is obvious when $\alpha = 0$. ■

Definition. If α is a real number and $\alpha > 0$, then

$\alpha^{-1} = \{x \text{ in } \mathbf{Q}: x \leq 0, \text{ or } x > 0 \text{ and } 1/x \text{ is not in } \alpha, \text{ but } 1/x \text{ is not the smallest member of } \mathbf{Q} - \alpha\};$

if $\alpha < 0$, then $\alpha^{-1} = -(|\alpha|^{-1})$.

THEOREM If α is a real number unequal to 0, then α^{-1} is a real number.

PROOF Clearly it suffices to consider only $\alpha > 0$. Four conditions must be checked.

- (1) Suppose $y < x$, and x is in α^{-1} . If $y \leq 0$, then y is in α^{-1} . If $y > 0$, then $x > 0$, so $1/x$ is not in α . Since $1/y > 1/x$, it follows that $1/y$ is not in α , and $1/y$ is clearly not the smallest element of $\mathbf{Q} - \alpha$, so y is in α^{-1} .
- (2) Clearly $\alpha^{-1} \neq \emptyset$.
- (3) Since $\alpha > 0$, there is some positive rational number x in α . Then $1/x$ is not in α^{-1} , so $\alpha^{-1} \neq \mathbf{Q}$.
- (4) Suppose x is in α^{-1} . If $x \leq 0$, there is clearly some y in α^{-1} with $y > x$ because α^{-1} contains some positive rationals. If $x > 0$, then $1/x$ is not in α . Since $1/x$ is not the smallest member of $\mathbf{Q} - \alpha$, there is a rational number y not in α , with $y < 1/x$. Choose a rational number z with $y < z < 1/x$. Then $1/z$ is in α , and $1/z > x$. Thus α^{-1} does not contain a largest member. ■

In order to prove that α^{-1} is really the multiplicative inverse of α , it helps to have another lemma, which is the multiplicative analogue of our first lemma.

LEMMA Let α be a real number with $\alpha > 0$, and z a rational number with $z > 1$. Then there are rational numbers x in α , and y not in α , such that $y/x = z$. Moreover, we can assume that y is not the least element of $\mathbf{Q} - \alpha$.

PROOF Suppose first that z is in α . Since $z - 1 > 0$ and

$$z^n = (1 + (z - 1))^n \geq 1 + n(z - 1),$$

it follows that the numbers

$$z, z^2, z^3, \dots$$

cannot all be in α . So there is some k such that $x = z^k$ is in α , and $y = z^{k+1}$ is not in α . Clearly $y/x = z$. Moreover, if y happens to be the least element of $\mathbf{Q} - \alpha$, let $x' > x$ be an element of α , and replace x by x' and y by yx'/x .

If z is not in α , there is a similar proof, based on the fact that the numbers $1/z^k$ cannot all fail to be in α . ■

THEOREM If α is a real number and $\alpha \neq 0$, then $\alpha \cdot \alpha^{-1} = 1$.

PROOF It obviously suffices to consider only $\alpha > 0$, in which case $\alpha^{-1} > 0$. Suppose that x is a positive rational number in α , and y is a positive rational number in α^{-1} . Then $1/y$ is not in α , so $1/y > x$; consequently $xy < 1$, which means that xy is in 1 . Since all rational numbers $x \leq 0$ are also in 1 , this shows that every member of $\alpha \cdot \alpha^{-1}$ is in 1 .

To prove the converse assertion, let z be in 1 . If $z \leq 0$, then clearly z is in $\alpha \cdot \alpha^{-1}$. Suppose $0 < z < 1$. According to the lemma, there are positive rational numbers x in α , and y not in α , such that $y/x = 1/z$; and we can assume that y is not the smallest element of $\mathbf{Q} - \alpha$. But this means that $z = x \cdot (1/y)$, where x is in α , and $1/y$ is in α^{-1} . Consequently, z is in $\alpha \cdot \alpha^{-1}$. ■

We are almost done! Only the proof of the distributive law remains. Once again we must consider many cases, but do not despair. The case when all numbers are positive contains an interesting point, and the other cases can all be taken care of very neatly.

THEOREM If α , β , and γ are real numbers, then $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$.

PROOF Assume first that $\alpha, \beta, \gamma > 0$. Then both numbers in the equation contain all rational numbers ≤ 0 . A positive rational number in $\alpha \cdot (\beta + \gamma)$ is of the form $x \cdot (y + z)$ for positive x in α , y in β , and z in γ . Since $x \cdot (y + z) = x \cdot y + x \cdot z$, where $x \cdot y$ is a positive element of $\alpha \cdot \beta$, and $x \cdot z$ is a positive element of $\alpha \cdot \gamma$, this number is also in $\alpha \cdot \beta + \alpha \cdot \gamma$. Thus, every element of $\alpha \cdot (\beta + \gamma)$ is also in $\alpha \cdot \beta + \alpha \cdot \gamma$.

On the other hand, a positive rational number in $\alpha \cdot \beta + \alpha \cdot \gamma$ is of the form $x_1 \cdot y + x_2 \cdot z$ for positive x_1, x_2 in α, y in β , and z in γ . If $x_1 \leq x_2$, then $(x_1/x_2) \cdot y \leq y$, so $(x_1/x_2) \cdot y$ is in β . Thus

$$x_1 \cdot y + x_2 \cdot z = x_2[(x_1/x_2)y + z]$$

is in $\alpha \cdot (\beta + \gamma)$. Of course, the same trick works if $x_2 \leq x_1$.

To complete the proof it is necessary to consider the cases when α, β , and γ are not all > 0 . If any one of the three equals 0 , the proof is easy and the cases involving $\alpha < 0$ can be derived immediately once all the possibilities for β and γ have been accounted for. Thus we assume $\alpha > 0$ and consider three cases: $\beta, \gamma < 0$, and $\beta < 0, \gamma > 0$, and $\beta > 0, \gamma < 0$. The first follows immediately from the case already proved, and the third follows from the second by interchanging β and γ . Therefore we concentrate on the case $\beta < 0, \gamma > 0$. There are then two possibilities:

(1) $\beta + \gamma \geq 0$. Then

$$\alpha \cdot \gamma = \alpha \cdot ([\beta + \gamma] + |\beta|) = \alpha \cdot (\beta + \gamma) + \alpha \cdot |\beta|,$$

so

$$\begin{aligned} \alpha \cdot (\beta + \gamma) &= -(\alpha \cdot |\beta|) + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma. \end{aligned}$$

(2) $\beta + \gamma \leq 0$. Then

$$\alpha \cdot |\beta| = \alpha \cdot (|\beta + \gamma| + \gamma) = \alpha \cdot |\beta + \gamma| + \alpha \cdot \gamma,$$

so

$$\alpha \cdot (\beta + \gamma) = -(\alpha \cdot |\beta + \gamma|) = -(\alpha \cdot |\beta|) + \alpha \cdot \gamma = \alpha \cdot \beta + \alpha \cdot \gamma. \blacksquare$$

This proof completes the work of the chapter. Although long and frequently tedious, this chapter contains results sufficiently important to be read in detail at least once (and preferably not more than once!). For the first time we know that we have not been operating in a vacuum—there is indeed a complete ordered field, the theorems of this book are not based on assumptions which can never be realized. One interesting and horrid possibility remains: there may be several complete ordered fields. If this is true, then the theorems of calculus are unexpectedly rich in content, but the properties P1–P13 are disappointingly incomplete. The last chapter disposes of this possibility; properties P1–P13 completely characterize the real numbers—anything that can be proved about real numbers can be proved on the basis of these properties alone.

PROBLEMS

There are only two problems in this set, but each asks for an entirely different construction of the real numbers! The detailed examination of another construction is recommended only for masochists, but the main idea behind these other constructions is worth knowing. The real numbers constructed in this

chapter might be called “the algebraist’s real numbers,” since they were purposely defined so as to guarantee the least upper bound property, which involves the ordering $<$, an algebraic notion. The real number system constructed in the next problem might be called “the analyst’s real numbers,” since they are devised so that Cauchy sequences will always converge.

1. Since every real number ought to be the limit of some Cauchy sequence of rational numbers, we might try to *define* a real number to be a Cauchy sequence of rational numbers. Since two Cauchy sequences might converge to the same real number, however, this proposal requires some modifications.
 - (a) Define two Cauchy sequences of rational numbers $\{a_n\}$ and $\{b_n\}$ to be *equivalent* (denoted by $\{a_n\} \sim \{b_n\}$) if $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$. Prove that $\{a_n\} \sim \{a_n\}$, that $\{a_n\} \sim \{b_n\}$ if $\{b_n\} \sim \{a_n\}$, and that $\{a_n\} \sim \{c_n\}$ if $\{a_n\} \sim \{b_n\}$ and $\{b_n\} \sim \{c_n\}$.
 - (b) Suppose that α is the set of all sequences equivalent to $\{a_n\}$, and β is the set of all sequences equivalent to $\{b_n\}$. Prove that either $\alpha \cap \beta = \emptyset$ or $\alpha = \beta$. (If $\alpha \cap \beta \neq \emptyset$, then there is some $\{c_n\}$ in both α and β . Show that in this case α and β both consist precisely of those sequences equivalent to $\{c_n\}$.)

Part (b) shows that the collection of all Cauchy sequences can be split up into disjoint sets, each set consisting of all sequences equivalent to some fixed sequence. We define a real number to be such a collection, and denote the set of all real numbers by $\mathbf{R}$.

- (c) If α and β are real numbers, let $\{a_n\}$ be a sequence in α , and $\{b_n\}$ a sequence in β . Define $\alpha + \beta$ to be the collection of all sequences equivalent to the sequence $\{a_n + b_n\}$. Show that $\{a_n + b_n\}$ is a Cauchy sequence and also show that this definition does not depend on the particular sequences $\{a_n\}$ and $\{b_n\}$ chosen for α and β . Check also that the analogous definition of multiplication is well defined.
 - (d) Show that $\mathbf{R}$ is a field with these operations; existence of a multiplicative inverse is the only interesting point to check.
 - (e) Define the positive real numbers P so that $\mathbf{R}$ will be an ordered field.
 - (f) Prove that every Cauchy sequence of real numbers converges. Remember that if $\{\alpha_n\}$ is a sequence of real numbers, then each α_n is itself a collection of Cauchy sequences of rational numbers.
2. This problem outlines a construction of “the high-school student’s real numbers.” We define a real number to be a pair $(a, \{b_n\})$, where a is an integer and $\{b_n\}$ is a sequence of natural numbers from 0 to 9, with the proviso that the sequence is not eventually 9; intuitively, this pair represents $a + \sum_{n=1}^{\infty} b_n 10^{-n}$. With this definition, a real number is a very concrete object, but the difficulties involved in defining addition and multi-

plication are formidable (how do you add infinite decimals without worrying about carrying digits infinitely far out?). A reasonable approach is outlined below; the trick is to use least upper bounds right from the start.

(a) Define $(a, \{b_n\}) < (c, \{d_n\})$ if $a < c$, or if $a = c$ and for some n we have $b_n < d_n$ but $b_j = d_j$ for $1 \leq j < n$. Using this definition, prove the least upper bound property.

(b) Given $\alpha = (a, \{b_n\})$, define $\alpha_k = a + \sum_{n=1}^k b_n 10^{-n}$; intuitively, α_k is the rational number obtained by changing all decimal places after the k th to 0. Conversely, given a rational number r of the form $a + \sum_{n=1}^k b_n 10^{-n}$, let r' denote the real number $(a, \{b_n'\})$, where $b_n' = b_n$ for $1 \leq n \leq k$ and $b_n' = 0$ for $n > k$. Now for $\alpha = (a, \{b_n\})$ and $\beta = (c, \{d_n\})$ define

$$\alpha + \beta = \sup \{(\alpha_k + \beta_k)': k \text{ a natural number}\}$$

(the least upper bound exists by part (a)). If multiplication is defined similarly, then the verification of all conditions for a field is a straightforward task, not highly recommended. Once more, however, existence of multiplicative inverses will be the hardest.

CHAPTER 29 UNIQUENESS OF THE REAL NUMBERS

We shall now revert to the usual notation for real numbers, reserving boldface symbols for other fields which may turn up. Moreover, we will regard integers and rational numbers as special kinds of real numbers, and forget about the specific way in which real numbers were defined. In this chapter we are interested in only one question: are there any complete ordered fields other than $\mathbf{R}$? The answer to this question, if taken literally, is “yes.” For example, the field F_3 introduced in Chapter 25 is a complete ordered field, and it is certainly not $\mathbf{R}$. This field is a “silly” example because the pair (a, a) can be regarded as just another name for the real number a ; the operations

$$\begin{aligned}(a, a) + (b, b) &= (a + b, a + b), \\ (a, a) \cdot (b, b) &= (a \cdot b, a \cdot b),\end{aligned}$$

are consistent with this renaming. This sort of example shows that any intelligent consideration of the question requires some mathematical means of discussing such renaming procedures.

If the elements of a field F are going to be used to rename elements of $\mathbf{R}$, then for each a in $\mathbf{R}$ there should correspond a “name” $f(a)$ in F . The notation $f(a)$ suggests that renaming can be formulated in terms of functions. In order to do this we will need a concept of function much more general than any which has occurred until now; in fact, we will require the most general notion of “function” used in mathematics. A function, in this general sense, is simply a rule which assigns to some things, other things. To be formal, a **function** is a collection of ordered pairs (of objects of any sort) which does not contain two distinct pairs with the same first element. The **domain** of a function f is the set A of all objects a such that (a, b) is in f for some b ; this (unique) b is denoted by $f(a)$. If $f(a)$ is in the set B for all a in A , then f is called a function **from** A **to** B . For example,

if $f(x) = \sin x$ for all x in $\mathbf{R}$ (and f is defined only for x in $\mathbf{R}$), then f is a function from $\mathbf{R}$ to $\mathbf{R}$; it is also a function from $\mathbf{R}$ to $[-1, 1]$;

if $f(z) = \sin z$ for all z in $\mathbf{C}$, then f is a function from $\mathbf{C}$ to $\mathbf{C}$;

if $f(z) = e^z$ for all z in $\mathbf{C}$, then f is a function from $\mathbf{C}$ to $\mathbf{C}$; it is also a function from $\mathbf{C}$ to $\{z \text{ in } \mathbf{C}: z \neq 0\}$;

θ is a function from $\{z \text{ in } \mathbf{C}: z \neq 0\}$ to $\{x \text{ in } \mathbf{R}: 0 \leq x < 2\pi\}$;

if f is the collection of all pairs $(a, (a, a))$ for a in $\mathbf{R}$, then f is a function from $\mathbf{R}$ to F_3 .

Suppose that F_1 and F_2 are two fields; we will denote the operations in F_1 by $\oplus$, $\odot$, etc. and the operations in F_2 by $+$, $\cdot$, etc. If F_2 is going to be considered as a collection of new names for elements of F_1 , then there should be a function from F_1 to F_2 with the following properties:

- (1) The function f should be one-one, that is, if $x \neq y$, then we should have $f(x) \neq f(y)$; this means that no two elements of F_1 have the same name.
- (2) The function f should be “onto,” that is, for every element z in F_2 there should be some x in F_1 such that $z = f(x)$; this means that every element of F_2 is used to name some element of F_1 .
- (3) For all x and y in F_1 we should have

$$\begin{aligned}f(x \oplus y) &= f(x) + f(y), \\f(x \odot y) &= f(x) \cdot f(y);\end{aligned}$$

this means that the renaming procedure is consistent with the operations of the field.

If we are also considering F_1 and F_2 as ordered fields, we add one more requirement:

- (4) If $x \odot y$, then $f(x) < f(y)$.

A function with these properties is called an *isomorphism* from F_1 to F_2 . This definition is so important that we restate it formally.

DEFINITION

If F_1 and F_2 are two fields, an **isomorphism** from F_1 to F_2 is a function f from F_1 to F_2 with the following properties:

- (1) If $x \neq y$, then $f(x) \neq f(y)$.
- (2) If z is in F_2 , then $z = f(x)$ for some x in F_1 .
- (3) If x and y are in F_1 , then

$$\begin{aligned}f(x \oplus y) &= f(x) + f(y), \\f(x \odot y) &= f(x) \cdot f(y).\end{aligned}$$

If F_1 and F_2 are ordered fields we also require:

- (4) If $x \odot y$, then $f(x) < f(y)$.

The fields F_1 and F_2 are called **isomorphic** if there is an isomorphism between them. Isomorphic fields may be regarded as essentially the same—any important property of one will automatically hold for the other. Therefore, we can, and should, reformulate the question asked at the beginning of the chapter; if F is a complete ordered field it is silly to expect F to equal $\mathbf{R}$ —rather, we would like to know if F is isomorphic to $\mathbf{R}$. In the following theorem, F will be a field, with operations $+$ and $\cdot$, and “positive elements” $\mathbf{P}$; we write $a < b$ to mean that $b - a$ is in $\mathbf{P}$, and so forth.

THEOREM If F is a complete ordered field, then F is isomorphic to $\mathbf{R}$.

PROOF Since two fields are defined to be isomorphic if there is an isomorphism between them, we must actually construct a function f from $\mathbf{R}$ to F which is an isomorphism. We begin by defining f on the integers as follows:

$$\begin{aligned} f(0) &= 0, \\ f(n) &= \underbrace{1 + \cdots + 1}_{n \text{ times}} \quad \text{for } n > 0, \\ f(n) &= -(\underbrace{1 + \cdots + 1}_{|n| \text{ times}}) \quad \text{for } n < 0. \end{aligned}$$

It is easy to check that

$$\begin{aligned} f(m+n) &= f(m) + f(n), \\ f(m \cdot n) &= f(m) \cdot f(n), \end{aligned}$$

for all integers m and n , and it is convenient to denote $f(n)$ by n . We then define f on the rational numbers by

$$f(m/n) = m/n = m \cdot n^{-1}$$

(notice that $1 + \cdots + 1 \neq 0$ if $n > 0$, since F is an ordered field). This definition makes sense because if $m/n = k/l$, then $ml = nk$, so $m \cdot l = k \cdot n$, so $m \cdot n^{-1} = k \cdot l^{-1}$. It is easy to check that

$$\begin{aligned} f(r_1 + r_2) &= f(r_1) + f(r_2), \\ f(r_1 \cdot r_2) &= f(r_1) \cdot f(r_2), \end{aligned}$$

for all rational numbers r_1 and r_2 , and that $f(r_1) < f(r_2)$ if $r_1 < r_2$.

The definition of $f(x)$ for arbitrary x is based on the now familiar idea that any real number is determined by the rational numbers less than it. For any x in $\mathbf{R}$, let A_x be the subset of F consisting of all $f(r)$, for all rational numbers $r < x$. The set A_x is certainly not empty, and it is also bounded above, for if r_0 is a rational number with $r_0 > x$, then $f(r_0) > f(r)$ for all $f(r)$ in A_x . Since F is a complete ordered field, the set A_x has a least upper bound; we define $f(x)$ as $\sup A_x$.

We now have $f(x)$ defined in two different ways, first for rational x , and then for any x . Before proceeding further, it is necessary to show that these two definitions agree for rational x . In other words, if x is a rational number, we want to show that

$$\sup A_x = f(x),$$

where $f(x)$ here denotes m/n , for $x = m/n$. This is not automatic, but depends on the completeness of F ; a slight digression is thus required.

Since F is complete, the elements

$$\underbrace{1 + \cdots + 1}_{n \text{ times}} \quad \text{for natural numbers } n$$

form a set which is not bounded above; the proof is exactly the same as the proof for $\mathbf{R}$ (Theorem 8-2). The consequences of this fact for $\mathbf{R}$ have exact analogues in F : in particular, if a and b are elements of F with $a < b$, then there is a rational number r such that

$$a < f(r) < b.$$

Having made this observation, we return to the proof that the two definitions of $f(x)$ agree for rational x . If y is a rational number with $y < x$, then we have already seen that $f(y) < f(x)$. Thus every element of A_x is $< f(x)$. Consequently,

$$\sup A_x \leq f(x).$$

On the other hand, suppose that we had

$$\sup A_x < f(x).$$

Then there would be a rational number r such that

$$\sup A_x < f(r) < f(x).$$

But the condition $f(r) < f(x)$ means that $r < x$, which means that $f(r)$ is in the set A_x ; this clearly contradicts the condition $\sup A_x < f(r)$. This shows that the original assumption is false, so

$$\sup A_x = f(x).$$

We thus have a certain well-defined function f from $\mathbf{R}$ to F . In order to show that f is an isomorphism we must verify conditions (1)–(4) of the definition. We will begin with (4).

If x and y are real numbers with $x < y$, then clearly A_x is contained in A_y . Thus

$$f(x) = \sup A_x \leq \sup A_y = f(y).$$

To rule out the possibility of equality, notice that there are rational numbers r and s with

$$x < r < s < y.$$

We know that $f(r) < f(s)$. It follows that

$$f(x) \leq f(r) < f(s) \leq f(y).$$

This proves (4).

Condition (1) follows immediately from (4): If $x \neq y$, then either $x < y$ or $y < x$; in the first case $f(x) < f(y)$, and in the second case $f(y) < f(x)$; in either case $f(x) \neq f(y)$.

To prove (2), let a be an element of F , and let B be the set of all rational numbers r with $f(r) < a$. The set B is not empty, and it is also bounded above, because there is a rational number s with $f(s) > a$, so that $f(s) > f(r)$ for r in B , which implies that $s > r$. Let x be the least upper bound of B ; we claim

that $f(x) = a$. In order to prove this it suffices to eliminate the alternatives

$$\begin{aligned} f(x) &< a, \\ a &< f(x). \end{aligned}$$

In the first case there would be a rational number r with

$$f(x) < f(r) < a.$$

But this means that $x < r$ and that r is in B , which contradicts the fact that $x = \sup B$. In the second case there would be a rational number r with

$$a < f(r) < f(x).$$

This implies that $r < x$. Since $x = \sup B$, this means that $r < s$ for some s in B . Hence

$$f(r) < f(s) < a,$$

again a contradiction. Thus $f(x) = a$, proving (2).

To check (3), let x and y be real numbers and suppose that $f(x + y) \neq f(x) + f(y)$. Then either

$$f(x + y) < f(x) + f(y) \quad \text{or} \quad f(x) + f(y) < f(x + y).$$

In the first case there would be a rational number r such that

$$f(x + y) < f(r) < f(x) + f(y).$$

But this would mean that

$$x + y < r.$$

Therefore r could be written as the sum of two rational numbers

$$r = r_1 + r_2, \quad \text{where } x < r_1 \text{ and } y < r_2.$$

Then, using the facts checked about f for rational numbers, it would follow that

$$f(r) = f(r_1 + r_2) = f(r_1) + f(r_2) > f(x) + f(y),$$

a contradiction. The other case is handled similarly.

Finally, if x and y are positive real numbers, the same sort of reasoning shows that

$$f(x \cdot y) = f(x) \cdot f(y);$$

the general case is then a simple consequence. ■

This theorem brings to an end our investigation of the real numbers, and resolves any doubts about them: There is a complete ordered field and, up to isomorphism, only one complete ordered field. It is an important part of a mathematical education to follow a construction of the real numbers in detail, but it is not necessary to refer ever again to this particular construction. It is utterly irrelevant that a real number happens to be a collection of rational numbers, and such a fact should never enter the proof of any important

theorem about the real numbers. Reasonable proofs should use only the fact that the real numbers are a complete ordered field, because this property of the real numbers characterizes them up to isomorphism, and any significant mathematical property of the real numbers will be true for all isomorphic fields. To be candid I should admit that this last assertion is just a prejudice of the author, but it is one shared by almost all other mathematicians.

PROBLEMS

1. Let f be an isomorphism from F_1 to F_2 .
 - (a) Show that $f(0) = 0$ and $f(1) = 1$. (Here 0 and 1 on the left denote elements in F_1 , while 0 and 1 on the right denote elements of F_2 .)
 - (b) Show that $f(-a) = -f(a)$ and $f(a^{-1}) = f(a)^{-1}$, for $a \neq 0$.
2. Here is an opportunity to convince yourself that any significant property of a field is shared by any field isomorphic to it. The point of this problem is to write out very formal proofs until you are certain that all statements of this sort are obvious. F_1 and F_2 will be two fields which are isomorphic; for simplicity we will denote the operations in both by $+$ and $\cdot$. Show that:
 - (a) If the equation $x^2 + 1 = 0$ has a solution in F_1 , then it has a solution in F_2 .
 - (b) If every polynomial equation $x^n + a_{n-1} \cdot x^{n-1} + \cdots + a_0 = 0$ with $a_0, \dots, a_{n-1}$ in F_1 has a root in F_1 , then every polynomial equation $x^n + b_{n-1} \cdot x^{n-1} + \cdots + b_0 = 0$ with $b_0, \dots, b_{n-1}$ in F_2 has a root in F_2 .
 - (c) If $1 + \cdots + 1$ (summed m times) $= 0$ in F_1 , then the same is true in F_2 .
 - (d) If F_1 and F_2 are ordered fields (and the isomorphism f satisfies $f(x) < f(y)$ for $x < y$) and F_1 is complete, then F_2 is complete.
3. Let f be an isomorphism from F_1 to F_2 and g an isomorphism from F_2 to F_3 . Define the function $g \circ f$ from F_1 to F_3 by $(g \circ f)(x) = g(f(x))$. Show that $g \circ f$ is an isomorphism.
4. Suppose that F is a complete ordered field, so that there is an isomorphism f from $\mathbf{R}$ to F . Show that there is actually only *one* isomorphism from $\mathbf{R}$ to F . Hint: In case $F = \mathbf{R}$, this is Problem 3-17. Now if f and g are two isomorphisms from $\mathbf{R}$ to F consider $g^{-1} \circ f$.
5. Find an isomorphism from $\mathbf{C}$ to $\mathbf{C}$ other than the identity function.

SUGGESTED READING

*A man ought to read
just as inclination leads him;
for what he reads as a task
will do him little good.*

SAMUEL JOHNSON

One purpose of this bibliography is to guide the reader to other sources, but the most important function it can serve is to indicate the variety of mathematical reading available. Consequently, there is an attempt to achieve diversity, but no pretense of being complete. The present plethora of mathematics books would make such an undertaking almost hopeless in any case, and since I have tried to encourage independent reading, the more standard a text, the less likely it is to appear here. In some cases, this philosophy may seem to have been carried to extremes, as some entries in the list cannot be read by a student just finishing a first course of calculus until several years have elapsed. Nevertheless, there are many selections which can be read now, and I can't believe that it hurts to have some idea of what lies ahead.

One of the most elementary unproved theorems mentioned in this book is the fact that every natural number can be written as a product of primes in only one way. A proof of this basic theorem will be found near the beginning of almost any book on elementary number theory. Few books have won so enthusiastic an audience as

- [1] *An Introduction to the Theory of Numbers* (third edition), by G. H. Hardy and E. M. Wright; Clarendon Press, Oxford, 1960.

The Pergamon Press publishes a series, Popular Lectures in Mathematics, with several worthwhile titles, among them

- [2] *A Selection of Problems in the Theory of Numbers*, by W. Sierpinski; Macmillan (Pergamon), New York, 1964.

Finally, I will mention a little book which I hope is still in print:

- [3] *Three Pearls of Number Theory*, by A. Khinchin; Graylock Press, Rochester, N.Y., 1952.

The subject of irrational numbers straddles the fields of number theory and analysis. An excellent introduction will be found in

- [4] *Irrational Numbers*, by I. M. Niven; Wiley, New York, 1956.

Together with many historical notes, there are references to some fairly elementary articles in journals. There is also a proof that π is transcendental (see also [53]) and, finally, a proof of the "Gelfond-Schneider theorem": If a and b are algebraic, with $a \neq 0$ or 1, and b is irrational, then a^b is transcendental.

All the books listed so far begin with natural numbers, but whenever necessary take for granted the irrational numbers, not to mention the integers and rational numbers. Several recent books present a construction of the rational numbers from the natural numbers, but it is hard to believe that there can be a more lucid treatment than the one to be found in

- [5] *Foundations of Analysis*, by E. Landau; Chelsea, New York, 1951.

Incidentally, the original German edition,

- [6] *Grundlagen der Analysis* (fourth edition), by E. Landau; Chelsea, New York, 1965.

is now available in paper back, together with a complete German-English dictionary (of about 300 words) for the whole book—an excellent way to begin

reading mathematical German. The basic idea for constructing the real numbers is derived from Dedekind, whose contributions can be found in

- [7] *Essays on the Theory of Numbers*, by R. Dedekind; Dover, New York, 1963.

While many mathematicians are content to accept the natural numbers as a natural starting point, numbers can be defined in terms of sets, the most basic starting point of all. A charming exposition of set theory can be found in a sophisticated little book called

- [8] *Naive Set Theory*, by P. R. Halmos; Van Nostrand, Princeton, N.J., 1961.

Another very good introduction is

- [9] *Theory of Sets*, by E. Kamke; Dover, New York, 1950.

Perhaps it is necessary to assure some victims of the “new math” that set theory does have some mathematical content (in fact, some very deep theorems). Using these deep results, Kamke proves that there is a discontinuous function f such that $f(x + y) = f(x) + f(y)$ for all x and y . For those who enjoy reading the classics, the most important notions of set theory were first introduced by Cantor, whose work is reproduced in

- [10] *Contributions to the Founding of the Theory of Transfinite Numbers*, by G. Cantor; Dover, New York, 1952.

Inequalities, which were treated as an elementary topic in Chapters 1 and 2, actually form a specialized field. A good elementary introduction is provided by

- [11] *Analytic Inequalities*, by N. Kazarinoff; Holt, Rinehart and Winston, New York, 1961.

Twelve different proofs that the geometric mean is less than or equal to the arithmetic mean, each based on a different principle, can be found in the beginning of the more advanced book

- [12] *Inequalities*, by E. Beckenbach and R. Bellman; Springer, New York, 1961.

The classic work on inequalities is

- [13] *Inequalities* (second edition), by G. H. Hardy, J. E. Littlewood, and G. Polya; Cambridge University Press, New York, 1952.

Each of the authors of this triple collaboration has provided his own contribution to the sparse literature about the nature of mathematical thinking, written from a mathematician’s point of view. My favorite is

- [14] *A Mathematician’s Apology*, by G. H. Hardy; Cambridge University Press, New York, 1940.

Littlewood's anecdotal selections are entitled

- [15] *A Mathematician's Miscellany*, by J. E. Littlewood; Methuen, 1953.

Polya's contribution is pedagogy at the highest level:

- [16] *Mathematics and Plausible Reasoning*, by G. Polya (Vol. I: *Induction and Analogy in Mathematics*; Vol. II: *Patterns of Plausible Inference*); Princeton University Press, Princeton, N.J., 1954.

Geometry is the other main field which can be considered as background for calculus. Euclid's *Elements* is still a masterful mathematical work, but should perhaps be postponed until some preparation has been made. Probably the best modern work on "classical geometry" is

- [17] *Elementary Geometry from an Advanced Standpoint*, by E. Moise; Addison-Wesley, Reading, Mass., 1963.

This beautiful book provides excellent historical perspectives and contains a thorough discussion of the role of the "Archimedean axiom" in geometry; in addition, Chapter 28 describes an ordered field in which the Archimedean axiom does not hold. Speaking of beautiful geometry books, all sorts of fascinating things can be found in

- [18] *Introduction to Geometry*, by H. S. Coxeter; Wiley, New York, 1961.

Almost all treatments of geometry at least mention convexity, which forms another specialized topic. I cannot imagine a better introduction to convexity, or a better mathematical experience in general, than reading and working through

- [19] *Convex Figures*, by I. M. Yaglom and W. G. Boltyanskii; Holt, Rinehart and Winston, New York, 1961.

This book contains a carefully arranged sequence of definitions and *statements* of theorems, whose proofs are to be supplied by the reader (worked-out proofs are supplied in the back of the book). Another geometry book has been modeled on the same principle:

- [20] *Combinatorial Geometry in the Plane*, by H. Hadwiger and H. Debrunner; Holt, Rinehart and Winston, New York, 1964.

Along with these two out-of-the-ordinary books, I might mention an extremely valuable little book, also of a specialized sort,

- [21] *Counterexamples in Analysis*, by B. Gelbaum and J. Olmsted; Holden-Day, San Francisco, 1964.

Many of the examples in this book come from more advanced topics in analysis, but quite a few can be appreciated by someone who knows calculus.

Of calculus books I will mention only two, each something of a classic:

[22] *A Course of Pure Mathematics* (tenth edition), by G. H. Hardy; Cambridge University Press, New York, 1952.

[23] *Differential and Integral Calculus*, by R. Courant; Vol. I, second edition, 1937; Vol. II, first edition, Wiley (Interscience), New York, 1936.

Courant is especially strong on applications. In addition, the latter parts of Volume I contain material usually found in advanced calculus, including differential equations and Fourier series. An introduction to Fourier series (requiring a little advanced calculus) will also be found in

[24] *An Introduction to Fourier Series and Integrals*, by R. Seeley; W. A. Benjamin, New York, 1966.

The second volume of Courant (advanced calculus in earnest) contains additional material on differential equations, as well as an introduction to the calculus of variations. I will not mention any books devoted solely to the calculus of variations, since they are (of necessity) quite difficult. Innumerable books on differential equations have been written, but none of the elementary ones seems to generate much enthusiasm. There is a somewhat more advanced book, however, which is universally admired:

[25] *Lectures on Ordinary Differential Equations*, by W. Hurewicz; M.I.T. Press, Cambridge, Mass., 1958.

I will bypass the more or less standard advanced calculus books (which can easily be found by the reader himself) since nowadays there is a movement to revise the whole presentation of advanced calculus, basing it upon linear algebra. One of the first, and still one of the nicest, treatments of advanced calculus using linear algebra is

[26] *Calculus of Vector Functions*, by R. H. Crowell and R. E. Williamson; Prentice-Hall, Englewood Cliffs, N.J., 1962.

Several recent books on advanced calculus attempt to acquaint undergraduates with very large areas of modern mathematics. My favorite, of course, is

[27] *Calculus on Manifolds*, by M. Spivak; W. A. Benjamin, New York, 1965.

There are three other topics which are somewhat out of place in this bibliography because they are rapidly becoming established as part of a standard undergraduate curriculum. The purposeful study of fields and related systems is the domain of "algebra." The first undergraduate text on modern algebra was

[28] *A Survey of Modern Algebra* (third edition), by G. Birkhoff and S. MacLane; Macmillan, New York, 1941.

One of the strongest competitors is now

- [29] *Topics in Algebra*, by I. N. Herstein; Ginn (Blaisdell), Boston, Mass., 1963.

Herstein's book is meant to be intermediate in difficulty between *A Survey of Modern Algebra* and one of the great classics:

- [30] *Modern Algebra*, by B. L. van der Waerden; Ungar, New York, 1953.

By the way, this book contains a proof of the partial fraction decomposition of a rational function.

One of the standard texts in complex analysis is

- [31] *Complex Analysis* (second edition), by L. V. Ahlfors; McGraw-Hill, New York, 1966.

I am personally looking forward to the publication of

- [32] *Lectures on the Theory of Functions of a Complex Variable*, by G. W. Mackey; Van Nostrand, Princeton, N.J., 1967.

Finally, the study of topology, although never mentioned before, has really been in the background of many discussions, since it is the natural generalization of the ideas about limits and continuity which play such a prominent role in Part II of this book. There are now many elementary books on topology, but I am dismayed by the prospect of reading and evaluating them all, so I will merely call attention to the fact that they exist, and warn that some are not very good.

The next few topics, ranging from elementary to very difficult, are included in this bibliography because they have been alluded to in the text. The proof that a nondecreasing function is differentiable at almost all points (and an explanation of just what this means) receives a beautiful exposition in

- [33] *Functional Analysis*, by F. Riesz and B. Sz.-Nagy; Ungar, New York, 1955.

(After this elementary beginning, the book moves on to quite advanced material.) The gamma function has an elegant little book devoted entirely to its properties, most of them proved by using the theorem of Bohr and Mollerup which was mentioned in Problem 18-25:

- [34] *The Gamma Function*, by E. Artin; Holt, Rinehart and Winston, New York, 1964.

The gamma function is only one of several important improper integrals in mathematics. In particular, the calculation of $\int_0^\infty e^{-x^2} dx$ (see Problem 18-27) is important in probability theory, where the "normal distribution function"

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}y^2} dy$$

plays a fundamental role. The following book has already become something of a classic:

- [35] *An Introduction to Probability Theory and Its Applications* (second edition), by W. Feller; Wiley, New York, 1957.

The impossibility of integrating certain functions in elementary terms (among them $f(x) = e^{-x^2}$) is one of the most esoteric subjects in mathematics. An interesting discussion of the possibilities of integrating in elementary terms, with an outline of the impossibility proofs, and references to the original papers of Liouville, will be found in

- [36] *The Integration of Functions of a Single Variable* (second edition), by G. H. Hardy; Cambridge University Press, New York, 1958.

A complete presentation of the impossibility proofs will be found in

- [37] *Integration in Finite Terms*, by J. Ritt; Columbia University Press, New York, 1948.

Oddly enough, a related but seemingly more difficult problem has a much neater solution. There are simple differential equations ($y'' + xy = 0$ is a specific example) whose solutions cannot be expressed even in terms of indefinite integrals of elementary functions. This fact is proved on page 43 of the (60-page) book:

- [38] *An Introduction to Differential Algebra*, by I. Kaplansky; Hermann, Paris, 1957.

To read this book you will need to know quite a bit of algebra, however. (Just recently, equally algebraic and polished treatments of Liouville's theorems have been obtained, and presumably they will be published soon.)

A few words should also be said in defense of the process of integrating in elementary terms, which many mathematicians look upon as an art (unlike differentiation, which is merely a skill). You are probably already aware that the process of integration can be expedited by tables of indefinite integrals. For those who enjoy perusing tables there is a really beautiful collection, that includes indefinite integrals, definite improper integrals, and a great deal more besides (if you should ever happen to need the value of the thirty-fourth Bernoulli number, this is the place to look):

- [39] *Tables of Integrals, Series, and Products*, by I. S. Gradshteyn and I. W. Ryzhik; Academic Press, New York, 1965.

For the thrifty, there is a paperback table of integrals:

- [40] *Tables of Indefinite Integrals*, by G. Petit Bois; Dover, New York, 1961.

The remaining references are of a somewhat different sort. They fall into three categories, of which the first is historical. The letter of H. A. Schwartz referred to in Problem 11-49 will be found in

- [41] *Ways of Thought of Great Mathematicians*, by H. Meschkowski; Holden-Day, San Francisco, 1964.

Some historical remarks, and an attempt to incorporate them into the teaching of calculus, will be found in

- [42] *The Calculus: A Genetic Approach*, by O. Toeplitz; University of Chicago Press, Chicago, Ill., 1963.

Most textbooks on the history of mathematics are both superficial and dull. An admirable exception is

- [43] *An Introduction to the History of Mathematics*, by H. Eves; Holt, Rinehart and Winston, New York, 1964.

Three good scholarly works are

- [44] *History of Analytic Geometry*, by C. Boyer; Academic Press, New York, 1965.
 [45] *A History of the Calculus, and Its Conceptual Development*, by C. Boyer; Dover, New York, 1959.
 [46] *The Mathematics of Great Amateurs*, by J. Coolidge; Dover, New York, 1963.

A recent book presents philosophical ruminations on the history of mathematics by an author whose exceptional erudition is all the more impressive in a first-rate mathematician:

- [47] *The Role of Mathematics in the Rise of Science*, by S. Bochner; Princeton University Press, Princeton, N.J., 1966.

Finally, extracts from original sources will be found in

- [48] *A Source Book in Mathematics* (2 vols.), by D. Smith; Dover, New York, 1959.

The large number of books listed here might easily give too hopeful a view of the availability of historical information. Although the very early origins of mathematics are receiving the most careful scrutiny, no one seems to care very much about the origins of calculus. For example, it is almost impossible to find out who first proved the Mean Value Theorem (according to the *Encyklopädie der Mathematischen Wissenschaften*, Volume II, it was O. Bonnet, whose name is familiar to students of differential geometry from the “Gauss-Bonnet Theorem”). Similarly, nearly every history book tells us that Wallis proved Wallis’s formula by a “complicated method of interpolation,” but almost none bothers to mention what it was, even though it inspired Euler’s investiga-

tions of the gamma function (a description is given in the answer book, along with the solution to Problem 18-26).

The second category in this final group of books might be described as “popularizations.” *The New York Times* editorials notwithstanding, there are a surprisingly large number of first-rate ones by real mathematicians:

- [49] *What is Mathematics?* (fourth edition), by R. Courant and H. Robbins; Oxford University Press, New York, 1947.
- [50] *Geometry and the Imagination*, by D. Hilbert and S. Cohn-Vossen. Chelsea, New York, 1956.
- [51] *The Enjoyment of Mathematics*, by H. Rademacher and O. Toeplitz; Princeton University Press, Princeton, N.J., 1957.
- [52] *Famous Problems of Mathematics* (second edition), by H. Tietze; Graylock Press, Rochester, N.Y., 1965.

One of the most renowned “popularizations” is especially concerned with the teaching of mathematics:

- [53] *Elementary Mathematics from an Advanced Standpoint*, by F. Klein (vol. 1: *Arithmetic, Algebra, Analysis*; vol. 2: *Geometry*); Dover, New York, 1948.

Volume 1 contains a proof of the transcendence of π which, although not so elementary as the one in [4], is a direct analogue of the proof that e is transcendental, replacing integrals with complex line integrals. It can be read as soon as the basic facts about complex analysis are known.

The third category is the very opposite extreme—original papers. The difficulties encountered here are formidable, and I have only had the courage to list one such paper, the source of the quotation for Part IV. It is not even in English, although you do have a choice of foreign languages. The article in the original French is in

- [54] *Oeuvres Complètes d'Abel*; Christiania. Johnson Reprint Corporation, New York, 1965.

It first appeared in a German translation in the *Journal für die reine und angewandte Mathematik* (1) 1826. To compound the difficulties, these references will usually be available only in university libraries. Yet the study of this paper will probably be as valuable as any other reading mentioned here. The reason is suggested by a remark of Abel himself, who attributed his profound knowledge of mathematics to the fact that he read the masters, rather than the pupils.

ANSWERS
TO SELECTED
PROBLEMS

CHAPTER 1

1. (i) $1 = a^{-1}a = a^{-1}(ax) = (a^{-1}a)x = 1 \cdot x = x$.
 (iii) If $x^2 = y^2$, then $0 = x^2 - y^2 = (x - y)(x + y)$, so either $x - y = 0$ or $x + y = 0$, that is, either $x = -y$ or $x = y$.
 (vi) Replace y by $-y$ in (iv).
2. One step requires dividing by $x - y = 0$.
3. (i) $a/b = ab^{-1} = (ac)(b^{-1}c^{-1}) = (ac)(bc)^{-1}$ (by (iii)) $= ac/bc$.
 (ii) $(ad + bc)/(bd) = (ad + bc)(bd)^{-1} = (ad + bc)(b^{-1}d^{-1})$ (by (iii)) $= ab^{-1} + cd^{-1} = a/b + c/d$.
 (iii) $ab(a^{-1}b^{-1}) = (a \cdot a^{-1})(b \cdot b^{-1}) = 1$, so $a^{-1} \cdot b^{-1} = (ab)^{-1}$.
 (v) $(a/b)/(c/d) = (a/b)(c/d)^{-1} = (a \cdot b^{-1})(c \cdot d^{-1})^{-1} = (a \cdot b^{-1})(c^{-1} \cdot d) = ad(b^{-1} \cdot c^{-1}) = ad(bc)^{-1} = (ad)/(bc)$.
4. (i) $x < -1$.
 (iii) $x > \sqrt{7}$ or $x < -\sqrt{7}$.
 (v) All x , since $x^2 - 2x + 2 = (x - 1)^2 + 1$.
 (vii) $x > 3$ or $x < -2$, since 3 and -2 are the roots of $x^2 - x - 6 = 0$.
 (ix) $x > \pi$ or $-5 < x < 3$.
 (xi) $x < 3$.
 (xiii) $x > 1$ or $0 < x < 1$.
5. (i) $b - a$ and $d - c$ are in P , so $(b - a) + (d - c) = (b + d) - (a + c)$ is in P . Thus, $b + d > a + c$.
 (iii) Using (ii), $-c < -d$; then (i) implies that $a + (-c) < b + (-d)$.
 (v) $(b - a)$ and $-c$ are in P , so $-c(b - a) = ac - bc$ is in P , that is, $ac > bc$.
 (vii) Using (iv), $a > 0$ and $a < 1$, so $a^2 < a$.
 (ix) Substitute a for c and b for d in (viii).
9. (i) $\sqrt{2} + \sqrt{3} - \sqrt{5} + \sqrt{7}$.
 (iii) $|a + b| + |c| - |a + b + c|$.
 (v) $\sqrt{2} + \sqrt{3} + \sqrt{5} - \sqrt{7}$.
10. (i) a if $a \geq -b$ and $b \geq 0$;
 $-a$ if $a \leq -b$ and $b \leq 0$;
 $a + 2b$ if $a \geq -b$ and $b \leq 0$;
 $-a - 2b$ if $a \leq -b$ and $b \geq 0$.
 (iii) $x - x^2$ if $x \geq 0$;
 $-x - x^2$ if $x \leq 0$.
11. (i) $x = 11, -5$.
 (iii) $-6 < x < -2$.
 (v) No x (the distance from x to 1 plus the distance from x to -1 is at least 2).
 (vii) $x = 1, -1$.
12. (i) $(|xy|)^2 = (xy)^2 = x^2y^2 = |x|^2|y|^2 = (|x| \cdot |y|)^2$; since $|xy|$ and $|x| \cdot |y|$ are both ≥ 0 , this proves that $|xy| = |x| \cdot |y|$.
 (iii) $|x|/|y| = |x| \cdot |y|^{-1} = |x| \cdot |y^{-1}|$ (by (ii)) $= |xy^{-1}|$ (by (i)) $= |x/y|$.

- (v) It follows from (iv) that $|x| = |y - (y - x)| \leq |y| + |y - x|$, so $|x| - |y| \leq |x - y|$.
- (vii) $|x + y + z| \leq |x + y| + |z| \leq |x| + |y| + |z|$. If equality holds, then $|x + y| = |x| + |y|$, so x and y have the same sign. Moreover, z must have the same sign as $x + y$, so x , y , and z must all have the same sign (unless one is 0).

CHAPTER 2

1. (i) Since $1^2 = 1 \cdot (2) \cdot (2 \cdot 1 + 1)/6$, the formula is true for $n = 1$. Suppose that the formula is true for k . Then

$$\begin{aligned} 1^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)] \\ &= \frac{(k+1)}{6} [(k+2)(2k+3)] \\ &= \frac{(k+1)(k+2)(2[k+1]+1)}{6}, \end{aligned}$$

so the formula is true for $k+1$.

2. (i)
$$\begin{aligned} \sum_{i=1}^n (2i-1) &= 1 + 3 + 5 + \cdots + (2n-1) \\ &= 1 + 2 + 3 + \cdots + 2n - 2(1 + \cdots + n) \\ &= \frac{(2n)(2n+1)}{2} - n(n+1) \\ &= n^2. \end{aligned}$$

4. (a) Since

$$1 + r = \frac{1 - r^2}{1 - r},$$

the formula is true for $n = 1$. Suppose that

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Then

$$\begin{aligned} 1 + r + \cdots + r^n + r^{n+1} &= \frac{1 - r^{n+1}}{1 - r} + r^{n+1} \\ &= \frac{1 - r^{n+1} + r^{n+1}(1 - r)}{1 - r} \\ &= \frac{1 - r^{n+1}}{1 - r}. \end{aligned}$$

(b)

$$\begin{aligned} S &= 1 + r + \cdots + r^n \\ rS &= r + \cdots + r^n + r^{n+1}. \end{aligned}$$

Thus

$$S(1 - r) = S - rS = 1 - r^{n+1},$$

so

$$S = \frac{1 - r^{n+1}}{1 - r}.$$

5. (i) From

$$(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1, \quad k = 1, \dots, n$$

we obtain

$$(n+1)^4 - 1 = 4 \sum_{k=1}^n k^3 + 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k + n,$$

so

$$\begin{aligned} \sum_{k=1}^n k^3 &= \frac{(n+1)^4 - 1 - 6 \frac{n(n+1)(2n+1)}{6} - 4 \frac{n(n+1)}{2} - n}{4} \\ &= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4}. \end{aligned}$$

(iii) From

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}, \quad k = 1, \dots, n$$

we obtain

$$1 - \frac{1}{n+1} = \sum_{k=1}^n \frac{1}{k(k+1)}.$$

7. 1 is either even or odd, in fact it is odd. Suppose n is either even or odd; then n can be written either as $2k$ or $2k+1$. In the first case $n+1 = 2k+1$ is odd; in the second case $n+1 = 2k+1+1 = 2(k+1)$ is even. In either case, $n+1$ is either even or odd. (Admittedly, this looks fishy, but it is really correct.)
8. Let B be the set of all natural numbers l such that $n_0 - 1 + l$ is in A . Then 1 is in B , and $l+1$ is in B if l is in B , so B contains all natural numbers, which means that A contains all natural numbers $\geq n_0$.
11. (a) Yes, for if $a+b$ were rational, then $b = (a+b) - a$ would be rational. If a and b are irrational, then $a+b$ could be rational, for b could be $r-a$ for some rational number a .
- (b) If $a = 0$, then ab is rational. But if $a \neq 0$, then ab could not be rational, for then $b = (ab) \cdot a^{-1}$ would be rational.

- (c) Yes; for example, $\sqrt[4]{2}$.
 (d) Yes; for example, $\sqrt{2}$ and $-\sqrt{2}$.
 12. (a) Since

$$\begin{aligned}(3n+1)^2 &= 9n^2 + 6n + 1 = 3(3n^2 + 2n) + 1, \\ (3n+2)^2 &= 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1,\end{aligned}$$

it follows that if k^2 is divisible by 3, then k must also be divisible by 3. Now suppose that $\sqrt{3}$ were rational, and let $\sqrt{3} = p/q$ where p and q have no common factor. Then $p^2 = 3q^2$, so p^2 is divisible by 3, so p must be. Thus, $p = 3p'$ for some natural number p' , and consequently $(3p')^2 = 3q^2$, or $3(p')^2 = q^2$. Thus, q is also divisible by 3, a contradiction.

The same proofs work for $\sqrt{5}$ and $\sqrt{6}$, because the equations

$$\begin{aligned}(5n+1)^2 &= 25n^2 + 10n + 1 = 5(5n^2 + 2n) + 1, \\ (5n+2)^2 &= 25n^2 + 20n + 4 = 5(5n^2 + 4n) + 4, \\ (5n+3)^2 &= 25n^2 + 30n + 9 = 5(5n^2 + 6n + 1) + 4, \\ (5n+4)^2 &= 25n^2 + 40n + 16 = 5(5n^2 + 8n + 3) + 1,\end{aligned}$$

and the corresponding equations for numbers of the form $6n+m$, show that if k^2 is divisible by 5 or 6, then k must be. The proof fails for $\sqrt{4}$, because $(4n+2)^2$ is divisible by 4. (For precisely this reason this proof cannot be used to show that in general $\sqrt{a}$ is irrational if a is not a perfect square—we have no guarantee that $(an+m)^2$ might not be a multiple of a for some $m < a$. Actually, this assertion is true, but the proof requires the information in Problem 16.)

- (b) Since

$$(2n+1)^3 = 8n^3 + 12n^2 + 6n + 1 = 2(4n^3 + 6n^2 + 3n) + 1,$$

it follows that if k^3 is even, then k is even. If $\sqrt[3]{2} = p/q$ where p and q had no common factors, then $p^3 = 2q^3$, so p^3 is divisible by 2, so p must be. Thus, $p = 2p'$ for some natural number p' , and consequently $(2p')^3 = 2q^3$, or $4(p')^3 = q^3$. Thus, q is also even, a contradiction.

The proof for $\sqrt[3]{3}$ is similar, using the equations

$$\begin{aligned}(3n+1)^3 &= 27n^3 + 27n^2 + 9n + 1 = 3(9n^3 + 9n^2 + 3n) + 1, \\ (3n+2)^3 &= 27n^3 + 54n^2 + 36n + 8 = 3(9n^3 + 18n^2 + 12n + 2) + 2.\end{aligned}$$

18. If $n = 1$, then $(1+h)^n = 1+nh$. Suppose that $(1+h)^n \geq 1+nh$. Then

$$\begin{aligned}(1+h)^{n+1} &= (1+h)(1+h)^n \geq (1+h)(1+nh), \quad \text{since } 1+h > 0 \\ &= 1 + (n+1)h + nh^2 \geq 1 + (n+1)h.\end{aligned}$$

For $h > 0$, the inequality follows directly from the binomial theorem, since all the other terms appearing in the expansion of $(1 + h)^n$ are positive.

CHAPTER 3

1. (i) $(x + 1)/(x + 2)$; the expression $f(f(x))$ makes sense only when $x \neq -1$ and $x \neq -2$.
 (iii) $1/(1 + cx)$ (for $x \neq -1/c$ if $c \neq 0$).
 (v) $(x + y + 2)/(x + 1)(y + 1)$ (for $x, y \neq -1$).
 (vii) Only $c = 1$, since $f(x) = f(cx)$ implies that $x = cx$, and this must be true for at least one $x \neq 0$.
2. (i) $y \geq 0$ and rational, or $y \geq 1$.
 (iii) 0.
 (v) $-1, 0, 1$.
3. (i) $\{x: -1 \leq x \leq 1\}$.
 (iii) $\{x: x \neq 1 \text{ and } x \neq 2\}$.
 (v) $\emptyset$.
4. (i) 2^{2y} .
 (iii) $2^{2 \sin t} + \sin(2^t)$.
5. (i) $P \circ S$.
 (iii) $S \circ S$.
 (v) $P \circ P$.
 (vii) $S \circ S \circ S \circ P \circ P \circ P \circ S$.
11. (a) y .
 (b) $H(y)$.
 (c) $H(y)$.

12. (a)

	even	odd
even	even	neither
odd	neither	odd

(b)

	even	odd
even	even	odd
odd	odd	even

(c)

	f even	f odd
g even	even	even
g odd	even	odd

- (d) Let $g(x) = f(x)$ for $x \geq 0$ and define g arbitrarily for $x < 0$.

21. (i) Let $g(x) = h(x) = 1$ and let f be a function for which $f(2) \neq f(1) + f(1)$. Then $f \circ (g + h) \neq f \circ g + f \circ h$.
- (ii) $[(g + h) \circ f](x) = (g + h)(f(x)) = g(f(x)) + h(f(x)) = (g \circ f)(x) + (h \circ f)(x) = [(g \circ f) + (h \circ f)](x)$.
- (iii) $\frac{1}{f \circ g}(x) = \frac{1}{f(g(x))} = \frac{1}{f}(g(x)) = \left(\frac{1}{f} \circ g\right)(x)$.
- (iv) Let $g(x) = 2$ and let f be a function for which $f(\frac{1}{2}) \neq 1/f(2)$. Then $1/(f \circ g) \neq f \circ (1/g)$.

CHAPTER 4

1. (i) $(2, 4)$.
 (iii) $[2, 4]$.
 (v) $(-2, 2)$.
 (vii) $(-\infty, 1] \cup [1, \infty)$.
2. (i) All points below the graph of $f(x) = x$.
 (iii) All points below the graph of $f(x) = x^2$.
 (v) All points between the graphs of $f(x) = x + 1$ and $f(x) = x - 1$.
 (vii) A collection of straight lines parallel to the graph of $f(x) = -x$, intersecting the horizontal axis at the points $(n, 0)$ for integers n .
 (ix) All points inside the circle of radius 1 around $(1, 2)$.
3. (i) A square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.
 (iii) The union of the graph of $f(x) = x$ and of $f(x) = 2 - x$.
 (v) The point $(0, 0)$.
 (vii) The circle of radius $\sqrt{5}$ around $(1, 0)$, since $x^2 - 2x + y^2 = (x - 1)^2 + y^2 - 1$.
5. (a) Simply observe that the graph of $f(x) = m(x - a) + b = mx + (b - ma)$ is a straight line with slope m , which goes through the point (a, b) . (The important point about this exercise is simply to remember the point slope form.)
 (b) The straight line through (a, b) and (c, d) has slope $(d - b)/(c - a)$, so the equation follows from part (a).
 (c) When $m = m'$ and $b \neq b'$. In that case, there is clearly no number x with $f(x) = g(x)$, while such a number x always exists if $m \neq m'$, namely, $x = (b' - b)/(m - m')$.
6. (a) If $B = 0$ and $A \neq 0$, then the set is the vertical straight line formed by all points (x, y) with $x = -C/A$. If $B \neq 0$, the set is the graph of $f(x) = (-A/B)x + (-C/A)$.
 (b) The points (x, y) on the vertical line with $x = a$ are precisely the ones which satisfy $1 \cdot x + 0 \cdot y + (-a) = 0$. The points (x, y) on the graph of $f(x) = mx + b$ are precisely the ones which satisfy $(-m)x + 1 \cdot y + (-b) = 0$.
10. (i) The graph of f is symmetric with respect to the vertical axis.
 (ii) The graph of f is symmetric with respect to the origin. Equivalently, the part of the graph to the left of the vertical axis is

obtained by reflecting first through the vertical axis, and then through the horizontal axis.

- (iii) The graph of f lies above or on the horizontal axis.
 (iv) The graph of f repeats the part between 0 and a over and over.
 19. (a) The square of the distance from (x, x^2) to $(0, \frac{1}{4})$ is

$$\begin{aligned} x^2 + \left(x^2 - \frac{1}{4}\right)^2 &= x^2 + x^4 - \frac{x^2}{2} + \frac{1}{16} \\ &= x^4 + \frac{x^2}{2} + \frac{1}{16} \\ &= \left(x^2 + \frac{1}{4}\right)^2, \end{aligned}$$

which is the square of the distance from (x, x^2) to the graph of g .

- (b) The point (x, y) satisfies this condition if and only if

$$(x - \alpha)^2 + (y - \beta)^2 = (y - \gamma)^2,$$

or

$$x^2 - 2\alpha x + \alpha^2 + y^2 - 2\beta y + \beta^2 = y^2 - 2\gamma y + \gamma^2,$$

or

$$y = \left(\frac{1}{2\beta - 2\gamma}\right)x^2 + \left(\frac{\alpha}{\gamma - \beta}\right)x + \left(\frac{\alpha^2 + \beta^2 - \gamma^2}{2\beta - 2\gamma}\right).$$

(This solution works only for $\beta \neq \gamma$, which is just the condition that P is not on L . If P is on L , then the solution is the vertical line through P .)

CHAPTER 5

1. (ii) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} (x^2 + 2x + 4) = 12.$

(iv) $\lim_{x \rightarrow y} \frac{x^n - y^n}{x - y} = \lim_{x \rightarrow y} x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1}$
 $= y^{n-1} + y^{n-1} + \cdots + y^{n-1} = ny^{n-1}.$

(vi) $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} = \lim_{h \rightarrow 0} \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})}$
 $= \lim_{h \rightarrow 0} \frac{1}{\sqrt{a+h} + \sqrt{a}} = \frac{1}{2\sqrt{a}}.$

2. (i) It is possible to find δ by beginning with the equation

$$x^4 - a^4 = (x - a)(x^3 + ax^2 + a^2x + a^3).$$

If $|x - a| < 1$, then $|x| < 1 + |a|$, so

$$\begin{aligned} |x^3 + ax^2 + a^2x + a^3| &\leq |x|^3 + |a| \cdot |x|^2 + |a|^2 \cdot |x| + |a|^3 \\ &< (1 + |a|)^3 + |a|(1 + |a|)^2 + |a|^2(1 + |a|) + |a|^3; \end{aligned}$$

therefore we can choose

$$\delta = \min \left(1, \frac{\epsilon}{(1 + |a|)^3 + |a|(1 + |a|)^2 + |a|^2(1 + |a|) + |a|^3} \right).$$

It is instructive, and probably easier, to use part (2) of the lemma. This shows that $|x^4 - a^4| < \epsilon$ when

$$|x^2 - a^2| < \min \left(1, \frac{\epsilon}{2(|a|^2 + 1)} \right),$$

which is true when

$$\begin{aligned} |x - a| &< \min \left(1, \frac{\min \left(1, \frac{\epsilon}{2(|a|^2 + 1)} \right)}{2(|a| + 1)} \right) \\ &= \min \left(1, \frac{\epsilon}{4(|a|^2 + 1)(|a| + 1)} \right) = \delta. \end{aligned}$$

(ii) By part (3) of the lemma, $|1/x - 1| < \epsilon$ when

$$|x - 1| < \min \left(\frac{1}{2}, \frac{\epsilon}{2} \right) = \delta.$$

(iii) By part (1) of the lemma, $|(x^4 + 1/x) - 2| < \epsilon$ when $|1/x - 1| < \epsilon/2$ and $|x^4 - 1| < \epsilon/2$. According to parts (i) and (ii) of this problem, this happens when

$$|x - 1| < \min \left(\frac{1}{2}, \frac{\epsilon}{4}, 1, \frac{\epsilon}{8 \cdot 2 \cdot 2} \right) = \min \left(\frac{1}{2}, \frac{\epsilon}{32} \right) = \delta.$$

(v) Let $\delta = \epsilon^2$, since $0 < |x| < \epsilon^2$ implies that $\sqrt{|x|} < \epsilon$.

5. (i) We need $|f(x) - 2| < \epsilon/2$ and $|g(x) - 4| < \epsilon/2$, so we need

$$0 < |x - 2| < \min \left(\sin^2 \left(\frac{\epsilon^2}{36} \right) + \frac{\epsilon}{2}, \frac{\epsilon^2}{4} \right) = \delta.$$

(iii) We need

$$|g(x) - 4| < \min \left(\frac{|4|}{2}, \frac{\epsilon|4|^2}{2} \right),$$

so we need

$$0 < |x - 2| < [\min(2, 8\epsilon)]^2 = \delta.$$

8. Let $l = \lim_{x \rightarrow a} f(x)$ and define $g(h) = f(a + h)$. Then for every $\epsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \epsilon$. Now, if $0 < |h| < \delta$, then $0 < |(h + a) - a| < \delta$, so $|f(a + h) - l| < \epsilon$. This inequality can be written $|g(h) - l| < \epsilon$. Thus, $\lim_{h \rightarrow 0} g(h) = l$, which can also be written $\lim_{h \rightarrow 0} f(a + h) = l$. The same sort of argument shows that if $\lim_{h \rightarrow 0} f(a + h) = m$, then $\lim_{x \rightarrow a} f(x) = m$. So either limit exists if the other does, and in this case they are equal.

9. (a) Intuitively, we can get $f(x)$ as close to l as we like if and only if we can get $f(x) - l$ as close to 0 as we like. The formal proof is so trivial that it takes a bit of work to make it look like a proof at all. To be very precise, suppose $\lim_{x \rightarrow a} f(x) = l$ and let $g(x) = f(x) - l$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. This last inequality can be written $|g(x) - 0| < \varepsilon$, so $\lim_{x \rightarrow a} g(x) = 0$. The argument in the other direction is similarly uninteresting.
- (b) Intuitively, making x close to a is the same as making $x - a$ close to 0. Formally: Suppose that $\lim_{x \rightarrow a} f(x) = l$, and let $g(x) = f(x - a)$. Then for all $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$. Now, if $0 < |y| < \delta$, then $0 < |(y + a) - a| < \delta$, so $|f(y + a) - l| < \varepsilon$. But this last inequality can be written $|g(y) - l| < \varepsilon$. So $\lim_{y \rightarrow 0} g(y) = l$. The argument in the reverse direction is similar.
- (c) Intuitively, x is close to 0 if and only if x^3 is. Formally: Let $\lim_{x \rightarrow 0} f(x) = l$. For every $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x| < \delta$, then $|f(x) - l| < \varepsilon$. Then if $0 < |x| < \min(1, \delta)$, we have $0 < |x^3| < \delta$, so $|f(x^3) - l| < \varepsilon$. Thus, $\lim_{x \rightarrow 0} f(x) = l$. On the other hand, if we assume that $\lim_{x \rightarrow 0} f(x^3)$ exists, say $\lim_{x \rightarrow 0} f(x^3) = m$, then for all $\varepsilon > 0$ there is a δ such that if $0 < |x| < \delta$, then $|f(x^3) - m| < \varepsilon$. Then if $0 < |x| < \delta^3$, we have $0 < |\sqrt[3]{x}| < \delta$, so $|f((\sqrt[3]{x})^3) - m| < \varepsilon$, or $|f(x) - m| < \varepsilon$. Thus $\lim_{x \rightarrow 0} f(x) = m$.
- (d) Let $f(x) = 1$ for $x \geq 0$, and $f(x) = -1$ for $x < 0$. Then $\lim_{x \rightarrow 0} f(x^2) = 1$, but $\lim_{x \rightarrow 0} f(x)$ does not exist.
15. (a) The function $f(x) = 1/x$ cannot approach a limit at 0, since it becomes arbitrarily large near 0. In fact, no matter what $\delta > 0$ may be, there is some x satisfying $0 < |x| < \delta$, but $1/x > |l| + \varepsilon$, namely, $x = \min(\delta, 1/(|l| + \varepsilon))$. This x does not satisfy $|1/x - l| < \varepsilon$.
- (b) No matter what $\delta > 0$ may be, there is some x satisfying $0 < |x - 1| < \delta$, but $1/(x - 1) > |l| + \varepsilon$, namely, $x = \min(1 + \delta, 1 + 1/(|l| + \varepsilon))$. This x does not satisfy $|1/(x - 1) - l| < \varepsilon$. (It is also possible to apply Problem 9(b): $\lim_{x \rightarrow 0} 1/x = \lim_{x \rightarrow 1} 1/(x - 1)$ if the latter exists, so this limit does not exist, because of part (a).)
23. (i) This is the usual definition, simply calling the numbers δ and ε , instead of ε and δ .
- (ii) This is a minor modification of (i): if the condition is true for all $\delta > 0$, then it applies to $\delta/2$, so there is an $\varepsilon > 0$ such that if $0 < |x - a| < \varepsilon$, then $|f(x) - l| \leq \delta/2 < \delta$.
- (iii) This is a similar modification: apply it to $\delta/5$ to obtain (i).

- (iv) This is also a modification: it says the same thing as (i), since $\epsilon/10 > 0$, and it is only the existence of *some* $\epsilon > 0$ that is in question.
27. If $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = l$, then for every $\epsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\begin{aligned} \text{if } a < x < a + \delta_1, \text{ then } |f(x) - l| < \epsilon, \\ \text{if } a - \delta_2 < x < a, \text{ then } |f(x) - l| < \epsilon. \end{aligned}$$

Let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then either $a - \delta < x - \delta < x < a$ or else $a < x < a + \delta < a + \delta_1$, so $|f(x) - l| < \epsilon$.

28. (i) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \epsilon$ for $0 < x < \delta$. If $-\delta < x < 0$, then $0 < -x < \delta$, so $|f(-x) - l| < \epsilon$. Thus $\lim_{x \rightarrow 0^-} f(-x) = l$. Similarly, if $\lim_{x \rightarrow 0^-} f(x)$ exists, then $\lim_{x \rightarrow 0^+} f(x)$ exists and has the same value. (Intuitively, x is close to 0 and positive if and only if $-x$ is close to 0 and negative.)
- (ii) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \epsilon$ for $0 < x < \delta$. So if $0 < |x| < \delta$, then $|f(|x|) - l| < \epsilon$. Thus $\lim_{x \rightarrow 0} f(|x|) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then $|x|$ is close to 0 and positive.)
- (iii) If $l = \lim_{x \rightarrow 0^+} f(x)$, then for all $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - l| < \epsilon$ for $0 < x < \delta$. If $0 < |x| < \sqrt{\delta}$, then $0 < x^2 < \delta$, so $|f(x^2) - l| < \epsilon$. Thus $\lim_{x \rightarrow 0} f(x^2) = l$. The reverse direction is similar. (Intuitively, if x is close to 0, then x^2 is close to 0 and positive.)
31. If $l = \lim_{x \rightarrow \infty} f(x)$, then for every $\epsilon > 0$ there is some N such that $|f(x) - l| < \epsilon$ for $x > N$, and we can clearly assume that $N > 0$. Now, if $0 < x < 1/N$, then $1/x > N$, so $|f(1/x) - l| < \epsilon$. Thus $\lim_{x \rightarrow 0^+} f(1/x) = l$. The reverse direction is similar.

CHAPTER 6

1. (i) $F(x) = x + 2$ for all x .
(iii) $F(x) = 0$ for all x .

CHAPTER 7

1. (i) Bounded above and below; minimum value 0; no maximum value.
(iii) Bounded below but not above; minimum value 0.
(v) Bounded above and below. It is understood that $a > -1$ (so that $-a - 1 < a + 1$). If $-1 < a \leq \frac{1}{2}$, then $a < -a - 1$, so $f(x) = a + 2$ for all x in $(-a - 1, a + 1)$, so $a + 2$ is the maximum and

- minimum value. If $-\frac{1}{2} < a \leq 0$, then f has the minimum value a^2 , and if $a \geq 0$, then f has the minimum value 0. Since $a + 2 > (a + 1)^2$ only for $[-1 - \sqrt{5}]/2 < a < [1 + \sqrt{5}]/2$, when $a \geq -\frac{1}{2}$ the function f has a maximum value only for $a \leq [1 + \sqrt{5}]/2$ (the maximum value being $a + 2$).
- (vii) Bounded above and below; maximum value 1; minimum value 0.
 - (ix) Bounded above and below; maximum value 1; minimum value -1 .
 - (xi) f has a maximum and minimum value, since f is continuous.
2. (i) $n = -2$, since $f(-2) < 0 < f(-1)$.
 (iii) $n = -1$, since $f(-1) = -1 < 0 < f(0)$.
 3. (i) If $f(x) = x^{179} + 163/(1 + x^2 + \sin^2 x)$, then f is continuous on $\mathbf{R}$ and $f(1) > 0$, while $f(-2) < 0$, so $f(x) = 0$ for some x in $(-2, 1)$.
 5. f is constant, for if f took on two different values, then f would take on all values in between, which would include irrational values.
 7. (1) $f(x) = x$;
 (2) $f(x) = -x$;
 (3) $f(x) = |x|$;
 (4) $f(x) = -|x|$.
 10. Apply Theorem 1 to $f - g$.
 11. If $f(0) = 0$ or $f(1) = 1$, choose $x = 0$ or 1 . If $f(0) > 0 = I(0)$ and $f(1) < 1 = I(1)$, then Problem 10 applied to f and I implies that $f(x) = x$ for some x .

CHAPTER 8

1. (i) 1 is the greatest element, and the greatest lower bound is 0, which is not in the set.
 (iii) 1 is the greatest element, and 0 is the least element.
 (v) Since $\{x: x^2 + x + 1 \geq 0\} = \mathbf{R}$, there is no least upper bound or greatest lower bound.
 (vii) Since $\{x: x < 0 \text{ and } x^2 + x - 1 < 0\} = ([-1 - \sqrt{5}]/2, 0)$, the greatest lower bound is $[-1 - \sqrt{5}]/2$, and the least upper bound is 0; neither belongs to the set.
2. (a) Since $A \neq \emptyset$, there is some x in A . Then $-x$ is in $-A$, so $-A \neq \emptyset$. Since A is bounded above, there is some y such that $y \geq x$ for all x in A . Then $-y \leq -x$ for all x in A , so $-y \leq z$ for all z in $-A$, so $-A$ is bounded below. Let $\alpha = \sup(-A)$. Then α is an upper bound for $-A$, so, reversing the argument just given, $-\alpha$ is a lower bound for A .
 Moreover, if β is any lower bound for A , then $-\beta$ is an upper bound for $-A$, so $-\beta \geq \alpha$, so $\beta \leq -\alpha$. Thus $-\alpha$ is the greatest lower bound for A .
5. (a) If l is the largest integer with $l \leq x$, then $l + 1 > x$, but $l + 1 \leq x + 1 < y$. So we can let $k = l + 1$. (Proof that a largest such

integer l exists: Since $\mathbf{N}$ is not bounded above, there is some natural number n with $-n < x < n$. There are consequently only a finite number of integers l with $-n \leq l \leq x$. Pick the largest.)

- (b) Since $y - x > 0$, there is some natural number n with $1/n < y - x$. Since $ny - nx > 1$, there is, by part (a), an integer k with $nx < k < ny$, which means that $x < k/n < y$.
- (c) Choose $r + \sqrt{2}(s - r)/2$.
- (d) By part (b), there is a rational number r with $x < r < y$, and therefore a rational number s with $x < r < s < y$. Apply part (c) to $r < s$.
10. Let k be the largest integer $\leq x/\alpha$ (the solution to Problem 5 shows that such a k exists), and let $x' = x - k\alpha \geq 0$. If $x - k\alpha = x' \geq \alpha$, then $x \geq (k+1)\alpha$, so $k+1 \leq x/\alpha$, contradicting the choice of k . So $0 \leq x' < \alpha$.
12. (a) Since any y in B satisfies $y \geq x$ for all x in A , any y in B is an upper bound for A , so $y \geq \sup A$.
- (b) Part (a) shows that $\sup A$ is a lower bound for B , so $\sup A \leq \inf B$.
13. Since $x \leq \sup A$ and $y \leq \sup B$ for every x in A , and y in B , it follows that $x + y \leq \sup A + \sup B$. Thus, $\sup A + \sup B$ is an upper bound for $A + B$, so $\sup(A + B) \leq \sup A + \sup B$. If x and y are chosen in A and B , respectively, so that $\sup A - x < \epsilon/2$ and $\sup B - y < \epsilon/2$, then $\sup A + \sup B - (x + y) < \epsilon$. Hence,

$$\sup(A + B) \geq x + y > \sup A + \sup B - \epsilon.$$

CHAPTER 9

1. (a)

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = -\frac{1}{a^2}. \end{aligned}$$

(b) The tangent line through $(a, 1/a)$ is the graph of

$$\begin{aligned} g(x) &= \frac{-1}{a^2}(x - a) + \frac{1}{a} \\ &= \frac{-x}{a^2} + \frac{2}{a}. \end{aligned}$$

If $f(x) = g(x)$, then

$$\frac{1}{x} = -\frac{x}{a^2} + \frac{2}{a}$$

or

$$x^2 - 2ax + a^2 = 0,$$

so $x = a$.

2. (a)

$$\begin{aligned}
 f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(a+h)^2} - \frac{1}{a^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(-2ah - h^2)}{ha^2(a+h)^2} = -\frac{2}{a^3}.
 \end{aligned}$$

(b) The tangent line through $(a, 1/a^2)$ is the graph of

$$\begin{aligned}
 g(x) &= -\frac{2}{a^3}(x-a) + \frac{1}{a^2} \\
 &= -\frac{2x}{a^3} + \frac{3}{a^2}.
 \end{aligned}$$

If $f(x) = g(x)$, then

$$\frac{1}{x^2} = \frac{-2x}{a^3} + \frac{3}{a^2},$$

or

$$2x^3 - 3ax^2 + a^3 = 0,$$

or

$$0 = (x-a)(2x^2 - ax - a^2) = (x-a)(2x+a)(x-a).$$

So $x = a$ or $x = -a/2$; the point $(-a/2, 4/a^2)$ lies on the opposite side of the vertical axis from $(a, 1/a^2)$.

3.

$$\begin{aligned}
 f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(\sqrt{a+h} - \sqrt{a})(\sqrt{a+h} + \sqrt{a})}{h(\sqrt{a+h} + \sqrt{a})} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{a+h} + \sqrt{a})} \\
 &= \frac{1}{2\sqrt{a}}.
 \end{aligned}$$

4. Conjecture: $S_n'(x) = nx^{n-1}$. Proof:

$$\begin{aligned}
 S_n'(x) &= \lim_{h \rightarrow 0} \frac{S_n(x+h) - S_n(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sum_{j=0}^n \binom{n}{j} x^{n-j} h^j - x^n}{h} \\
 &= \lim_{h \rightarrow 0} \sum_{j=1}^n \binom{n}{j} x^{n-j} h^{j-1} \\
 &= \binom{n}{1} x^{n-1} = nx^{n-1}, \quad \text{since } \lim_{h \rightarrow 0} h^{j-1} = 0 \text{ for } j > 1.
 \end{aligned}$$

5. $f'(x) = 0$ for x not an integer, and $f'(x)$ is not defined if x is an integer.

6. (a)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{[f(x+h) + c] - [f(x) + c]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x). \end{aligned}$$

(b)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= c \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = cf'(x). \end{aligned}$$

7. (a) $f'(9) = 3 \cdot 9^2$; $f'(25) = 3 \cdot (25)^2$; $f'(36) = 3 \cdot (36)^2$.

(b) $f'(3^2) = f'(9) = 3 \cdot 9^2$; $f'(5^2) = f'(25) = 3 \cdot (25)^2$;
 $f'(6^2) = f'(36) = 3 \cdot (36)^2$.

(c) $f'(a^2) = 3(a^2)^2 = 3a^4$; $f'(x^2) = 3(x^2)^2 = 3x^4$.

(d) $f'(x^2) = 3x^4$; but $g(x) = x^6$, so $g'(x) = 6x^5$.

8. (a)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h+c) - f(x+c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f([x+c] + h) - f(x+c)}{h} = f'(x+c). \end{aligned}$$

(b)

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{f(cx+ch) - f(cx)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c[f(cx+ch) - f(cx)]}{ch} = \lim_{k \rightarrow 0} \frac{c[f(cx+k) - f(cx)]}{k} \\ &= c \cdot \lim_{k \rightarrow 0} \frac{f(cx+k) - f(cx)}{k} = c \cdot f'(cx). \end{aligned}$$

(Compare the manipulations in this calculation with Problem 5-13.)

(c) If $g(x) = f(x+a)$, then $g'(x) = f'(x+a)$, by part (a). But $g = f$, so $f'(x) = g'(x) = f'(x+a)$ for all x , which means that f' is periodic, with period a .

9. (i) If $g(x) = x^5$, then $g'(x) = 5x^4$. Now $f(x) = g(x+5)$, so by Problem 8(a), $f'(x) = g'(x+5) = 5(x+5)^4$.

(ii) $f(x) = (x-3)^5$, so $f'(x) = 5(x-3)^4$, as in part (i).

(iii) $f(x) = (x+2)^7$, so $f'(x) = 7(x+2)^6$, as in part (i).

10. If $f(x) = g(t+x)$, then $f'(x) = g'(t+x)$, by Problem 8(a). If $f(t) = g(t+x)$, then $f'(t) = g'(t+x)$, by Problem 8(a), so $f'(x) = g'(2x)$.

11. (a) If $s(t) = ct^2$, then $s'(t) = 2ct$, and there is no number k such that $s'(t) = ks(t)$ [that is, $2ct = kct^2$] for all t .

(By the way, at this point we do not know any nonzero function f for

- which f' is proportional to f . After Chapter 17 it might be amusing to determine what the world would be like if Galileo were correct.)
- (b) (i) If $s(t) = (a/2)t^2$, then $s'(t) = at$, so $s''(t) = a$.
 (ii) $[s'(t)]^2 = (at)^2 = 2a \cdot (a/2)t^2 = 2as(t)$.
- (c) The chandelier falls $s(t) = 16t^2$ feet in t seconds, so it falls 400 feet in t seconds if $400 = 16t^2$, or $t = 5$. After 5 seconds the velocity will be $s'(5) = 5a = 5 \cdot 16 = 80$ feet per second. The speed was half this amount when $40 = s'(t) = 16t$, or $t = \frac{5}{2}$.
21. (a) This is another way of writing the definition (see Problem 5-8).
 (b) This follows from Problem 5-10, applied to the functions $\alpha(h) = [f(a+h) - f(a)]/h$ and $\beta(h) = [g(a+h) - g(a)]/h$.
26. (i) $f''(x) = 6x$.
 (iii) $f''(x) = 4x^3$.
30. (i) means that $f'(a) = na^{n-1}$ if $f(x) = x^n$.
 (iii) means that $g'(a) = f'(a)$ if $g(x) = f(x) + c$.
 (v) means the same as (iii).
 (vii) means that $g'(b) = f'(b+a)$ if $g(x) = f(x+a)$.
 (ix) means that $g'(b) = cf'(cb)$ if $g(x) = f(cx)$.

CHAPTER 10

1. (i) $(1 + 2x) \cdot \cos(x + x^2)$.
 (iii) $(-\sin x) \cdot \cos(\cos x)$.
 (v) $\cos\left(\frac{\cos x}{x}\right) \cdot \frac{-x \sin x - \cos x}{x^2}$.
 (vii) $(\cos(x + \sin x)) \cdot (1 + \cos x)$.
2. (i) $(\cos((x+1)^2(x+2))) \cdot [2(x+1)(x+2) + (x+1)^2]$.
 (iii) $[2 \sin((x + \sin x)^2) \cdot \cos((x + \sin x)^2)] \cdot 2(x + \sin x)(1 + \cos x)$.
 (v) $(\cos(x \sin x)) \cdot (\sin x + x \cos x) + (\cos(\sin x^2)(\cos x^2)) \cdot 2x$.
 (vii) $(2 \sin x \cos x \sin x^2 \sin^2 x^2) + (2x \cos x^2 \sin^2 x \sin^2 x^2) + (\sin x^2 \cos x^2 \sin^2 x \sin x^2)$.
 (ix) $6(x + \sin^5 x)^5(1 + 5 \sin^4 x \cos x)$.
 (xi) $\cos(\sin^7 x^7 + 1)^7 \cdot 7(\sin^7 x^7 + 1)^6 \cdot (7 \sin^6 x^7 \cdot \cos x^7 \cdot 7x^6)$.
 (xiii) $\cos(x^2 + \sin(x^2 + \sin x^2)) \cdot [(2x + \cos(x^2 + \sin x^2)) \cdot (2x + 2x \cos x^2)]$.
 (xv) $\frac{(1 + \sin x)(2x \cos x^2 \cdot \sin^2 x + \sin x^2 \cdot 2 \sin x \cos x) - \cos x \sin x^2 \sin^2 x}{(1 + \sin x)^2}$.
- (xvii) $\cos\left(\frac{x^3}{\sin\left(\frac{x^3}{\sin x}\right)}\right)$

$$\frac{3x^2 \sin\left(\frac{x^3}{\sin x}\right) - x^3 \cos\left(\frac{x^3}{\sin x}\right) \cdot \left(\frac{3x^2 \sin x - x^3 \cos x}{\sin^2 x}\right)}{\sin^2\left(\frac{x^3}{\sin x}\right)}$$

4. (i) $-\frac{(x+1)^2}{(x+2)^2}$.
 (iii) $2x^2$.
5. (i) $-x^2$.
 (iii) 17.
6. (i) $f'(x) = g'(x + g(a))$.
 (iii) $f'(x) = g'(x + g(x)) \cdot (1 + g'(x))$.
 (v) $f'(x) = g(a)$.
7. (a) $A'(t) = 2\pi r(t)r'(t)$. Since $r'(t) = 6$ for that t with $r(t) = 4$, it follows that $A'(t) = 2\pi \cdot 4 \cdot 6 = 48\pi$ when $r(t) = 4$.
 (b) If $V(t)$ is the volume at time t , then $V(t) = 4\pi r(t)^3/3$, so $V'(t) = 4\pi r(t)^2 r'(t) = 4\pi \cdot 4^2 \cdot 6 = 384\pi$ when $r(t) = 4$.
 (c) First method: Since $A'(t) = 2\pi r(t)r'(t)$, and $A'(t) = 5$ for $r(t) = 3$, it follows that

$$r'(t) = \frac{A'(t)}{2\pi r(t)} = \frac{5}{6\pi} \quad \text{when } r(t) = 3.$$

Thus

$$\begin{aligned} V'(t) &= 4\pi r(t)^2 r'(t) \\ &= 4\pi \cdot 9 \cdot \frac{5}{6\pi} \\ &= 30 \quad \text{when } r(t) = 3. \end{aligned}$$

To apply the second method, we first note that if

$$f(t) = A(t)^{3/2} = \sqrt{A(t)^3},$$

then, using Problem 9-3 and the Chain Rule,

$$\begin{aligned} f'(t) &= \frac{1}{2\sqrt{A(t)^3}} \cdot 3A(t)^2 A'(t) \\ &= \frac{1}{2A(t)^{3/2}} \cdot 3A(t)^2 A'(t) \\ &= \frac{3}{2} A(t)^{1/2} A'(t) \quad (\text{just as we might have guessed}). \end{aligned}$$

Now

$$\begin{aligned} V(t) &= \frac{4\pi r(t)^3}{3} = \frac{4\pi [r(t)^2]^{3/2}}{3} \\ &= \frac{4[\pi r(t)^2]^{3/2}}{3\pi^{1/2}} \\ &= \frac{4A(t)^{3/2}}{3\pi^{1/2}}. \end{aligned}$$

So

$$\begin{aligned}
 V'(t) &= \frac{4}{3\pi^{1/2}} \cdot \frac{3}{2} \sqrt{A(t)} A'(t) \\
 &= \frac{2}{\pi^{1/2}} \cdot \pi^{1/2} t A'(t) \\
 &= 2 \cdot 3 \cdot 5 = 30.
 \end{aligned}$$

8. (i) $(f \circ h)'(0) = f'(h(0)) \cdot h'(0) = f'(3) \cdot \sin^2(\sin 1)$
 $= 9(\sin \frac{1}{3})\sin^2(\sin 1).$
 (iii) $\alpha'(x^2) = h'(x^4) \cdot 2x^2 = \sin^2(\sin(x^4 + 1)) \cdot 2x^2.$
 10. The Chain Rule implies that

$$\begin{aligned}
 \left(\frac{1}{g}\right)'(x) &= (f \circ g)'(x) = f'(g(x)) \cdot g'(x) \\
 &= -\frac{1}{g(x)^2} \cdot g'(x).
 \end{aligned}$$

30. (i) $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} = (\cos y) \cdot (1 + 2x) = (\cos(x + x^2)) \cdot (1 + 2x).$
 (iii) $\frac{dz}{dx} = \frac{dz}{du} \cdot \frac{du}{dx} = (-\sin u) \cdot (\cos x) = (-\cos(\sin x)) \cdot (\cos x).$

CHAPTER 11

1. (i) $0 = f'(x) = 3x^2 - 2x - 8$ for $x = 2$ and $x = -\frac{4}{3}$, both of which are in $[-2, 2]$;
 $f(-2) = 5$, $f(2) = -11$, $f(-\frac{4}{3}) = \frac{208}{27}$;
 maximum $= \frac{208}{27}$, minimum $= -11.$
 (iii) $0 = f'(x) = 12x^3 - 24x^2 + 12x = 12x(x^2 - 2x + 1)$ for $x = 0$ and $x = 1$, of which only 0 is in $[-\frac{1}{2}, \frac{1}{2}]$;
 $f(-\frac{1}{2}) = \frac{43}{16}$, $f(\frac{1}{2}) = \frac{11}{16}$, $f(0) = 0$;
 maximum $= \frac{43}{16}$, minimum $= 0.$
 (v) $0 = f'(x) =$

$$\frac{x^2 + 1 - (x + 1)2x}{(x^2 + 1)^2} = \frac{1 - 2x - x^2}{(x^2 + 1)^2}$$

for $x = -1 + \sqrt{2}$ and $x = -1 - \sqrt{2}$, of which only $-1 + \sqrt{2}$ is in $[-1, \frac{1}{2}]$;

$$\begin{aligned}
 f(-1) &= 0, f(\frac{1}{2}) = \frac{6}{5}, f(-1 + \sqrt{2}) = (1 + \sqrt{2})/2; \\
 \text{maximum} &= (1 + \sqrt{2})/2, \text{minimum} = 0.
 \end{aligned}$$

2. (i) $-\frac{4}{3}$ is a local maximum point, and 2 is a local minimum point.
 (iii) 0 is a local minimum point, and there are no local maximum points.
 (v) $-1 + \sqrt{2}$ is a local maximum point, and $-1 - \sqrt{2}$ is a local minimum point.

3. (a) Notice that f actually has a minimum value, since f is a polynomial function of even degree. The minimum occurs at a point x with

$$0 = f'(x) = 2 \sum_{i=1}^n (x - a_i),$$

so $x = (a_1 + \cdots + a_n)/n$.

4. (i) 3 and 7 are local maximum points, and 5 and 9 are local minimum points.
 (iii) All irrational $x > 0$ are local minimum points, and all irrational $x < 0$ are local maximum points.
 (v) x is a local maximum (minimum) point if the decimal expansion contains (does not contain) a 5.
 5. If $f(x)$ is the total length of the path, then

$$f(x) = \sqrt{x^2 + a^2} + \sqrt{(1-x)^2 + b^2}.$$

The positive function f clearly has a minimum, since $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = \infty$, and f is differentiable everywhere, so the minimum occurs at a point x with $f'(x) = 0$. Now, $f'(x) = 0$ when

$$\frac{x}{\sqrt{x^2 + a^2}} - \frac{(1-x)}{\sqrt{(1-x)^2 + b^2}} = 0.$$

This equation says that $\arctan \alpha = \arctan \beta$.

It is also possible to notice that $f(x)$ is equal to the sum of the lengths of the dashed line segment and the line segment from $(x, 0)$ to $(1, b)$. This is shortest when the two line segments lie along a line (because of Problem 4-8(b), if a rigorous reason is required); a little plane geometry shows that this happens when $\alpha = \beta$.

6. If x is the length of one side of a rectangle of perimeter P , then the length of the other side is $(P - 2x)/2$, so the area is

$$A(x) = \frac{x(P - 2x)}{2}.$$

So the rectangle with greatest area occurs when x is the maximum point for f on $(0, P/2)$. Since A is continuous on $[0, P/2]$, and $A(0) = A(P/2) = 0$, and $A(x) > 0$ for x in $(0, P/2)$, the maximum exists. Since A is differentiable on $(0, P/2)$, the minimum point x satisfies

$$\begin{aligned} 0 = A'(x) &= \frac{P - 2x}{2} - x \\ &= \frac{P - 4x}{2}, \end{aligned}$$

so $x = P/4$.

7. Let $S(r)$ be the surface area of the right circular cylinder of volume V with radius r . Since

$$V = \pi r^2 h \quad \text{where } h \text{ is the height,}$$

we have $h = V/\pi r^2$, so

$$\begin{aligned} S(r) &= 2\pi r^2 + 2\pi r h \\ &= 2\pi r^2 + \frac{2V}{r}. \end{aligned}$$

We want the minimum point of S on $(0, \infty)$; this exists, since $\lim_{r \rightarrow 0} S(r) = \infty$ and $\lim_{r \rightarrow \infty} S(r) = \infty$. Since S is differentiable on $(0, \infty)$, the minimum point r satisfies

$$\begin{aligned} 0 = S'(r) &= 4\pi r - \frac{2V}{r^2} \\ &= \frac{4\pi r^3 - 2V}{r^2}, \end{aligned}$$

or

$$r = \sqrt[3]{\frac{V}{2\pi}}.$$

8. 1 is a local maximum point, and 3 is a local minimum point.
14. (a) We have

$$\begin{aligned} \frac{f(b) - f(a)}{b - a} &= f'(x) \quad \text{for some } x \text{ in } (a, b) \\ &\geq M, \end{aligned}$$

$$\text{so } f(b) - f(a) \geq M(b - a).$$

(b) We have

$$\begin{aligned} \frac{f(b) - f(a)}{b - a} &= f'(x) \quad \text{for some } x \text{ in } (a, b) \\ &\leq m, \end{aligned}$$

$$\text{so } f(b) - f(a) \leq m(b - a).$$

- (c) If $|f(x)| \leq M$ for all x in $[a, b]$, then $-M \leq f(x) \leq M$, so

$$f(a) - M(b - a) \leq f(b) \leq f(a) + M(b - a),$$

or

$$|f(b) - f(a)| \leq M(b - a).$$

16. (a) $f(x) = -\cos x + a$ for some number a (because $f(x) = -\cos x$ is one such function, and any two such functions differ by a constant function).

- (b) $f'(x) = x^4/4 + a$ for some number a , so $f(x) = x^5/20 + ax + b$ for some numbers a and b .
- (c) $f''(x) = x^2/2 + x^3/3 + a$ for some a , so $f'(x) = x^3/6 + x^4/12 + ax + b$ for some a and b , so $f(x) = x^4/24 + x^5/60 + ax^2/2 + bx + c$ for some numbers a , b , and c . Equivalently, and more simply, $f(x) = x^4/24 + x^5/60 + ax^2 + bx + c$ for some numbers a , b , and c .
17. (a) Since $s''(t) = -32$, we have $s'(t) = -32t + \alpha$ for some α , so $s(t) = -16t^2 + \alpha t + \beta$ for some α and β .
- (b) Clearly, $s(0) = 0 + 0 + \beta$ and $s'(0) = 0 + \alpha$. Thus, $\alpha = v_0$ and $\beta = s_0$.
- (c) In this case, $s_0 = 0$ and $v_0 = v$, so $s(t) = -16t^2 + vt$. The maximum value of s occurs when $0 = s'(t) = -32t + v$, or $t = v/32$, so the maximum value is

$$\begin{aligned} s\left(\frac{v}{32}\right) &= -16\left(\frac{v}{32}\right)^2 + v\left(\frac{v}{32}\right) \\ &= \frac{-v^2}{64} + \frac{v^2}{32} \\ &= \frac{v^2}{64}. \end{aligned}$$

At that moment the velocity is clearly 0, but the acceleration is -32 (as at any time). The weight hits the ground at time $t > 0$ when

$$0 = s(t) = -16t^2 + vt,$$

or $t = v/16$ (it takes as long to fall back down as it took to reach the top). The velocity is then

$$\begin{aligned} s'(v/16) &= -32\left(\frac{v}{16}\right) + v \\ &= -v \end{aligned}$$

(the same velocity with which it was initially moving upward).

29. Apply the Mean Value Theorem to $f(x) = \sqrt{x}$ on $[64, 66]$:

$$\frac{\sqrt{66} - \sqrt{64}}{66 - 64} = f'(x) = \frac{1}{2\sqrt{x}} \quad \text{for some } x \text{ in } [64, 66].$$

Since $64 < x < 81$, we have $8 < \sqrt{x} < 9$, so

$$\frac{1}{2 \cdot 9} < \frac{\sqrt{66} - 8}{2} < \frac{1}{2 \cdot 8}.$$

33. L'Hôpital's Rule does not lead to the equation

$$\lim_{x \rightarrow 1} \frac{3x^2 + 1}{2x - 3} = \lim_{x \rightarrow 1} \frac{6x}{2}$$

because $\lim_{x \rightarrow 1} 3x^2 + 1 \neq 0$.

34. (i)

$$\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \frac{1}{\sec^2 x} = \lim_{x \rightarrow 0} \cos^2 x = 1.$$

(ii)

$$\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{2x} = 1.$$

CHAPTER 12

1. (i) $f^{-1}(x) = (x - 1)^{1/3}$. (If $y = f^{-1}(x)$, then $x = f(y) = y^3 + 1$, so $y = (x - 1)^{1/3}$.)
 (iii) $f^{-1} = f$. (If $y = f^{-1}(x)$, then

$$x = f(y) = \begin{cases} y, & y \text{ rational} \\ -y, & y \text{ irrational;} \end{cases}$$

since $\pm y$ is rational or irrational if and only if y is, we have $y = x$ if x is rational and $y = -x$ if x is irrational, so $y = f(x)$.)

(v)

$$f^{-1}(x) = \begin{cases} x, & x \neq a_1, \dots, a_n \\ a_{i-1}, & x = a_i, \quad i = 2, \dots, n \\ a_n, & x = a_1. \end{cases}$$

- (vii) $f^{-1} = f$.
 2. (i) f^{-1} is increasing and $f^{-1}(x)$ is not defined for $x \leq 0$.
 (iii) f^{-1} is decreasing and $f^{-1}(x)$ is not defined for $x \leq 0$.
 4. Suppose f is increasing. Let $a < b$. Then $f^{-1}(a) \neq f^{-1}(b)$, since f^{-1} is one-one. So either $f^{-1}(a) < f^{-1}(b)$ or $f^{-1}(a) > f^{-1}(b)$. But if $f^{-1}(a) > f^{-1}(b)$, then

$$b = f(f^{-1}(b)) < f(f^{-1}(a)) = a,$$

a contradiction. The proof is similar for decreasing f , or one can consider $-f$ instead.

5. Clearly, $f + g$ is increasing, for if $f(a) < f(b)$ and $g(a) < g(b)$, then $(f + g)(a) = f(a) + g(a) < f(b) + g(b) = (f + g)(b)$.
 $f \cdot g$ is not necessarily increasing; for example, if $f(x) = g(x) = x$. (But $f \cdot g$ is increasing if $f(x) \geq 0$ for all x .)
 $f \circ g$ is increasing, for if $a < b$, then $g(a) < g(b)$, so $f(g(a)) < f(g(b))$.
 6. (a) If $(f \circ g)(x) = (f \circ g)(y)$, so that $f(g(x)) = f(g(y))$, then $g(x) = g(y)$, since f is one-one, so $x = y$, since g is one-one.
 $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$: for if $y = (f \circ g)^{-1}(x)$, then $x = (f \circ g)(y) = f(g(y))$, so $g(y) = f^{-1}(x)$, so $y = g^{-1}(f^{-1}(x))$.
 7. If $f(x) = f(y)$, then

$$\frac{ax + b}{cx + d} = \frac{ay + b}{cy + d}$$

so

$$acxy + bcy + adx + bd = acxy + ady + bcx + bd,$$

or

$$ad(x - y) = bc(x - y).$$

If $ad \neq bc$, this implies that $x - y = 0$. (But if $ad = bc$, then $f(x) = f(y)$ for all x and y in the domain of f .)

If $y = f^{-1}(x)$, then $x = f(y)$, so

$$x = \frac{ay + b}{cy + d},$$

so

$$f^{-1}(x) = y = \frac{-dx + b}{cx - a}, \quad \text{for } x \neq a/c.$$

8. (i) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, 2]$ or $[2, \infty)$, since f is increasing on $(-\infty, 0]$ and $[2, \infty)$, and decreasing on $[0, 2]$.
- (iii) Those intervals $[a, b]$ which are contained in $(-\infty, 0]$ or $[0, \infty)$, since f is increasing on $(-\infty, 0]$ and decreasing on $[0, \infty)$.
14. The formula for the derivative reads:

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

(In this formula, it is understood that dx/dy means $(f^{-1})'(y)$, while dy/dx is an "expression involving x ," and in the final answer x must be replaced by y , by means of the equation $y = f(x)$.)

The computation in Problem 14, when completed, shows that

$$\begin{aligned} \frac{dx^{1/n}}{dx} &= \frac{1}{n(x^{1/n})^{n-1}} = \frac{1}{nx^{1-(1/n)}} \\ &= \frac{1}{n} x^{(1/n)-1}. \end{aligned}$$

$$\begin{aligned} 15. \quad G'(x) &= x(f^{-1})'(x) + f^{-1}(x) - F'(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - f(f^{-1}(x)) \cdot (f^{-1})'(x) \\ &= x(f^{-1})'(x) + f^{-1}(x) - x(f^{-1})'(x) \\ &= f^{-1}(x). \end{aligned}$$

16. (i)

$$(h^{-1})'(3) = \frac{1}{h'(h^{-1}(3))} = \frac{1}{h'(0)} = \frac{1}{\sin^2(\sin 1)}.$$

17. Since

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))},$$

we have

$$\begin{aligned}(f^{-1})''(x) &= \frac{-f''(f^{-1}(x)) \cdot (f^{-1})'(x)}{[f'(f^{-1}(x))]^2} \\ &= \frac{-f''(f^{-1}(x))}{[f'(f^{-1}(x))]^3}.\end{aligned}$$

CHAPTER 13

1. If $P_n = \{t_0, \dots, t_n\}$ is the partition with $t_i = ib/n$, then

$$\begin{aligned}L(f, P_n) &= \sum_{i=1}^n (t_{i-1})^3 \cdot (t_i - t_{i-1}) \\ &= \sum_{i=1}^n (i-1)^3 \cdot \frac{b^3}{n^3} \cdot \frac{b}{n} \\ &= \frac{b^4}{n^4} \sum_{j=0}^{n-1} j^3 \\ &= \frac{b^4}{n^4} \left[\frac{(n-1)^4}{4} + \frac{(n-1)^3}{2} + \frac{(n-1)^2}{4} \right],\end{aligned}$$

and similarly

$$\begin{aligned}U(f, P_n) &= \frac{b^4}{n^4} \sum_{j=1}^n j^3 \\ &= \frac{b^4}{n^4} \left[\frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} \right].\end{aligned}$$

Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large, so $U(f, P_n) - L(f, P_n)$ can be made as small as desired, by choosing n large enough. This shows that f is integrable. Moreover, there is only one number a with $L(f, P_n) \leq a \leq U(f, P_n)$ for all n ; since $\int_0^b x^3 dx$ has this property, the proof that $\int_0^b x^3 dx = b^4/4$ will be complete once we show that $L(f, P_n) \leq b^4/4 \leq U(f, P_n)$ for all n . This can be done by a straightforward computation, but it actually follows from the fact that $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^4/4$ as desired by choosing n sufficiently large. In fact, if it were true that $b^4/4 < \int_0^b x^3 dx$, then it would not be possible to make $U(f, P_n)$ as close as desired to $b^4/4$ by choosing n large enough, since each $U(f, P_n) \geq \int_0^b x^3 dx$, and similarly we cannot have $b^4/4 > \int_0^b x^3 dx$.

2. We have

$$L(f, P_n) = \frac{b^5}{n^5} \left[\frac{(n-1)^5}{5} + \frac{(n-1)^4}{2} + \frac{(n-1)^3}{3} - \frac{(n-1)}{30} \right],$$

$$U(f, P_n) = \frac{b^5}{n^5} \left[\frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30} \right].$$

Clearly $L(f, P_n)$ and $U(f, P_n)$ can be made as close to $b^5/5$ as desired by choosing n large enough. As in Problem 1, this implies that

$$\int_0^b x^4 dx = b^5/5.$$

4. (i) $\int_0^2 f = 0.$

(iii) $\int_0^2 f = 3.$

(v) f is not integrable.

(vii) $\int_0^2 f = 1.$

(For a rigorous proof that the functions in (i), (iii), and (vii) are integrable, see Problem 15. The values of the integrals, which are clear from the geometric picture, can also be deduced rigorously by using the ideas in the proof of Problem 15, together with known integrals.)

5. (i)

$$\int_{-2}^2 \left[\left(\frac{x^2}{2} + 2 \right) - x^2 \right] dx = \frac{16}{3}.$$

(iii)

$$\int_{-\sqrt{2}/2}^{\sqrt{2}/2} [(1 - x^2) - x^2] dx = \frac{2\sqrt{2}}{3}.$$

(v)

$$\int_0^2 [(x^2 - 2x + 4) - x^2] dx = 4.$$

6.

$$\begin{aligned} \int_a^b \left(\int_c^d f(x)g(y) dy \right) dx &= \int_a^b \left(f(x) \int_c^d g(y) dy \right) dx \quad (\text{here } f(x) \text{ is the constant}) \\ &= \int_c^d g(y) dy \cdot \int_a^b f(x) dx \\ &\quad (\text{here } \int_c^d g(y) dy \text{ is the constant}). \end{aligned}$$

10. (a) Clearly $L(f, P) \geq 0$ for every partition P .

(b) Apply part (a) to $f - g$, and use the fact that

$$\int_a^b (f - g) = \int_a^b f - \int_a^b g.$$

24. (a) 0.

(b) $\frac{1}{2}$.

27. (a) Clearly

$$m(b - a) \leq L(f, P) \leq U(f, P) \leq M(b - a)$$

for all partitions P of $[a, b]$. Consequently,

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

Thus

$$\mu = \frac{\int_a^b f(x) dx}{b-a}$$

satisfies $m \leq \mu \leq M$.

- (b) Let m and M be the minimum and maximum values of f on $[a, b]$. Since f is continuous, it takes on the values m and M , and consequently the number μ of part (a).

31. Since

$$-|f| \leq f \leq |f|,$$

we have

$$-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|,$$

so

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

(Problem 30 implies that $\int_a^b |f|$ makes sense.)

CHAPTER 14

1. (i) $(\sin^3 x^3) \cdot 3x^2.$

(iii) $\int_8^x \frac{1}{1+t^2+\sin^2 t} dt.$

(v) $\int_a^b \frac{1}{1+t^2+\sin^2 t} dt.$

(vii) $(F^{-1})'(x) = \frac{1}{F'(F^{-1}(x))} = F^{-1}(x).$

2. (i) All $x \neq 1.$

(iii) All $x \neq 1.$

(v) All $x.$

(vii) All $x \neq 0.$ (F is not differentiable at 0 because $F(x) = 0$ for $x \leq 0$, but there are $x > 0$ arbitrarily close to 0 with $F(x) = \frac{1}{2}.$)

3. (i)

$$\begin{aligned} (f^{-1})'(0) &= \frac{1}{f'(f^{-1}(0))} = \frac{1}{1 + \sin(\sin(f^{-1}(0)))} \\ &= \frac{1}{1 + \sin(\sin 0)} = 1. \end{aligned}$$

4. $F(x) = x \int_0^x f(t) dt$, so

$$F'(x) = xf(x) + \int_0^x f(t) dt.$$

7. $f(x) = \int_0^x \left(\int_0^y \left(\int_0^z \frac{1}{\sqrt{1+\sin^2 t}} dt \right) dz \right) dy.$

9. We can choose

$$f(x) = \frac{x^{(1/n)+1}}{\frac{1}{n} + 1}.$$

Then

$$\int_0^b \sqrt[n]{x} \, dx = f(b) - f(0) = \frac{b^{(1/n)+1}}{\frac{1}{n} + 1}.$$

CHAPTER 15

1. (i)

$$\frac{1}{1 + \arctan^2(\arctan x)} \cdot \frac{1}{1 + \arctan^2 x} \cdot \frac{1}{1 + x^2}.$$

(iii)

$$\frac{1}{1 + (\tan x \arctan x)^2} \cdot \left(\sec^2 x \arctan x + \frac{\tan x}{1 + x^2} \right).$$

2. (i) 0.

(iii) 0.

(v) 0.

6. (a) $\sin 2x = \sin(x + x) = \sin x \cos x + \cos x \sin x = 2 \sin x \cos x$.

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x.$$

$$\sin 3x = \sin(2x + x) = \sin 2x \cos x + \cos 2x \sin x$$

$$= 2 \sin x \cos^2 x + (\cos^2 x - \sin^2 x) \sin x$$

$$= 3 \sin x \cos^2 x - \sin^3 x.$$

$$\cos 3x = \cos(2x + x) = \cos 2x \cos x - \sin 2x \sin x$$

$$= (\cos^2 x - \sin^2 x) \cos x = 2 \sin^2 x \cos x$$

$$= \cos^3 x - \sin^2 x \cos x - 2 \sin^2 x \cos x$$

$$= \cos^3 x - 3 \sin^2 x \cos x$$

$$= 4 \cos^3 x - 3 \cos x.$$

(b) Since $\cos \pi/4 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 2 \cdot \frac{\pi}{4} = 2 \cos^2 \frac{\pi}{4} - 1,$$

we have $\cos \pi/4 = \sqrt{2}/2$. It follows, since $\sin \pi/4 > 0$ and $\sin^2 + \cos^2 = 1$, that $\sin \pi/4 = \sqrt{2}/2$, and consequently $\tan \pi/4 = 1$. Similarly, since $\cos \pi/6 > 0$ and

$$0 = \cos \frac{\pi}{2} = \cos 3 \cdot \frac{\pi}{6} = 4 \cos^3 \frac{\pi}{6} - 3 \cos \frac{\pi}{6},$$

we have $\cos \pi/6 = \sqrt{3}/2$. It follows, since $\sin \pi/6 > 0$, that $\sin \pi/6 = \sqrt{1 - (\sqrt{3}/2)^2} = \frac{1}{2}$.

8. (a)

$$\begin{aligned}\tan(x+y) &= \frac{\sin(x+y)}{\cos(x+y)} \\ &= \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y} \\ &= \frac{\frac{\sin x \cos y}{\cos x \cos y} + \frac{\cos x \sin y}{\cos x \cos y}}{\frac{\cos x \cos y}{\cos x \cos y} - \frac{\sin x \sin y}{\cos x \cos y}} \\ &= \frac{\tan x + \tan y}{1 - \tan x \tan y}\end{aligned}$$

(b) From part (a) we have

$$\begin{aligned}\tan(\arctan x + \arctan y) &= \frac{\tan(\arctan x) + \tan(\arctan y)}{1 - \tan(\arctan x)\tan(\arctan y)} \\ &= \frac{x + y}{1 - xy},\end{aligned}$$

provided that $\arctan x$, $\arctan y$, and $\arctan x + \arctan y \neq k\pi + \pi/2$. Since $-\pi/2 < \arctan x$, $\arctan y < \pi/2$, this is always the case except when $\arctan x + \arctan y = \pm\pi/2$, which is equivalent to $xy = 1$. From this equation we can conclude that

$$\arctan x + \arctan y = \arctan\left(\frac{x+y}{1-xy}\right)$$

provided that $\arctan x + \arctan y$ lies in $(-\pi/2, \pi/2)$, which is true whenever $xy < 1$. (If $x, y > 0$ and $xy > 1$, so that $\arctan x + \arctan y > \pi/2$, then we must add π to the right side, and if $x, y < 0$ and $xy > 1$, so that $\arctan x + \arctan y < -\pi/2$, then we must subtract π .)

10. The first formula is derived by subtracting the second of the following two equations from the first:

$$\begin{aligned}\cos(m-n)x &= \cos(mx - nx) = \cos mx \cos(-nx) - \sin mx \sin(-nx) \\ &= \cos mx \cos nx + \sin mx \sin nx, \\ \cos(m+n)x &= \cos mx \cos nx - \sin mx \sin nx.\end{aligned}$$

The other formulas are derived similarly.

11. It follows from Problem 10 that if $m \neq n$, then

$$\begin{aligned}\int_{-\pi}^{\pi} \sin mx \sin nx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} [\cos(m-n)x - \cos(m+n)x] \, dx \\ &= \frac{1}{2} \left\{ \left[\frac{\sin(m-n)\pi}{m-n} - \frac{\sin(m+n)\pi}{m+n} \right] \right. \\ &\quad \left. - \left[\frac{\sin(m-n)\pi}{m-n} - \frac{\sin(m+n)\pi}{m+n} \right] \right\} \\ &= 0.\end{aligned}$$

But if $m = n$, then

$$\begin{aligned}\int_{-\pi}^{\pi} \sin mx \sin nx \, dx &= \frac{1}{2} \int_{-\pi}^{\pi} 1 - \cos(m+n)x \, dx \\ &= \frac{1}{2} \{ [\pi - \cos(m+n)\pi] - [-\pi - \cos(m+n)\pi] \} \\ &= \pi.\end{aligned}$$

The other formulas are proved similarly.

14. (a) We have

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - 2 \sin^2 x \\ &= 2 \cos^2 x - 1.\end{aligned}$$

So

$$\begin{aligned}\sin^2 x &= \frac{1 - \cos 2x}{2}, \\ \cos^2 x &= \frac{1 + \cos 2x}{2}.\end{aligned}$$

(b) These formulas follow from part (a), because $\cos x/2 \geq 0$ and $\sin x/2 \geq 0$ (since $0 \leq x \leq \pi/2$).

(c)

$$\begin{aligned}\int_a^b \sin^2 x \, dx &= \int_a^b \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2} (b - a) - \frac{1}{4} (\sin 2b - \sin 2a). \\ \int_a^b \cos^2 x \, dx &= \int_a^b \frac{1 + \cos 2x}{2} \, dx = \frac{1}{2} (b - a) + \frac{1}{4} (\sin 2b - \sin 2a).\end{aligned}$$

18. (a) $\arctan 1 - \arctan 0 = \pi/4$.

(b) $\lim_{x \rightarrow \infty} \arctan x - \arctan 0 = \pi/2$.

19. $\lim_{x \rightarrow \infty} x \sin \frac{1}{x} = \lim_{x \rightarrow 0^+} \frac{1}{x} \sin x = 1$.

20. (a) $(\sin^\circ)'(x) = \frac{\pi}{180} \cos\left(\frac{\pi x}{180}\right) = \frac{\pi}{180} \cos^\circ(x)$.

$$(\cos^\circ)'(x) = \frac{\pi}{180} \cdot -\sin\left(\frac{\pi x}{180}\right) = \frac{-\pi}{180} \sin^\circ(x).$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin^\circ x}{x} = \lim_{x \rightarrow 0} \frac{\sin(\pi x/180)}{x} = \lim_{x \rightarrow 0} \frac{\pi}{180} \cdot \frac{\sin(\pi x/180)}{\pi x/180} = \frac{\pi}{180}$$

$$\lim_{x \rightarrow \infty} x \sin^\circ \frac{1}{x} = \lim_{x \rightarrow 0^+} \frac{\sin^\circ x}{x} = \frac{\pi}{180}$$

CHAPTER 17

$$1. (i) e^{e^{e^x}} \cdot e^{e^{e^x}} \cdot e^{e^x} \cdot e^x.$$

$$(iii) (\sin x)^{\sin(\sin x)} [(\log(\sin x)) \cdot \cos(\sin x) \cdot \cos x + (\cos x/\sin x) \cdot \sin(\sin x)].$$

$$(v) \sin x^{\sin x^{\sin x}} [(\log(\sin x)) \cdot \sin x^{\sin x} + \{(\log(\sin x)) \cdot \cos x + (\cos x/\sin x) \cdot \sin x\} + (\cos x/\sin x) \cdot \sin x^{\sin x}].$$

$$(vii) \left[\arcsin\left(\frac{x}{\sin x}\right) \right]^{\log(\sin e^x)} \left[\left(\log\left(\arcsin\left(\frac{x}{\sin x}\right)\right) \right) \cdot \frac{(\cos e^x)e^x}{\sin e^x} + \log(\sin e^x) \cdot \frac{\sin x - \cos x}{\arcsin\left(\frac{x}{\sin x}\right) \sqrt{1 - \left(\frac{x}{\sin x}\right)^2 \cdot \sin^2 x}} \right].$$

$$(ix) (\log x)^{\log x} \cdot \left[\log(\log x) \cdot \frac{1}{x} + \log x \cdot \frac{1}{\log x} \cdot \frac{1}{x} \right].$$

$$3. (i) 0.$$

$$(iii) \frac{1}{8}.$$

$$(v) \frac{1}{8}.$$

$$5. (a)$$

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \left[\frac{e^{2x}}{4} + \frac{1}{2} + \frac{e^{-2x}}{4} \right] - \left[\frac{e^{2x}}{4} - \frac{1}{2} + \frac{e^{-2x}}{4} \right] \\ &= 1. \end{aligned}$$

(c)

$$\begin{aligned} \sinh x \cosh y + \cosh x \sinh y &= \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) + \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^y - e^{-y}}{2} \right) \\ &= \left[\frac{e^{x+y}}{4} + \frac{e^{-x-y}}{4} - \frac{e^{-x+y}}{4} + \frac{e^{x-y}}{4} \right] + \left[\frac{e^{x+y}}{4} + \frac{e^{-x-y}}{4} + \frac{e^{-x+y}}{4} - \frac{e^{x-y}}{4} \right] \\ &= \frac{e^{x+y} + e^{-(x+y)}}{2} = \sinh(x+y). \end{aligned}$$

(e) Since

$$\sinh x = \frac{e^x + e^{-x}}{2},$$

we have

$$\sinh'(x) = \frac{e^x - e^{-x}}{2} = \cosh x.$$

(g) Since

$$\tanh x = \frac{\sinh x}{\cosh x},$$

we have

$$\begin{aligned}\tanh'(x) &= \frac{(\cosh x)^2 - (\sinh x)^2}{\cosh^2 x} \\ &= \frac{1}{\cosh^2 x} \quad \text{by part (a).}\end{aligned}$$

6. (a) If $y = \arg \cosh x$, then $x \geq 0$ and

$$x = \cosh y = \sqrt{1 + \sinh^2 y} \quad \text{by part (a).}$$

So

$$\sinh(\arg \cosh x) = \sinh y = \sqrt{x^2 - 1} \quad \text{since } \sinh y \geq 0 \text{ for } y \geq 0.$$

(c)

$$\begin{aligned}(\arg \sinh)'(x) &= \frac{1}{\sinh'(\arg \sinh(x))} \\ &= \frac{1}{\cosh(\arg \sinh(x))} \\ &= \frac{1}{\sqrt{1 + x^2}} \quad \text{by part (b).}\end{aligned}$$

$$\begin{aligned}\text{(e)} \quad (\arg \tanh)'(x) &= \frac{1}{\tanh'(\arg \tanh(x))} \\ &= \cosh^2(\arg \tanh(x)).\end{aligned}$$

Now,

$$\tanh^2 y + \frac{1}{\cosh^2 y} = 1 \quad \text{by Problem 5(b),}$$

so

$$\tanh^2(\arg \tanh(x)) + \frac{1}{\cosh^2(\arg \tanh(x))} = 1,$$

or

$$\cosh^2(\arg \tanh(x)) = \frac{1}{1 - x^2}.$$

7. (a) If $y = \arg \sinh x$, then

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

so

$$\begin{aligned}e^y - e^{-y} &= 2x, \\e^{2y} - 2xe^y - 1 &= 0, \\e^y &= \frac{2x \pm \sqrt{4x^2 + 4}}{2}\end{aligned}$$

so

$$e^y = x + \sqrt{1 + x^2} \quad \text{since } e^y > 0$$

or

$$y = \arg \sinh x = \log(x + \sqrt{1 + x^2}).$$

Similarly,

$$\begin{aligned}\arg \cosh x &= \log(x + \sqrt{x^2 - 1}), \\ \arg \tanh x &= \frac{1}{2} \log(1 + x) - \frac{1}{2} \log(1 - x).\end{aligned}$$

(b)

$$\begin{aligned}\int_a^b \frac{1}{\sqrt{1 + x^2}} dx &= \arg \sinh b - \arg \sinh a \quad \text{by Problem 6(c)} \\ &= \log(b + \sqrt{1 + b^2}) - \log(a + \sqrt{1 + a^2}). \\ \int_a^b \frac{1}{\sqrt{x^2 - 1}} dx &= \log(b + \sqrt{b^2 - 1}) - \log(a + \sqrt{a^2 - 1}). \\ \int_a^b \frac{1}{1 - x^2} dx &= \frac{1}{2} [\log(1 + b) - \log(1 - b) - \log(1 + a) + \log(1 - a)].\end{aligned}$$

8. (a) $\lim_{x \rightarrow \infty} a^x = \lim_{x \rightarrow \infty} e^{x \log a}$. Since $\log a < 0$, we have $\lim_{x \rightarrow \infty} x \log a = -\infty$, so $\lim_{x \rightarrow \infty} e^{x \log a} = 0$.

(c) $\lim_{x \rightarrow \infty} \frac{(\log x)^n}{x} = \lim_{y \rightarrow \infty} \frac{y^n}{e^y} = 0$.

(e) $\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \log x}$. Now, $\lim_{x \rightarrow 0^+} x \log x = 0$ by part (d), so $\lim_{x \rightarrow 0^+} x^x = 1$.

12. (a) $\lim_{y \rightarrow 0} \log(1 + y)/y = \log'(1) = 1$.

(b) $\lim_{x \rightarrow \infty} x \log(1 + 1/x) = \lim_{y \rightarrow 0^+} \log(1 + y)/y = 1$.

(c)

$$\begin{aligned}e &= \exp(1) = \exp(\lim_{x \rightarrow \infty} x \log(1 + 1/x)) \\ (*) &= \lim_{x \rightarrow \infty} \exp(x \log(1 + 1/x)) \\ &= \lim_{x \rightarrow \infty} (1 + 1/x)^x.\end{aligned}$$

(The starred equality depends on the continuity of $\exp$ at 1, and can be justified as follows. For every $\varepsilon > 0$ there is some $\delta > 0$ such that $|e - \exp y| < \varepsilon$ for $|y - 1| < \delta$. Moreover, there is some N such that $|x \log(1 + 1/x) - 1| < \delta$ for $x > N$. So $|e - x \log(1 + 1/x)| < \varepsilon$ for $x > N$.

(d)

$$\begin{aligned}
 e^a &= [\lim_{x \rightarrow \infty} (1 + 1/x)^x]^a = \lim_{x \rightarrow \infty} (1 + 1/x)^{ax} \\
 &= \lim_{ax \rightarrow \infty} (1 + 1/x)^{ax} \\
 &= \lim_{y \rightarrow \infty} (1 + a/y)^y.
 \end{aligned}$$

14. After one year the number of dollars yielded by an initial investment of one dollar will be

$$\lim_{x \rightarrow \infty} (1 + a/100x)^x = e^{a/100}.$$

15. Clearly $f'(x) = 1/x$ for $x > 0$. If $x < 0$, then $f(x) = \log(-x)$, so $f'(x) = (-1) \cdot 1/(-x) = 1/x$.
16. Let $g(x) = f(x)/e^{cx}$. Then

$$\begin{aligned}
 g'(x) &= \frac{e^{cx}f'(x) - f(x)ce^{cx}}{e^{2cx}} \\
 &= 0,
 \end{aligned}$$

so there is some number k such that $g(x) = k$ for all x .

17. (a) According to Problem 16, there is some k such that $A(t) = ke^{ct}$. Then $k = ke^{0t} = A_0$. So $A(t) = A_0e^{ct}$.
- (b) If $A(t + \tau) = A(t)/2$, then

$$A_0e^{ct+c\tau} = \frac{A_0e^{ct}}{2},$$

so $e^{ct}e^{c\tau} = e^{ct}/2$ or $e^{c\tau} = \frac{1}{2}$, so $\tau = -(\log 2)/c$. It is easy to check that this τ does work.

18. Newton's law states that, for a certain (positive) number c ,

$$T'(t) = c(T - M),$$

which can be written

$$(T - M)' = c(T - M).$$

So by Problem 16 there is some number k such that

$$T(t) - M = ke^{ct},$$

and $k = ke^{0t} = T(0) - M = T_0 - M$. So $T(t) = M + (T_0 - M)e^{ct}$.

CHAPTER 18

1. (i) $(\sqrt[5]{x^3} + \sqrt[6]{x}/\sqrt{x} = x^{1/10} + x^{-1/3}).$
- (ii) $\frac{1}{\sqrt{x-1} + \sqrt{x+1}} = \frac{\sqrt{x-1} - \sqrt{x+1}}{-2}.$
- (iii) $(e^x + e^{2x} + e^{3x})/e^{4x} = e^{-3x} + e^{-2x} + e^{-x}.$
- (iv) $a^x/b^x = (a/b)^x = e^{x \log(a/b)}.$

$$(v) \quad \tan^2 x = \sec^2 x - 1.$$

$$(vi) \quad \frac{1}{a^2 + x^2} = \frac{1/a^2}{1 + \left(\frac{x}{a}\right)^2}.$$

$$(vii) \quad \frac{1}{\sqrt{a^2 - x^2}} = \frac{1/a}{\sqrt{1 - (x/a)^2}}.$$

$$(viii) \quad \frac{1}{1 + \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x.$$

$$(ix) \quad \frac{8x^2 + 6x + 4}{x + 1} = 8x - 2 + \frac{6}{x + 1}.$$

$$(x) \quad \frac{1}{\sqrt{2x - x^2}} = \frac{1}{\sqrt{1 - (x - 1)^2}}.$$

$$2. (i) \quad -\cos e^x. \text{ (Let } u = e^x \text{.)}$$

$$(iii) \quad (\log x)^2/2. \text{ (Let } u = \log x \text{.)}$$

$$(v) \quad e^{e^x}. \text{ (Let } u = e^{e^x} \text{.)}$$

$$(vii) \quad 2e\sqrt{x}. \text{ (Let } u = \sqrt{x} \text{.)}$$

$$(ix) \quad -(\log(\cos x))^2/2. \text{ (Let } u = \log(\cos x) \text{.)}$$

$$3. (i) \quad \int x^2 e^x dx = x^2 e^x - \int 2x e^x dx = x^2 e^x - [2x e^x - \int e^x dx] \\ = x^2 e^x - 2x e^x + 2e^x.$$

$$(iii) \quad \text{We have}$$

$$\begin{aligned} \int e^{ax} \sin bx dx &= \frac{e^{ax} \sin bx}{a} - \frac{b}{a} \int e^{ax} \cos bx dx \\ &= \frac{e^{ax} \sin bx}{a} - \frac{b}{a} \left[\frac{e^{ax} \cos bx}{a} - \frac{b}{a} \int e^{ax} (-\sin bx) dx \right], \end{aligned}$$

so

$$\int e^{ax} \sin bx dx = \frac{a}{a^2 + b^2} e^{ax} \sin bx - \frac{b}{a^2 + b^2} e^{ax} \cos bx.$$

$$(v) \quad \text{Using the result } \int (\log x)^2 dx = x(\log x)^2 - 2x(\log x) + 2x \text{ from the text, we have}$$

$$\begin{aligned} \int (\log x)^3 dx &= [x(\log x)^2 - 2x(\log x) + 2x] \log x \\ &\quad - \int \frac{1}{x} [x(\log x)^2 - 2x(\log x) + 2x] dx \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x \log x \\ &\quad - \int (\log x)^2 dx + 2[x \log x - x] - 2x \\ &= x(\log x)^3 - 2x(\log x)^2 + 2x \log x \\ &\quad - [x(\log x)^2 - 2x(\log x) + 2x] + 2[x \log x - x] - 2x \\ &= x(\log x)^3 - 3x(\log x)^2 + 6x \log x - 6x. \end{aligned}$$

$$\begin{aligned}
 \text{(vii)} \quad \int \sec^3 x \, dx &= \int (\sec^2 x)(\sec x) \, dx = \tan x \sec x \\
 &\quad - \int (\tan x)(\sec x \tan x) \, dx \\
 &= \tan x \sec x - \int \sec x (\sec^2 x - 1) \, dx \\
 &= \tan x \sec x - \int \sec^3 x \, dx + \int \sec x \, dx,
 \end{aligned}$$

so

$$\int \sec^3 x \, dx = \frac{1}{2}[\tan x \sec x + \log(\sec x + \tan x)].$$

$$\begin{aligned}
 \text{(ix)} \quad \int \sqrt{x} \log x \, dx &= \frac{2x^{3/2}}{3} \log x - \frac{2}{3} \int x^{3/2} \cdot \frac{1}{x} \, dx \\
 &= \frac{2x^{3/2}}{3} \log x - \frac{2}{3} \int x^{1/2} \, dx \\
 &= \frac{2x^{3/2}}{3} \log x - \frac{4}{9} x^{3/2}.
 \end{aligned}$$

4. (i) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\int \frac{\cos u \, du}{\sqrt{1 - \sin^2 u}} = \int 1 \, du = u = \arcsin x.$$

- (iii) Let $x = \sec u$, $dx = \sec u \tan u \, du$. The integral becomes

$$\begin{aligned}
 \int \frac{\sec u \tan u \, du}{\sqrt{\sec^2 u - 1}} &= \int \sec u \, du = \log(\sec u + \tan u) \\
 &= \log(x + \sqrt{x^2 - 1}).
 \end{aligned}$$

- (v) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\begin{aligned}
 \int \frac{\cos u \, du}{\sin u \sqrt{1 - \sin^2 u}} &= \int \csc u \, du = -\log(\csc u + \cot u) \\
 &= -\log\left(\frac{1}{x} + \frac{\sqrt{1 - x^2}}{x}\right).
 \end{aligned}$$

- (vii) Let $x = \sin u$, $dx = \cos u \, du$. The integral becomes

$$\begin{aligned}
 \int (\sin^3 u \cos u) \cos u \, du &= \int \sin^3 u \cos^2 u \, du = \int (\sin u)(1 - \cos^2 u) \cos^2 u \, du \\
 &= \int (\sin u)(\cos^2 u - \cos^4 u) \, du = -\frac{\cos^3 u}{3} + \frac{\cos^5 u}{5} \\
 &= -\frac{(1 - x^2)^{3/2}}{3} + \frac{(1 - x^2)^{5/2}}{5}.
 \end{aligned}$$

- (ix) Let $x = \tan u$, $dx = \sec^2 u \, du$. The integral becomes

$$\begin{aligned}
 \int \sec u \sec^2 u \, du &= \int \sec^3 u \, du \\
 &= \frac{1}{2}[\tan u \sec u + \log(\sec u + \tan u)] \quad \text{by Problem 3(vii)} \\
 &= \frac{1}{2}[x \sqrt{1 + x^2} + \log(x + \sqrt{1 + x^2})].
 \end{aligned}$$

5. (i) Let $u = \sqrt{x+1}$, $x = u^2 - 1$, $dx = 2u du$. The integral becomes

$$\begin{aligned}\int \frac{2u du}{1+u} &= \int \left(2 + \frac{-2}{1+u}\right) du \\ &= 2u - 2 \log(1+u) = 2\sqrt{x+1} - 2 \log(1+\sqrt{x+1}).\end{aligned}$$

- (iii) Let $u = x^{1/6}$, $x = u^6$, $dx = 6u^5 du$. The integral becomes

$$\begin{aligned}\int \frac{6u^5 du}{u^3 + u^2} &= 6 \int \left(u^2 - u + 1 - \frac{1}{u+1}\right) du = 2u^3 - 3u^2 + 6u - 6 \log(u+1) \\ &= 2\sqrt[6]{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6 \log(\sqrt[6]{x} + 1).\end{aligned}$$

- (v) Let $u = \tan x$, $x = \arctan u$, $dx = du/(1+u^2)$. The integral becomes

$$\begin{aligned}\int \frac{du}{(1+u^2)(2+u)} &= \frac{1}{5} \int \left(\frac{1}{2+u} - \frac{u-2}{1+u^2}\right) du \\ &= \frac{1}{5} \int \frac{du}{2+u} - \frac{1}{10} \int \frac{2u}{1+u^2} du + \frac{2}{5} \int \frac{du}{1+u^2} \\ &= \frac{1}{5} \log(2+u) - \frac{1}{10} \log(1+u^2) + \frac{2}{5} \arctan u \\ &= \frac{1}{5} \log(2+\tan x) - \frac{1}{10} \log(1+\tan^2 x) + \frac{2}{5} x.\end{aligned}$$

- (vii) Let $u = 2^x$, $x = (\log u)/(\log 2)$, $dx = du/(u \log 2)$. The integral becomes

$$\begin{aligned}\frac{1}{\log 2} \int \frac{u^2 + 1}{(u+1)u} du &= \frac{1}{\log 2} \int \left(1 + \frac{1-u}{u(u+1)}\right) du \\ &= \frac{1}{\log 2} \int \left(1 + \frac{1}{u} - \frac{2}{u+1}\right) du \\ &= \frac{1}{\log 2} [u + \log u - 2 \log(u+1)] \\ &= \frac{1}{\log 2} [2^x + x \log 2 - 2 \log(2^x + 1)].\end{aligned}$$

- (ix) Let $u = \sqrt{x}$, $x = u^2$, $dx = 2u du$. The integral becomes

$$\int \frac{\sqrt{1-u^2} 2u du}{1-u}.$$

Now let $u = \sin y$, $du = \cos y \, dy$. The integral becomes

$$\begin{aligned}
 \int \frac{2 \cos y \sin y \cos y}{1 - \sin y} dy &= 2 \int \frac{(1 - \sin^2 y) \sin y}{1 - \sin y} dy \\
 &= 2 \int (1 + \sin y) \sin y \, dy \\
 &= 2 \int \sin y \, dy + \int 1 - \cos 2y \, dy \\
 &= -2 \cos y + y - \frac{\sin 2y}{2} = -2 \cos y + y - \sin y \cos y \\
 &= -2 \sqrt{1 - u^2} + \arcsin u - u \sqrt{1 - u^2} \\
 &= -2 \sqrt{1 - x} + \arcsin \sqrt{x} - \sqrt{x} \sqrt{1 - x}.
 \end{aligned}$$

The substitution $u = \sqrt{1 - x}$, $x = 1 - u^2$, $dx = -2u \, du$ leads to

$$\int \frac{-2u^2 \, du}{1 - \sqrt{1 - u^2}},$$

and the substitution $u = \sin y$ then leads to

$$\begin{aligned}
 \int \frac{-2 \sin^2 y \cos y \, dy}{1 - \cos y} &= -2 \sin y - y - \sin y \cos y \\
 &= -2u - \arcsin u - u \sqrt{1 - u^2} \\
 &= -2 \sqrt{1 - x} - \arcsin \sqrt{1 - x} - \sqrt{1 - x} \sqrt{x}.
 \end{aligned}$$

These answers agree, since

$$\arcsin \sqrt{x} = \frac{\pi}{2} - \arcsin \sqrt{1 - x}$$

(check this by comparing their derivatives and their values for $x = 0$).

6. In these problems I will denote the original integral.

(i)

$$\begin{aligned}
 I &= \int \frac{2}{x-1} dx + \int \frac{3}{(x+1)^2} dx \\
 &= 2 \log(x-1) - \frac{3}{x+1}.
 \end{aligned}$$

(iii)

$$\begin{aligned}
 I &= \int \frac{1}{(x-1)^2} dx + \int \frac{4}{(x+1)^3} dx \\
 &= -\frac{1}{(x-1)} - \frac{2}{(x+1)^2}.
 \end{aligned}$$

(v)

$$\begin{aligned}
 I &= \frac{1}{2} \int \frac{2x}{x^2 + 1} dx + \int \frac{4}{x^2 + 1} dx \\
 &= \frac{1}{2} \log(x^2 + 1) + 4 \arctan x.
 \end{aligned}$$

(vii)

$$\begin{aligned}
 I &= \int \frac{1}{(x+1)} dx + \int \frac{2x}{(x^2+x+1)} dx \\
 &= \int \frac{1}{x+1} dx + \int \frac{2x+1}{x^2+x+1} dx - \int \frac{1}{x^2+x+1} dx.
 \end{aligned}$$

Now

$$\begin{aligned}
 \int \frac{1}{x^2+x+1} dx &= \int \frac{1}{(x+\frac{1}{2})^2 + \frac{3}{4}} dx \\
 &= \frac{4}{3} \int \frac{1}{\left[\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right]^2 + 1} dx \\
 &= \frac{4}{3} \cdot \frac{\sqrt{3}}{2} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right) \\
 &= \frac{2\sqrt{3}}{3} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right),
 \end{aligned}$$

so

$$I = \log(x+1) + \log(x^2+x+1) - \frac{2\sqrt{3}}{3} \arctan\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right).$$

(ix)

$$\begin{aligned}
 I &= \int \frac{2x+1}{(x^2+x+1)^2} dx - \int \frac{1}{(x^2+x+1)^2} dx \\
 &= \int \frac{2x+1}{(x^2+x+1)^2} dx - \frac{16}{9} \int \frac{1}{\left(\left[\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right]^2 + 1\right)^2} dx.
 \end{aligned}$$

Now the substitution

$$u = \frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right), \quad dx = \frac{\sqrt{3}}{2} du$$

changes the second integral to

$$-\frac{16}{9} \cdot \frac{\sqrt{3}}{2} \int \frac{du}{(u^2+1)^2}.$$

Using the reduction formula, this can be written

$$-\frac{8\sqrt{3}}{9} \left[\frac{u}{2(u^2+1)} + \frac{1}{2} \int \frac{du}{u^2+1} \right] = -\frac{8\sqrt{3}}{9} \left[\frac{\log(u^2+1)}{4} + \frac{1}{2} \arctan u \right],$$

so

$$I = -\frac{1}{x^2+x+1} - \frac{2\sqrt{3}}{9} \log \left(\frac{4}{3} (x^2+x+1) \right) - \frac{4\sqrt{3}}{9} \arctan \left(\frac{2}{\sqrt{3}} \left(x + \frac{1}{2} \right) \right).$$

11. The equation $\int e^x \sin x \, dx = e^x \sin x - e^x \cos x - \int e^x \sin x \, dx$ means that any function F with $F'(x) = e^x \sin x$ can be written $F(x) = e^x \sin x - e^x \cos x - G(x)$ where G is another function with $G'(x) = e^x \sin x$. Of course, $G = F + c$ for some number c , but it is not necessarily true that $F = G$.

12. (a)

$$\begin{aligned} \int \arcsin x \, dx &= \int 1 \cdot \arcsin x \, dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \, dx \\ &= x \arcsin x + \sqrt{1-x^2}. \end{aligned}$$

13. (a)

$$\begin{aligned} \int \sin^4 x \, dx &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx \\ &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cos x}{2} + \frac{1}{2} \int 1 \, dx \right] \\ &= -\frac{\sin^3 x \cos x}{4} - \frac{3 \sin x \cos x}{8} + \frac{3}{8} x. \\ \int \sin^4 x \, dx &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \int \left(\frac{1}{4} - \frac{\cos 2x}{2} + \frac{\cos^2 2x}{4} \right) dx \\ &= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \int \frac{1 + \cos 4x}{2} dx \\ &= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \left[\frac{x}{2} + \frac{\sin 4x}{8} \right] \\ &= \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32}. \end{aligned}$$

- (b) It follows that these two answers are the same, since they have the same value for $x = 0$.

17. (a)

$$\begin{aligned}\sin^n x \, dx &= \int (\sin x)(\sin^{n-1} x) \, dx \\ &= -\cos x \sin^{n-1} x + (n-1) \int (\cos x)(\sin^{n-2} x)(\cos x) \, dx \\ &= -\cos x \sin^{n-1} x + (n-1) \int (\sin^{n-2} x - \sin^n x) \, dx,\end{aligned}$$

so

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

(b)

$$\begin{aligned}\int \cos^n x \, dx &= \int (\cos x)(\cos^{n-1} x) \, dx \\ &= \sin x \cos^{n-1} x + (n-1) \int \sin x (\cos^{n-2} x) \sin x \, dx \\ &= \sin x \cos^{n-1} x + (n-1) \int (\cos^{n-2} x - \cos^n x) \, dx,\end{aligned}$$

so

$$\int \cos^n x \, dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

(c)

$$\begin{aligned}\int \frac{dx}{(1+x^2)^n} &= \int \frac{dx}{(1+x^2)^{n-1}} - \int \frac{x^2 \, dx}{(1+x^2)^n} \\ &= \int \frac{dx}{(1+x^2)^{n-1}} - \int x \cdot \frac{x}{(1+x^2)^n} \, dx \\ &= \int \frac{dx}{(1+x^2)^{n-1}} - \left[\frac{x}{2(1-n)(1+x^2)^{n-1}} \right. \\ &\quad \left. - \int \frac{dx}{2(1-n)(1+x^2)^{n-1}} \right]\end{aligned}$$

so

$$\int \frac{dx}{(1+x^2)^n} = \frac{1}{2(n-1)} \frac{x}{(x^2+1)^{n-1}} - \frac{(2n-3)}{2(n-1)} \int \frac{1}{(x^2+1)^{n-1}} \, dx.$$

We can also use the substitution $x = \tan u$, $dx = \sec^2 u \, du$, which changes the integral to

$$\begin{aligned}\int \frac{\sec^2 u \, du}{\sec^{2n} u} &= \int \cos^{2n-2} u \, du \\ &= \frac{1}{2n-2} \cos^{2n-3} u \sin u + \frac{2n-3}{2n-2} \int \cos^{2n-4} u \, du \\ &= \frac{1}{2n-2} \cdot \frac{1}{(\sqrt{1+x^2})^{2n-3}} \cdot \frac{x}{\sqrt{1+x^2}} + \frac{2n-3}{2n-2} \int \frac{dx}{(1+x^2)^{n-1}} \\ &= \frac{1}{2(n-1)} \frac{x}{(1+x^2)^{n-1}} + \frac{2n-3}{2n-2} \int \frac{dx}{(1+x^2)^{n-1}}.\end{aligned}$$

CHAPTER 19

1. (i) $P_{3,0}(x) = e + ex + ex^2 + (5e/3!)x^3$.
- (iii) $P_{2n,\pi/2}(x) = 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \cdots + \frac{(-1)^n(x - \pi/2)^{2n}}{(2n)!}$.
- (v) $P_{n,1}(x) = e + e(x - 1) + \frac{e(x - 1)^2}{2!} + \cdots + \frac{e(x - 1)^n}{n!}$.
- (vii) $P_{4,0}(x) = x + x^3$.
- (ix) $P_{2n+1,0}(x) = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$.
2. If f is a polynomial function of degree n , then $f^{(n+1)} = 0$. It follows from Taylor's Theorem that $R_{n,a}(x) = 0$, so $f(x) = P_{n,a}(x)$.
 - (i) $-12 + 2(x - 3) + (x - 3)^2$.
 - (iii) $243 + 405(x - 3) + 270(x - 3)^2 + 90(x - 3)^3 + 15(x - 3)^4 + (x - 3)^5$.
3. (i) $\sum_{i=0}^9 \frac{(-1)^i}{(2i+1)!} \left(\text{since } \frac{1}{(2n+2)!} < 10^{-17} \text{ for } 2n+2 \geq 19, \text{ or } n \geq 9 \right)$.
- (iii) $\sum_{i=0}^8 \frac{(-1)^i}{2^i(2i+1)!} \left(\text{since } \frac{1}{2^{2n+2}(2n+2)!} < 10^{-20} \text{ for } 2n+2 \geq 18, \right.$
 $\left. \text{or } n \geq 8 \right)$.
- (v) $\sum_{i=0}^{11} \frac{2^i}{i!} \left(\text{since } \frac{3^2 \cdot 2^{n+1}}{(n+1)!} < 10^{-5} \text{ for } n+1 \geq 12, \text{ or } n \geq 11 \right)$.
7. (i) $c_i = a_i + b_i$.
- (iii) $c_i = (i+1)a_i$.
- (v) $c_0 = \int_0^a f(t) dt$; $c_i = a_{i-1}/i$ for $i > 0$.

CHAPTER 21

1. (i) $1 - n/(n+1) = 1/(n+1) < \varepsilon$ for $n+1 > 1/\varepsilon$.
- (iii) $\lim_{n \rightarrow \infty} \sqrt[n]{n^2 + 1} - \sqrt[n]{n+1} = \lim_{n \rightarrow \infty} (\sqrt[n]{n^2 + 1} - \sqrt[n]{n^2}) + \lim_{n \rightarrow \infty} (\sqrt[n]{n} - \sqrt[n]{n+1}) = 0 + 0 = 0$. (Each of these two limits can be proved in the same way that $\lim_{n \rightarrow \infty} (\sqrt[n]{n+1} - \sqrt[n]{n}) = 0$ was proved in the text.)
- (v) Clearly $\lim_{n \rightarrow \infty} (\log a)/n = 0$. So $\lim_{n \rightarrow \infty} \sqrt[n]{a} = \lim_{n \rightarrow \infty} e^{(\log a)/n} = e^0$ (by Theorem 1) = 1.
- (vii) $\sqrt[n]{n^2} \leq \sqrt[n]{n^2 + n} \leq \sqrt[n]{2n^2}$, so $(\sqrt[n]{n})^2 \leq \sqrt[n]{n^2 + n} \leq \sqrt[n]{2} (\sqrt[n]{n})^2$, and $\lim_{n \rightarrow \infty} (\sqrt[n]{n})^2 = \lim_{n \rightarrow \infty} \sqrt[n]{2} (\sqrt[n]{n})^2 = 1$ by parts (v) and (vi).
- (ix) Clearly $\alpha(n) \leq \log_2 n$, and $\lim_{n \rightarrow \infty} (\log_2 n)/n = 0$.

4. (a) If $0 < a < 2$, then $a^2 < 2a < 4$, so $a < \sqrt{2a} < 2$.
 (b) Part (a) shows that

$$\sqrt{2} < \sqrt{2\sqrt{2}} < \sqrt{2\sqrt{2\sqrt{2}}} < \cdots < 2,$$

so the sequence converges by Theorem 2.

- (c) If this sequence is denoted by $\{a_n\}$, then the sequence $\{\sqrt{2a_n}\}$ is the same as $\{a_{n+1}\}$. So the hint shows that $l = \sqrt{2l}$, or $l = 2$.
 5. If x is rational, then $n!\pi x$ is a multiple of π for sufficiently large n , so $(\cos n!\pi x)^{2k} = 1$ for all such n , so $\lim_{n \rightarrow \infty} (\lim_{k \rightarrow \infty} (\cos n!\pi x)^{2k}) = 1$. If x is irrational, then $n!\pi x$ is not a multiple of π for any n , so $|\cos n!\pi x| < 1$, so $\lim_{k \rightarrow \infty} (\cos n!\pi x)^{2k} = 0$, so $f(x) = 0$.
 6. (i) $\int_0^1 e^x dx = e - 1$. (Use partitions of $[0, 1]$ into n equal parts.)
 (iii) $\int_0^1 \frac{1}{1+x} dx = \log 2$.
 (v) $\int_0^1 \frac{1}{(1+x)^2} dx = \frac{1}{2}$.

CHAPTER 22

1. (i) (Absolutely) convergent, since $|(\sin n\theta)/n^2| \leq 1/n^2$.
 (iii) Divergent, since the first $2n$ terms have sum $\frac{1}{2} + \cdots + 1/n$. (Leibnitz's Theorem does not apply since the terms are not decreasing in absolute value.)
 (v) Convergent, since

$$\frac{1}{\sqrt[3]{n^2 - 1}} \leq \frac{2}{n^{2/3}}$$

for sufficiently large n .

- (vii) Convergent, since

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2/(n+1)!}{n^2/n!} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^2 \cdot \frac{1}{n+1} = 0.$$

- (ix) Divergent, since $1/(\log n) > 1/n$.
 (xi) Convergent, since $1/(\log n)^n < \frac{1}{2^n}$ for $n > 9$.
 (xiii) Divergent, since

$$\frac{n^2}{n^3 + 1} > \frac{1}{2n}$$

for large enough n .

- (xv) Divergent, since

$$\int_3^N \frac{1}{x \log x} dx = \log(\log N) - \log(\log 3) \rightarrow \infty \text{ as } N \rightarrow \infty.$$

(Notice that $f(x) = 1/(x \log x)$ is decreasing on $[3, \infty)$, since

$$f'(x) = \frac{-[1 + \log x]}{(x \log x)^2} < 0 \quad \text{for } x > e.$$

(xvii) Convergent, since $1/n^2(\log n) < 1/n^2$ for $n > 2$.

(xix) Convergent, since

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2^{n+1}(n+1)!/(n+1)^{n+1}}{2^n n!/n^n} &= \lim_{n \rightarrow \infty} \frac{2(n+1)n^n}{(n+1)^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{2}{\left(1 + \frac{1}{n}\right)^n} = \frac{2}{e}, \end{aligned}$$

by Problem 17-12.

4. (a) For each N we clearly have

$$0 \leq \sum_{n=1}^N a_n 10^{-n} < 9 \sum_{n=1}^{\infty} 10^{-n} = 1,$$

so $\sum_{n=1}^{\infty} a_n 10^{-n}$ converges by the boundedness criterion, and lies between 0 and 1. (Actually, this number is denoted by $0.a_1 a_2 a_3 a_4 \dots$ only when the sequence $\{a_n\}$ is not eventually 0.)

15. The area of the shaded region is $\frac{1}{2}$. The integral is

$$\begin{aligned} &\frac{1}{2} \left(\left[1 - \frac{1}{2} \right] + \left[\frac{1}{4} - \frac{1}{8} \right] + \left[\frac{1}{16} - \frac{1}{32} \right] + \cdots \right) - \frac{1}{2} \left(\left[\frac{1}{2} - \frac{1}{4} \right] + \left[\frac{1}{8} - \frac{1}{16} \right] + \cdots \right) \\ &= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} + \cdots \right) - \frac{1}{2} \left(\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots \right) \\ &= \frac{1}{4} \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \right) - \frac{1}{8} \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \right) \\ &= \frac{1}{8} \left(1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \cdots \right) \\ &= \frac{1}{8} \cdot \frac{1}{1 - \frac{1}{4}} \\ &= \frac{1}{6}. \end{aligned}$$

CHAPTER 23

1. (i)

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & x = 0 \\ 1, & 0 < x \leq 1. \end{cases}$$

$\{f_n\}$ does not converge uniformly to f .

(iii) $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$ (since $\lim_{n \rightarrow \infty} x^n = 0$ for $x < 1$). The sequence $\{f_n\}$ does not converge uniformly to f ; in fact, for any n we have $f_n(x)$ large for sufficiently large x .

- (v) $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$, and $\{f_n\}$ converges uniformly to f , since $|f_n(x)| \leq 1/n$ for all x .
2. (i) $-\frac{1}{a} - \frac{x}{a^2} - \frac{x^2}{a^3} - \dots$
- (iii) $\sum_{k=0}^{\infty} (-1)^k \binom{-\frac{1}{2}}{k} x^k$
- (v) $\sum_{k=0}^{\infty} \frac{(-1)^k \binom{-\frac{1}{2}}{k}}{2k+1} x^{2k+1}$
3. (i) e^{-x}
- (iii) If

$$f(x) = \frac{x^2}{2} - \frac{x^3}{3 \cdot 2} + \frac{x^4}{4 \cdot 3} - \dots, \quad |x| \leq 1$$

then

$$\begin{aligned} f'(x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \\ &= \log(1+x) \quad |x| < 1, \end{aligned}$$

so for $|x| < 1$ we have $f(x) = (1+x)\log(1+x) - (1+x) + c$ for some number c . Since $f(0) = 0$, we have $c = 1$, so $f(x) = (1+x)\log(1+x) - x$ for $|x| < 1$.

4. Since

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

we have

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}$$

(notice that the right side is 1 for $x = 0$). So

$$f^{(k)}(0) = \begin{cases} \frac{(-1)^n}{(2n+1)!}, & k = 2n \\ 0, & k \text{ odd.} \end{cases}$$

CHAPTER 24

1. (i) $|3 + 4i| = 5$; $\theta = \arctan \frac{4}{3}$.
- (iii) $|(1+i)^5| = (|1+i|)^5 = 1$; since $\pi/4 = \arctan 1/1$ is an argument for $1+i$, an argument for $(1+i)^5$ is $5\pi/4$.
- (v) $|(|3+4i|)| = |5| = 5$; $\theta = 0$.

2. (i)

$$\begin{aligned}
 x &= \frac{-i \pm \sqrt{-1-4}}{2} \\
 &= \frac{-i \pm \sqrt{5}i}{2} \\
 &= \frac{(-1 + \sqrt{5})i}{2} \text{ or } \frac{(-1 - \sqrt{5})i}{2}.
 \end{aligned}$$

(iii) $x^2 + 2ix - 1 = (x + i)^2$, so the only solution is $x = -i$.(v) $x^3 - x^2 - x - 2 = (x - 2)(x^2 + x + 1)$. The solutions are

$$2, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

3. (i) All $z = iy$ with y real.(iii) All z on the perpendicular bisector of the line segment between a and b .(v) No z , since $|x + iy| < 1 - x$ implies that $x \geq 1$, and that $x^2 + y^2 < 1 - 2x + x^2$, so $y^2 < 1 - 2x$, which is impossible, since $1 - 2x < 0$ for $x \geq 1$.

4. $|x + iy| = x^2 + y^2 = x^2 + (-y)^2 = |x - iy|$.

$$(z + \bar{z})/2 = [(x + iy) + (x - iy)]/2 = x.$$

$$(z - \bar{z})/2i = [(x + iy) - (x - iy)]/2i = y.$$

5. $|z + w|^2 + |z - w|^2 = (z + w)(\bar{z} + \bar{w}) + (z - w)(\bar{z} - \bar{w}) = 2z\bar{z} + 2w\bar{w} = 2(|z|^2 + |w|^2)$. Geometrically, this says that the sum of the squares of the diagonals of a parallelogram equal the sum of the squares of the sides.**CHAPTER 26**1. (i) Converges absolutely, since $|(1 + i)^n/n!| \leq (\sqrt{2})^n/n!$, and

$$\sum_{n=1}^{\infty} (\sqrt{2})^n/n! \text{ converges.}$$

(iii) Converges, but not absolutely, since the real terms form the series

$$-\frac{1}{2} + \frac{1}{4} - \frac{1}{6} + \frac{1}{8} - \cdots$$

and the imaginary terms form the series

$$i(\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots).$$

(v) Diverges, since the real terms form the series

$$\frac{\log 3}{3} + 2\frac{\log 4}{4} + \frac{\log 5}{5} + \frac{\log 7}{7} + 2\frac{\log 8}{8} + \frac{\log 9}{9} + \cdots$$

2. (i) The limit

$$\lim_{n \rightarrow \infty} \frac{|z|^{n+1}/(n+1)^2}{|z|^n/n^2} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^2 |z| = |z|$$

is < 1 for $|z| < 1$, but > 1 for $|z| > 1$.

- (iii) The limit

$$\lim_{n \rightarrow \infty} \frac{|z|^{n+1}}{|z|^n} = |z|$$

is < 1 for $|z| < 1$ but > 1 for $|z| > 1$.

- (v) The limit

$$\lim_{n \rightarrow \infty} \frac{2^{n+1}|z|^{(n+1)!}}{2^n|z|^{n!}} = \lim_{n \rightarrow \infty} 2|z|^{(n+1)!-n!}$$

is 0 for $|z| < 1$, but ∞ for $|z| > 1$.

3. (i) The limits

$$\lim_{n \rightarrow \infty} \sqrt[2n]{\frac{|z|^{2n}}{3^n}} = \frac{|z|}{\sqrt{3}} \quad \text{and} \quad \lim_{n \rightarrow \infty} \sqrt[2n+1]{\frac{|z|^{2n+1}}{2^{n+1}}} = \frac{|z|}{\sqrt{2}}$$

are < 1 for $|z| < \sqrt{2}$, so the series converges absolutely for $|z| < \sqrt{2}$. But the series does not converge absolutely for $|z| > \sqrt{2}$, so the radius of convergence is $\sqrt{2}$.

- (iii) Since

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n!|z|^n}{n^n}} = \lim_{n \rightarrow \infty} \frac{|z|\sqrt[n]{n!}}{n} \leq \lim_{n \rightarrow \infty} \frac{|z|}{n} = 0,$$

the series converges absolutely for all z , so the radius of convergence is ∞ .

- (v) The limit

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^n z^{n!}} = 2 \lim_{n \rightarrow \infty} z^{(n-1)!}$$

is 0 for $|z| < 1$, but ∞ for $|z| > 1$, so the radius of convergence is 1.

GLOSSARY OF SYMBOLS

P	9	$(-\infty, a)$	55	$L \int_a^b f$	249
$ a $	11	$(-\infty, a]$	55	$U \int_a^b f$	249
$\sqrt{x}$	12	$(-\infty, \infty)$	55	$\int_a^\infty f$	254
$\max(x, y)$	16	$[x]$	70	$\int_a^\infty f(x) dx$	254
$\min(x, y)$	16	$\{x\}$	70	$\int_{-\infty}^a f$	254
ε ("epsilon")	19	δ ("delta")	78	$\int_{-\infty}^\infty f$	254
N	21	$\lim_{x \rightarrow a} f(x)$	81	$\sin^\circ$	257
$\emptyset$	23	$\lim_{x \rightarrow a^+} f$	81	$\sin^r$	257
$n!$	23	$\lim_{x \downarrow a} f(x)$	86	π	258
$\sum_{i=1}^n a_i$	24	$\lim_{x \rightarrow a} f(x)$	86	$A(x)$	259
Z	25	$\lim_{x \uparrow a} f(x)$	87	$\cos$	259, 261
Q	25	$\lim_{x \rightarrow \infty} f(x)$	87	$\sin$	259, 261
R	25, 495	$\lim_{x \rightarrow -\infty} f(x)$	92	$\sec$	263
$\binom{n}{k}$	27	$\lim_{x \rightarrow a} f(x) = \infty$	92	$\tan$	263
$f(x)$	38, 45, 507	$\lim_{x \rightarrow \infty} f(x) = \infty$	92	$\csc$	263
I	41	$\lim_{x \rightarrow a} f(x)$	92	$\cot$	263
$f + g$	41	$\sup A$	112	$\arcsin$	263
$A \cap B$	41	$\text{lub } A$	112	$\arccos$	264
$f \cdot g$	41	$\inf A$	112	$\arctan$	265
f/g	41	$\text{glb } A$	112	$\ell(f, P)$	275
$c \cdot g$	41	$\lim A$	121	$\mathfrak{L}(x)$	276
$\{x: \dots\}$	41	$\limsup A$	121	$\log$	285
$\{a_1, \dots, a_n\}$	42	$\lim A$	121	$\exp$	287
$f + g + h$	42	$\overline{\lim} A$	121	e	287
$f \cdot g \cdot h$	42	$f'(a)$	127	e^x	288
$f \circ g$	42	f'	127	a^x	288
$f \circ g \circ h$	43	$\frac{df(x)}{dx}$	130	$\log_a$	289
$x \rightarrow f(x)$	43	$\left. \frac{df(x)}{dx} \right _{x=a}$	131	$\sinh$	296
$\prod_{i=1}^n a_i$	47	f''	137	$\cosh$	296
a^{b^c}	47	f'''	137	$\tanh$	296
C_A	48	$f^{(k)}$	137	$\arg \sinh$	296
$A \cup B$	48	$\frac{d^2 f(x)}{dx^2}$	138	$\arg \tanh$	296
$R - A$	48	f^{-1}	201	$\arg \cosh$	296
$ f $	49	$R(f, a, b)$	214	Nap log	299
$\max(f, g)$	49	$L(f, P)$	215	$F(x) \Big _a^b$	302
$\min(f, g)$	49	$U(f, P)$	215	$\int f$	303
$f < g$	51	$\int_a^b f$	219	$\int f(x) dx$	303
the pair (a, b)	52	$\int_a^b f(x) dx$	225	$\Gamma(x)$	327
the open interval (a, b)	54			$P_{n,a}$	334
$[a, b]$	55			$P_{n,a,f}$	334
(a, ∞)	55			$R_{n,a}$	343
$[a, \infty)$	55			$\binom{\alpha}{n}$	357
				$\{a_n\}$	372

$\lim_{n \rightarrow \infty} a_n$ 373 $\lim_{n \rightarrow \infty} a_n = \infty$ 376 γ 382 $\overline{\lim_{n \rightarrow \infty} x_n}$ 385 $\limsup_{n \rightarrow \infty} x_n$ 385 $\overline{\lim_{n \rightarrow \infty} x_n}$ 385 $\liminf_{n \rightarrow \infty} x_n$ 385 $N(n; a, b)$ 386 $\sum_{n=1}^{\infty} a_n$ 389 i 433, 439 $\mathbf{C}$ 438 $\bar{z}$ 441 $|z|$ 441 Re 448 Im 448 θ 449 $\lim_{z \rightarrow a} f(z)$ 449 $f'(a)$ 458 $\sin$ 470 $\cos$ 470 $\exp$ 470 b_n 478 B_n 479 D 479 D^k 479 e^D 479 Δ 479 φ_n 481 ψ_n 482 $+$ 487, 497 $\cdot$ 487, 501 $\mathbf{0}$ 487, 498 $\mathbf{1}$ 488, 502 $-a$ 488, 498 a^{-1} 488, 502 $\mathbf{P}$ 489 $>$ 490, 496 $<$ 490, 496 $\geq$ 490, 496 $\leq$ 490, 496 $|\alpha|$ 501

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GLOSSARY OF SYMBOLS

P 9	$(-\infty, a)$ 55	$L \int_a^b f$ 249
$ a $ 11	$(-\infty, a]$ 55	$U \int_a^b f$ 249
$\sqrt{x}$ 12	$(-\infty, \infty)$ 55	$\int_a^\infty f$ 254
$\max(x, y)$ 16	$[x]$ 70	$\int_a^\infty f(x) dx$ 254
$\min(x, y)$ 16	$\{x\}$ 70	$\int_{-\infty}^a f$ 254
ϵ ("epsilon") 19	δ ("delta") 78	$\int_{-\infty}^\infty f$ 254
N 21	$\lim_{x \rightarrow a} f(x)$ 81	$\sin^\circ$ 257
$\emptyset$ 23	$\lim_a f$ 81	$\sin^r$ 257
$n!$ 23	$\lim_{x \rightarrow a^+} f(x)$ 86	π 258
$\sum_{i=1}^n a_i$ 24	$\lim_{x \downarrow a} f(x)$ 86	$A(x)$ 259
Z 25	$\lim_{x \rightarrow a^-} f(x)$ 86	$\cos$ 259, 261
Q 25	$\lim_{x \uparrow a} f(x)$ 87	$\sin$ 259, 261
R 25, 495	$\lim_{x \rightarrow \infty} f(x)$ 87	$\sec$ 263
$\binom{n}{k}$ 27	$\lim_{x \rightarrow -\infty} f(x)$ 92	$\tan$ 263
$f(x)$ 38, 45, 507	$\lim_{x \rightarrow a} f(x) = \infty$ 92	$\csc$ 263
I 41	$\lim_{x \rightarrow \infty} f(x) = \infty$ 92	$\cot$ 263
$f + g$ 41	$\sup A$ 112	$\arcsin$ 263
$A \cap B$ 41	$\text{lub } A$ 112	$\arccos$ 264
$f \cdot g$ 41	$\inf A$ 112	$\arctan$ 265
f/g 41	$\text{glb } A$ 112	$\ell(f, P)$ 275
$c \cdot g$ 41	$\lim A$ 121	$\mathfrak{L}(x)$ 276
$\{x: \dots\}$ 41	$\limsup A$ 121	$\log$ 285
$\{a_1, \dots, a_n\}$ 42	$\lim A$ 121	$\exp$ 287
$f + g + h$ 42	$f'(a)$ 127	e 287
$f \cdot g \cdot h$ 42	f' 127	e^x 288
$f \circ g$ 42	$\frac{df(x)}{dx}$ 130	a^x 288
$f \circ g \circ h$ 43	$\left. \frac{df(x)}{dx} \right _{x=a}$ 131	$\log_a$ 289
$x \rightarrow f(x)$ 43	f'' 137	$\sinh$ 296
$\prod_{i=1}^n a_i$ 47	f''' 137	$\cosh$ 296
a^{b^c} 47	$f^{(k)}$ 137	$\tanh$ 296
C_A 48	$\frac{d^2 f(x)}{dx^2}$ 138	$\arg \sinh$ 296
$A \cup B$ 48	f^{-1} 201	$\arg \tanh$ 296
$R - A$ 48	$R(f, a, b)$ 214	$\arg \cosh$ 296
$ f $ 49	$L(f, P)$ 215	Nap log 299
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(a, ∞) 55		$R_{n,a}$ 343
$[a, \infty)$ 55		$\binom{\alpha}{n}$ 357
		$\{a_n\}$ 372

Glossary of Symbols

$\lim_{n \rightarrow \infty} a_n$ 373
 $\lim_{n \rightarrow \infty} a_n = \infty$ 376
 γ 382
 $\overline{\lim_{n \rightarrow \infty} x_n}$ 385
 $\lim_{n \rightarrow \infty} \sup x_n$ 385
 $\lim_{n \rightarrow \infty} x_n$ 385
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e^D 479
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